Introductory

ROTORDYNAMICS

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Preface

In the design of rotating machinery, it is more and more important to be familiar with and make intensive use of (numerical) models which can accurately predict the dynamic behavior of the system. In many cases, a linear model for the bending vibrations of the rotor bearing system can be used to predict the behavior of a certain design or to understand and improve the behavior of systems already in operation . The main part of these lecture notes are dealing with bending vibrations. However, the influence of torsion -which in general is somewhat easier to predict- will also be treated in one section. An essential, but practically more than acceptable, limitation is that only the **linear** dynamic behavior will be studied. Nonlinear analyses of rotor bearing systems can become important in extreme situations (damaged rotors, rubbing, compliant journal bearings etc.) but the analysis is rather complicated and will not be treated here - see for example [Vorst-96].

In many cases the equations of motion of the rotor system can be derived systematically by writing down the kinetic and potential energy of the system parts and looking at the virtual work of the nonconservative forces. Then, *Lagrange's equations* can be used to generate the set of equations

of motion in the selected generalized coordinates. Near that, in many cases some background knowledge of the fundamentals of rigid body dynamics will be necessary, therefore a short resume is given in Appendix A. In a few cases, the *Rayleigh-Ritz* method will be applied to generate an approximate numerical model for a rotor system. But, it will be shown that in general the *Finite Element Method* will be the obvious approach for an application in a design environment.

In Chapter 1 we start with the main questions which are relevant in almost any rotordynamic analysis. The underlying concepts and basic principles of rotordynamics will be introduced and illustrated by using very simple models for which solutions still can be generated by hand calculations. The general structure of the equations of motion will be outlined and important topics like forward- and backward whirl, instability, critical speeds, unbalance response and the Campbell plot will be introduced. Bearings are perhaps the most important elements in any rotating system. Therefore, in Chapter 2, the different bearing types and their dynamic characteristics will be elucidated. Due to their wide area of application, hydrodynamic bearings will be discussed rather detailed and in a more superficial way, attention will also be paid to roller bearings, hydrostatic bearings, gas bearings, electro magnetic bearings and squeeze film dampers.
Chapter 3 gives an introduction into the Finite Element Method. After resuming the general procedure, the typical elements used in rotordynamics (besides bearings) are derived: the rigid disk element and a Timoshenko beam element. Finally the element-assembling procedure will be described. With that, this chapter can also be seen as the theoretical foundation of the rotordynamic $\boldsymbol{M_{AT}L_{AB}}$ toolbox called $\boldsymbol{R_O D_Y}$.

Chapter 4 deals with several illustrative examples: a rotor, supported in two fluid-film bearings, illustrating the existence of an onset speed of instability, a 20 degree of freedom model for a large industrial fan for which the Campbell plot and 3-dimensional plots of the eigenmodes will be evaluated and finally a three-disk rotor whose free response and unbalance response will be discussed. In the last two examples, we are dealing with roller-bearing supports. The attention will not only be paid to the generation of response data in different forms, but especially also the interpretation of these results will be the subject of evaluation.

The important aspect of balancing rotorsystems in order to minimize vibrations and bearing forces or to allow for a safe operation near critical

speeds, will be covered in Chapter 5. The basic strategy of single-plane and two-plane balancing and the role of rotor flexibility will be considered.

Chapter 6 is a chapter devoted to several miscellaneous topics such as: some points of interest when applying a Rayleigh-Ritz approach as semi-analytical modeling tool, the destabilizing effect of internal (viscous) damping, non-rotationally symmetric rotors, approximation of the rotor response during the crossing of a critical speed and the analysis of rotors supported in compressible (for example gas) bearings.

Next, Chapter 7 gives a short introduction in the analysis of torsional problems. By switching to another, specific degree of freedom (the rotation around the central axis), the formulations for the rigid disk element and the beam element can be updated, and in addition to that, by introducing a special gear transmission element with tooth flexibility, a relative large variety of torsional problems can be analyzed systematically.

Finally, in Chapter 8 a mix of relatively simple and complicated problems is presented, followed by some solution. These solutions are not intended to be the one and only answer to the problem. They can help the reader in checking certain numerical steps and/or $\boldsymbol{R_O D_Y}$ outcomes but also can support the reader in developing the necessary creativity and modeling experience for solving complex rotordynamic problems.

It is my opinion that in getting familiar with rotordynamics, the study of theoretical concepts should go hand in hand with the analysis of (simple) practical examples and the interpretation of response phenomena. For this second part, the toolbox $\boldsymbol{R_O D_Y}$ can be a powerful and user-friendly option. In some examples, the necessary input to run this programme for a specific problem has been outlined. However, for more information on preparing inputs and running the programme see Appendix C.

Positioning

This introduction in the field of Rotordynamics is not intend to present the latest, state-of-the-art research results in this field, especially not the dedicated research on non-linear Rotordynamics. It also does not cover the existing professional numerical tools which are available on the market and in industry today.
The introduction is especially meant to become acquainted with the basic concepts in the field and to facilitate the step to the advanced literature or software.
However, in the majority of design problems where rotordynamic influences have to be tackled, linear models of limited complexity -as covered in this book- will satisfy.

Finally it is also my believe that the quality of predictions from a computer code greatly depends on the soundness of the basic model and the physical insight of the analyst. Superior algorithms or computer codes will never cure bad models or a lack of engineering judgment.

Bram de Kraker, Nuenen, March 2009

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Basic Principles of Rotordynamics

The descendant of 17th century French migrs, **Gustaf de Laval** was born in the Swedish province of Dalecarlia (Dalarna) in 1845. Even as a child he was considered to be unusually clever and innovative. A very successful student, he received a degree in engineering and continued on to become a Ph.D. At the time he patented the centrifugal separator in 1878, he was employed at the Kloster Works in Dalecarlia, an important industrial estate for mechanical engineering. Clearly de Laval possessed manifold technical talents, because he was able to produce a great number of inventions.
Apart from the centrifugal separator and the milking machine, the steam turbine is among the best-known. As a result of these exploits he acquired an impressive reputation as both an inventor and industrialist, and through these channels he was also able to influence many other contemporary inventors.
During his lifetime, Gustaf de Laval acquired 92 Swedish patents and founded 37 companies. His inventions have survived and evolved to create work for millions. He died February 2, 1913.

`http://www.delaval.com/About_DeLaval/TheCompany/History/default.htm`

1.1 Introduction

Rotating machinery as pumps, turbines, fans, etc. are more and more characterized by the need for high power transmission in smaller machines. Typical examples are aircraft engines, turbine compressors in chemical plants and large flywheels in a hybrid transmission of motorcars.
The most important factor in this high power-level is the increasingly high rotor-speed of the rotating system. The logical consequences of this high speed are a number of specific problems such as large inertial forces in the system, unexpected effects due to so-called whirl motions of the shaft, unacceptable large vibration levels and finally rotordynamic instabilities.
Very often the rotordynamic behavior of a new system is not investigated too detailed because one has the idea that it will not be important or the analysis is too complex. However, it regularly appears that after installing the machine it cannot meet the specifications formulated in its design phase or it does not work at all. The problem to assure that a rotating system under the relevant operating conditions will work reliable and with a low-vibration level is not an easy job.
In general, a rotordynamic analysis is characterized by two typical aspects, namely on the one hand the frequently good correspondence between predictive results from a numerical model (if at least correct parameter-values haven been used) and practical measurement data, and on the other hand the fact that sometimes the numerical prediction of the rotordynamic behavior seems not to correlate at all with our practical experience or intuition.
A simple illustration of the latter characteristic is the fact that an unbalanced, supercritical operating rotor will approach its centric position for an increasing rotorspeed, instead of showing some increasing motion, which might be our first impression.
Another remarkable, incorrect assumption is that energy-dissipation (damping) in a system will *always* stabilize the dynamic behavior. In case of so-called internal damping a rotor can become completely unstable leading to an unacceptable dynamic behavior.

Looking at the predictive power of sometimes rather simple numerical rotordynamic models and our obviously misleading intuition, we can only conclude that it is very important to be familiar with the basic principles of rotordynamics and with up-to-date numerical tools for the analysis of the dynamic behavior of realistic rotor-bearing systems as we meet in practice.

1.2 Intentions of a Rotordynamic Analysis

In designing new rotordynamic systems or during the running or trouble-shooting of existing ones as they show up in engineering practice, the following basic questions (individually or in any combination) can be relevant:

* **Prediction of critical speeds**. These are specific rotorspeeds at which the system shows (too) large response amplitudes, probably due to a small, but practically unavoidable, rotor-unbalance. Determining these critical speeds generally is seen as the first step leading to a safe operating condition for the system.
* **Searching for the most relevant design modification(s) to manipulate the existing critical speeds**. If the critical speeds cannot be avoided by choosing a proper working range for the system, the structure might have to be redesigned in order to shift them out of the operating range, or if that appears to be impossible, to reduce there negative effects.
* **Calculation of the eigenfrequencies of torsional vibrations**. If subsystems have to coöperate, for example an elektromotor, gear-box, coupling and compressor, unacceptable torsional vibrations can show up in the total power-transmission line due to a coincidental interference between eigenfrequencies of the coupled system and possible excitation frequencies.
* **Calculation of size and location of balancing weights from measurement data**. Once a rotorsystem has been installed and in operation, this procedure normally has to be applied and can be used to reduce the vibration level for all frequencies but especially for speeds close to the system-eigenfrequencies.
* **Calculation of the limit-speed of instability**. Nowadays, this probably is the most difficult question in rotordynamics, because the physical models for some types of destabilizing forces in real mechanical systems are still too inaccurate. An example is the complex interaction between a rotating impeller and the medium in modern fluid-pumps. However, instabilities due to journal bearings and seals can be analyzed rather accurate at this moment.
* **Investigation of possible design-modifications to suppress these dynamic instabilities**. Numerical simulations with rotor-dynamic

models can give very worthwhile information on the *relative* stabilizing effect of possible design modifications even in the case where the models for the destabilizing forces are not yet understood so very well.

In general, the mathematical models which are used to analyse realistic rotordynamic problems are complex, for example they have many degrees of freedom and nonsymmetric system matrices. This makes a rotordynamic analysis somewhat more complicated than a regular linear dynamic analysis as presented in [Kraker/Campen-01]. Consequently, satisfying the objectives as formulated above will only be possible with the use of advanced numerical tools, dedicated to this field of application.
However, many of the phenomena and new concepts can also be illustrated to some extend by analyzing simple models which still can be solved *by hand calculations*. In the remaining part of this chapter, simple one- or two-degree-of-freedom models will be formulated and analyzed for a first introduction of the most important rotordynamic phenomena.

1.3 Basic Theory

The most simple dynamic model for a rotor-bearing system consists of a mass-spring system with only one degree of freedom (dof). (see Fig. 1.1). The lowest critical speed of a rotor-bearing system can be approximated by

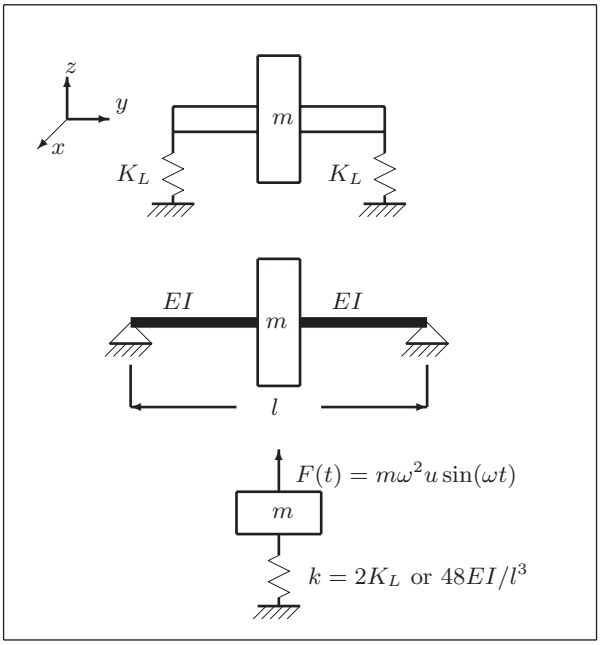


Fig. 1.1. Simple rigid and flexible rotors

the eigenfrequency of such a system, namely

$$N_1 = \frac{60}{2\pi}\sqrt{\frac{k}{m}} \qquad [rev/min] \tag{1.1}$$

where k $[N/m]$ is the *effective spring-stiffness* of the first vibration mode and m $[kg]$ is the *effective mass*. For rotors which may assumed to be rigid with respect to the supporting bearing stiffness, for the effective mass the total rotor-mass can be taken and for the effective stiffness only the bearing stiffness. (Fig. 1.1, top).

For rotors with a flexible shaft (Fig. 1.1, middle), the effective stiffness is

dominated by the bending stiffness of the shaft. In that case only a part of the total rotor-mass has to be taken as the effective mass due to the smaller displacements in the neighborhood of the bearings.
Such simple rotor models of course have some serious limitations. First of all they will only allow for displacements in one direction (in Fig. 1.1 the z-direction) whereas a rotor-system in general can perform a 2-dimensional, so-called **whirl-motion** in the bearings with a complicated behavior. This disadvantage can be removed by extending the system to 2 dof's, namely for displacements in x- and z-direction, and the introduction of **cross-coupling terms** (arising from gyroscopy, journal bearings, seals, etc.) in the differential equations. Although such a 2-dof system in principle can

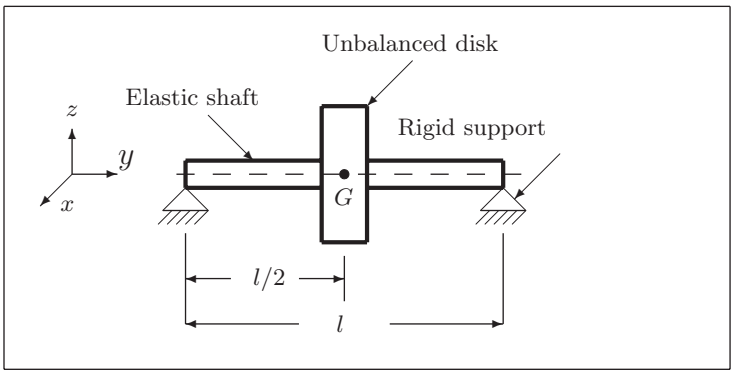


Fig. 1.2. The **Laval-Jeffcott** rotor

carry out a whirl motion of a rotor-bearing system it still does not give a realistic picture because especially a rotating unbalance is missing. Perfectly balanced rotors do not exist in reality and it is particularly this unbalance which is, in most cases, the dominant source of excitation of the system, leading to so-called synchronous vibrations.

Such a single disc rotor model was first presented by A. Föppl in Germany in 1895. Föppl named this elementary model the Laval rotor in recognition of the contributions made in the area of turbomachinery by the Swedish engineer Gustav Patric De Laval. In 1919 the English engineer H.H. Jeffcott subsequently presented a study using the same elementary model. As the result of references

made to these two pioneering works over the past several decades, this single-disc rotor model is referred to as the Laval rotor in some sections of the world and as the Jeffcott rotor in other. In order to recognize the early work of both Föppl and Jeffcott, the name **Laval-Jeffcott rotor** *will be adopted here*

The model consists of a rigid, unbalanced disc, symmetrically mounted on a flexible, massless shaft between two rigid bearings. We assume some viscous damping in the system related to the absolute velocity of the disc. Due to the rotating unbalance forces the geometrical center of the disc (G) can trace an orbit in the x-z plane, the so-called whirl-orbit. This whirling motion has the same angular frequency (magnitude as well as direction) as the rotorfrequency and therefore is called a **synchronous motion**. Inversely, if the system shows a large synchronous dynamic vibration, it very often will be the result of a rotor-unbalance. Not all rotor-vibrations are synchronous vibrations, especially the more destructive rotordynamic problems are associated with **nonsynchronous vibrations**. Fig. 1.3 illustrates the

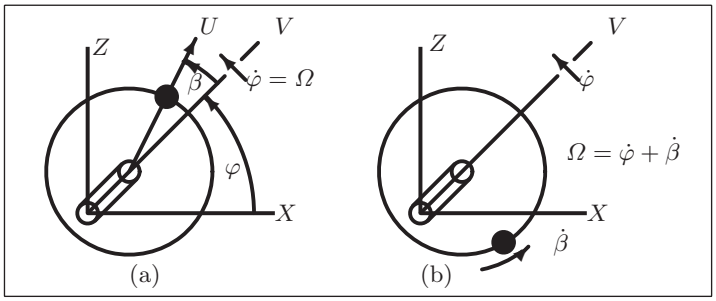


Fig. 1.3. Difference between synchronous whirl (a) and nonsynchronous whirl (b)

difference between a synchronous and a nonsynchronous motion. The black dot represents the unbalance. In Fig. 1.3a, $\dot{\varphi}$ is the whirl speed. The angle β remains constant, so that the whirl-speed $\dot{\varphi}$ and the rotor-speed Ω are equal. In Fig. 1.3b, $\dot{\beta}$ is the angular velocity of the rotor, *relative to the rotating whirl-axis* V, so, the rotor-speed Ω is the sum of $\dot{\beta}$ and $\dot{\varphi}$. In this case the whirl-speed and the rotor-speed are different leading to a nonsynchronous motion. To distinguish between these two types of motion is very

important for the correct interpretation of measured frequency-spectra of machines.

1.4 Analysis of a Laval-Jeffcott Rotor

Fig. 1.4 gives a side-view of a whirling Laval-Jeffcott rotor with coordinates to define the motions. The mass-center of the unbalanced disc is called M. The point G is the geometric center of the disc. The unbalance is therefore given by the distance e between G and M and the bending of the shaft is indicated by r. Gravity effects are ignored.
The bending stiffness of the shaft is called k $[N/m]$ and the disc-mass m $[kg]$, whereas we also assume that all the energy dissipation in the system can be modeled by a viscous damping term with damping constant b $[Ns/m]$. In principle the system has 3 dof's (2 translations and 1 rotation). We assume a **constant rotorspeed** Ω, so only the 2 translations remain. The use of polar coordinates $(r,\ \varphi,\ \beta)$ has some advantages for

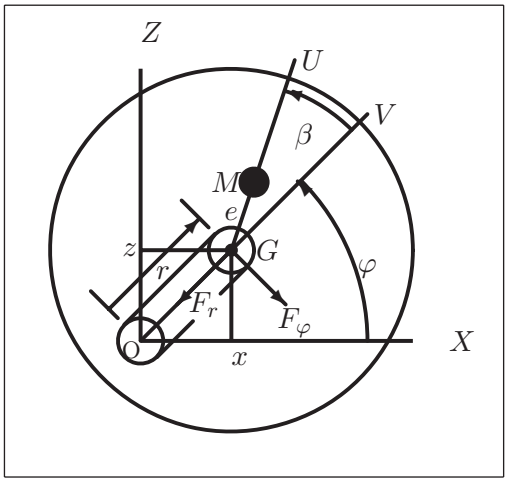


Fig. 1.4. Side-view of a Laval-Jeffcott rotor, synchronous motion

the description of the synchronous whirl, but the corresponding differential equations will become nonlinear and therefore less appropriate for the analysis of rotordynamic instabilities. Cartesian coördinates $(x,\ y,\ \beta)$ however produce linear equations of motion and also correspond better with the quantities which normally are measured directly in practice. Here we will discuss both sets of equations of motion and their solutions for a constant rotorspeed Ω.

Polar coordinates

We consider the rotor given in Fig. 1.4 with mass m, center of mass M and geometrical center G. For the derivation of the differential equations we will use the method of **Lagrange**. Therefore we need the absolute velocity of the center of mass of the rigid disc and the rotational velocity around this center of mass. For a synchronous motion we have $\Omega\ t = \varphi + \beta$, where Ω is the rotorspeed and β=arbitrary, but constant. Using polar coordinates (r, φ), the position $\underline{x}_M$ of the center of mass in the fixed {X,Y,Z} coordinate system can be written as

$$\underline{x}_M = \begin{bmatrix} x_M \\ y_M \end{bmatrix} = \begin{bmatrix} r\cos(\varphi) + e\cos(\Omega t) \\ r\sin(\varphi) + e\sin(\Omega t) \end{bmatrix} \tag{1.2}$$

For the velocity $\underline{v}_M$, we have to differentiate (1.2) to time leading to

$$\underline{v}_M = \begin{bmatrix} -r\dot{\varphi}\sin(\varphi) + \dot{r}\cos(\varphi) - e\Omega\sin(\Omega t) \\ r\dot{\varphi}\cos(\varphi) + \dot{r}\sin(\varphi) + e\Omega\cos(\Omega t) \end{bmatrix} \tag{1.3}$$

Now the kinetic energy T for the rigid disc can be written as

$$T = \frac{1}{2}m\ \underline{v}_M^T \underline{v}_M + \frac{1}{2}J\ \Omega^2 \tag{1.4}$$

where J is the mass-moment of inertia of the disc with respect of center of mass. This gives

$$\begin{aligned} T = \tfrac{1}{2}m[\dot{r}^2 + r^2\dot{\varphi}^2 + e^2\Omega^2 - \dot{r}e\Omega\sin(\Omega t - \varphi) \\ + re\dot{\varphi}\Omega\cos(\Omega t - \varphi)] + \tfrac{1}{2}J\Omega^2 \end{aligned} \tag{1.5}$$

We assume a constant rotorspeed Ω, so only 2 generalized cooördinates remain, namely $\underline{q}^T = [r, \varphi]$. For the elastic energy due to bending of the shaft we find

$$U = \frac{1}{2}k\ r^2 \tag{1.6}$$

Further we have a nonconservative force resulting from the assumed viscous damping term, acting in the geometric center of the disc and proportional with its velocity, so leading to a radial force $F_r = b\ \dot{r}$ and a tangential force $F_\varphi = b\ r\ \dot{\varphi}$ (see Fig. 1.4). Using the virtual displacements $\delta\underline{q}^T = [\delta r\ ,\ \delta\varphi]$ the virtual work resulting from these forces can be written as

$$\delta A = -b\dot{r}(\delta r) - br\dot{\varphi}(r\delta\varphi) = \underline{Q}^{*T}\delta\underline{q} \tag{1.7}$$

where $\underline{Q}^{*T} = [-b\dot{r}\ ,\ \ -br^2\dot{\varphi}]$ are the so-called generalized forces. Using Lagrange's equations

$$\frac{d}{dt}\left(T_{,\dot{\underline{q}}}\right) - T_{,\underline{q}} + U_{,\underline{q}} = \underline{Q}^* \tag{1.8}$$

then leads to the (nonlinear) differential equations in polar coordinates

$$\begin{aligned} m\ddot{r} + b\dot{r} + (k - m\dot{\varphi}^2)r &= m\Omega^2 e\cos(\Omega t - \varphi) \\ mr^2\ddot{\varphi} + (2mr\dot{r} + br^2)\dot{\varphi} &= m\Omega^2 re\sin(\Omega t - \varphi) \end{aligned} \tag{1.9}$$

For getting a **synchronous motion** we assume $r = r_s =$ constant and $\varphi = \varphi_s = \Omega t - \beta_s,$ with $\beta_s =$ constant, which leads us to

$$(k - m\Omega^2)r_s = m\Omega^2 e\cos(\beta_s) \qquad \text{and} \qquad br_s\Omega = m\Omega^2 e\sin(\beta_s) \tag{1.10}$$

If we on the one hand take the sum of the square of both relations and on the other hand divide the second one by the first one, we get

$$\begin{aligned} (k - m\Omega^2)^2 r_s^2 + b^2 r_s^2\Omega^2 &= m^2\Omega^4 e^2 \\ \tan(\beta_s) &= \frac{b\Omega}{(k - m\Omega^2)} \end{aligned} \tag{1.11}$$

This directly gives the solution for the synchronous motion

$$\begin{aligned} r_s &= \frac{\Omega^2 e}{\sqrt{(k/m - \Omega^2)^2 + (b\Omega/m)^2}}, \\ \Omega t - \varphi_s = \beta_s &= \tan^{-1}\left[\frac{b\Omega}{m(k/m - \Omega^2)}\right] \end{aligned} \tag{1.12}$$

So we are dealing with a circular whirl orbit in the x-z plane with amplitude r_s and fase-angle β_s with respect to the V-axis (rotating with rotorspeed Ω) (see Fig. 1.4). Typical plots of r_s and β_s are given in Fig. 1.5 respectively Fig. 1.6 for 2 different values for the dimensionless damping factor $\xi = b/(2\sqrt{km})$.

Cartesian coordinates

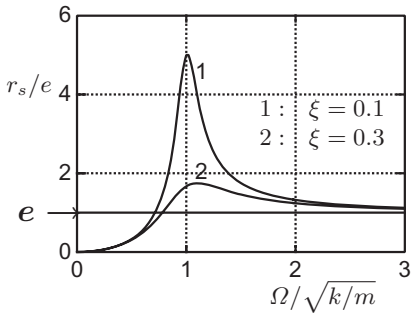


Fig. 1.5. $|r_s|$ versus $(\Omega/\sqrt{k/m})$

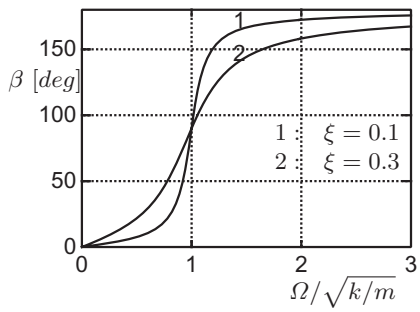


Fig. 1.6. β_s versus $(\Omega/\sqrt{k/m})$

The equations of motion in the Cartesian coordinates can be derived in the same way by using Lagrange's equations or directly from (1.9) by using the coordinate transformation

$$\begin{bmatrix} r \\ \varphi \end{bmatrix} = \begin{bmatrix} \sqrt{x^2 + z^2} \\ \arctan(z/x) \end{bmatrix} \tag{1.13}$$

These **linear** equations of motion now become

$$\begin{aligned} m\ddot{x} + b\dot{x} + kx &= m\Omega^2 e \cos(\Omega t) \\ m\ddot{z} + b\dot{z} + kz &= m\Omega^2 e \sin(\Omega t) \end{aligned} \tag{1.14}$$

The solution for a synchronous whirl will be

$$\begin{aligned} x_s &= \frac{\Omega^2 e}{\sqrt{(k/m - \Omega^2)^2 + (b\Omega/m)^2}} \cos(\Omega t - \beta_s) \\ z_s &= \frac{\Omega^2 e}{\sqrt{(k/m - \Omega^2)^2 + (b\Omega/m)^2}} \sin(\Omega t - \beta_s) \\ \beta_s &= \tan^{-1}\left[\frac{b\Omega}{m(k/m - \Omega^2)}\right] \end{aligned} \tag{1.15}$$

Finally it can easy be shown that

$$r_s = \sqrt{x_s^2 + z_s^2} \tag{1.16}$$

1.4.1 Physical interpretation of the response

Fig. 1.5 shows that the amplitude of the synchronous whirl grows with an increasing rotorspeed until a maximum is reached at the so-called *critical rotorspeed*. For supercritical speeds the whirl amplitude decreases again and finally, it will settle down on a stationary value which is equal to the unbalance (e). This implies that for high rotorspeeds the synchronous whirl-amplitude can be made arbitrary small by reducing this unbalance, so by balancing a rotor. For speeds around the critical speed the most important parameter for whirl-amplitude reduction is the damping in the system. From Fig. 1.5, we can also extract an obvious definition of what generally is called a **critical speed**, namely *the rotorspeed where the synchronous response due to an unbalance has a (clear) maximum*. Damping does not influence this critical speed significantly.
The investigation of Fig. 1.4 together with the solution for the fase-angle β in Fig. 1.6 can explain the asymptotic behavior of the whirl-amplitude for high rotorspeeds. For small rotor-speeds the angle β starts at zero (mass center M in the direction of OZ), passes 90 degrees at the critical speed en finally goes to 180 degrees for high, supercritical speeds. Thus for high rotorspeeds the mass center M falls inside the orbit of the geometrical center G en converges to zero for very high rotorspeeds. It looks like the geometrical center G is rotating around the mass center M. This phenomenon is called **critical speed inversion**. From the analysis of the Laval-Jeffcott rotor, 3 possible ways to reduce a synchronous whirl amplitude can be derived, namely

- balancing the rotor,
- changing the operating range (sufficiently far away from a critical speed),
- adding damping to the rotor-bearing system.

Although the Laval-Jeffcott rotor is only a very simple model of reality, these actions to reduce synchronous vibrations effects are very often also effective in much more complicated situations. When critical speeds have to be passed very slowly (for fast passages special formulations have been developed, see o.a. Childs [3]), of when the machine has to operate close to a critical speed, the most effective rule is to add damping to the system. Fortunately, many rotating machines have flexible bearing supports where additional damping can be introduced or where fluid film bearings (sometimes specially designed with respect to damping characteristics) can be

applied. It is very important to note that so-called internal damping (for example hysteresis in a shaft or friction in a shrink-fit of a disc on the shaft) is a completely different type of damping. It can be shown that this type of damping can have a negative influence and that it can lead to self-excited, nonsynchronous vibrations and even can make the rotor-bearing system unstable (see Section 6.2).

1.4.2 A Flexible Bearing Support

The supports of a rotor-shaft of real machines will always have some flexibility. In practice it is sometimes even advisable that the support is more flexible than the shaft for two main reasons

- A low support-stiffness can reduce the bearing forces considerably and consequently extend the life-time of the bearing
- Some support-stiffness makes relative displacements possible so that a dampingmechanism can be more effective (no relative velocity means no damping)

To illustrate these effects we look at a system of a rigid rotor on flexible supports as indicated in Fig. 1.7. In principle the differential-equations and

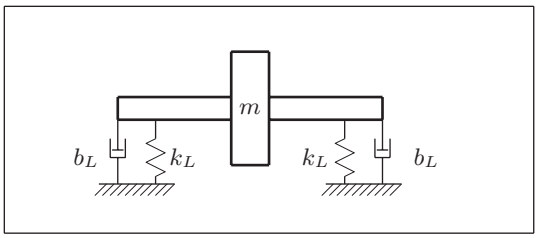


Fig. 1.7. Rigid rotor on flexible supports

therefore also the solutions are the same as for the Laval-Jeffcottt rotor discussed before. In this case only somewhat different definitions for stiffness and damping are used. The damping might be due to the application of journal bearings, so-called squeeze-film dampers or other specially designed damping elements.

The force which in case of a synchronous whirl has to be transmitted by the bearings is the vector-sum of the radial spring-force $F_k = k_L r_s$ and the

tangential damper-force $F_c = b_L \Omega r_s$.
The amplitude F_L of this bearingforce becomes

$$F_L = \sqrt{F_k^2 + F_c^2} = r_s \sqrt{k_L^2 + (b_L \Omega)^2} \tag{1.17}$$

with r_s given by (1.12).
For a rigid support the bearingforce would be given by $F_\infty = m\Omega^2 e/2$. For high rotorspeeds this force will grow unlimited which is very undesirable. By introduction of a well chosen bearing flexibility this bearingforce can be reduced considerable in comparison with a rigid support.
To illustrate this the normalized bearingforce (ratio of F_L and F_∞) as a function of the nondimensional rotorspeed is given in Fig. 1.8 for 2 different values for the nondimensional damping ξ.

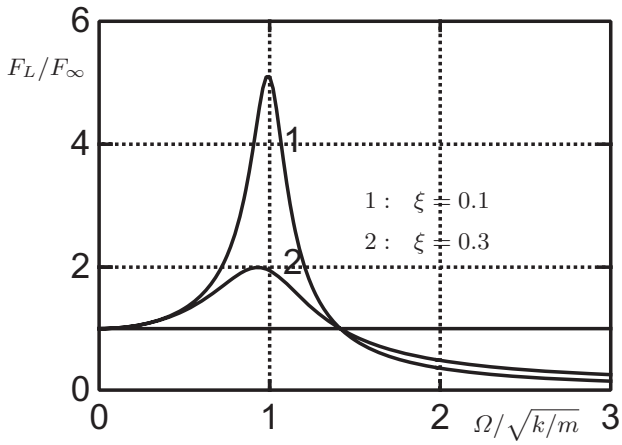


Fig. 1.8. Normalized bearingforce versus nondimensional rotorspeed for 2 damping-levels

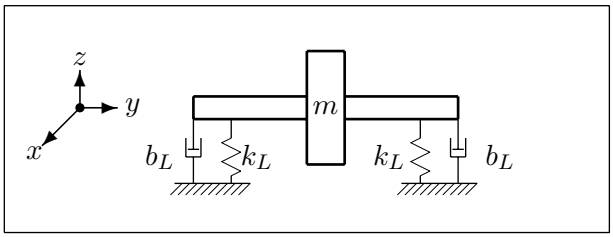
Some important conclusions can be drawn:

- Bearing-flexibility can reduce the bearingforce considerably if the support is designed such that the system can run super-critically.
- Damping in the support is only slightly negative for high rotorspeeds.
- For passing critical speeds damping in the supports is essential for limiting the bearing forces.
- Bearing-flexibility is not always positive. In a bad design, bearing forces can appear which are larger than in the case of a rigid support.

Note

Goals: Simple $\boldsymbol{M_{AT}L_{AB}}$ *-use; dealing with complex numbers.*

Additionally, we look at a system of a rigid rotor on non-symmetric flexible supports as indicated in the next figure.



We assume that the bearing stiffness k_L and -damping b_L values are different for $x-$ and $z-$ direction, but are the same for left- and right bearing. If we call these values: $k_x,\ k_z,\ b_x,\ b_z$, we can get the equations of motion directly from (1.14)

$$m\ddot{x} + 2b_x\dot{x} + 2k_x x = m\Omega^2 e\cos(\Omega t)$$
$$m\ddot{z} + 2b_z\dot{z} + 2k_z z = m\Omega^2 e\sin(\Omega t)$$

or:

$$\underline{M}\ \underline{\ddot{y}} + \underline{B}\ \underline{\dot{y}} + \underline{K}\ \underline{y} = \underline{f}(t) \tag{1.18}$$

with:

$$\underline{M} = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \quad \underline{B} = \begin{bmatrix} 2b_x & 0 \\ 0 & 2b_z \end{bmatrix} \quad \underline{K} = \begin{bmatrix} 2k_x & 0 \\ 0 & 2k_z \end{bmatrix}$$

$$\underline{y}(t) = \begin{bmatrix} x(t) \\ z(t) \end{bmatrix} \quad \underline{f}(t) = m\Omega^2 e \begin{bmatrix} \cos(\Omega t) \\ \sin(\Omega t) \end{bmatrix}$$

To do some numerical evaluation of this system we assume:
$m = 1;\quad k_x = 50;\quad k_z = 100;\quad b_x = b_z = 2;\quad e = 0.01$
This gives the undamped eigenfrequencies $\omega_x = 10.0\ \ [rad/s]$ and $\omega_z = 14.4\ [rad/s]$ and additionally, the dimensionale damping factors $\xi_x = 0.14$ and $\xi_z = 0.1$.
If we write the excitation column $f(t)$ as:

$$\underline{f}(t) = Re[\underline{\hat{f}}\ e^{j\Omega t}]$$

we should write for the complex amplitude of the excitation $\underline{\hat{f}}(\Omega)$:

$$\underline{\hat{f}}(\Omega) = m\Omega^2 e \begin{bmatrix} 1 \\ -j \end{bmatrix}$$

Then we also substitute:

$$\underline{y}(t) = Re[\underline{\hat{y}}\ e^{j\Omega t}]$$

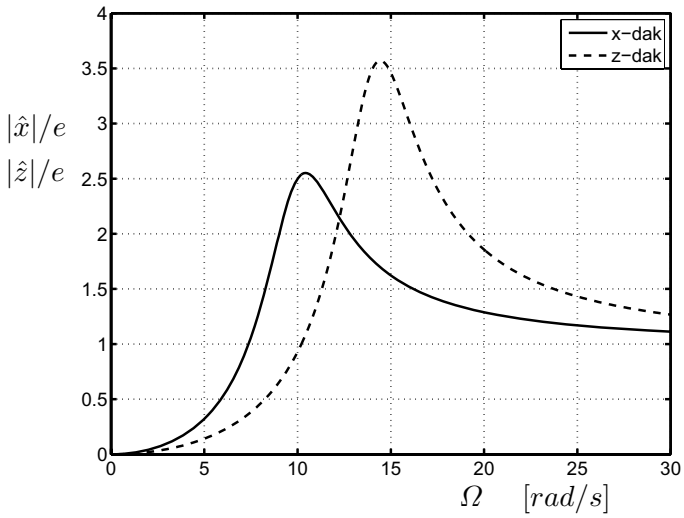
which gives the complex amplitude of the response:

$$\underline{\hat{y}}(\Omega) = \begin{bmatrix} \hat{x}(\Omega) \\ \hat{z}(\Omega) \end{bmatrix} = [-\Omega^2 \underline{M} + j\Omega \underline{B} + \underline{K}]^{-1} \underline{\hat{f}}$$

So we find the complex amplitudes of the response $\hat{x}(\Omega)$ and $\hat{z}(\Omega)$. The corresponding time-functions can then be found from:

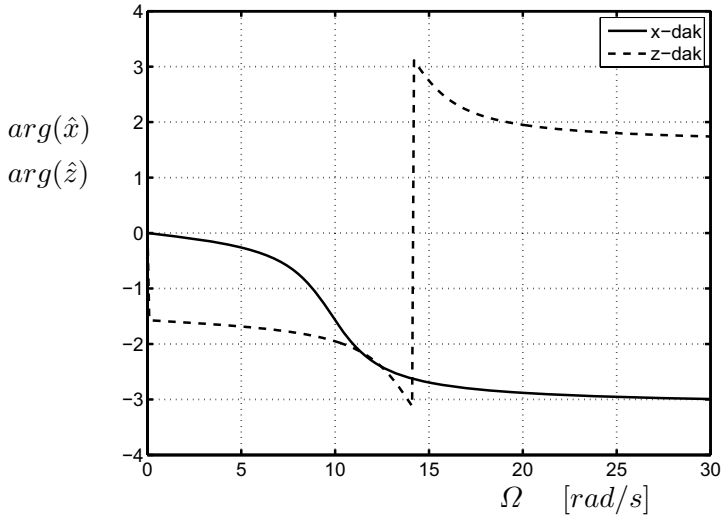
$$x(t) = Re[\hat{x}\ e^{j\Omega t}]; \qquad z(t) = Re[\hat{z}\ e^{j\Omega t}]$$

The magnitudes of the complex amplitudes $\hat{x}(\Omega)$ and $\hat{z}(\Omega)$ are shown as function of the rotorspeed in the next figure.



(scaled) Magnitude of $\hat{x}(\Omega)$ and $\hat{z}(\Omega)$ as function of Ω

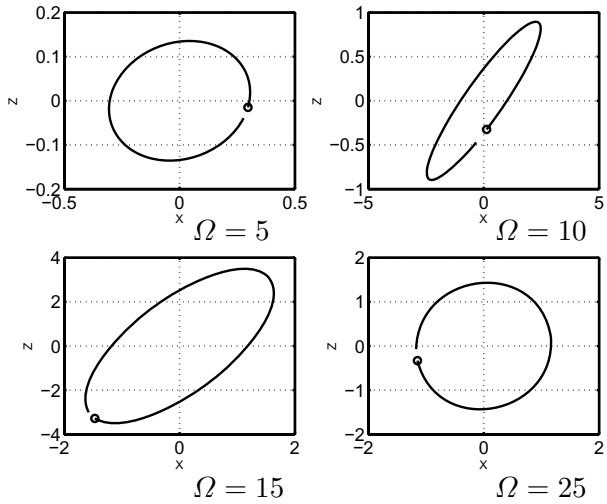
We can see that we have two different eigenfrequencies, one at $\approx 10 \quad [rad/s]$, the x-mode, and a second one at $\approx 14.4 \quad [rad/s]$, the z-mode. The system indeed looks to be weakly damped. For $\Omega \Rightarrow 0$, the amplitudes also go to zero and for $\Omega \Rightarrow \infty$, both amplitudes go to the eccentricity e.
In the next figure, the phase angles of these complex amplitudes are shown.



Phase-angles of $\hat{x}(\Omega)$ and $\hat{z}(\Omega)$ as function of Ω

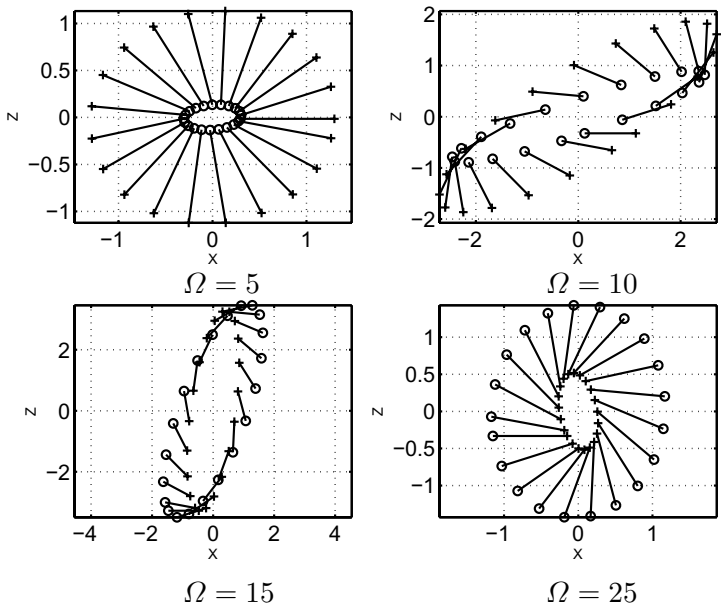
Phase-angles of complex amplitudes very often are difficult to understand. We see that for small Ω-values, the x-coördinate has almost a zero phase-angle and the z-coördinate a phase-angle of $-\pi/2$.
For very high Ω-values, the x-coördinate is going asymptotically to a phase-angle of $-\pi$, and the z-coördinate is going asymptotically to a phase-angle of $+\pi/2$, whereas the magnitudes go to the eccentricity e. This means that for large Ω the shaft-center point is moving to a position, opposite of the eccentricity center point, and that the latter is moving to the origin. This means that the shaft-center point will finally rotate in a circle around the origin with radius e. This was already explained in the analytical treatment of the symmetric case.
In order to understand this complex dynamic behavior better, we might also look at the real temporal behavior. The displacements $x(t)$ and $z(t)$ have been calculated for 4 specific rotorspeeds. The results are shown next. In the plots, the point for $t = 0$ is marked

by a small circle and the orbit is calculated for almost 1 period. This allows for understanding the direction of rotation of the shaft-center point.



(scaled) Orbits of the shaft-center point for different rotorspeeds.

For $\Omega = 5$ we see a small circular motion which is in phase with the rotation of the eccentricity center (which is a circular, counterclockwise rotation, starting from the x-axis).
Through $\Omega = 10$ and $\Omega = 15$ the amplitudes are getting large (eigenfrequencies) and the shaft-center point is moving more and more out of phase with the eccentricity center point. Finally, for $\Omega = 25$, the orbit of the shaft-center point is almost circular, around the almost (fixed) eccentricity center point.
Another way of looking at dynamic motion is looking at the evolution of the shaftcenter-excentricitycenter connection line. This is shown in the next figure. The shaft-center points are indicated with (o) and the eccentricity-center points with a (+). This also clearly shows the difference between subcritical and supercritical (self-balancing) motion of this system.



Motion of the disc in the x-z plane;
(o) is shaft-center, (+) is the eccentricity point

Question: What happens with the orbit ar $\Omega \approx 12$ [rad/s] ?

End of note

1.4.3 Rotordynamic Instability

Many rotordynamic problems are related to synchronous whirl motions, for example due to unbalance forces. A second important group of problems is related to the occurrence of *unstable, nonsynchronous* whirl motions which typically show up when a specific limit-value for the rotorspeed (the threshold speed of instability) has been reached. To illustrate the background of this instability we look at a simple rigid, symmetrical rotor supported by 2

fluid-film bearings as given in Fig. 1.7.
However, we now assume that the bearing-model leads to so-called **cross coupling terms**, which means that a displacement in x-direction will not only result in a force in x-direction (through the direct stiffness component k), but also in a force in z-direction and vice-versa. The associated stiffness-cross-terms are indicated by k_{xz} respectively k_{zx}, (See also Chapter 2).
Further we assume the presence of direct damping terms b.
The differential equations for this situation can be written as

$$\begin{aligned} m\ddot{x} + b\dot{x} + kx + k_{xz}z &= 0 \\ m\ddot{z} + b\dot{z} + kz + k_{zx}x &= 0 \end{aligned} \tag{1.19}$$

We do not have any additional external excitation. The solutions of this set of linear, homogeneous differential equations can now be written as

$$x = A_1\ e^{st} \quad \text{and} \quad z = A_2\ e^{st} \tag{1.20}$$

where s is the eigenvalue (see further) and A_1, A_2 can be determined from the initial conditions of the system. Using (1.20), (1.19) can be written in matrix-form

$$\begin{bmatrix} (ms^2 + bs + k) & k_{xz} \\ k_{zx} & (ms^2 + bs + k) \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{1.21}$$

Because we are dealing with a homogeneous set of equations, a nontrivial solution (A_1 en A_2 both nonzero) can only exist if the determinant of the matrix in (1.21) will be made zero.
This leads to the so-called **characteristic equation**

$$(ms^2 + bs + k)^2 - k_{xz}k_{zx} = 0 \tag{1.22}$$

The eigenvalues for the rotor-system are the roots of this polynomial function in s. In principle they will be complex numbers and they can be written as

$$s = \nu + j\omega_d \tag{1.23}$$

where ν (the real part of the eigenvalue) is a measure for the damping and ω_d is called the damped-eigenfrequency. The general solution (1.20) can then be written as

$$x(t) = A_1 e^{\nu t}(\cos(\omega_d t) + j\sin(\omega_d t))$$
$$z(t) = A_2 e^{\nu t}(\cos(\omega_d t) + j\sin(\omega_d t)) \tag{1.24}$$

For $\nu > 0$, this motion grows exponentially with time and is therefore said to be **unstable**. The sign of ν is thus the essential element in the stability of the solution. For simplicity, we continue with $k_{xz} = -k_{zx} = \kappa$ (which in some cases is also in practice the case), which leads to

$$(ms^2 + bs + k)^2 + \kappa^2 = 0 \tag{1.25}$$

so that

$$(ms^2 + bs + k) = \pm j\kappa \Longrightarrow ms^2 + bs + (k \pm j\kappa) = 0 \tag{1.26}$$

This equation has two solutions, namely

$$s_{1,2} = \frac{-b}{2m} \pm \sqrt{\{\frac{b^2}{4m^2} - \frac{k}{m}\} \pm j\frac{\kappa}{m}} \tag{1.27}$$

We want to write these roots $s_{1,2}$ as $s_i = s_R + js_I$ and to find an expression for the real part s_R respectively the imaginary part s_I.

Note

In (1.27) we are dealing with the square root of a complex number, so for a given complex number $a + cj$ we search for the complex number $e + fj$ for which

$$\sqrt{a + cj} = e + fj, \text{ or} \tag{1.28}$$
$$e^2 - f^2 + 2efj = a + cj, \text{ so} \tag{1.29}$$
$$e^2 - f^2 = a, \text{ and} \tag{1.30}$$
$$2ef = c \tag{1.31}$$

Substitution of $f = c/(2e)$ in (1.30) gives

$$e = \pm\sqrt{\frac{1}{2}(a \pm \sqrt{a^2 + c^2})} \tag{1.32}$$

We only want real values for e, so:

$$e = \pm\sqrt{\frac{1}{2}(a + \sqrt{a^2 + c^2})} \tag{1.33}$$

And by substitution of $e = c/(2f)$ in (1.30) we get:

$$f = \pm\sqrt{\frac{1}{2}(-a \pm \sqrt{a^2 + c^2})} \tag{1.34}$$

and only looking for real f:

$$f = \pm\sqrt{\frac{1}{2}(-a + \sqrt{a^2 + c^2})} \tag{1.35}$$

So we find the solutions $(e + fj)$ and $(-e - fj)$.

End of note

Going back to (1.27) and using

$$a = \frac{b^2}{4m^2} - \frac{k}{m} \qquad \text{and} \qquad c = \pm\frac{\kappa}{m} \tag{1.36}$$

gives the square roots $s_{1,2}$

$$s_{1,2} = \nu_{1,2} \pm j(\omega_d)_{1,2} \tag{1.37}$$

This leads to:

$$\begin{aligned} \nu_{1,2} &= -\frac{b}{2m} \pm \sqrt{\left(\frac{b}{2m}\right)^2 + (\omega_d^2 - \frac{k}{m})} \\ (\omega_d^2)_{1,2} &= \frac{k}{2m} - \frac{b^2 - \sqrt{(b^2 - 4km)^2 + 16\kappa^2 m^2}}{8m^2} \end{aligned} \tag{1.38}$$

We introduce the nondimensional damping ξ

$$\xi = \frac{b}{2\sqrt{km}} \tag{1.39}$$

Notice that the sign of κ does not influence these values.
Fig. 1.9 and Fig. 1.10 give a picture of the whirl frequency ω_d and the (damping) coefficient ν as function of the cross-coupling term κ for 3 different values for the nondimensional damping ξ.

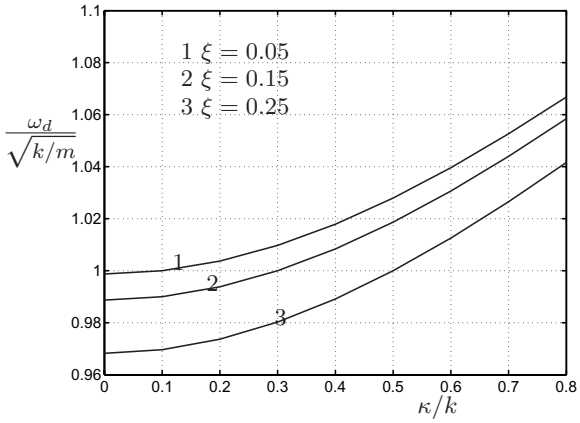


Fig. 1.9. $(\omega_d/\sqrt{k/m})$ versus κ

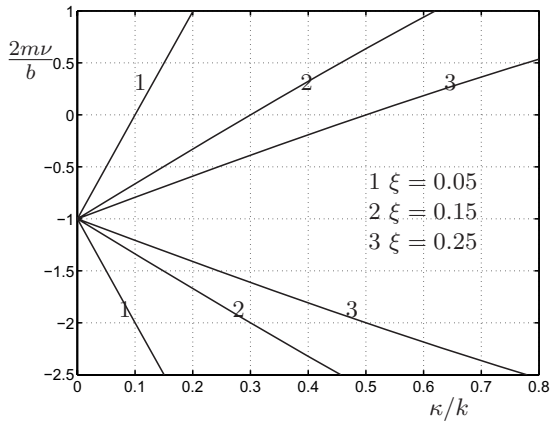


Fig. 1.10. $(2m\nu)/b$ versus κ

Fig. 1.10 shows that the system has one eigenvalue with a negative real part (becoming more negative for larger κ), giving a **stable** motion. There is however also an eigenvalue with a real part which will become positive for a certain κ-value. Then the system will become **unstable**. Fig. 1.9 shows that the whirl-frequency ω_d corresponding to an unstable motion is normally somewhat larger than the critical speed as introduced before. Fig. 1.10 shows that a rotor-bearing system with for example 5 % damping (a very regular value), will become already unstable when the cross-coupling term κ reaches a value of only 10% of the direct stiffness term k. Adding

damping to the system will have a positive effect on the stability (can sometimes make an unstable rotor stable again). Finally it should be remarked that these cross-coupling terms normally are a function of the rotorspeed which makes the analysis rather complicated. The use of some computer programme will then be necessary in general.

Once the set of differential-equations has been solved, so a whirl motion has been found (solutions for $x(t)$ en $z(t)$), the orbit of the shaft-center can be evaluated. Now 2 situations are possible, namely

* The whirl-rotation has the same direction as the rotor-speed (the magnitude does not has to be the same). This situation is called a **forward whirl**.
* The whirl-rotation is in the opposite direction of the rotorspeed, This is called a **backward whirl**. For example in case of a Laval-Jeffcott rotor where the disc is not symmetrically mounted between the 2 supports, gyroscopic terms will show up. Then 2 possible whirling motions will be possible, namely a forward whirl with a whirling frequency which grows with increasing rotorspeed and a backward whirl with a whirling frequency which decreases with increasing rotorspeed. (see Chapter 3)

Note

Goals: Simple $\boldsymbol{M_{AT}L_{AB}}$ -use; whirling evaluation

Let us again look at the simpel system with cross-coupling stiffness terms:

$$\underline{M}\ \underline{\ddot{x}} + \underline{B}\ \underline{\dot{x}} + \underline{K}\ \underline{x} = \underline{0}$$

with:

$$\underline{M} = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \quad \underline{B} = \begin{bmatrix} b & 0 \\ 0 & b \end{bmatrix} \quad \underline{K} = k \begin{bmatrix} 1 & \kappa \\ -\kappa & 1 \end{bmatrix}$$

$$\underline{x}(t) = \begin{bmatrix} x(t) \\ z(t) \end{bmatrix}$$

To do some numerical evaluation of this system we assume:
$m = 1; \quad k = 100; \quad b = 4.$

This gives (for $\kappa = 0$) two undamped eigenfrequencies $\omega_u = 10.0 \quad [rad/s]$ and additionally, the dimensionale damping factors $\xi = 0.2$.
To evaluate the free vibrations of this system, we switch to the **state-space formulation**

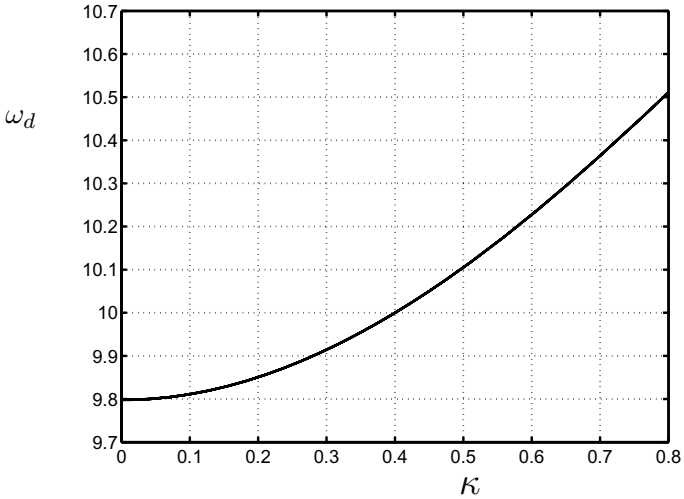
$$\underline{C}\ \underline{\dot{y}} + \underline{D}\ \underline{y} = \underline{0}$$

with:

$$\underline{C} = \begin{bmatrix} \underline{B} & \underline{M} \\ \underline{M} & \underline{0} \end{bmatrix} \quad \underline{D} = \begin{bmatrix} \underline{K} & \underline{0} \\ \underline{0} & -\underline{M} \end{bmatrix}$$

$$\underline{y}(t) = \begin{bmatrix} \underline{x}(t) \\ \underline{\dot{x}}(t) \end{bmatrix}$$

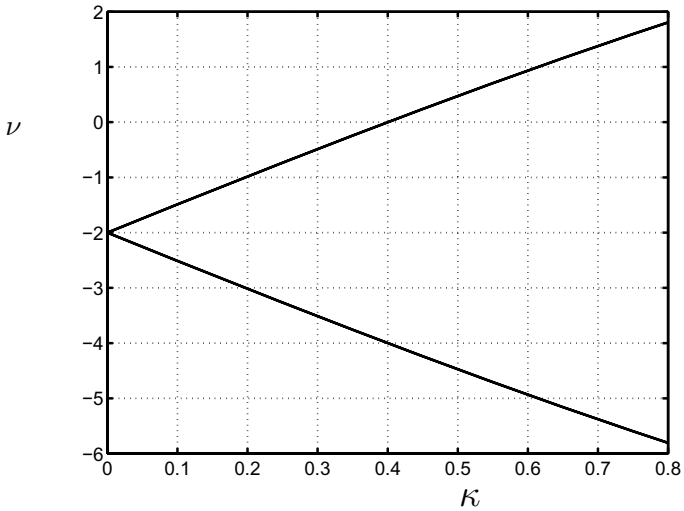
(This is in more detail treated in Section 3.5)
With the help of $\boldsymbol{M_{AT}L_{AB}}$, we can easily solve the eigenvalueproblem $[\lambda\underline{C} + \underline{D}]\underline{v} = \underline{0}$, giving 4 eigenvalues λ_i and corresponding eigenvectors $\underline{v}_i$, for $i = 1...4$. The system is undercritically damped which means that the eigenvalues and -vectors show up in complex conjugate pairs.
The real parts of these eigenvalues correspond with the symbol ν as used before and the (positive) imaginary parts with the damped eigenfrequency ω_d.
The imaginary parts as function of the cross coupling stiffness-parameter κ are shown in the next figure.



Damped eigenfrequencies as function of cross-coupling parameter κ

We see that for $\kappa = 0$ the damped eigenfrequency is starting somewhat below the undamped eigenfrequencies $\omega_o = 10.0$, because we know that $\omega_d = \omega_0\sqrt{1-\xi^2}$. For growing κ the damped eigenfrequency also is increasing as we have seen before in Fig. 1.9

The two real parts of the eigenvalues are shown next.

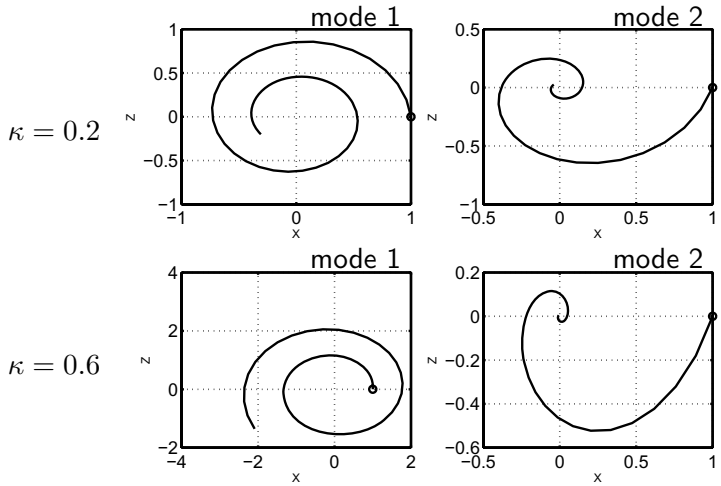


Real parts lof eigenvalues as function of cross-coupling parameter κ

We see that both real parts start with a negative value (stable system). For increasing κ, one is getting more and more negative, the other is increasing and becomes positive at $\kappa = 0.4$. There the system is becoming unstable.
We can also see this when we plot the free motion of the system. For this free motion we may write:

$$\underline{x}(t) = \begin{bmatrix} x(t) \\ z(t) \end{bmatrix} = Re[\underline{u}_i e^{\lambda_i t}]$$

We have two different modes and the eigenmodes $\underline{u}_i$ are scaled such that $\underline{u}_i(1)$=1.0.
We look at two different values for κ, namely $\kappa = 0.2$ and $\kappa = 0.6$. In the next figure the results for the two modes are shown for both values of κ. The starting point is marked by a small circle.



Orbits of free motion for two typical κ-values

For the stable system ($\kappa = 0.2$), we see on the left a moderately damped, counterclockwise spiral -a forward whirl- and on the right a a more strongly damped clockwise spiral -a backward whirl-. Both go to zero for $t \Rightarrow \infty$.
For $\kappa = 0.6$, the unstable system, we see on the left a diverging counterclockwise spiral - forward whirl- underlining the instability of the system. Additionally we again see a strongly damped clockwise spiral

-a backward whirl-.

Question: We have seen that the sign of the cross-coupling term κ does not influence the onset to instability or the damped eigenfrequency. But does it perhaps influence the whirling direction?

End of note

1.5 Critical Speeds and the Campbell Plot

A critical speed has been introduced before by analyzing the dynamic behavior of the Laval-Jeffcott rotor. In this paragraph this phenomenon will be studied in somewhat more detail, using a simple rotor-system with 2 degrees of freedom, including a gyroscopic term. This makes an analytic evaluation still possible.
We consider a massless shaft with length l with on the right-hand side a rigid disc and fixed at the left-hand side (only the rotation Ω around the longitudinal axis is possible). At $y = a$ the shaft is supported by a bearing, modeled by two linear springs with spring stiffnesses k_x respectively k_z for x- respectively z-direction. (see Fig. 1.11). We apply the Rayleigh-

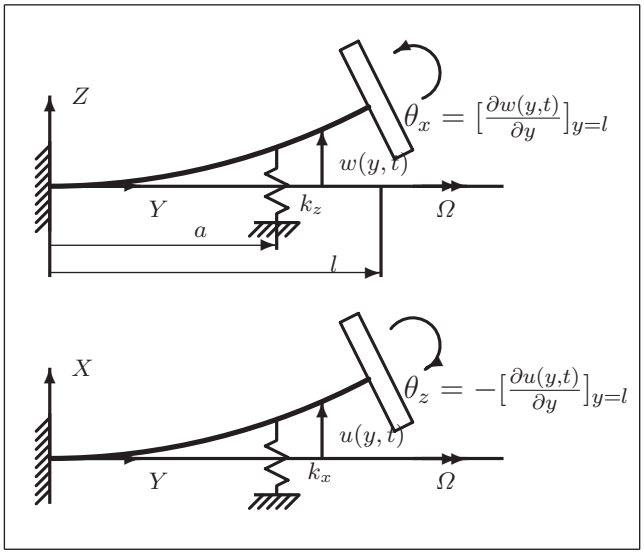


Fig. 1.11. Simple rotor with bearing and gyroscopic effects

Ritz method for the creation of a discrete model for this rotor. For the displacement-functions in x-direction and z-direction we choose

$$u(y,t) = \frac{y^2}{l^2}\, u_l(t) \qquad w(y,t) = \frac{y^2}{l^2}\, w_l(t) \tag{1.40}$$

These functions fulfil the necessary kinematic boundary conditions.

> The choice of displacementfuntion is important and can be crucial for the predictive value of the model results. An example is given in Section 6.1.

For the elastic energy we get

$$\begin{aligned} U = \tfrac{1}{2}EI \int_0^l [\tfrac{\partial^2 u}{\partial y^2}]^2 dx + \tfrac{1}{2}k_x u^2(a,t) + \\ \tfrac{1}{2}EI \int_0^l [\tfrac{\partial^2 w}{\partial y^2}]^2 dx + \tfrac{1}{2}k_z w^2(a,t) \end{aligned} \tag{1.41}$$

Substitution of the displacement functions (1.40) gives

$$U = \frac{1}{2}[\frac{4EI}{l^3} + k_x \frac{a^2}{l^4}]u_l^2 + \frac{1}{2}[\frac{4EI}{l^3} + k_z \frac{a^2}{l^4}]w_l^2 \tag{1.42}$$

which can be written as

$$\begin{aligned} U &= \frac{1}{2}[k_x^* \; u_l^2 + k_z^* \; w_l^2] \quad \text{with} \\ k_x^* &= \frac{4EI}{l^3} + k_x \frac{a^4}{l^4} \\ k_z^* &= \frac{4EI}{l^3} + k_z \frac{a^4}{l^4} \end{aligned} \tag{1.43}$$

For the kinetic energy of the rigid disc we can write (this will be handled in Section 3.2)

$$T = \frac{1}{2}[m(\dot{u})^2 + m(\dot{w})^2 + J_t(\dot{\theta}_x)^2 + J_t(\dot{\theta}_z)^2 + J_p(\Omega^2 - 2\Omega\theta_z\dot{\theta}_x)]_{y=l} \tag{1.44}$$

where Ω is the constant rotorspeed around the y-as, m the mass of the disc, J_t the mass-moment of inertia of the disc around an axis in the plane of the disc and J_p the mass-moment of inertia around an axis perpendicular to the disc-plane. Substitution of the displacement-functions now leads to

$$T = \frac{1}{2}[(m + \frac{4J_t}{l^2})\dot{u}_l^2 + (m + \frac{4J_t}{l^2})\dot{w}_l^2 + J_p(\Omega^2 + 8\Omega\frac{u_l\dot{w}_l}{l^2})] \tag{1.45}$$

or

$$T = \frac{1}{2}[m^*(\dot{u}_l^2 + \dot{w}_l^2) + 2m_p\Omega u_l\dot{w}_l] \qquad \text{with} \tag{1.46}$$
$$m^* = m + \frac{4J_t}{l^2} \qquad m_p = \frac{4J_p}{l^2}$$

The constant term with Ω^2 has been omitted because it will disappear. The application of Lagrange's equations will give

$$\begin{aligned} m^*\ddot{u}_l - m_p\Omega\dot{w}_l + k_x^* u_l &= 0 \\ m^*\ddot{w}_l + m_p\Omega\dot{u}_l + k_z^* w_l &= 0 \end{aligned} \tag{1.47}$$

Special attention should be paid to the terms in $\dot{w}_l$ and $\dot{u}_l$, coming from the gyroscopy and leading to a **skew-symmetric damping matrix**, proportional to the rotorspeed Ω.
Secondly, it should be remarked that if $k_x = k_z$ we are dealing with a symmetrical rotorsystem and if $k_x \neq k_z$ we have a so-called nonsymmetrical system.
The equations (1.47) are the basis for the further evaluations. For simplicity we write them as

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \ddot{u} \\ \ddot{w} \end{bmatrix} + \Omega \begin{bmatrix} 0 & -c \\ c & 0 \end{bmatrix} \begin{bmatrix} \dot{u} \\ \dot{w} \end{bmatrix} + \begin{bmatrix} k_x & 0 \\ 0 & k_z \end{bmatrix} \begin{bmatrix} u \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{1.48}$$

or

$$\underline{M}\ \underline{\ddot{q}} + \underline{B}\ \underline{\dot{q}} + \underline{K}\ \underline{q} = \underline{o} \tag{1.49}$$

First we look at the symmetrical case ($k_x = k_z = k$). The homogeneous part has the solution

$$\underline{q} = \underline{u}\ e^{\lambda\ t}$$

For $\underline{u} \neq \underline{o}$ this gives the characteristic equation

$$\lambda^4 m^2 + (2mk + c^2\Omega^2)\lambda^2 + k^2 = 0 \tag{1.50}$$

When $\Omega = 0$ we have 2 equal roots

$$\lambda_1 = \lambda_2 = j\ \omega_o; \lambda_3 = \lambda_4 = -j\ \omega_o$$

where $\omega_o = \sqrt{k/m}$. So we find two equal eigenfrequencies, one for the x-direction and one for the z-direction (the equations (1.49) are decoupled

for $\Omega = 0$).
In the case $\Omega \neq 0$ we get the solutions

$$\begin{aligned}
\lambda_1 &= \pm j\ \omega_1 \\
\lambda_2 &= \pm j\ \omega_2 \\
\omega_1 &= \sqrt{\omega_o^2 + \frac{c^2\Omega^2}{2m^2}(1 - \sqrt{1 + \frac{4m^2\omega_o^2}{c^2\Omega^2}})} \\
\omega_2 &= \sqrt{\omega_o^2 + \frac{c^2\Omega^2}{2m^2}(1 + \sqrt{1 + \frac{4m^2\omega_o^2}{c^2\Omega^2}})}
\end{aligned} \tag{1.51}$$

Now we have two free vibrations with purely imaginary eigenvalues which means that we are dealing with undamped motions of the rotor. It also follows that

$$\omega_1 \leq \omega_o \leq \omega_2$$

A plot of these eigenfrequencies ω_1 en ω_2 as a function of Ω usually is called a **Campbell plot**. Before we look at this graphical interpretation we first look at the corresponding eigenmodes. For these modes it follows directly from the eigenvalue-problem with $\underline{u}^T = [u_1, u_2]$

$$u_1 = \frac{c\Omega\lambda}{k + m\lambda^2} u_2$$

So that

$$\begin{aligned}
\lambda_1 &= j\ \omega_1; \Longrightarrow \underline{u}_1^T = \alpha[j, 1] \\
\lambda_2 &= -j\ \omega_1; \Longrightarrow \underline{u}_2^T = \alpha[-j, 1] \\
\lambda_3 &= j\ \omega_2; \Longrightarrow \underline{u}_3^T = \alpha[-j, 1] \\
\lambda_4 &= -j\ \omega_2; \Longrightarrow \underline{u}_4^T = \alpha[j, 1]
\end{aligned} \tag{1.52}$$

where α is an arbitrary constant. For the general solution of free vibrations we can now write

$$\underline{q} = A_1\underline{u}_1 e^{j\omega_1 t} + B_1\underline{u}_2 e^{-j\omega_1 t} + A_2\underline{u}_3 e^{j\omega_2 t} + B_2\underline{u}_4 e^{-j\omega_2 t} \tag{1.53}$$

The 4 constants A_1, A_2, B_1, B_2 are depending on the initial conditions. First we assume that these initial conditions can be chosen such that only the eigenfrequency ω_1 remains. At $t = 0$ we therefore demand

$$\underline{q}(t=0) = \begin{bmatrix} 0 \\ q_{2o} \end{bmatrix} \qquad \text{and} \qquad \dot{\underline{q}}(t=0) = \begin{bmatrix} -\omega_1 \; q_{2o} \\ 0 \end{bmatrix}$$

which gives

$$\underline{q} = \frac{q_{2o}}{2} \begin{bmatrix} je^{j\omega_1 t} - je^{-j\omega_1 t} \\ e^{j\omega_1 t} + e^{-j\omega_1 t} \end{bmatrix} \tag{1.54}$$

This can be transformed into

$$\underline{q} = q_{2o} \begin{bmatrix} -\sin(\omega_1 t) \\ \cos(\omega_1 t) \end{bmatrix} \tag{1.55}$$

So the disc-center is following a cylindrical orbit in the X-Z plane with radius $r = q_{2o}$, (see Fig. 1.12, left figure). From this figure also follows that the

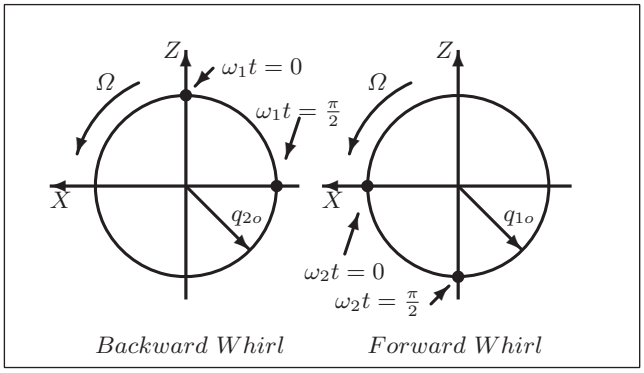


Fig. 1.12. Free vibrations of symmetric rotor

whirl-motion is in the opposite direction of the rotorspeed, we call this a **backward whirl**.
Taking another set of initial conditions we can visualize the second mode. We now start from

$$\underline{q}(t=0) = \begin{bmatrix} q_{1o} \\ 0 \end{bmatrix} \qquad \text{and} \qquad \dot{\underline{q}}(t=0) = \begin{bmatrix} 0 \\ -\omega_2 \; q_{1o} \end{bmatrix}$$

which gives

$$\underline{q} = q_{1o} \begin{bmatrix} \cos(\omega_2 t) \\ -\sin(\omega_2 t) \end{bmatrix} \tag{1.56}$$

This second mode in indicated in Fig. 1.12, right figure. In this case the whirl direction is the same as the rotor-speed direction, we call this a **forward whirl**.

Note

The type of motion (forward- or backward whirl) can be determined systematically in the following way. We make use of (see also appendix A)

$$\vec{r} * \vec{v} = \vec{r} * (\vec{\omega} * \vec{r}) = (\vec{r}.\vec{r})\vec{\omega} - (\vec{r}.\vec{\omega})\vec{r}$$

The second term in the right hand side disappears in our case, so we get

$$\vec{r} * \vec{v} = |r|^2 \vec{\omega}$$

For illustration we look at the first eigenvalue and corresponding eigenmode, $\lambda_1 = j\omega_1$, $\underline{u}_1^T = [j, 0, 1]$ (extended to 3 dimensions). This gives the real position vector $\vec{r}(t)$ with matrix-representation $\underline{r}(t)$

$$\underline{r}(t) = Re[\underline{u}_1 \; e^{\lambda_1 t}] = \begin{bmatrix} -\sin(\omega_1 t) \\ 0 \\ \cos(\omega_1 t) \end{bmatrix}$$

and the corresponding velocity-vector $\vec{v}(t)$ has as matrix-representation

$$\underline{v}(t) = \dot{\underline{r}} = Re[\lambda_1 \; \underline{u}_1 \; e^{\lambda_1 \; t}] = \omega_1 \begin{bmatrix} -\cos(\omega_1 t) \\ 0 \\ -\sin(\omega_1 t) \end{bmatrix}$$

The corresponding whirl-rotation vector $\vec{\omega}_w$ can be found from (see also appendix A)

$$\vec{\omega}_w = \vec{r} * \vec{v} \Longrightarrow \underline{\omega}_w = (\underline{r}) \; \underline{v} =$$

$$\omega_1 \begin{bmatrix} 0 & -\cos(\omega_1 t) & 0 \\ \cos(\omega_1 t) & 0 & \sin(\omega_1 t) \\ 0 & -\sin(\omega_1 t) & 0 \end{bmatrix} \begin{bmatrix} -\cos(\omega_1 t) \\ 0 \\ -\sin(\omega_1 t) \end{bmatrix} \tag{1.57}$$

So, we get

$$\underline{\omega}_w = \omega_1 \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} \tag{1.58}$$

De whirl-rotation vector is also pointing in the negative Y-direction (and opposite to the rotorspeed vector); it is therefore called a backward whirl.

End of note

For the further evaluation of the dynamic behavior of this rotor we consider the special case $m = 1$, $c = 0.5$ and $k = 1$.
A so-called **Campbell-plot** gives the imaginary parts of the system eigenvalues as a function of the rotorspeed, (see Fig. 1.13). In this plot also the

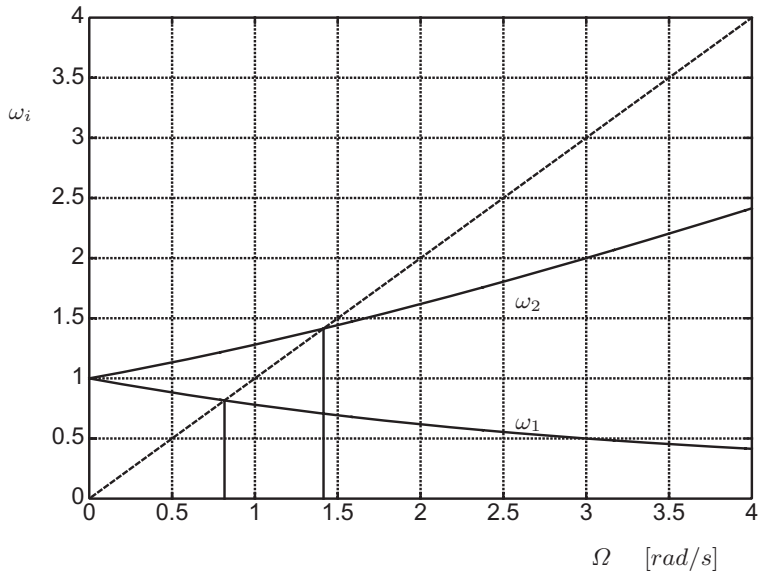


Fig. 1.13. Campbell plot

line $\omega_i = \Omega$ is drawn. The intersections of this line with the ω_1 respectively ω_2 -graphs give the **critical speeds** of this system. At these intersections the eigenfrequencies (undamped in this specific situation) are exactly equal to the rotorspeed. Even the smallest unbalance (this gives a harmonic excitation synchronous with the rotorspeed) will lead to resonance and in this case an unbounded system response. This also follows from Fig. 1.14. In this figure the frequency-response of the dofs u and w corresponding to a *nonrotating* harmonic excitation with angular frequency ω, and acting only

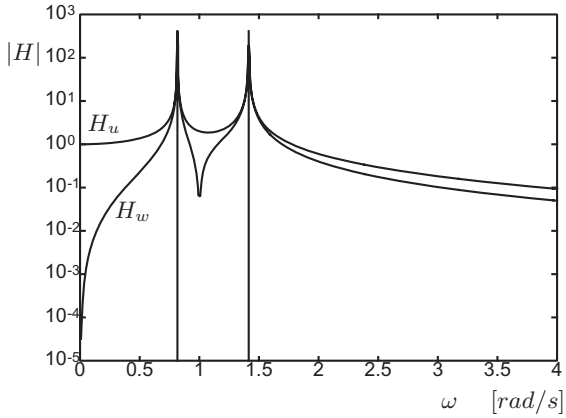


Fig. 1.14. Frequency response functions for a unit harmonic force in x-direction

in the x-direction of the disc, is shown. Two sharp resonance peaks can be seen for each of the critical speeds.
For this simpel system the critical speeds can also be calculated directly by assuming an eigenvalue with zero real part

$$\lambda = \pm j\omega$$

and substituting this in the characteristic equation (1.50), giving

$$\begin{aligned} &m^2\omega^2 - (2mk + c^2\omega^2)\omega^2 + k^2 = 0 \quad \text{or} \\ &(m^2 - c^2)\omega^4 - 2mk\omega^2 + k^2 = 0 \end{aligned} \tag{1.59}$$

This equation has the solutions

$$\omega_{c1} = \sqrt{\frac{k}{m+c}}, \qquad \omega_{c2} = \sqrt{\frac{k}{m-c}} \tag{1.60}$$

For the special case this leads to $\omega_{c1} = 0.8165\ [rad/s]$ and $\omega_{c2} = 1.4142\ [rad/s]$ which fits perfectly with the results from the Campbell plot. In Fig. 1.14 we can see a resonance at both the eigenfrequencies of the system in case of an excitation in the x-direction of the disc. This means that forward as well as backward whirl will be excited, which however is not always the case. The Frequency-response-function matrix $H(\Omega)$ can in general be written as shown in [Kraker/Campen-01]

$$\underline{H}(\Omega) = \sum_{i=1}^{4} \frac{\underline{u}_i \underline{v}_i^T}{c_i[j\Omega - \lambda_i(\Omega)]} \tag{1.61}$$

with

$\underline{u}_i$ the i^e **right**-eigenvector,
$\underline{v}_i$ the i^e **left** -eigenvector, namely the eigenvectors corresponding to the transposed system matrices $\underline{M}^T, \underline{B}^T, \underline{K}^T$,
c_i a normalization constant,
λ_i the corresponding eigenvalue,
Ω the excitation frequency, in this case the rotorspeed.

Simply can be seen that the left-eigenvectors can be written as

$$\begin{aligned}
\lambda_1 &= j\ \omega_1; \Longrightarrow \underline{v}_1^T = \alpha[-j, 1] \\
\lambda_2 &= -j\ \omega_1; \Longrightarrow \underline{v}_2^T = \alpha[j, 1] \\
\lambda_3 &= j\ \omega_2; \Longrightarrow \underline{v}_3^T = \alpha[j, 1] \\
\lambda_4 &= -j\ \omega_2; \Longrightarrow \underline{v}_4^T = \alpha[-j, 1]
\end{aligned} \tag{1.62}$$

It is important to note that in this case the eigenvalues (not the eigenvectors) are a function of the rotorspeed Ω. The complex amplitude of the respons $\hat{\underline{q}}$ follows from the complexe amplitude of the excitation $\hat{\underline{f}}$ via

$$\hat{\underline{q}} = \underline{H}(\Omega)\ \hat{\underline{f}} \tag{1.63}$$

In case of an excitation due to a mass-unbalance ϵ we can write the excitation as

$$\begin{aligned}
\underline{f} = \begin{bmatrix} f_x \\ f_z \end{bmatrix} = (m\ \epsilon\ \Omega^2) \begin{bmatrix} \sin(\Omega t) \\ \cos(\Omega t) \end{bmatrix} = \\
f_o\ Re\{\begin{bmatrix} -j \\ 1 \end{bmatrix} e^{j\Omega t}\} = Re\{\hat{\underline{f}}\ e^{j\Omega t}\}
\end{aligned} \tag{1.64}$$

For the complex amplitude of the excitation we find with $f_o = m\ \epsilon\ \Omega^2$

$$\hat{\underline{f}} = f_o \begin{bmatrix} -j \\ 1 \end{bmatrix} \tag{1.65}$$

For the calculation of the complex amplitude of the response via (1.61) we are dealing with products of the type $\underline{u}_i \ (\underline{v}_i^T \ \hat{\underline{f}})$. Looking at the left eigenvectors according to (1.62) we can see that

$$\underline{v}_1^T \ \hat{\underline{f}} = 0, \quad \underline{v}_2^T \ \hat{\underline{f}} = 2, \quad \underline{v}_3^T \ \hat{\underline{f}} = 2, \quad \underline{v}_4^T \ \hat{\underline{f}} = 0$$

Surprising is that in the summation (1.63) only the eigenvalues $\lambda_2 = -j\omega_1$ and $\lambda_3 = j\omega_2$ participate. In the numerator of (1.61) we are dealing with $\{j\Omega - \lambda_i\}$ for $0 \leq \Omega < \infty$. This can only become zero (so lead to resonance) for λ_3 namely for $\Omega = \omega_2$. The eigenvalue $\lambda_2 = -j\omega_1$ only contributes marginally to the response. In Fig. 1.15 the unbalance-response of

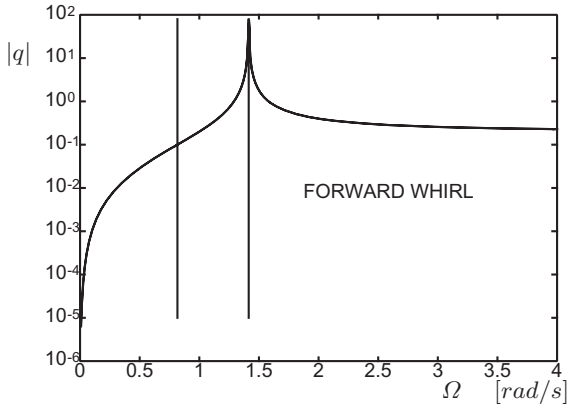


Fig. 1.15. Unbalance-response for $\epsilon = 0.1$

this system is given for a mass-eccentricity of $\epsilon = 0.1$. Also from this figure it is clear that in case of an unbalance-excitation only the forward whirl will be excited and that the backward whirl will not be activated by a rotor-unbalance. This is one of the reasons why the backward whirl very often has been ignored. However, we derived before that for example a non-rotating harmonic excitation will activate both the forward whirl motion as well as the backward whirl motion as shown in Fig. 1.14.

The Campbell plot appears to be a transparent method to get information on the critical speeds if the system parameters are a function of the rotor-speed such as in case of gyroscopy or models for journal bearings or seals. This however is only true when there is not too much damping in the system. To illustrate this we look again at the same single disc model (1.48),

but now with an additional external viscous damping term, so the system

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \ddot{u} \\ \ddot{w} \end{bmatrix} + \begin{bmatrix} b & -\omega c \\ \omega c & b \end{bmatrix} \begin{bmatrix} \dot{u} \\ \dot{w} \end{bmatrix} + \begin{bmatrix} k_x & 0 \\ 0 & k_z \end{bmatrix} \begin{bmatrix} u \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{1.66}$$

where we again assume that $m = 1$, $c = 0.5$ and $k = 1$, and for the additional dampingterm b we will apply 3 characteristic values, namely leading to a weakly damped- $(b = 0.1)$, moderately damped- $(b = 0.5)$ and a heavily damped system $(b = 1.4)$. For each situation 3 graphs will be presented, namely

graph a: the relevant Campbell plot with the critical speeds,
graph b: the frequency response function corresponding to a nonrotating, harmonic excitation in only the x-direction,
graph c: the unbalance response due to a mass eccentricity of $\epsilon = 0.1$.

In all the sets of 3 plots, the 2 critical speeds, given by the intersections in the Campbell plot, are shown by two vertical lines.

Fig. 1.16 illustrates that also for a weakly damped system the Campbell

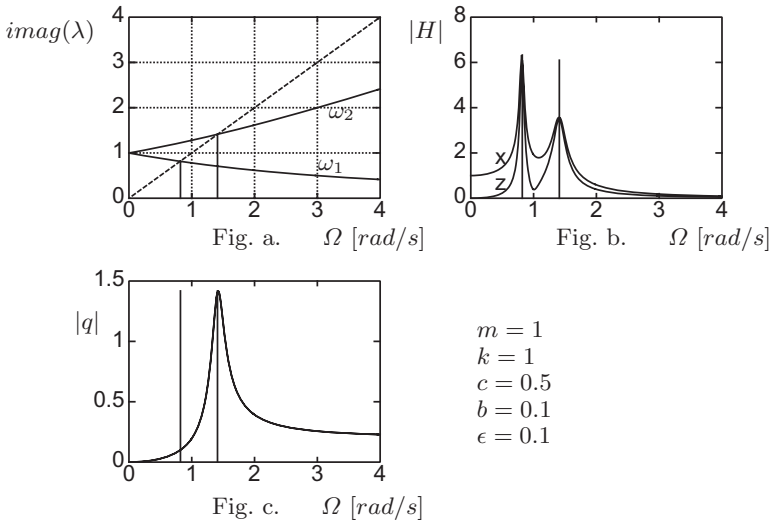


Fig. 1.16. Weakly damped system

plot accurately gives the critical speeds. Graph b gives a clear picture of

the resonance at these two frequencies and graph c gives a clear resonance at only the forward whirl mode. Fig. 1.17 shows the same results for the

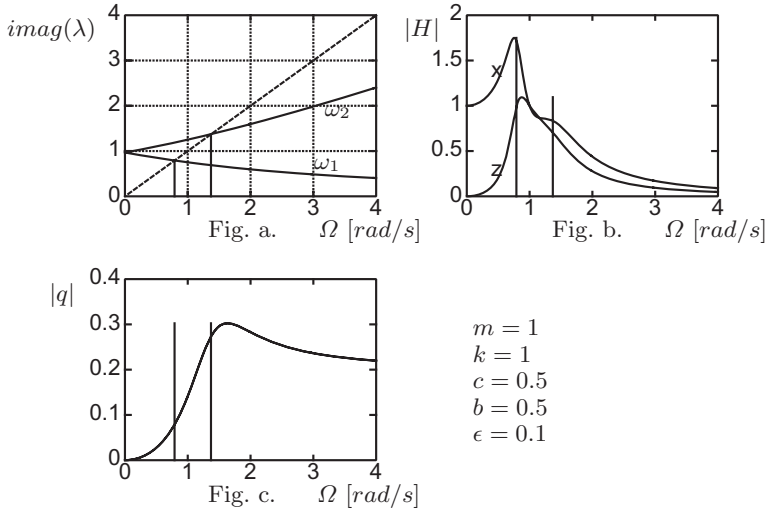


Fig. 1.17. Moderately damped system

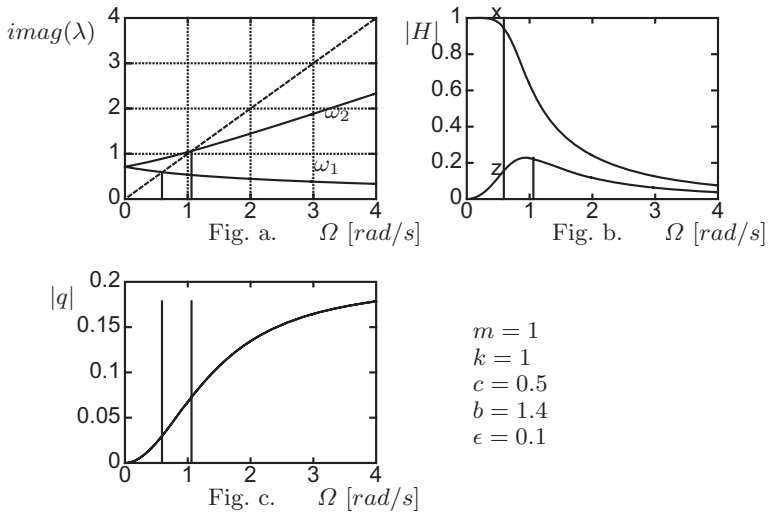


Fig. 1.18. Heavily damped system

moderately damped system. From the Campbell plot, two critical speeds might be identified but the graphs b and c show that no explicit resonance peaks can be distinguished.
This is even more the case for the heavily damped system (Fig. 1.18). The Campbell plot still suggests that we have two critical speeds but if we resume the definition of a critical speed according to Section 1.4.1, *the speed where the synchronous response due to an unbalance has a maximum*, we can see from graphs b and c that resonance is out of the question.
It is therefore injudicious only to use the Campbell plot (what many textbooks do) in order to draw conclusions about the existence of critical speeds. Also the real parts of the corresponding eigenvalues or a plot of the frequency response functions will give important information in order to conclude whether a rotorspeed is really a critical speed.
Strictly speaking, the use of a Campbell plot to find critical speeds will only give accurate results if the rotor system is relatively weakly damped. This will not be the case in many situations, for example in the presence of fluid-film bearings, seals, fluid-structure interaction, etc.

Note

The problem of using only the imaginary part of the eigenvalues to identify resonances (critical speeds) can also be illustrated by looking at a simple 1-dof mass-spring-damper system

$$\ddot{x} + 2\omega_o\xi\dot{x} + \omega_o^2 x = f(t) \tag{1.67}$$

The evaluation of this simple system shows that:

- the eigenvalues are $\lambda_{1,2} = -\xi\ \omega_o \pm j\omega_o\sqrt{1-\xi^2}$, so the imaginary part is $\omega_o\sqrt{1-\xi^2}$,
- the direct frequency response function reads $H(\Omega) = 1/\{-\Omega^2 + 2j\omega_o\Omega\xi + \omega_o^2\}$. This function has a maximum for $\Omega = \omega_o\sqrt{1-\mathbf{2}\ \xi^2}$,
- the unbalance response function reads $U(\Omega) = \Omega^2/\{-\Omega^2 + 2j\omega_o\Omega\xi + \omega_o^2\}$, this function has its maximum for $\Omega = \omega_o\sqrt{1+\mathbf{2}\ \xi^2}$.

For different levels of damping, the relative difference between the imaginary part and the frequency for which we find a maximum is shown in the following table.

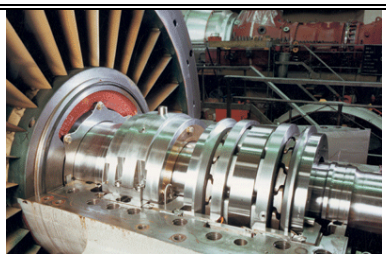
Dimensionless damping factor ξ	Imaginary part $\sqrt{1-\xi^2}$	Direct FRF $\sqrt{1-2\xi^2}$	Unbalance response $\sqrt{1+2\xi^2}$
0.01	0.99995	0.9999	1.0001
0.1	0.995	0.99	1.01
0.3	0.95	0.90	1.09
0.5	0.87	0.71	1.22

So we see that for weakly damped systems $(\xi < 0.2)$, the differences in practical situations can be ignored, but that for higher damping levels one should be careful.

End of note

2

Bearings



Plain Journal Bearing

Ball Bearing

Squeeze Film Damper

Gas Bearing

Spiral Groove Bearing

Magnetic Bearing

2.1 Introduction

In many cases, the dynamic behavior of rotating machinery will be determined primarily by the presence and the specific properties of bearings and/or seals. The most important reason for this is that the stiffness of a rotor-bearing system is mainly determined by the bearing stiffness in series with the shaft bending stiffness and that the system damping is coming mainly from the bearing- and or seal damping.
In many rotating machinery we moreover are dealing with some interaction of a medium (fluid) and the rotor leading to considerable forces acting on the rotor as for example in case of centrifugal pumps. We will not discuss this type of interaction.
In Section 1.4.2 it is demonstrated, that by a proper choice for the bearings-stiffness the forces which have to be transmitted through these bearings can be minimized or that by creating sufficient external damping, a rotorsystem can operate stable, despite the presence of some unavoidable internal damping.
Additional to the introduction of stiffness and damping in a rotorsystem one of the most important aspects of the application of fluid-film bearings is the introduction of so-called **cross-coupling terms** in the numerical model. This has far-reaching consequences for the stability of the system, as already shortly introduced in Section 1.4.3.
Looking at the design or the analysis of rotating machinery, rotor supports (bearings) can best be classified by the way the bearing force is generated by the bearing. The most important bearing types are

- Rolling-element bearings,
- Fluid-film (incompressible) bearings,
  - Hydrostatic bearings,
  - Hydrodynamic bearings,
  - Squeeze-film dampers,
- Elektro-magnetic bearings,
- Gas (compressible) bearings.

Most of the bearings in aircraft engines, gas-turbines, etc. are rolling-element bearings. The most important reason for this is that such bearings, far before they are going to fail, already produce some signals indicating the decreasing condition, for example by generating high-frequent components in the vibration spectrum. On the other hand fluid film bearings sometimes can

fail very unexpectedly with all kinds of disastrous consequences.
In less critical industrial circumstances, fluid film bearings are widely used, mainly because of their very long lifetime en the relatively low power-loss due to friction.
Gas bearings in general have the same properties as fluid film bearings with the main difference that the bearing-data (stiffness and damping) will not only change with the rotorspeed but will also change with the excitation frequency. This latter property makes the analysis of systems with compressible bearings rather complicated.
Electro magnetic bearings (**EMB**), are an important research topic for the last 5 years. The application of this type of bearing gives the designer very much flexibility in optimizing the dynamic behavior, although the analysis and application is still very complex. However, in principle, the stiffness, damping, cross coupling terms etc. can be selected (by programming the digital controller) in such a way that certain design demands can be fulfilled in an optimal way.
In the following bearing types mentioned before will shortly be discussed with an exception for hydrodynamic bearings which will be discussed in somewhat more detail because of their wide area of application.

2.2 Rolling-element Bearings

In the case of rolling-element bearings we are dealing with bearingforces which result from (in general elastic) deformation of spherical or cylindrical elements and the inner- respectively outer cage, following the theory of Hertz. So, in rotordynamics a frequently used numerical model generally consists of two linear springs with constant stiffness in two perpendicular directions (so no cross-coupling terms) en generally with only very little damping.
In some cases such a simple model can not represent reality sufficiently, and than the model has to be extended to a nonlinear stiffness or perhaps a rotorspeed dependent stiffness. So the application of rolling-element bearings does not necessarily lead to the instability problems discussed in Chapter 1. The lack of damping generally is very disadvantageous. The non-rotating outer-ring of the rolling-element bearing therefore will often be supported by an oil-filled chamber with a small clearance in order to realize the necessary damping. This is called a Squeeze-Film-Damper (SFD), see Section 2.5.
Sometimes the manufacturer of the bearing can give the necessary stiffness- and/or damping data, sometimes they can be found in literature, (see for example [Harris-91]) or can be calculated by means of some computerprogramma (see [Roosmalen-94]). Normally this information will be incomplete (no damping data) or not so accurate and specially designed experiments will have to be carried out as discussed in [Kraker-99]. Roughly one can say that the stiffness of ball bearings is in the order of $k = 1.0\ 10^8\ [N/m]$ and the stiffness for straight rolling-element bearings is a factor of 5-10 higher. Axial pre-loading of rolling-element bearings can influence the stiffness very much (see [Kraker-99]).

2.3 Hydrostatic Bearings

In hydrostatic bearings the bearing-fluid is pressed to the bearing through a number of orifices, equally distributed over the circumference. The necessary presence of equipment to generate and control this oil pressure and the probability of failure of it are the most important disadvantage of these type of bearings. Due to a displacement of the shaft center in the bearing

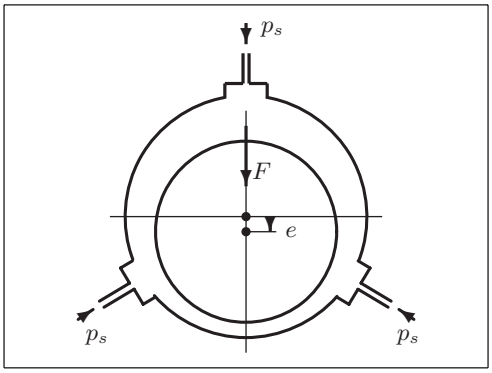


Fig. 2.1. Schematic representation of a hydrostatic bearing

as shown in Fig 2.1 the pressure on one side of the bearing wil increase and decrease on the other side. This pressure-difference will generate a reaction force with the same line of action as the prescribed displacement but in opposite direction. A hydrostatic bearing therefore may be modeled as a pair of orthogonal (linear) springs.
Also a prescribed velocity will lead to a reaction force due to viscous- or compressibility effects which means that this type of bearing can also generate damping. For more information see [Bassani-92].

2.4 Hydrodynamic Bearings

In case of hydrodynamic bearings (see e.g. ([Szeri-80]) the bearing capacity is originating simply and solely from the motion of the shaft and will be dependent on the viscosity of the medium. The medium can enter the bearing easily through some supply-holes, present on places where the hydrodynamic pressure is low.
In case of heavily loaded bearings or insufficient fluid supply the hydrodynamic film can break down and the journal can contact the bearing surface. This is called boundary lubrication. We assume that we only have to deal with that in special circumstances such as in starting-up a machine. We will only look at so-called full-film bearings.
The hydrodynamic pressure generation in a cylindrical bearing is illustrated

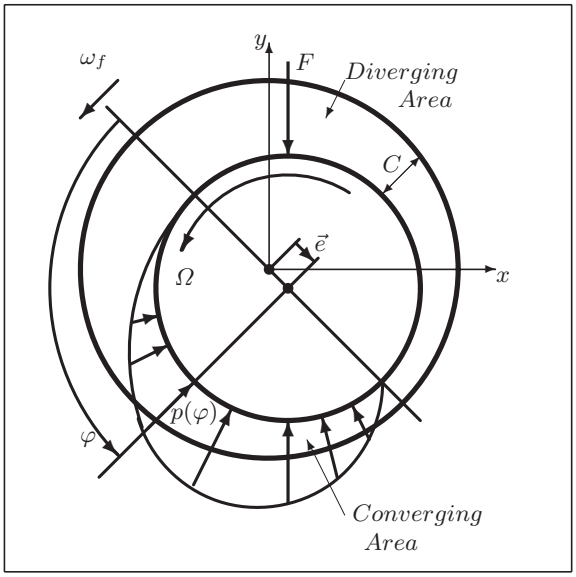


Fig. 2.2. Hydrodynamic pressure distribution in a cylindrical bearing

in Fig 2.2. Viscous effects are responsible for pulling the fluid into the converging wedge which is created by the exocentric position of the shaft in the bearing. Therefore the pressure in the fluid on this side will rise and vice-versa the pressure in the diverging wedge on the opposite side will decrease.

The net effect of the pressure distribution, integrated over the bearing circumference gives the bearing reactionforce which should compensate the bearing load.
The reactionforce increases with increasing eccentricity (e), so that again an equilibrium will be created. This means that at first the bearing will operate as a (linear) spring. The presence of a translational velocity $(\dot{e})$ of the shaft in the journal will also generate a reaction force which means that the bearing will also reacts as a (viscous) damper.
This damping behavior is one of the most pleasant aspects of fluid film bearings as structural elements in rotordynamic applications.
The basic cylindrical bearing, as given in Fig 2.2, is the cheapest and most elementary of all hydrodynamic bearing types. In practice however rotordynamic instability problems will arise when using these bearings which means that alternative bearing types have been developed such as tilting-pad bearings, 3-lobe bearings etc. (see Fig 2.3). For extensive information on the

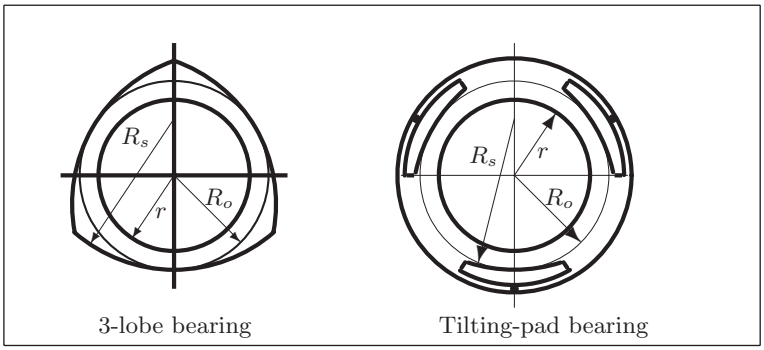


Fig. 2.3. Examples of non-cylindrical fluid-film bearing types

many possibilities which can be met in practice see e.g. [Someya-89].
From a rotordynamics point of view the following aspects are important when applying hydrodynamic bearings

- magnitude and direction of the bearing force,
- static equilibrium position of the bearing,
- bearing stiffness and -damping (the so-called bearing coefficients) for small perturbations around this equilibrium position.

In order to answer these questions the pressure-distribution in the bearing must be known which gives the equilibrium position and the bearing coefficients.

2.4.1 Pressure Distribution in the Bearing

The basic problem of evaluating a hydrodynamic bearing is the calculation of the pressure distribution for a given bearing geometry. In practice this means that specific solutions (with more or less severe assumptions) for the **Reynolds equation** have to be developed.
The basic equation is the Reynolds equation for a dynamically loaded, full-film bearing geometry with an **incompressible** medium

$$\frac{\partial}{\partial x}(h^3\frac{\partial p}{\partial x}) + \frac{\partial}{\partial y}(h^3\frac{\partial p}{\partial y}) = 6\eta\{\frac{\partial}{\partial x}[(U_o + U_1)h] + 2\frac{\partial h}{\partial t}\} \tag{2.1}$$

where

- $p = p(x, y)$ the pressure distribution,
- x, y the coordinates of a point in the thin film,
- h, the local film thickness as a function of x and z-coordinate,
- η, the viscosity,
- $U_o,\ U_1$ the tangential velocity of the two walls making the film,
- $\partial h/\partial t$, the time derivative of the local filmthickness,

This Reynolds equation is based on some assumptions and simplifications. Here we only resume these because they are important in order to be alert on the limits of application of this model

1. viscous effects are dominant, the viscosity η is the only important fluid parameter,
2. inertia terms may be neglected,
3. the fluid is incompressible,
4. the fluid film is very thin, no pressure variation across the thickness of the film and the curvature of the film can be neglected,
5. the viscosity is constant for the whole film,
6. there is no slip at the walls.

Most of the bearings have a cylindrical geometry. In cylindrical coordinates the Reynolds equation will be

$$\frac{\partial}{\partial\varphi}[(1+\epsilon\cos(\varphi))^3\frac{\partial p}{\partial\varphi}] + R^3\ \frac{\partial}{\partial y}[(1+\epsilon\cos(\varphi))^3\frac{\partial p}{\partial y}] =$$
$$-6\eta(\frac{R}{C})^3[(\Omega-2\omega_f)\epsilon\sin(\varphi) - 2\dot{\epsilon}\cos(\varphi)] \tag{2.2}$$

where

- φ is the angle to replace coordinate x to appoint a point in the film
- $\partial/\partial\varphi = R\ \partial/\partial x$,
- $\epsilon = e/C$,
- $e =$ eccentricity,
- $C =$ radial clearance,
- $R =$ journal radius,
- $h\ = C(1+\epsilon\cos(\varphi))$, the local film thickness,
- $\Omega =$ the angular velocity of the journal (the rotorspeed),
- $\omega_f =$ the angular velocity of the rotating frame in which the angle φ is defined.

For a plain journal bearing with atmospheric pressure at both sides of the journal ($z = 0$ and $z = L$) and without any cavitation the boundary conditions for the partial differential-equation (2.2) will be

$$p(\varphi, 0) = p(\varphi, L) = p_a$$
$$p(0, y) = p(2\pi, y) = p_o \tag{2.3}$$

Here p_a is the atmospheric pressure and p_o the supply pressure of the bearing. Solutions in closed form of this equation are not available, except for the case of very small eccentricities. In order to obtain useful solutions, two very successful approaches have been developed

1. simplify the Reynolds equation for special cases, so that approximate solution easily can be generated,
2. apply the Finite Element Method (FEM) or Finite Difference Method (FDM) in order to generate accurate numerical solutions,

The first approach has produced two important theories for plain journal bearings, namely the *long bearing theory (lbt)*, also called *Sommerfeld* bearing theory and the *short bearing theorie (sbt)* or *Ocvirk* bearing theory.

For the long bearing theory the most important assumption is that the second term in equation (2.2) may be neglected with respect to the first

term, which means that the pressure distribution is no longer depending on the axial coordinate z but only depends on the circumferential coordinate φ. The simplified equation can now be integrated leading to the pressure distribution

$$p_{lbt}(\varphi) = p_o + 6\eta(\frac{R}{C})^2\{(\Omega - 2\omega_f)\frac{\epsilon 2 + \epsilon\cos\varphi\sin\varphi}{(2+\epsilon^2)(1+\epsilon\cos(\varphi))^2} + \dot{\epsilon}\frac{1}{\epsilon}[\frac{1}{(1+\epsilon\cos(\varphi))^2} - \frac{1}{(1+\epsilon)^2}]\} \tag{2.4}$$

where p_o is the known pressure for $\varphi = 0$.
For the short bearing theory, the first term of (2.2) will be neglected with respect to the other terms. Practically, this is a very logical assumption when the length to diameter ratio of the bearing is sufficiently small $(L/D) < 0.25$. In that case the pressure in the film will become a parabolic function of the axial coordinate z. The pressure distribution for a constant bearing load ($\dot{\epsilon} = 0$) now will be (see e.g. [Vance-88] and [Childs-93])

$$p_{sbt}(\varphi) = p_a + \frac{3\eta\Omega}{C^2}(\frac{L^2}{4} - y^2)\frac{\epsilon\sin(\varphi)}{(1+\epsilon\cos(\varphi))^3} \tag{2.5}$$

where p_a is the pressure at the edges of the bearing.
Fig 2.4 gives this pressure distribution in the bearing (for a specific situation) for the midplane ($y = 0$) for 3 values for the eccentricity. As mentioned

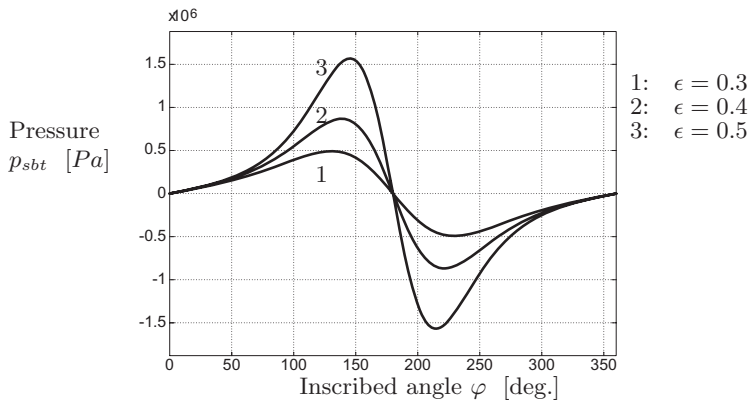


Fig. 2.4. Stationary pressure distribution for the short bearing theory for 3 excentricities

earlier, the pressure distribution in a bearing (see 2.4) is dependent on the

supply pressure p_o or atmospheric pressure p_a (see 2.5). From Fig 2.4 it appears, that the oil film pressure (for $p_a = 0$) in the diverging area will be highly negative. In general this supply pressure, respectively atmospheric pressure, will be relatively small (especially for highly loaded bearings) which should mean very negative fluid film pressures. This is not possible due to cavitation of the fluid by which only slightly negative pressure will be possible.
Heavily loaded bearings with a considerable eccentricity therefore are cavitating over approximately one half of the circumference. These are referred to as π-*film bearings*.
For a static bearing position ($\dot{\epsilon} = 0;\ \omega_f = 0$) the cavitation-area is $\pi < \varphi < 2\pi$. In a nonstationary position also the cavitation-area will change. A cavitating π-film bearing sometimes will be preferred above a non-cavitating 2π-film bearing (which can be realized by a sufficiently high supply pressure), because the stability behavior would be superior.
For a considerable bearing load the cavitation-area will be smaller than half the circumference. Generally, it is then acceptable just to set the negative pressures equal to zero.

2.4.2 Static Equilibrium Position

The static equilibrium position of the bearing under a unidirectional load (for example the own rotor weight) can be calculated by equating the two components of this load with two integrals of the oil film pressure over the bearing surface. This force is then decomposed in a component in the direction of the eccentricity vector $\vec{e}_r$ and in a component perpendicular to that $\vec{e}_t$, as indicated in Fig 2.5. The two integrals mentioned before represent the reaction forces generated by the oil film pressure on the bearing. They are a function of the eccentricity ϵ and the attitude angle ψ, and are given by

$$F_r = F\cos\psi = \int_0^l R \int_0^{2\pi} p(\varphi, y)\ \cos\varphi\ d\varphi\ dy$$

$$F_t = F\sin\psi = \int_0^l R \int_0^{2\pi} p(\varphi, y)\ \sin\varphi\ d\varphi\ dy \tag{2.6}$$

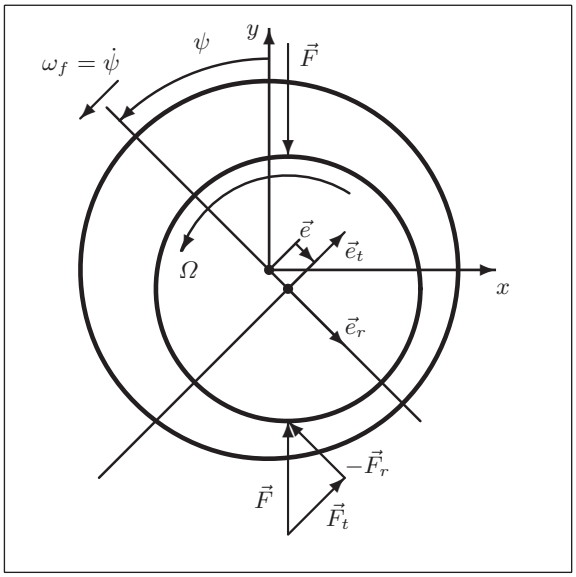


Fig. 2.5. Components of the bearing force $\vec{F}$

The pressure distribution $p(\varphi, z)$ in general is a non-linear function of the eccentricity. The integrals can be evaluated for specific bearing positions (given by φ en ϵ) if the corresponding pressure $p(\varphi, y)$ is known in closed form or in tabular form.
Fig 2.6 gives the trajectory of the bearing equilibrium position for a fully cavitating (π-film) short bearing. Special attention should be paid to the part of this trajectory for very lightly loaded bearings ($\epsilon \approx 0,\ \psi \approx \pi/2$) and on the other hand the part for very heavily loaded bearings ($\epsilon \approx 1,\ \psi \approx 0$). In the analysis of fluid film bearings, very often a dimensionless number, the so-called **Sommerfeld number** S, is used, which is defined as

$$S = \frac{\eta N}{P}\ (\frac{R}{C})^2 \qquad \text{with} \qquad N = \frac{\Omega}{2\pi},\ \ P = \frac{F}{LD} \tag{2.7}$$

The variable P is the load per unit of projected area and for the rotor-speed in [rev/s] or [Hz], we introduced the variable N. Substitution of the *non-cavitating* solution for the pressure in a long bearing in de integrals (2.6) (under the assumption $\dot{\epsilon} = 0$) gives the following expression for the Sommerfeld number as a function of the eccentricity ϵ

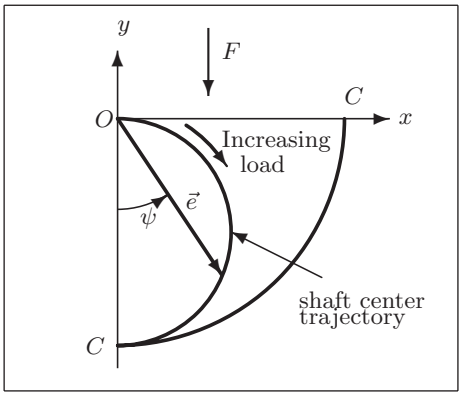


Fig. 2.6. Trajectory of the shaft center equilibrium position for cavitating short bearing

$$S = \frac{(2+\epsilon^2)\sqrt{1-\epsilon^2}}{12\pi^2\epsilon}, \qquad [\text{lbt}, \quad \dot{\epsilon} = 0, \quad \text{no cavitation}] \tag{2.8}$$

The angle ψ appears to be $\pi/2$ for all loads and the trajectory of shaft center positions for an exactly vertical bearing load will be a horizontal straight line. !!

However, we should realize that to prevent cavitation an extremely high supply pressure will be necessary, which generally is not a realistic assumption. The use of (2.8) therefore should practically be limited to $\epsilon < 0.2$. For a *fully cavitating*, long bearing the load-eccentricity relation becomes

$$S = \frac{(2+\epsilon^2)(1-\epsilon^2)}{6\pi\epsilon}\sqrt{\frac{1}{\pi^2+(4-\pi^2)\epsilon^2}} \qquad \{\text{lbt}, \quad \dot{\epsilon} = 0, \quad \text{fully cavitating}\} \tag{2.9}$$

with the angle ψ given by

$$\tan\psi = \frac{\pi\sqrt{1-\epsilon^2}}{2\epsilon} \tag{2.10}$$

For the *fully cavitating*, short bearing we finally get

$$S = \frac{(1-\epsilon^2)}{\pi(L/D)^2\epsilon}\sqrt{\frac{1}{\pi^2+(16-\pi^2)\epsilon^2}} \qquad \{\text{sbt}, \quad \dot{\epsilon} = 0, \quad \text{fully cavitating}\} \tag{2.11}$$

and for ψ

$$\tan\psi = \frac{\pi\sqrt{1-\epsilon^2}}{4\epsilon} \tag{2.12}$$

2.4.3 Dynamic Force Components

The relations given before are only valid for stationary conditions, so for $\dot{\epsilon} = 0; \quad \omega_f = \dot{\psi} = 0$. In rotordynamics, this equilibrium state is an important starting point for a dynamic analysis, but in such a dynamic analysis particularly the forces due to (small) deviations around this equilibrium state are important, so especially the effects for $\dot{\epsilon} \neq 0; \quad \omega_f = \dot{\psi} \neq 0$. The force components F_r and F_t corresponding to this situation of small deviations around the equilibrium state are

Non-cavitating, long bearing

$$\begin{aligned} F_r &= -12\eta RL(\frac{R}{C})^2 \frac{\pi\dot{\epsilon}}{(1-\epsilon^2)^{3/2}} \\ F_t &= 12\eta RL(\frac{R}{C})^2 \frac{(\Omega - 2\omega_f)\pi\epsilon}{(2+\epsilon^2)(1-\epsilon^2)^{1/2}} \end{aligned} \tag{2.13}$$

Cavitating (π-film) long bearing

$$\begin{aligned} F_r &= -6\eta RL(\frac{R}{C})^2[\frac{|\Omega - 2\omega_f|2\epsilon^2}{(2+\epsilon^2)(1-\epsilon^2)} + \frac{\pi\dot{\epsilon}}{(1-\epsilon^2)^{3/2}}] \\ F_t &= 6\eta RL(\frac{R}{C})^2[\frac{(\Omega - 2\omega_f)\pi\epsilon}{(2+\epsilon^2)(1-\epsilon^2)^{1/2}} + \frac{4\dot{\epsilon}}{(1+\epsilon)(1-\epsilon^2)}] + 2RLp_o \end{aligned} \tag{2.14}$$

Non-cavitating short bearing

$$\begin{aligned} F_r &= -\eta RL(\frac{L}{C})^2 \frac{\pi(1+2\epsilon^2)\dot{\epsilon}}{(1-\epsilon^2)^{5/2}} \\ F_t &= \eta RL(\frac{L}{C})^2 \frac{(\Omega - 2\omega_f)\epsilon}{2(1-\epsilon^2)^{3/2}} \end{aligned} \tag{2.15}$$

Cavitating (π-film) short bearing

$$F_r = -\eta RL(\frac{L}{C})^2[\frac{|\Omega - 2\omega_f|\epsilon^2}{(1-\epsilon^2)^2} + \frac{\pi(1+2\epsilon^2)\dot{\epsilon}}{2(1-\epsilon^2)^{5/2}}]$$
$$F_t = \eta RL(\frac{l}{C})^2[\frac{(\Omega - 2\omega_f)\pi\epsilon}{4(1-\epsilon^2)^{3/2}} + \frac{2\epsilon\dot{\epsilon}}{(1-\epsilon^2)^2}] + 2RLp_o \tag{2.16}$$

2.4.4 Linearized Bearing Coefficients

In Fig. 2.6 the stationary position is shown of the shaft center (indicated by the eccentricity ϵ_o and the angle ψ_o) as function of the vertical load on the bearing. Due to additional dynamic forces (for example due to unbalances or unstable motions) the rotor will carry out (small) motions around this equilibrium state. For the analysis of these motions we need to know the (small) variations of the fluid film pressure due to incremental variations of the shaft position and -velocity around the equilibrium. The so-called **bearing coefficients** are normally defined in the following way

$$\begin{bmatrix} F_x \\ F_y \end{bmatrix} = -\begin{bmatrix} K_{xx} & K_{xy} \\ K_{yx} & K_{yy} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} B_{xx} & B_{xy} \\ B_{yx} & B_{yy} \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} \tag{2.17}$$

or

$$\underline{F} = -\underline{K}\ \underline{q} - \underline{B}\ \underline{\dot{q}} \tag{2.18}$$

where

- $\underline{F}^T = [F_x, F_y]$ the bearing reaction forces in the fixed $[X, Z]$ frame
- $\underline{q}^T = [x, y]$ the displacements from the equilibrium state (ϵ_o, ψ_o)
- $\underline{K}$ the bearing stiffness matrix
- $\underline{B}$ the bearing damping matrix

In the equilibrium state also the rotating frame $(\vec{e}_r, \vec{e}_t)$ has a fixed position, so the bearing forces can also be defined in this frame

$$\begin{bmatrix} F_r \\ F_t \end{bmatrix} = -\begin{bmatrix} K_{rr} & K_{rt} \\ K_{tr} & K_{tt} \end{bmatrix} \begin{bmatrix} r \\ t \end{bmatrix} - \begin{bmatrix} B_{rr} & B_{rt} \\ B_{tr} & B_{tt} \end{bmatrix} \begin{bmatrix} \dot{r} \\ \dot{t} \end{bmatrix} \tag{2.19}$$

or

$$\underline{F}_e = -\underline{K}_e\ \underline{q}_e - \underline{B}_e\ \underline{\dot{q}}_e \tag{2.20}$$

where

- $\underline{F}_e^T = [F_r, F_t]$ the bearing forces in the rotating $[\vec{e}_r, \vec{e}_t]$ frame

- $\underline{q}_e^T = [r, t]$ the displacements from the equilibrium state (ϵ_o, ψ_o)
- $\underline{K}_e$ the bearing stiffness matrix
- $\underline{B}_e$ the bearing damping matrix

Using the coordinate transformation matrix $\underline{R}$

$$\begin{bmatrix} r \\ t \end{bmatrix} = \underline{q}_e = \underline{R}\ \underline{q} = \begin{bmatrix} \sin\psi_o & -\cos\psi_o \\ \cos\psi_o & \sin\psi_o \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \tag{2.21}$$

we can then write

$$\begin{aligned} \underline{K} &= \underline{R}^T \underline{K}_e \underline{R} \\ \underline{B} &= \underline{R}^T \underline{B}_e \underline{R} \end{aligned} \tag{2.22}$$

The stiffness terms and damping terms in the rotating frame can now easily be found from a Taylor series expansion of the bearing forces F_r en F_t around the chosen equilibrium

$$\begin{aligned} F_r(\epsilon, \psi, \dot{\epsilon}, \dot{\psi}) = F_r(\epsilon_o, \psi_o, 0, 0) + \frac{1}{C}\frac{\partial F_r}{\partial \epsilon} C d\epsilon + \frac{1}{C\epsilon}\frac{\partial F_r}{\partial \psi} C\epsilon d\psi + \\ \frac{1}{C}\frac{\partial F_r}{\partial \dot{\epsilon}} C d\dot{\epsilon} + \frac{1}{C\epsilon}\frac{\partial F_r}{\partial \dot{\psi}} C\epsilon d\dot{\psi} \end{aligned} \tag{2.23}$$

$$\begin{aligned} F_t(\epsilon, \psi, \dot{\epsilon}, \dot{\psi}) = F_t(\epsilon_o, \psi_o, 0, 0) + \frac{1}{C}\frac{\partial F_t}{\partial \epsilon} C d\epsilon + \frac{1}{C\epsilon}\frac{\partial F_t}{\partial \psi} C\epsilon d\psi + \\ \frac{1}{C}\frac{\partial F_t}{\partial \dot{\epsilon}} C d\dot{\epsilon} + \frac{1}{C\epsilon}\frac{\partial F_t}{\partial \dot{\psi}} C\epsilon d\dot{\psi} \end{aligned} \tag{2.24}$$

where second and higher order terms have been neglected. For the stiffness- and damping terms we then get

$$\begin{aligned} K_{rr} &= -\frac{1}{C}\frac{\partial F_r}{\partial \epsilon}, & K_{rt} &= -\frac{1}{C\epsilon}\frac{\partial F_r}{\partial \psi} \\ K_{tr} &= -\frac{1}{C}\frac{\partial F_t}{\partial \epsilon}, & K_{rt} &= -\frac{1}{C\epsilon}\frac{\partial F_t}{\partial \psi} \\ B_{rr} &= -\frac{1}{C}\frac{\partial F_r}{\partial \dot{\epsilon}}, & B_{rt} &= -\frac{1}{C\epsilon}\frac{\partial F_r}{\partial \dot{\psi}} \\ B_{tr} &= -\frac{1}{C}\frac{\partial F_t}{\partial \dot{\epsilon}}, & B_{rt} &= -\frac{1}{C\epsilon}\frac{\partial F_t}{\partial \dot{\psi}} \end{aligned} \tag{2.25}$$

The partial derivatives should be evaluated in the equilibrium state. For a number of bearing types these bearing coefficients can be found as formules,

in tables or in the form of graphs. The results are frequently presented in some dimensionless form to facilitate a universal application for a class of bearing types. Careful attention should be paid to the way this dimensionless making has been done because different method are used for this.
We are using the definition

$$K_{ij} = \frac{F_o}{C}\ \hat{K}_{ij}, \qquad B_{ij} = \frac{F_o}{C\ \Omega}\ \hat{B}_{ij} \tag{2.26}$$

where F_o is the static load of the bearing, following from (2.11), namely

$$F_o = \frac{\eta \Omega R L^3}{4C^2}\left[\frac{\epsilon_o\sqrt{\pi^2 + (16-\pi^2)\epsilon_o^2}}{1-\epsilon_o^2}\right] \tag{2.27}$$

For a cavitating, short bearing (π-film) we then find (see e.g. [Lund/Saibel-67])

$$\begin{aligned}
\hat{K}_{xx} &= 4\{2\pi^2 + (16-\pi^2)\epsilon_o^2\}\ Q(\epsilon_o)\\
\hat{K}_{xy} &= -\frac{\pi\{-\pi^2 + 2\pi^2\epsilon_o^2 + (16-\pi^2)\epsilon_o^4\}}{\epsilon_o(1-\epsilon_o^2)^{1/2}}\ Q(\epsilon_o)\\
\hat{K}_{yx} &= -\frac{\pi\{\pi^2 + (32+\pi^2)\epsilon_o^2 + 2(16-\pi^2)\epsilon_o^4\}}{\epsilon_o(1-\epsilon_o^2)^{1/2}}\ Q(\epsilon_o)\\
\hat{K}_{yy} &= \frac{4\{\pi^2 + (32+\pi^2)\epsilon_o^2 + 2(16-\pi^2)\epsilon_o^4\}}{(1-\epsilon_o^2)}\ Q(\epsilon_o)\\
\hat{B}_{xx} &= \frac{2\pi(1-\epsilon_o^2)^{1/2}\{\pi^2 + 2(\pi^2-8)\epsilon_o^2\}}{\epsilon_o}\ Q(\epsilon_o)\\
\hat{B}_{xy} &= -8\{\pi^2 + 2(\pi^2-8)\epsilon_o^2\}\ Q(\epsilon_o)\\
\hat{B}_{yx} &= \hat{B}_{xy}\\
\hat{B}_{yy} &= \frac{2\pi\{\pi^2 + 2(24-\pi^2)\epsilon_o^2 + \pi^2\epsilon_o^4\}}{\epsilon_o(1-\epsilon_o^2)^{1/2}}\ Q(\epsilon_o)
\end{aligned} \tag{2.28}$$

where

$$Q(\epsilon_o) = \frac{1}{\{\pi^2 + (16-\pi^2)\epsilon_o^2\}^{3/2}} \tag{2.29}$$

In Fig 2.7, the dimensionless coefficients (2.28) are shown graphically. From (2.27) one can see that the static equilibrium position goes to zero when the rotorspeed Ω is increasing or when the bearing load is decreasing (increasing Sommerfeld number). For the asymptotic situation $\epsilon_o \longrightarrow 0$, we get from (2.28)

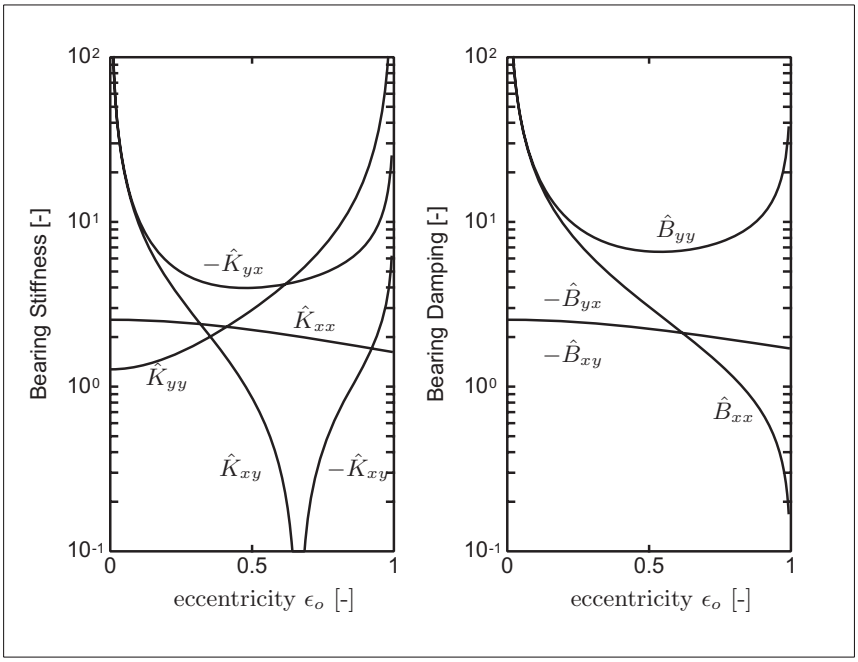


Fig. 2.7. Dimensionless stiffness- and damping terms for a fully cavitating, short bearing

$$\hat{K}_{xx}(0) = \frac{8}{\pi}; \quad \hat{K}_{yy}(0) = \frac{4}{\pi}; \quad \hat{K}_{xy}(0) = -\hat{K}_{yx}(0) \Longrightarrow \frac{1}{\epsilon_o} \tag{2.30}$$

$$\hat{B}_{xx}(0) = \hat{B}_{yy}(0) \Longrightarrow \frac{2}{\epsilon_o}, \quad \hat{B}_{xy}(0) = \hat{B}_{yx}(0) = -\frac{8}{\pi} \tag{2.31}$$

which gives

$$\frac{K_{xy}}{\Omega B_{xx}} \Longrightarrow \frac{1}{2} \tag{2.32}$$

Intermezzo

The factor $\frac{1}{2}$ found in (2.32) is directly related to the so-called " $\frac{1}{2}\Omega$-whirl " for slightly loaded fluid film bearings. This can be demon-

strated by looking at a single bearing with a single (shaft) mass m (1 dof system), governed by the set of equations of motion

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{y} \end{bmatrix} + \begin{bmatrix} B_{xx} & B_{xy} \\ B_{yx} & B_{yy} \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} + \begin{bmatrix} K_{xx} & K_{xy} \\ K_{yx} & K_{yy} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{2.33}$$

We use the dimensionless coordinates

$$X = x/C; \quad Y = y/C; \quad \tau = \Omega t; \quad p^2 = \frac{m\Omega^2 C}{F_o}$$

so

$$\begin{bmatrix} p^2 & 0 \\ 0 & p^2 \end{bmatrix} \begin{bmatrix} \ddot{X} \\ \ddot{Y} \end{bmatrix} + \begin{bmatrix} \hat{B}_{xx} & \hat{B}_{xy} \\ \hat{B}_{yx} & \hat{B}_{yy} \end{bmatrix} \begin{bmatrix} \dot{X} \\ \dot{Y} \end{bmatrix} + \begin{bmatrix} \hat{K}_{xx} & \hat{K}_{xy} \\ \hat{K}_{yx} & \hat{K}_{yy} \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{2.34}$$

We are interested in a solution exactly at the onset of instability, so the rotorspeed Ω_s for which at least one of the eigenvalues of the homogeneous system (2.33) has a purely imaginary value, thus a solution of the type

$$\begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} A \\ B \end{bmatrix} e^{\,j\omega_s \tau} = \begin{bmatrix} A \\ B \end{bmatrix} e^{\,j\omega_s \Omega_s t} \tag{2.35}$$

Here ω_s is the dimensionless whirl frequency, showing up for the rotorspeed Ω_s on the boundary of instability. So we are searching for a solution of

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} A \\ B \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{2.36}$$

with

$$\begin{aligned} Z_{11} &= (-\omega_s^2 p^2 + \hat{K}_{xx}) + j\omega_s \hat{B}_{xx} \\ Z_{12} &= \hat{K}_{xy} + j\omega_s \hat{B}_{xy} \\ Z_{21} &= \hat{K}_{yx} + j\omega_s \hat{B}_{yx} \\ Z_{22} &= (-\omega_s^2 p^2 + \hat{K}_{yy}) + j\omega_s \hat{B}_{yy} \end{aligned} \tag{2.37}$$

A solution $[A, B] \neq [0, 0]$ is only possible when the determinant of this matrix becomes zero. This gives

$$\begin{aligned}&[(-\omega_s^2 p^2 + \hat{K}_{xx})(-\omega_s^2 p^2 + \hat{K}_{yy}) - \omega_s^2 \hat{B}_{xx}\hat{B}_{yz} - \hat{K}_{xy}\hat{K}_{yx}\\ &+\omega_s^2 \hat{B}_{xy}\hat{B}_{yx}] + j\ \omega_s[(-\omega_s^2 p^2 + \hat{K}_{xx})\hat{B}_{yz} +\\ &(-\omega_s^2 p^2 + \hat{K}_{yz})\hat{B}_{xx} - \hat{K}_{xy}\hat{B}_{yx} - \hat{K}_{yx}\hat{B}_{xy}] = 0\end{aligned} \tag{2.38}$$

The real part as well as the imaginary part of this expression should be zero, giving

$$\begin{aligned}&D = (\omega_s^2 p^2) =\\ &\{\hat{B}_{xx}\hat{K}_{yy} + \hat{B}_{yy}\hat{K}_{xx} - \hat{K}_{xy}\hat{B}_{yx} - \hat{K}_{yx}\hat{B}_{xy}\}/\{\hat{B}_{xx} + \hat{B}_{yy}\}\\ &\omega_s^2 = \{(D - \hat{K}_{xx})(D - \hat{K}_{yy}) - \hat{K}_{xy}\hat{K}_{yx}\}/\{\hat{B}_{xx}\hat{B}_{yy} - \hat{B}_{xy}\hat{B}_{yx}\}\end{aligned} \tag{2.39}$$

Substitution of the limit situation (2.31) now gives

$$D = \frac{6}{\pi}; \qquad \omega_s = \frac{1}{2} \tag{2.40}$$

The whirl frequency in this limit situation becomes $\omega_s \Omega_s = \frac{1}{2}\Omega_s$, which is exactly half the frequency for which the rotor becomes unstable. Therefore this is called a "$\frac{1}{2} - \Omega$ whirl". In this limit situation only the terms proportional with $\frac{1}{\epsilon_o}$ are relevant, which are $\hat{K}_{xy} = -\hat{K}_{yx}$ and $\hat{B}_{xx} = \hat{B}_{yy}$, so that this factor $\frac{1}{2}$ also follows directly from (2.32). In the same way the dimensionless frequency

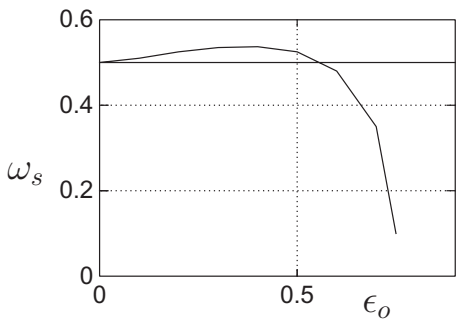


Fig. 2.8. Dimensionless frequency ω_s as function of the static eccentricity ϵ_o

ω_s can be determined from (2.38) for arbitrary eccentricity ϵ_o. The results are given in Fig 2.8. One can see that even for rather high excentricities ($\epsilon_o \approx 0.5$) the whirl frequency is approximately equal to half the rotorspeed. This "$\frac{1}{2} - \Omega$ whirl" therefore can show up in many practical situations.

For hydrodynamic bearings as frequently used in practice, bearing coefficients can also be found in graphical or tabular form in handbooks like ([Someya-89])).
An alternative and more accurate method is based on the use of impedances for the description of the bearing force. By using the vector-sum of Ocvirk-impedance and Sommerfeld-impedance, a bearing coefficient formulation can be obtained which gives reliable bearing-data for a large class of bearings, see e.g. [Childs-93] en [Childs-77].
We will not treat this method in detail here. However the method is programmed in the procedure $volcyl2m.m$ of the toolbox $\boldsymbol{R_O D_Y}$ (see also Section 4.2).
The input for this procedure consists of the bearing parameters: radius R, length L and clearance C, viscosity η, static load F_o and the rotorfrequency Ω. The output consists of the static equilibrium position $[\epsilon_o, \psi_o]$ and the bearing coefficients K_{xx}, K_{xy}, K_{yx}, K_{yy}, B_{xx}, B_{xy}, B_{yx} and B_{yy}.

To illustrate the application of this method we consider a shaft, symmetrically supported in two identical, cylindrical bearings, as shown in Fig. 2.9.

Fig. 2.9. Rotor model in two fluid film bearings

- Shaft $L_{shaft} = 0.12\ [m],\ D_{shaft} = 0.04\ [m],\ \rho = 7800\ [kgm^{-3}]$
- Bearings $R = 0.006\ [m],\ L = 0.008\ [m],\ \eta = 5.0\ 10^{-3}\ [Nsm^{-2}],\ C = 2.0\ 10^{-5}\ [m]$

For the static load we take the own weight of the rotor shaft (11.539 [N]), equally distributed over the 2 bearings. For the rotorspeed Ω we assume Ω=50...750 [Hz].
Fig 2.10 gives the dimensionless (static) eccentricity ϵ_o as a function of the rotorspeed for the impedance-method (procedure $volcyl2m$) and for the short bearing theory sbt, (2.27).

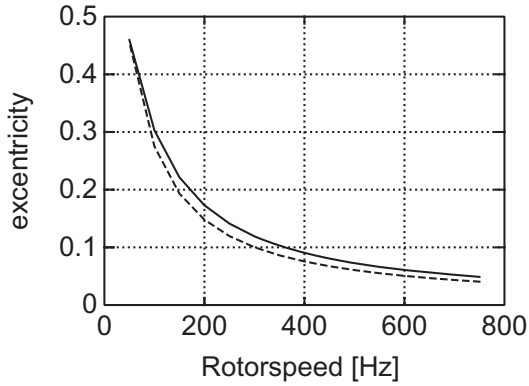


Fig. 2.10. Eccentricity ϵ_o as function of the rotorspeed in [Hz], using *volcyl2m*(—), respectively *sbt*(- - -)

The bearing coefficients for both methods are given in Fig 2.11. For this relatively short bearing $(L/D = 0.66)$ the differences (eccentricity as well as bearing coefficients) between the two approaches appear to be small. For the application of these models in a real rotorsystem and a discussion of the dynamic characteristics which will then be met, we refer to Section 4.2.

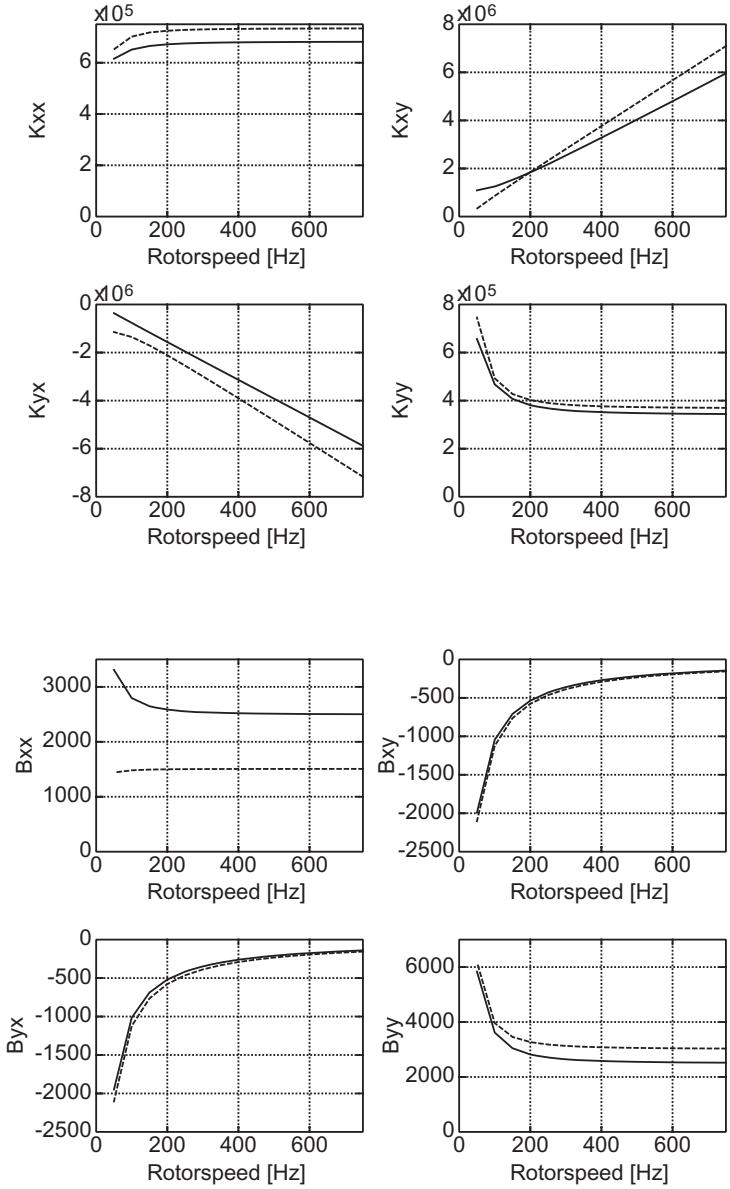


Fig. 2.11. Stiffness- and Damping coefficients as function of rotorspeed ([Hz]) using *volcyl2m*(—) and *sbt*(- - -)

2.5 Squeeze Film Dampers

As mentioned before squeeze film dampers (**SFD**) are important structural elements in aircraft engines, turbines, etc. for the creation of the necessary damping in rolling-element bearing supports. Fig 2.12 shows the basic idea

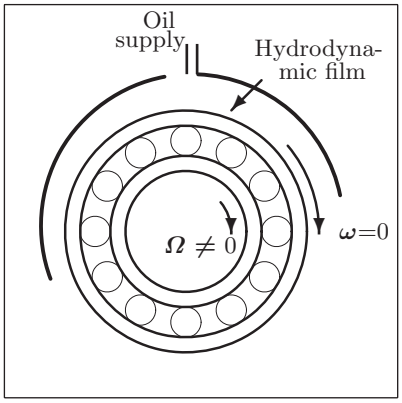


Fig. 2.12. Squeeze Film Damper principle

of a squeeze film damper. In principle the squeeze-film damper can be seen as a cylindrical fluid film bearing without rotational velocity, so $\Omega = 0$. The outer cage of the rolling-element bearing is supported elastically (for example by o-ring seals) in the SFD and the rotation is prevented by a locking pen.

The lack of rotation of a SFD assures that this element can not introduce instabilities for any rotorspeed. Under certain circumstances, nonlinear effects will become essential in the model for the SFD and this can result in a non-synchronous whirl, but it will be stable, if at least no other destabilizing forces are acting in the system. The main reason for this is that the SFD has no stiffness ($K_{ij} = 0$, for all i,j) and the damping is always positive.

The translational motion of a SFD also generates a complete periodic flow reversal which is not the case in normal bearings. The assumption in the Reynolds equation that inertia effects may be neglected is then not allowed anymore for most of the practical applications (characterized by $Re \gg 1$). Here Re is the squeeze film Reynolds number defined as

$$Re = \frac{\rho \ \Omega \ C^2}{\eta} \tag{2.41}$$

where

- Ω is the whirl speed
- ρ is the fluid density
- C is the radial clearance
- η is the viscosity.

Concluding, it can be stated that cavitation effects in squeeze film dampers can play an important role. They can be responsible for a large difference between the predictions based on a theoretical model and the practical experience. For more information we refer to ([Childs-93]) and ([Vance-88]).

2.6 Gas Bearings

Gas bearings (or air bearings) are based on the same principle as fluid film bearings (see also [Geerts-95a]). The fundamental difference is that in fluid film bearings the medium is assumed to be incompressible whereas in case of gas bearings this assumption has to be released. Therefore the classification in compressible- and incompressible bearings is also frequently usual.
So, essentially, not only the transport of the medium from one place to another is relevant but also the volumechanges when the bearing is loaded dynamically. The compression of the gas is assumed to be ideal, so the density will be proportional to the pressure in the gas.
For more information about the applicability of gas bearings see e.g. [Wang-93].
In general, gas bearings have a lower load carrying capacity, they need higher rotor speeds and have a smaller bearing clearance than as usual in fluid film bearings which makes the fabrication more complicated. Due to the much lower viscosity of the medium however, the friction moment will be very small. Other advantages of gas bearings are

- they operate very accurate,
- the temperature range is larger than for fluid film bearings,
- in general they are very environment-friendly , especially gas bearings.

In the following we will only look at externally pressurized, gas bearings sometimes also called aero-static bearings. Just as for fluid film bearings, the pressure distribution in the bearing can be calculated by (numerically) solving the Reynolds equation, taking into account the compressibility of the medium. In general these calculations (for example in case of an externally pressurized spiral groove bearing) are very complex and time-consuming.
For the evaluation of the bearing coefficients usually the central position is taken as the equilibrium state. This can be seen as some kind of "*worst case situation*" and in addition, it can be shown that the cross-coupling terms of the bearing coefficients (which may be responsible for a possible rotor-instability) are maximal for this position.
In the simulation, the shaft in the bearing is forced to prescribe a small harmonic perturbation around this central position and the corresponding gas pressure, and after integration the bearing force, can be calculated. This bearing force will now not only depend on the rotor speed but will also be a function of the excitation frequency which can be different from this rotorspeed. This makes the analysis of rotor systems with compressible

bearings rather difficult. The linearized bearing force for a specific rotor speed and excitation frequency can now be written as

$$\begin{bmatrix} F_x \\ F_y \end{bmatrix} = \begin{bmatrix} K_{xx} & K_{xy} \\ K_{yx} & K_{yy} \end{bmatrix} \begin{bmatrix} u_x \\ u_y \end{bmatrix} + \begin{bmatrix} B_{xx} & B_{xy} \\ B_{yzx} & B_{yy} \end{bmatrix} \begin{bmatrix} \dot{u}_x \\ \dot{u}_y \end{bmatrix} \tag{2.42}$$

or

$$\underline{F} = \underline{K}\ \underline{u} + \underline{B}\ \underline{\dot{u}} \tag{2.43}$$

The bearing stiffness matrix and -damping matrix are a function of the rotor speed Ω as well as the excitation frequency ω, so $\underline{K} = \underline{K}(\Omega, \omega)$ and $\underline{B} = \underline{B}(\Omega, \omega)$.
We assume that for a specific, prescribed harmonic bearing displacement $u_i = \hat{u}_i\ e^{j\omega t}$ and a **fixed rotor speed** Ω the bearing force $F_j = \hat{F}_j\ e^{j\omega t}$, $(i, j = x, y)$, can be calculated or measured if an experimental set-up is available. This gives the complex **dynamic stiffness** Z_{ij}, namely $\hat{F}_j = Z_{ji}\ \hat{u}_i$. The real part of this quantity gives the bearing stiffness coefficient K_{ji} and from the imaginary part the bearing damping coefficient B_{ji} can be derived.

Generally, the trend in the stiffness variation is to switch from some small level for low excitation frequencies to some higher level for higher excitation frequencies. The trend in the damping variation is to go from some constant level for low frequencies to a small value for high frequencies. These observations can be understood by realizing that for low excitation frequencies, the compressibility of the medium can be ignored and the bearing acts like a journal bearing (the gas is moving from one side of the bearing to the other and vice versa, giving the energy dissipation). For high excitation frequencies we mainly are dealing with gas compression giving a higher stiffness and less damping.
The cross-coupling terms of the bearing matrices show a completely different behavior. In the central position the stiffness matrix and damping matrix are skew symmetric and for the non rotating rotor they theoretically should be zero. In Fig 2.13 some representative characteristics of some gas bearing are shown. In this case we are dealing with a specific, externally pressurized spiral groove bearing. So, in case of gas bearings, it is common to use the dynamic stiffness which relates the complex amplitude of the excitation $\hat{F}$ to the complex amplitude $\hat{u}$ of the response, namely

$$\hat{F} = Z(\omega)\ \hat{u} \quad \Longrightarrow \quad Z(\omega) = Z_k + j\ Z_b = K(\omega) + j\ \omega B(\omega) \tag{2.44}$$

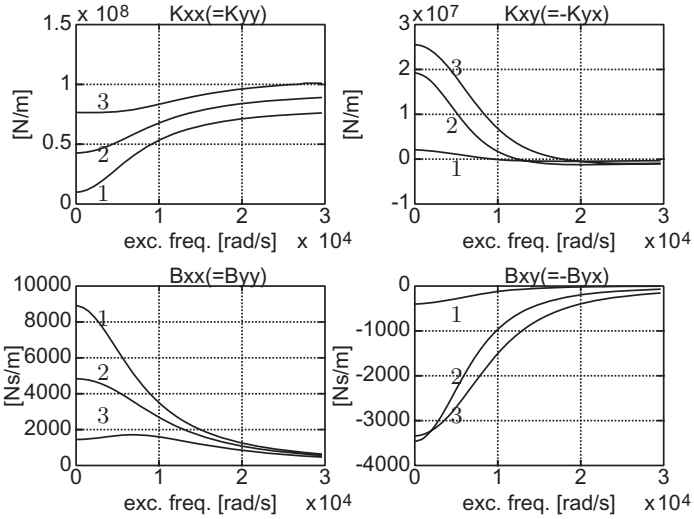


Fig. 2.13. Representative bearing coefficients as function of the excitation frequency ω for a gas bearing at 3 different rotor speeds Ω, (1) 1.0 [RPM], (2)50000 [RPM], (3)100000 [RPM]

Instead of the damping coefficient $B(\omega)$, sometimes the so-called **damping-stiffness** $\{\omega B(\omega)\}$ is used. Just as the regular stiffness it has the dimension $[Nm^{-1}]$.

Fig 2.14(b) gives a representative picture of this (complex) dynamic stiffness (for a fixed rotor speed Ω, namely $\Omega = 1.0$ [rad/s]). Part (a) shows the corresponding (real) stiffness $K = Z_k$ and the (real) damping-stiffness $Z_b = \omega B$. The figures (c) and (d) show the amplitude and argument of this complex stiffness.

The analysis of the dynamic behavior of rotor systems with excitation-frequency-dependent elements such as gas bearings is not an easy job. Especially the investigation of rotor dynamic instability can not simply be done as already illustrated in Chapter 1. A reasonably straightforward and efficient procedure which in that case can be applied will be treated in Section 6.5.

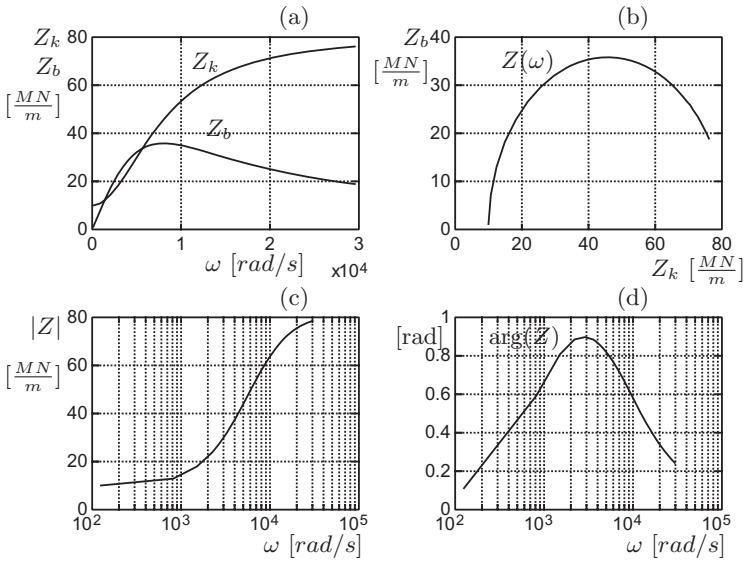


Fig. 2.14. (a) Stiffness-part Z_k and damping-part Z_b; (b) complex dynamic stiffness $Z(\omega)$; (c) amplitude $|Z(\omega)|$; (d) $arg[Z(\omega)]$. Situation for a fixed rotor speed $\Omega = 1.0$ $[rad/s]$

2.7 Electro Magnetic Bearings

In Fig 2.15 the principle of a simple electro magnetic bearing (EMB) is shown. A sensor measures the displacement from the reference position of the rotor shaft. Based on this displacement, a controller (micro processor) generates a control signal and an amplifier transforms this control signal into a control current which is activating the magnet. This control current generates electro magnetic forces which have to push back the rotor to the reference position.
The programmed control action has to take care for stiffness and damping of the rotor support as well as to guarantee the dynamic stability of the rotor system.
This stiffness and damping can be chosen to meet specific design criteria

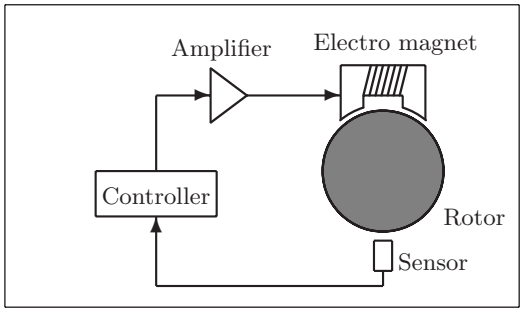


Fig. 2.15. Functional principle of an active electro magnetic bearing

for the dynamic behavior of the system or even changed during running the system in order to anticipate on fluctuating operation conditions.
Electro magnetic bearings are used more and more in industrial situations, for the following reasons

- The contactless support without any need for lubrication enables us to apply them under very extreme operating conditions (e.g. in vacuum)
- Very high rotor speeds are possible
- Very low bearing losses
- A very long lifetime with little maintenance
- By using a (digital) controller a wide range of possibilities for stiffness- and damping selection is possible

- With an accurate sensor an accurate rotor position can be achieved
- The bearing can also give information for a condition monitoring system giving an optimal reliability
- The bearing can even be used for the introduction of additional excitation forces for example for a parameter identification procedure

The following disadvantages should also be mentioned

- The bearing complexity and consequently the high prize
- There is not yet much experience with the application of these bearings in an industrial environment
- The turbulent developments in this area on one side and the the lack of overview on the other side.
- Much research must still be done with respect to safety, reliability, etc.

The basic properties of magnetic bearings such as stiffness, damping, rotor positioning, stability, etc. are strongly determined by the controller which will be used. In the next paragraph some basic principles will be treated. For an in-depth treatment of EMB's see ([Schweitzer at al.-94]).

2.7.1 Magnetic Bearing in a Control Loop

For simplicity the rotor is modeled as a single mass with only one degree of freedom.
In an EMB the magnet in fact is an element of a control loop as shown in Fig 2.16. The rotor can move only in vertical direction with degree of

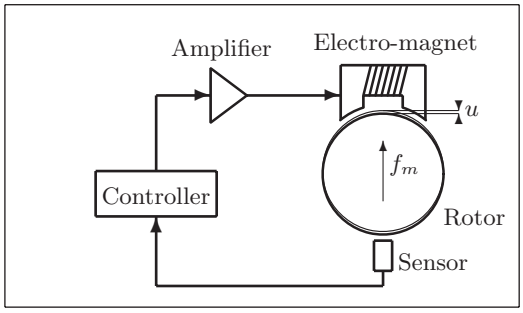


Fig. 2.16. Simplified AMB model

freedom u. For the bearing load we only take into account the weight mg of

the rotor. The primary goal is the stabilizing of the rotor in its equilibrium position $u = 0$. In that case the magnetic force f_m should be in equilibrium with the weight mg, so

$$f_m - mg = 0 \tag{2.45}$$

Each displacement from the equilibrium position is measured with the sensor (frequently from the inductive type). The controller should generate a control signal in such a way that this equilibrium will be stable. The whole control loop is called the magnetic bearing. Of course the rotor material in the neighborhood of the bearing should be ferromagnetic material which sometimes makes special adaptations necessary.
In the EMB the magnetic force plays a central role. This force is completely different from a normal elastic spring force as indicated in Fig 2.17. in this

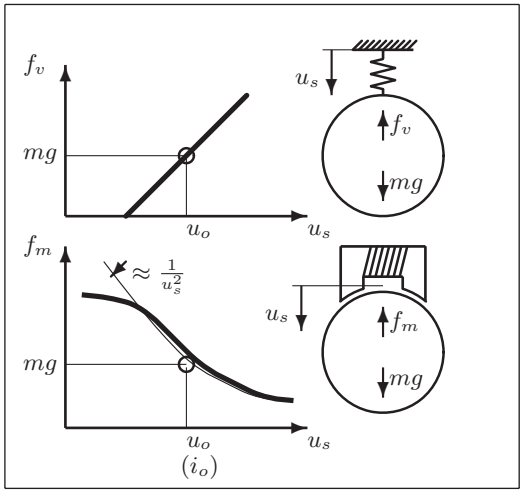


Fig. 2.17. Classical mechanical stiffness versus magnetic stiffness (constant current)

figure the force-displacement relation for an EMB with a constant current, is compared with an ordinary linear spring.
In the equilibrium u_o we have $f_m = mg$ respectively $f_v = mg$. For the (linear) spring the force increases with increasing displacement. Each perturbation from the equilibrium generates a restoring force which makes the equilibrium a stable one.
For the EMB we see a completely different behavior. An increasing displace-

ment will lead to a decreasing force which makes the equilibrium unstable. The sign of the derivative of the force-displacement relation (positive for the spring and negative for the EMB) is determining the stability of the equilibrium.
This force-displacement derivative is called the dynamic stiffness where the equilibrium position is also called the operating point, sometimes defined in terms of the magnet current i_o, see Fig 2.17. This figure gives the force-displacement relation of an EMB for a constant magnet current $i_m = i_o$. The force appears to be approximately proportional to the inverse of the displacement squared. Only for very small air gaps the force will reach a maximum depending on the magnetic properties of the materials and the magnet current.
This magnetic force can now be controlled by changing the magnet current. This force will approximately be proportional to the square of the magnet current (as long as there is no saturation) as shown in Fig 2.18. This means that we are dealing with two non-linear relations, namely the

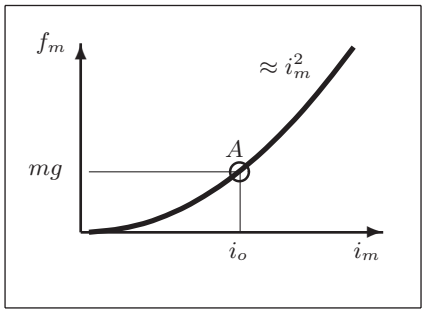


Fig. 2.18. Magnet force f_m as function of the magnet current i_m for a constant air gap

force-displacement relation for a constant magnet current and the force-current relation for a constant air gap.
For the design of a controller it is sufficient to use the linearized relations, so to use the derivatives in the operating point of the bearing. Fig 2.19 shows the linearization of the force f_m as function of the magnet current i_m. We introduce new variables, defined as (small) variations from the operating point namely $f = f_m - f_o$, and $i = i_m - i_o$, so that the linearized function can be written as

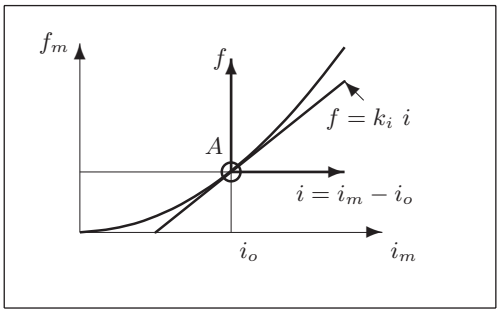


Fig. 2.19. Linearization of the force f_m as function of the magnet current i_m at constant air gap $u_s = u_o$, force f and current i are (small) variations from the operating point

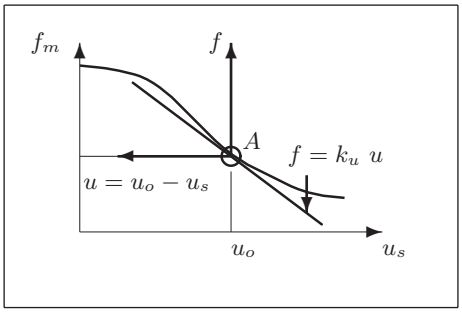


Fig. 2.20. Linearization of the force f_m as function of the displacement u_s at constant magnet current $i_m = i_o$, force f and displacement u are (small) variations from the operating point

$$f = k_i\ i; \qquad k_i = \left[\frac{\partial f_m}{\partial i_m}\right]_{i_m=i_o} \tag{2.46}$$

The quantity k_i is called the force-current factor $[NA^{-1}]$. The same can be done for the force f_m as function of the the displacement u_s such as given in Fig 2.20. In this case we introduce the new variable $u = u_o - u_s$ and we than get

$$f = k_u\ u; \qquad k_u = \left[\frac{-\partial f_m}{\partial u_s}\right]_{u_s=u_o} \tag{2.47}$$

De displacement u from the operating point is called positive when the air gap u_s is decreasing. The positive direction of displacement u and force f are therefore the same. The gradient k_u is called the force-displacement

factor $[Nm^{-1}]$.
Altogether now we get the linearized relation for the force as a function of the displacement as well as the current

$$f(u,i) = k_u\ u + k_i\ i \tag{2.48}$$

It should be remarked again that this relation is only valid for relatively small variations from the operating point. However in practice only in special situations the real nonlinear functions will have to be used.

2.7.2 The Controlled Magnetic Bearing

One of the most important questions of the controller design is the stabilization of the operating point by generating an appropriate linear spring and also some damping to eliminate operational vibrations. In should be remarked that in practice, an active controlled magnetic bearing is much more than just an electronic reproduction of a linear spring and viscous damper, but for the illustration of the principle we can look at this simplified model. We assume that the EMB can simulate a spring-damper pair as given in the

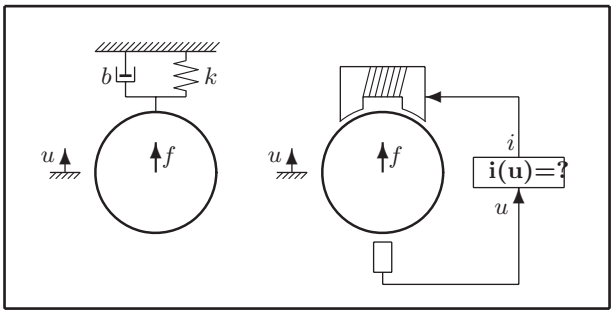


Fig. 2.21. Simple EMB controller simulating a linear spring and viscous damper combination

left part of Fig 2.21, so

$$f = -k\ u - b\ \dot{u} \tag{2.49}$$

Combination of this relation with (2.48) gives

$$i(u) = -\frac{(k+k_u)u + b\dot{u}}{k_i} \tag{2.50}$$

Assuming that all components behave like ideal systems (sensor, amplifier, no noise, etc.) then the dynamic behavior of the left system from Fig 2.21 will be equivalent with the dynamic behavior of the right system, which means it will show a decreasing, oscillating behavior after a certain disturbance.
Some important differences can be recognized between such a passive and active bearing system, namely

- In an EMB the stiffness and damping can be chosen more or less arbitrary or can even be changed during operation
- The operating point u_o can be chosen independent from the stiffness
- Under varying (quasi-static) loads the operating point does not have to change

The basic EMB-controller as defined before (2.50) in fact is just a simple PD-controller with a proportional- (P) and a differentiating feedback (D)

$$P = \frac{k + k_u}{k_i}, \qquad D = \frac{b}{k_i} \tag{2.51}$$

The parameters P and D depend on the stiffness k and damping b. The maximal load to be carried by the EMB and the maximum acceptable displacement for this load give a first estimate for this stiffness.
In general the stiffness k is chosen the same order as the (negative) bearing stiffness k_u. The eigenvalue magnitude in the open and closed loop will than be equal so that the controller will preserve the "speed" of the open-loop system.
The choice for the damping b depends on the choice for the stiffness k. A sufficiently high value for the damping will assure that oscillations will disappear very fast, when the damping is too high we will get a very slow respons. Because in general the velocity signal has a worse signal-to-noise ratio than the displacement signal and a high value for b will amplify this velocity noise, the value for b should not be chosen much larger than corresponding to the critical damping $\xi = b/2\sqrt{km} = 1$. As a nominal value for the dimensionless damping it is often advised to take $\xi \approx 0.5$.

In an EMB with PD-controller a change in the bearing load Δf will result in a change in position $\Delta u = k\Delta f$. In contrast with the mass-spring-damper analogon the EMB can completely compensate for this effect by using a reference-input (also called set-point value). With this set-point the rotor

position can be manipulated in the operational area of the bearing. Fig 2.22 gives a schematic representation of the control scheme and the variables in it. The error-signal being the difference between the measured displacement

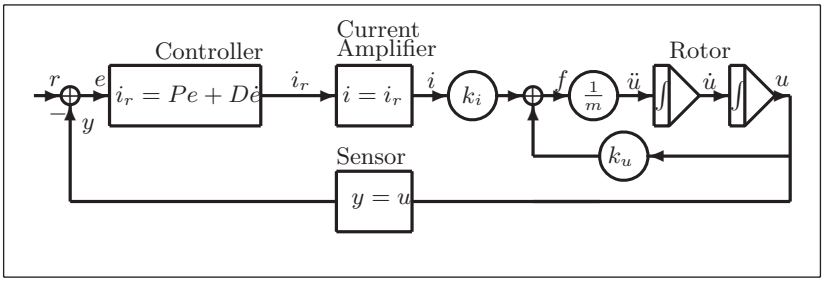


Fig. 2.22. EMB - PD - control scheme with set-point r

y and the set-point value e will be the input for the controller

$$e = r - y \tag{2.52}$$

Just as in (2.50) the output of the controller will be considered as the reference-input for the current i_r, the amplifier therefore should be a current-amplifier. At this moment this so-called magnet-current controller is the most widely used type of controller. In some situations alternative, so-called magnet-voltage control will be applied.
We assume that amplifier and sensor will behave ideal ($y = u,\ i = i_r$). For an unloaded bearing and $r = 0$ all the variables from Fig 2.22 will have a zero value in the set-point. For $r \neq 0$ a different operating point will result, however the velocity $\dot{u}$, acceleration $\ddot{u}$ and $\dot{e}$ will remain zero, so

$$k_u\ u = -k_i\ i \qquad \text{and} \qquad i = P\ e = P(r - u) \tag{2.53}$$

Using (2.51), we can find for the new operating point u_r corresponding to reference r

$$u_r = r\ \frac{k + k_u}{k_i} \tag{2.54}$$

Each change in u_r will lead to another operating point and other values for k_u and k_i.
In real practice it is often desired that the rotor position does not change

with the bearing-load. This means that we need an integrating action in the controller also. Each position-change u will than be integrated in time and added to the feedback until u has become zero again. (see Fig 2.23). The time-constant of the integrating action is T_i and this compensating

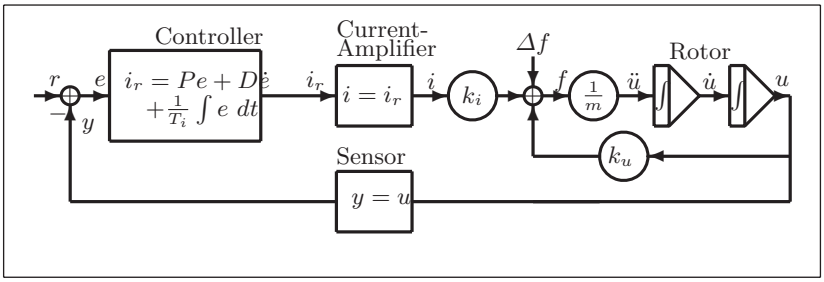


Fig. 2.23. EMB - PID - control scheme and load variation Δf

action can not be chosen arbitrary fast without also changing the stiffness and damping values.
It will be clear that this load-independency will be more and more difficult to maintain when the bearing load is changing rapidly in magnitude and/or direction. In that case the Frequency Respons Function (FRF) of the rotorsystem with controller, together with the reference input and noise signals will have to be analyzed in order to generate the optimal control-concept. (see e.g. [Doyle et al.-92]) In practice we normally are dealing with flexible rotors (an example is shown in Fig 2.24) with several degrees of freedom, several bearings, imperfect measurement data, etc. The given approach with 1 input and 1 output will than fail. A multivariate control methodology with state-reconstruction will then have to be applied (see e.g. [Kok-90]).

Finally it can be concluded that the application of active magnetic bearings for the dynamic control of rotorsystems is only possible by integration of dynamics and control. For a comprehensive treatment we refer to ([Schweitzer at al.-94]).

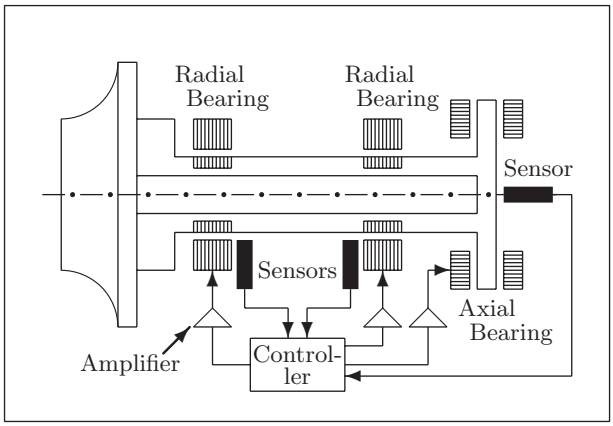
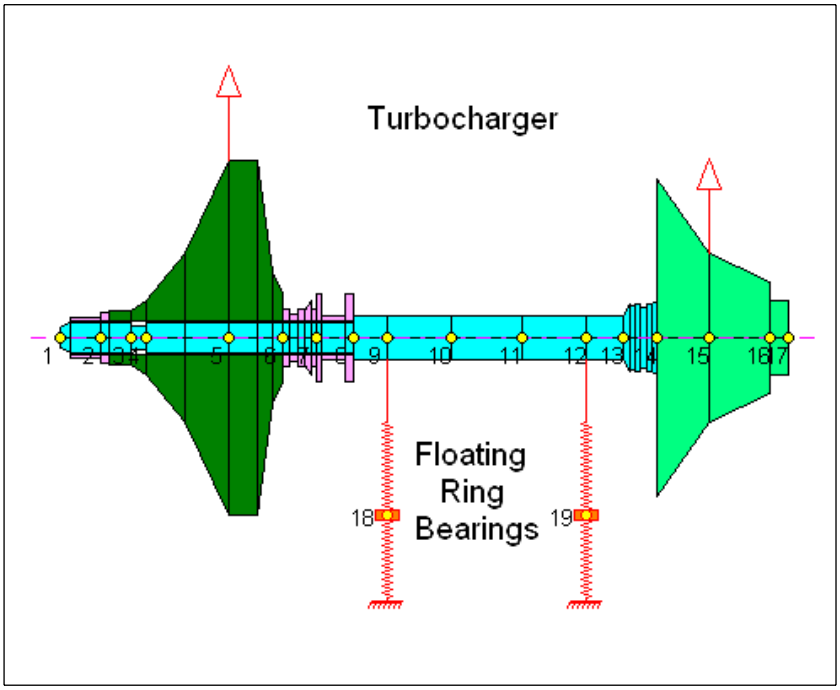


Fig. 2.24. Example of a realistic rotor system with 2 magnetic bearings

3

Finite Element Method



3.1 Introduction

In Chapter 1, the basic principles of a rotordynamic analysis have been introduced. This was done by studying very simple models (generally two-degree-of-freedom models of a Jeffcott-rotor type system). The models consisted of a rigid mass element, supported by linear springs and linear viscous dampers to represented shaft flexibility or fluid film bearings. For such bearings, more accurate models as used very frequently in the real day practise of rotordynamics, have been presented in Section 2.4. As could be seen in Chapter 1, the analysis of these fundamental, two-degree-of-freedom, rotor-bearing models already was not so easy and in some cases even a special approach had to be followed to get results. When dealing with practical rotor-dynamic problems, such extremely simplified models in many cases will not be capable of describing or understanding of the phenomena as met in the operation of the machine. This means that more complicated models will have to be developed and analyzed. The evaluation of such models will only be possible by using a computer and up to date numerical procedures. In Mechanical Engineering the Finite Element Method **FEM** is a well-known approach to evaluate linear (and also non-linear) complex structural mechanics problems. Also in this chapter this approach will be followed for the development of a numerical method for the analysis of complex (= multi-degree-of-freedom) rotor-bearing systems. We will restrict ourselves to **linear** models. For a large class of practical situations these linear models will appear to be sufficiently representative for understanding the rotor-dynamic problem. For high-speed rotors under extreme load conditions such an assumption of system linearity (especially in the bearings) might not be allowed anymore. Then a (much more complicated) non-linear analysis will have to be performed. (see for example [Vorst et al.-95] and [Vrande et al.-99]).

In the following sections, two fundamental element types used in rotor dynamic models will be presented (in addition to the bearing elements from Chapter 2). These elements are the rigid disk element and a conical shaft element (based on a Timoshenko beam model). Finally the assemblage proces to get a multi-degree-of-freedom model for the system will be presented. For the analysis of **linear** rotor bearing models, different studies have been carried out the last 10 years within a $\boldsymbol{M_{AT}L_{AB}}$ -environment (see for example [Jacobs-89], [Bot-93], [Geerts-95a], [Geerts-95b], [Bartholomeus-93], [Schie-91]). This gradually has come into being a $\boldsymbol{M_{AT}L_{AB}}$ -Toolbox

$\boldsymbol{R_O D_Y}$ for rotor-bearing systems. The theory in this chapter is closely related to this toolbox. Also in the next chapter the toolbox has been used intensively for the evaluation of some practical problems.

3.2 Rigid Disk Element

In rotating machinery such as turbines, pumps or gear-boxes, fan- and propeller-type structural parts are always present. In general such parts might be assumed to be rigid elements, introducing dynamic forces such as gyroscopic ones into the dynamical model. Because they never will be perfectly rotationally symmetric, they will also introduce things like unbalance forces into the model. Not only such an unbalance will be an important source of forces which will excite the system but for example also a skewness of a disk with respect to a shaft can lead to unexpected responses.
In this section we will derive a model for such a rigid disk element including mass unbalance and disk-skewness. The mass unbalance will be related to an eccentricity vector $\vec{\epsilon}$ with amplitude ϵ. This vector defines the center of mass M of the disk with respect to the geometrical center G of the disk. This vector rotates with a constant rotational velocity (the rotorspeed Ω) in in the $X - Z$ plane of the fixed basis. At $t = 0$ the angle between the eccentricity vector and the fixed Z-axis is called α as shown in Fig. 3.1. A

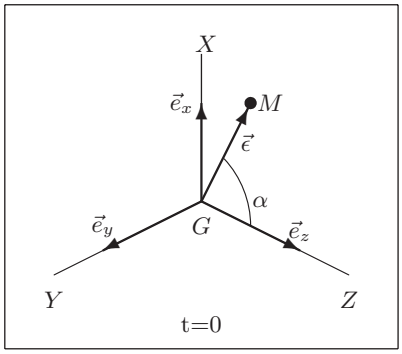


Fig. 3.1. Disk center-of-mass at t=0

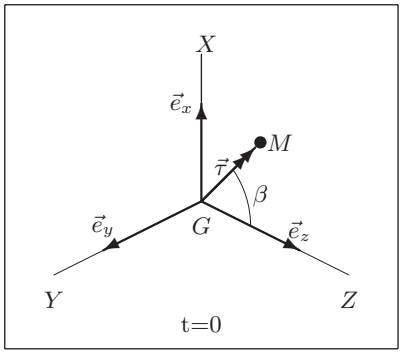


Fig. 3.2. Disk skewness at t=0

possible disk skewness will be introduced by the skewness vector $\vec{\tau}$. Also this vector rotates with the constant rotorspeed Ω. The amplitude of the skewness vector is called τ and at $t = 0$, the angle between the vector $\vec{\tau}$ and the Z-axis is defined as β, as indicated in Fig. 3.2. The matrix representations $\underline{\epsilon}$ and $\underline{\tau}$ of the vectors $\vec{\epsilon}$ and $\vec{\tau}$ for arbitrary time t with respect to the fixed basis can be written as

$$\underline{\epsilon} = \begin{bmatrix} \epsilon \sin(\Omega t + \alpha) \\ 0 \\ \epsilon \cos(\Omega t + \alpha) \end{bmatrix}; \qquad \underline{\tau} = \begin{bmatrix} \tau \sin(\Omega t + \beta) \\ 0 \\ \tau \cos(\Omega t + \beta) \end{bmatrix} \tag{3.1}$$

For the derivation of the equations of motion of the rigid disk we assume that the disk is rotating around the fixed Y-axis with a rotorspeed Ω, where this rotorspeed Ω in general will be large. First we will look at the rotations of the rigid disk and after that we will look at the simpler translational equations.

For the derivation of the equations of motion we use the fixed reference frame $\{X, Y, Z\}$ with the vector basis $\{\vec{e}_x, \vec{e}_y, \vec{e}_z\}$ but also a second coordinate system $\{X^L, Y^L, Z^L\}$ with the vector basis $\{\vec{\gamma}_x, \vec{\gamma}_y, \vec{\gamma}_z\}$. This second vector basis has the same origin as the fixed basis and the X^L- and Z^L-axes stay in the plane of the disk but do not participate in the rotation with the constant rotorspeed Ω. This second vector basis is called the **body reference frame**. The inertia tensor of the rigid disk with respect to this frame will be constant due to the assumed rotational symmetry of the disk. We

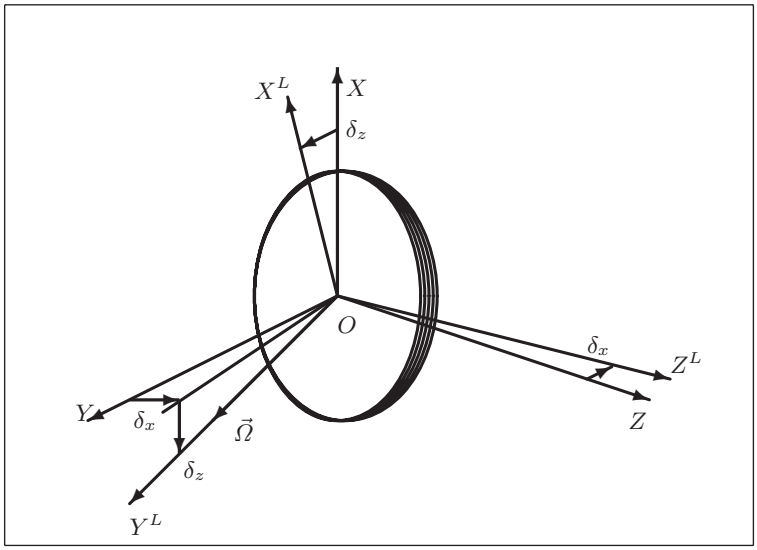


Fig. 3.3. Fixed reference frame and body reference frame

assume that the rotations of the disk (except for the large rotor-speed Ω)

are small enough to allow for a linear theory. For the total rotation of the disk we use the scalded **Bryant angles** δ_x and δ_z. (see also appendix A). Easily can be seen that

$$\begin{aligned} \delta_x &= \theta_x + \tau \sin(\Omega t + \beta) \\ \delta_z &= \theta_z + \tau \cos(\Omega t + \beta) \end{aligned} \tag{3.2}$$

where θ_x respectively θ_z are the rotation of the shaft cross section where the disk is connected (the deformation rotations). The total rotation of the disk except for the constant rotational speed Ω around the Y-axis is called the *frame-rotation*. This frame rotation is defined by the rotation vector $\vec{\delta}$ and its matrix representation $\underline{\delta}$ with respect to the fixed basis is:

$$\underline{\delta} = \begin{bmatrix} \delta_x \\ 0 \\ \delta_z \end{bmatrix} \tag{3.3}$$

This rotation vector can also be associated with a rotation matrix $\underline{R}$

$$\underline{R} = \begin{bmatrix} 1 & -\delta_z & 0 \\ \delta_z & 1 & -\delta_x \\ 0 & \delta_x & 1 \end{bmatrix} = \underline{I} + \underline{\Delta} \tag{3.4}$$

where $\underline{I}$ is the unity matrix and $\underline{\Delta}$ a **skew-symmetric** matrix

$$\underline{\Delta} = \begin{bmatrix} 0 & -\delta_z & 0 \\ \delta_z & 0 & -\delta_x \\ 0 & \delta_x & 0 \end{bmatrix} \tag{3.5}$$

For an arbitrary vector $\vec{x}$ with its matrix representations $\underline{x}$ and $\underline{x}^L$ with respect to the fixed and rotating frame we can write

$$\underline{x} = \underline{R}\ \underline{x}^L \qquad \underline{x}^L = \underline{R}^T\ \underline{x} \qquad \underline{R}^T\ \underline{R} = \underline{I} \tag{3.6}$$

Next we write the absolute rotational velocity vector $\vec{\omega}_a$ as the sum of the frame rotation vector $\vec{\omega}_f$ and the relative rotation vector $\vec{\omega}_r$. The latter then prescribes the rotation with the (large) constant rotational velocity Ω around the $Y^L - as$. So,

$$\vec{\omega}_a = \vec{\omega}_f + \vec{\omega}_r \tag{3.7}$$

where

$$\underline{\omega}_f = \begin{bmatrix} \dot{\delta}_x \\ 0 \\ \dot{\delta}_z \end{bmatrix} = \begin{bmatrix} \Omega\tau\cos(\Omega t + \beta) + \dot{\theta}_x \\ 0 \\ -\Omega\tau\sin(\Omega t + \beta) + \dot{\theta}_z \end{bmatrix} \quad \text{and} \quad \underline{\omega}_r = \begin{bmatrix} 0 \\ \Omega \\ 0 \end{bmatrix} \tag{3.8}$$

For the derivation of the equations of motion we start with the *angular momentum equation* with respect to the fixed point O of the disk. The matrix representation of this vector equation with respect to the fixed frame reads

$$\underline{M} = \frac{d}{dt}(\underline{D}) = \frac{d}{dt}(\underline{J}\ \underline{\omega}_a) \tag{3.9}$$

In this equation the symbol $\underline{J}$ is used for the matrix representation of the inertia tensor $\mathcal{J}$ with respect to the fixed frame. The problem is that this inertia-matrix is time-dependent due to the rotation of the disk. We therefore switch to a formulation where the matrix representation $\underline{J}^L$ of this tensor with respect to the body-fixed frame is playing a role because this matrix is time-independent. (The constant rotational velocity Ω does not change the principal moments of inertia of the rotational symmetric disk). For the angular momentum vector $\vec{D}$ the transformation holds

$$\underline{D} = \underline{R}\ \underline{D}^L \tag{3.10}$$

Substitution of this relation in (3.9) gives

$$\underline{M} = \dot{\underline{R}}\ \underline{D}^L + \underline{R}\ \frac{d}{dt}(\underline{D}^L) \tag{3.11}$$

For $\underline{D}^L$ we can write

$$\underline{D}^L = \underline{J}^L\ \underline{\Omega}_a^L \tag{3.12}$$

and because $\underline{J}^L$ is time-independent

$$\underline{M} = \dot{\underline{R}}\ \underline{D}^L + \underline{R}\ \underline{J}^L\ \dot{\underline{\omega}}_a^L \tag{3.13}$$

For switching back to the fixed frame we use

$$\underline{D}^L = \underline{R}^T\ \underline{D} \quad \text{and} \quad \dot{\underline{\omega}}_a^L = \underline{R}^T\ \dot{\underline{\omega}}_a \tag{3.14}$$

which leads to

$$\underline{M} = \dot{\underline{R}}\ \underline{R}^T\ \underline{D} + \underline{R}\ \underline{J}^L\ \underline{R}^T\ \dot{\underline{\omega}}_a \tag{3.15}$$

The matrix $\underline{\Omega}_f := \dot{\underline{R}}\ \underline{R}^T$ appears to be a skew-symmetric matrix, and it can be associated with a column $\underline{\omega}_f$

$$\underline{\dot{R}}\ \underline{R}^T := \underline{\Omega}_f = \begin{bmatrix} 0 & -\dot{\delta}_z & 0 \\ \dot{\delta}_z & 0 & -\dot{\delta}_x \\ 0 & \dot{\delta}_x & 0 \end{bmatrix}; \quad \text{and} \quad \underline{\omega}_f = \begin{bmatrix} \dot{\delta}_x \\ 0 \\ \dot{\delta}_z \end{bmatrix} \tag{3.16}$$

The inertia matrix $\underline{J}$ transforms as

$$\underline{J} = \underline{R}\ \underline{J}^L\ \underline{R}^T \tag{3.17}$$

so that we can write for matrix $\underline{J}^L$

$$\underline{J}^L = \begin{bmatrix} J_t & 0 & 0 \\ 0 & J_p & 0 \\ 0 & 0 & J_t \end{bmatrix} \tag{3.18}$$

where J_t is the diametral mass-moment of inertia of the disk to any diameter in the plane of the disk through the center of gravity and J_p the axial or polar mass-moment of inertia around an axis perpendicular to the plane of the disk. It is now easy to see that

$$\underline{J} = \underline{J}^L \tag{3.19}$$

for sufficiently small angles δ_x and δ_z. (quadratic terms have been neglected). So, finally for (3.15) we get

$$\underline{M} = \underline{\Omega}_f \underline{J}\ \underline{\omega}_a + \underline{J}\ \underline{\dot{\omega}}_a \tag{3.20}$$

Using (3.2),(3.7),(3.8) and (3.16) leads to

$$\begin{aligned} M_x &= J_t \ddot{\theta}_x - J_p \Omega \dot{\theta}_z - \Omega^2 \tau \sin(\Omega t + \beta)(J_t - J_p) \\ M_y &= 0 \\ M_z &= J_t \ddot{\theta}_z + J_p \Omega \dot{\theta}_x - \Omega^2 \tau \cos(\Omega t + \beta)(J_t - J_p) \end{aligned} \tag{3.21}$$

The middle (almost trivial) equation can also be removed because it does not give any information except that for a constant rotorspeed of this rigid body around the y-axis no external moment is necessary. In (3.21) the first terms of the right-hand-side are the usual terms for the rotational acceleration, the second terms take account for the gyroscopic effect of this body while the third terms can be seen as additional moments acting on the disk due to the initial skewness of the disk with respect to the shaft. If $J_t = J_p$ such as in case of a sphere-type rigid element, this skewness effect disappears.

An alternative way to generate the equations of motion for the rotation of the disk is to start from the expression for the kinetic energy of the disk and then applying Lagrange's equations. For this kinetic energy we can write

$$T = \frac{1}{2}\,\underline{\omega}_a^T \underline{J}\,\underline{\omega}_a = \frac{1}{2}\,\underline{\omega}_a^{L^T} \underline{J}^L \underline{\omega}_a^L \tag{3.22}$$

Because in the application of Lagrange's equations we will be dealing with a time differentiation we use the second expression because then the inertia matrix is time independent. We again will use the Bryant-angles $\{\delta_x, \delta_z\}$. Then the rotation matrix R becomes (see appendix A)

$$\underline{R} = \begin{bmatrix} c_z & -s_z & 0 \\ c_x s_z & c_x c_z & -s_x \\ s_x s_z & s_x c_z & c_x \end{bmatrix} \tag{3.23}$$

where we used the abbreviations

$$s_x = \sin(\delta_x); \quad c_x = \cos(\delta_x); \quad s_z = \sin(\delta_z); \quad c_z = \cos(\delta_z)$$

We have seen before that we can find $\underline{\omega}_f$ from the skew-symmetric matrix $\underline{\Omega}_f = \underline{\dot{R}}\,\underline{R}^T$, giving

$$\underline{\Omega}_f^L = \underline{R}^T \underline{\Omega}_f \underline{R} = \underline{R}^T \underline{\dot{R}}\,\underline{R}^T \underline{R} = \underline{R}^T \underline{\dot{R}} \tag{3.24}$$

Using (3.23) then leads to

$$\underline{\Omega}_f^L = \begin{bmatrix} 0 & -\dot{\delta}_z & -\dot{\delta}_x s_z \\ \dot{\delta}_z & 0 & -\dot{\delta}_x c_z \\ \dot{\delta}_x s_z & \dot{\delta}_x c_z & 0 \end{bmatrix} \tag{3.25}$$

The column $\underline{\omega}_f^L$ associated with this skew-symmetric matrix is

$$\underline{\omega}_f^L = \begin{bmatrix} \dot{\delta}_x \cos(\delta_z) \\ -\dot{\delta}_x \sin(\delta_z) \\ \dot{\delta}_z \end{bmatrix} \tag{3.26}$$

We again assume that the angles δ_x respectively δ_z are small which then leads to the absolute rotational velocity vector $\underline{\omega}_a^L$ with respect to the body-fixed frame

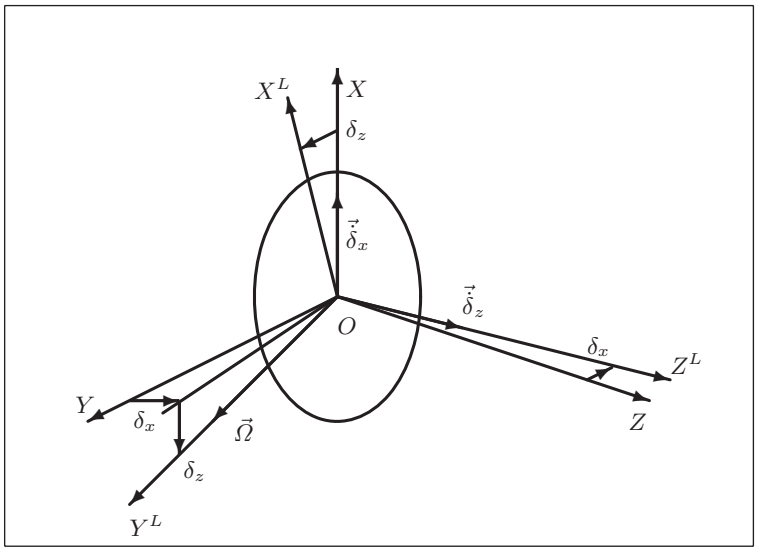


Fig. 3.4. Angular velocity vectors for the rigid disk

$$\underline{\omega}_a^L = \begin{bmatrix} \dot{\delta}_x \\ \Omega - \dot{\delta}_x \delta_z \\ \dot{\delta}_z \end{bmatrix} \tag{3.27}$$

The result of (3.27) can also directly be derived from Fig. 3.4, by looking at the components of the rotational velocity vectors $\vec{\dot{\delta}}_x$, $\vec{\dot{\delta}}_z$ and $\vec{\Omega}$ in the body-fixed frame. If we next use (3.18) we get for the kinetic energy of the disk

$$T = \frac{1}{2}[J_t(\dot{\delta}_x)^2 + J_t(\dot{\delta}_z)^2 + J_p(\Omega^2 - 2\Omega\delta_z\dot{\delta}_x)] \tag{3.28}$$

where the higher order terms $(\delta_z\dot{\delta}_x)^2$ have been ignored. The application of Lagrange's equations then leads to the following contribution

$$\frac{d}{dt}(T,_{\dot{q}}) - (T,_q) = \begin{bmatrix} J_t\ddot{\delta}_x - J_p\Omega\dot{\delta}_z \\ 0 \\ J_t\ddot{\delta}_z + J_p\Omega\dot{\delta}_x \end{bmatrix} \tag{3.29}$$

If we introduce (3.8) in this expression we arrive at the terms as already given before in (3.21).

The equations of motion for the rigid disk for the translations are much easier. Starting with Newton's second law

$$\underline{F} = m\ \underline{\ddot{u}}_a \tag{3.30}$$

where $\underline{\ddot{u}}_a$ are the absolute accelerations of the center of gravity of the disk with respect to the fixed frame. Then we can write

$$\underline{u}_a = \begin{bmatrix} u + \epsilon \sin(\Omega t + \alpha) \\ 0 \\ w + \epsilon \cos(\Omega t + \alpha) \end{bmatrix} \tag{3.31}$$

where u and v are the translations of the geometrical center of the disk. This leads to

$$\begin{aligned} F_x &= m\ddot{u} - m\Omega^2 \epsilon \sin(\Omega t + \alpha) \\ F_y &= 0 \\ F_z &= m\ddot{w} - m\Omega^2 \epsilon \cos(\Omega t + \alpha) \end{aligned} \tag{3.32}$$

If we define the column of degrees of freedom of the disk as $\underline{q}_d^T = [u, w, \theta_x, \theta_z]$, the total set of equations of motion for the rigid disk finally can be written as

$$\underline{M}_d\ \underline{\ddot{q}}_d + \underline{B}_d\ \underline{\dot{q}}_d = \underline{F}_d + \underline{f}_{unbalance} \tag{3.33}$$

with

$$\underline{M}_d = \begin{bmatrix} m & 0 & 0 & 0 \\ 0 & m & 0 & 0 \\ 0 & 0 & J_t & 0 \\ 0 & 0 & 0 & J_t \end{bmatrix}; \quad \underline{B}_d = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\Omega J_p \\ 0 & 0 & \Omega J_p & 0 \end{bmatrix}$$

$$\underline{F}_d = \begin{bmatrix} F_x \\ F_z \\ M_x \\ M_z \end{bmatrix}; \ \underline{f}_{unbalance} = \begin{bmatrix} m\Omega^2 \epsilon \sin(\Omega t + \alpha) \\ m\Omega^2 \epsilon \cos(\Omega t + \alpha) \\ (J_t - J_p)\Omega^2 \tau \sin(\Omega t + \beta) \\ (J_t - J_p)\Omega^2 \tau \cos(\Omega t + \beta) \end{bmatrix} \tag{3.34}$$

Note

Goal: some understanding about Gyroscopic effect

Let us look at a simple situation where the gyroscopic effects plays an essential role, namely in staying upright when riding a bicycle. Although it is not the only factor in the *Biking Stability* it is certainly an important one.
Let us assume that our front wheel can be seen as a rigid disk with a fixed shaft (no translations in x- and z-direction) as shown in next picture. We further assume a large rotational speed Ω around the

Fig. 3.5. Frontwheel of a bike as gyroscopic system

y-axis (assuming we are driving forward).
For the equations of motion we then get:

$$\begin{bmatrix} J_t & 0 \\ 0 & J_t \end{bmatrix} \begin{bmatrix} \ddot{\theta}_x \\ \ddot{\theta}_z \end{bmatrix} + \begin{bmatrix} 0 & -\Omega J_p \\ \Omega J_p & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_x \\ \dot{\theta}_z \end{bmatrix} = \begin{bmatrix} M_x \\ M_z \end{bmatrix}$$

Suppose we apply a small harmonic moment M_z around the Z-axis with small amplitude a and low frequency ω_a:

$$M_z = a \sin(\omega_a \ t) = Im[a \ e^{j \ \omega_a \ t}]$$

then we may write for the respons $\underline{q}^T(t) = [\theta_x(t); \theta_z(t)]$:

$$\underline{q}(t) = Im \left[\begin{bmatrix} \hat{\theta_x} \\ \hat{\theta_z} \end{bmatrix} e^{j\omega_a t} \right]$$

The solution follows from:

$$\begin{bmatrix} \hat{\theta_x} \\ \hat{\theta_z} \end{bmatrix} = \frac{1}{\omega_a^4 J_t^2 - \omega_a^2 \Omega^2 J_p^2} \begin{bmatrix} -\omega_a^2 J_t & ; & j\Omega\omega_a J_p \\ -j\Omega\omega_a J_p & ; & -\omega_a^2 J_t \end{bmatrix} \begin{bmatrix} 0 \\ a \end{bmatrix}$$

For simplicity we assume that the front wheel can be seen as a rigid ring with mass m and radius R for which $J_p = 2J_t = mR^2$. The complex amplitudes then become:

$$\begin{bmatrix} \hat{\theta_x} \\ \hat{\theta_z} \end{bmatrix} = \frac{2a}{\omega_a J_p[\omega_a^2 - 4\Omega^2]} \begin{bmatrix} 2j\Omega \\ -\omega_a \end{bmatrix}$$

Some remarks:

- we get an undefined solution (going to infinity) when $\omega_a = \pm 2\Omega$.
- for $\Omega \gg \omega_a$ we see that as approximation $\hat{\theta_x} \approx -aj/(J_p\Omega\omega_a)$ and $\hat{\theta_z} \approx 0$. This means that a harmonic moment $M_z(t) = a\sin(\omega_a\ t)$ around the z-axis will give a rotation around the x-axis:

  $$\theta_x(t) = \frac{-a}{J_p\Omega\omega_a}\cos(\omega_a\ t)$$

  So (for large Ω and J_p) a harmonic moment around the Z-axis will lead to a small harmonic motion around the X-axis, which is 90 degrees out of fase with the excitation moment. This allows us in keeping our upright position.

End of note

3.3 Conical Beam Element

Classical rotor dynamic models usually consist of long slender shafts with uniform cross section, a number of rigid disks and supported by bearings. For the shaft elements a constant cross-section, Euler-beam type element without shear deformation was used. If the cross section was not constant over some part of the shaft one could take some average diameter or introduce a staircase-type part consisting of very short beam-elements with changing diameter. This resulted in inaccurate results and/or an inefficient large number of degrees of freedom. Therefore in this section a Timoshenko beam-element will be presented with linearly changing cross-section. The formulation contains the translational as well as the rotational inertial effects, shear deformation and also the gyroscopic effects of the beam-element. In order to be able to use the more or less standard Finite Element formulation as used in rotor dynamics, the 4 additional degrees of freedom which will show up due to the introduction of shear deformation will be removed using a so-called **Guyan** reduction technique. This will lead to an efficient and accurate beam-element with two nodes. Each node contains two translations and two rotations, so the number of degrees of freedom for a single element is 8.

3.3.1 Degrees of Freedom and Interpolationfunctions

The conical beam element has been shown schematically in Fig 3.6. The

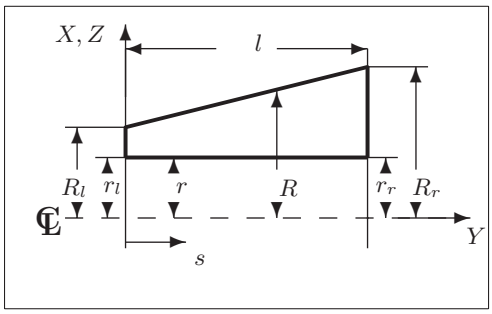


Fig. 3.6. Axial cross-section of the conical beam-element

circular cross section of the element is fully determined by the inner- and outer radius of the left end (r_l and R_l), respectively the inner and outer radius of the right end (r_r and R_r). We will use a dimensionless axial coordinate ξ, defined as $\xi = s/l$. The inner- and outer radius (r resp. R) of an arbitrary cross section can then be written as

$$\begin{aligned} r &= r_l(1-\xi) + r_r\xi \\ R &= R_l(1-\xi) + R_r\xi \end{aligned} \tag{3.35}$$

Additionally, we introduce dimensionless scalars for the ratio of inner- and outer radii, namely $\nu = r_r/r_l$, respectively $\sigma = R_r/R_l$. (3.35) can then be written as

$$r = r_l[1 + (\nu - 1)\xi] \quad \text{and} \quad R = R_l[1 + (\sigma - 1)\xi] \tag{3.36}$$

The area of an arbitrary cross section now reads

$$A = \pi(R^2 - r^2) = A_l(1 + \alpha_1\xi + \alpha_2\xi^2) \tag{3.37}$$

where

$$\begin{aligned} A_l &= \pi(R_l^2 - r_l^2) \\ \alpha_1 &= 2[R_l^2(\sigma-1) - r_l^2(\nu-1)]/(R_l^2 - r_l^2) \\ \alpha_2 &= [R_l^2(\sigma-1)^2 - r_l^2(\nu-1)^2]/(R_l^2 - r_l^2) \end{aligned} \tag{3.38}$$

Similarly, an expression for the moments of inertia of the cross section can be derived, leading to a 4th-order polynomial in ξ

$$I = \pi(R^4 - r^4)/4 = I_l[1 + \delta_1\xi + \delta_2\xi^2 + \delta_3\xi^3 + \delta_4\xi^4] \tag{3.39}$$

with the coefficients

$$\begin{aligned} I_l &= \pi(R_l^4 - r_l^4)/4 \\ \delta_1 &= 4[R_l^4(\sigma-1) - r_l^4(\nu-1)]/(R_l^4 - r_l^4) \\ \delta_2 &= 6[R_l^4(\sigma-1)^2 - r_l^4(\nu-1)^2]/(R_l^4 - r_l^4) \\ \delta_3 &= 4[R_l^4(\sigma-1)^3 - r_l^4(\nu-1)^3]/(R_l^4 - r_l^4) \\ \delta_4 &= [R_l^4(\sigma-1)^4 - r_l^4(\nu-1)^4]/(R_l^4 - r_l^4) \end{aligned} \tag{3.40}$$

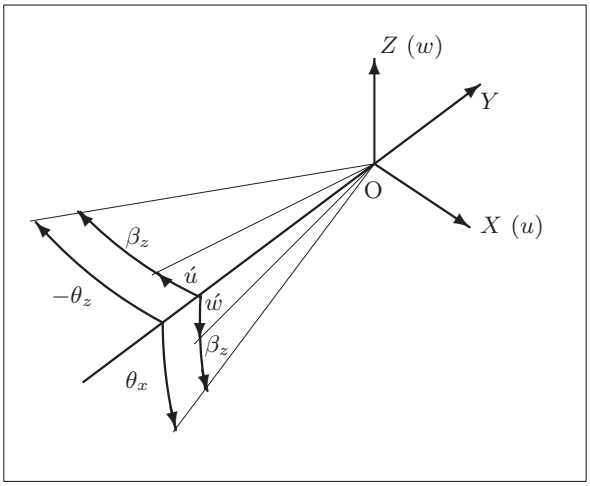


Fig. 3.7. The fixed reference frame and the chosen degrees of freedom

For the description of the deformation of the beam we use the fixed X,Y,Z-frame as shown in Fig. 3.7. The Y-axis coincides with the undeformed center line of the beam. For the deformation of the originally straight element we apply 12 degrees of freedom, namely two displacements (u, w), two rotations (θ_x, θ_z) and finally two shear angles (β_x, β_z) for each of the two nodal points (ends) of the beam element. The column of degrees of freedom then reads

$$\underline{q}^T = [u_l, w_l, \theta_{xl}, \theta_{zl}, u_r, w_r, \theta_{xr}, \theta_{zr}, \beta_{xl}, \beta_{zl}, \beta_{xr}, \beta_{zr}] \tag{3.41}$$

The first 8 degrees of freedom are identical to the set of degrees of freedom as normally used in beam elements for rotordynamic applications. The last 4 degrees of freedom represent the shear deformation of the beam. They are put at the end so that finally they can easily be removed using the Guyan elimination process. The rotation of the cross section of the beam can be written as (see Fig 3.8)

$$-\theta_z = \acute{u} + \beta_x \qquad \text{and} \qquad \theta_x = \acute{w} + \beta_z \tag{3.42}$$

where the $\acute{}$ symbol has been introduced to define the differentiation with respect to the axial coordinate s, so

$$\acute{u} = \frac{d\ u}{ds} \qquad \acute{w} = \frac{d\ w}{ds} \tag{3.43}$$

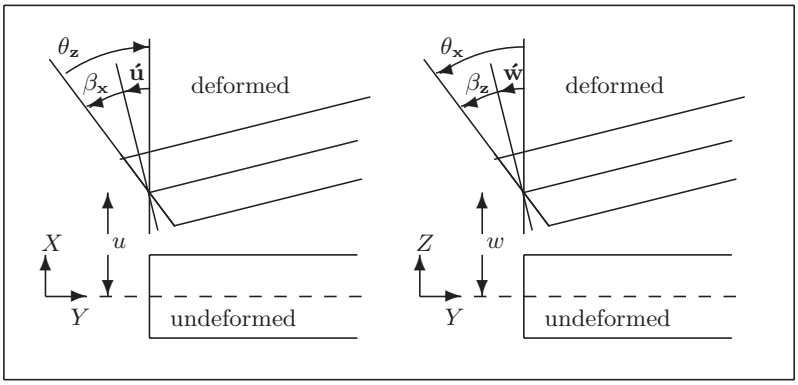


Fig. 3.8. Deformed cross-section of the beam

In Fig 3.8 the additional rotation of the beam cross section due to the shear deformation is clearly shown. The transversal displacements of an arbitrary point of the neutral axis can be expressed in the nodal degrees of freedom using the interpolation functions

$$\begin{bmatrix} u(s,t) \\ w(s,t) \end{bmatrix} = \underline{\psi}(\xi)\ \underline{q}(t) \tag{3.44}$$

where the interpolation matrix $\underline{\psi}$ has the structure

$$\underline{\psi} = \begin{bmatrix} \psi_1 & 0 & 0 & -\psi_2 & \psi_3 & 0 & 0 & -\psi_4 & -\psi_2 & 0 & -\psi_4 & 0 \\ 0 & \psi_1 & \psi_2 & 0 & 0 & \psi_3 & \psi_4 & 0 & 0 & -\psi_2 & 0 & -\psi_4 \end{bmatrix} \tag{3.45}$$

with

$$\begin{aligned} \psi_1 &= 1 - 3\xi^2 + 2\xi^3 \\ \psi_2 &= l(\xi - 2\xi^2 + \xi^3) \\ \psi_3 &= 3\xi^2 - 2\xi^3 \\ \psi_4 &= l(-\xi^2 + \xi^3) \end{aligned}$$

It will be clear that these interpolation functions are based on approximating the beam center line displacement by a cubic function in the axial coordinate. Using (3.44) and (3.45) and additionally assuming that the shear deformation will be approximated by a linear interpolation, the rotation of the cross section can be written as

$$\begin{bmatrix} \theta_x(s,t) \\ \theta_z(s,t) \end{bmatrix} = \underline{\phi}(\xi)\ \underline{q}(t) \tag{3.46}$$

with

$$\underline{\phi} = \begin{bmatrix} 0 & -\phi_1 & \phi_2 & 0 & 0 & \phi_1 & \phi_3 & 0 & 0 & \phi_4 & 0 & \phi_4 \\ \phi_1 & 0 & 0 & \phi_2 & -\phi_1 & 0 & 0 & \phi_3 & -\phi_4 & 0 & -\phi_4 & 0 \end{bmatrix} \tag{3.47}$$

with

$$\begin{aligned} \phi_1 &= 6(\xi - \xi^2) \\ \phi_2 &= 1 - 4\xi + 3\xi^2 \\ \phi_3 &= -2\xi + 3\xi^2 \\ \phi_4 &= 3\xi - 3\xi^2 \end{aligned}$$

For the shear deformation we then can write

$$\begin{bmatrix} \beta_x(s,t) \\ \beta_z(s,t) \end{bmatrix} = \underline{\chi}(\xi)\ \underline{q}(t) \tag{3.48}$$

with

$$\underline{\chi} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \chi_1 & 0 & \chi_2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \chi_1 & 0 & \chi_2 \end{bmatrix} \tag{3.49}$$

and

$$\begin{aligned} \chi_1 &= 1 - \xi \\ \chi_2 &= \xi \end{aligned}$$

3.3.2 The Element-equations

The element equations will be determined using Lagrange's equations. Therefore we need the expressions for the potential and kinetic energy of the beam section.

The potential energy

For a Timoshenko beam the potential energy can be written as:

$$E_p = \frac{1}{2} \int_0^l [EI(\acute{\theta}_x^2 + \acute{\theta}_z^2) + kGA(\beta_x^2 + \beta_z^2)] ds \tag{3.50}$$

where A and I are the area and moment of inertia of the beam cross section which will be a function of the axial coordinate. The modulus of elasticity E, shear-modulus G and shear coefficient k are assumed to be constants. Using the given interpolation functions and carrying out the integration over the length of the beam leads to a kinetic energy expression

$$E_p = \frac{1}{2}\ \underline{q}^T [\underline{K}_b + \underline{K}_s]\ \underline{q} \tag{3.51}$$

where $\underline{K}_b$ is the (12×12) stiffness matrix for bending and $\underline{K}_s$ the (12×12) stiffness matrix for shear. For these matrices we can write

$$\underline{K}_b = \int_0^l EI(\xi)\ \underline{\acute{\phi}}^T\ \underline{\acute{\phi}}\ ds \tag{3.52}$$

$$\underline{K}_s = \int_0^l kGA(\xi)\ \underline{\chi}^T\ \underline{\chi}\ ds \tag{3.53}$$

The kinetic energy

The total kinetic energy consists of the translational kinetic energy and the rotational kinetic energy including the gyroscopic contribution as discussed before in the treatment of the rigid disk. For this total kinetic energy we can write

$$E_k = \frac{1}{2}\int_0^l [\rho A(\dot{v}^2 + \dot{w}^2) + I_d(\dot{\theta}_x^2 + \dot{\theta}_z^2) + I_p\Omega^2 - 2\Omega I_p\dot{\theta}_x\theta_z]\ ds \tag{3.54}$$

where $I_d = \rho\ I(\xi)$ and $I_p = 2\rho\ I(\xi)$ for a circular cross section. Integrating over the length of the beam then gives

$$E_k = \frac{1}{2}\ \dot{\underline{q}}^T\ [\underline{M}_t + \underline{M}_r]\ \dot{\underline{q}} + \Omega\ \dot{\underline{q}}^T \underline{H}\ \underline{q} \tag{3.55}$$

where

$$\underline{M}_t = \int_0^l \rho A(\xi)\ \underline{\psi}^T \underline{\psi}\ ds \tag{3.56}$$

$$\underline{M}_r = \int_0^l \rho I(\xi)\ \underline{\phi}^T \underline{\phi}\ ds \tag{3.57}$$

$$\underline{H} = \int_0^l 2\rho I(\xi)\ \underline{\phi}^T \underline{N}\ \underline{\phi}\ ds \tag{3.58}$$

The (2×2) matrix $\underline{N}$ is defined as

$$\underline{N} = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}$$

3.3.3 The System equations

Using Lagrange's equation gives the equations of motion of the beam element with respect to the fixed frame. They can be written as

$$[\underline{M}_t + \underline{M}_r]\ \ddot{\underline{q}} + [\Omega\ \underline{G}]\ \dot{\underline{q}} + [\underline{K}_b + \underline{K}_s]\ \underline{q} = \underline{O} \tag{3.59}$$

where

$$\underline{G} = \underline{H} - \underline{H}^T$$

So, this matrix $\underline{G}$ can also be derived directly from the matrix $\underline{H}$ as given in (3.58) if we use the alternative, skew symmetric matrix $\underline{N}$, redefined as

$$\underline{N} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

All these matrices are given in detail in appendix B.

The matrices in equation (3.59) are all of order (12×12) and symmetric, except from the matrix $\underline{G}$. One way of continuation would be to accept this set of (12) degrees of freedom for each beam element but here we choose the alternative of eliminating the shear degrees of freedom using a Guyan (or static) reduction technique. This has the advantage that the number of degrees of freedom will reduce, but more important it will allow us to apply the element in an existing environment (the $\boldsymbol{R_O D_Y}$ toolbox), where each beam element has only 8 degrees of freedom (2 translations and 2 rotations. Numerical tests also have been shown that this can be done without significant loss of accuracy. Therefore we write (3.59) as

$$\underline{M}\ \ddot{\underline{q}} + \Omega\ \underline{G}\ \dot{\underline{q}} + \underline{K}\ \underline{q} = \underline{O} \tag{3.60}$$

The column of degrees of freedom $\underline{q}$ can be partitioned in a sub-column $\underline{q}_b$ containing the 8 *bending-dofs* and a sub-column $\underline{q}_s$ containing the 4 *shear-dofs*, so $\underline{q}^T = [\underline{q}_b^T\ ,\ \underline{q}_s^T]$. The system matrices $\underline{M}$, $\underline{Q}$ and $\underline{K}$ are partitioned accordingly, so that we can write

$$\begin{bmatrix} \underline{M}_{bb} & \underline{M}_{bs} \\ \underline{M}_{sb} & \underline{M}_{ss} \end{bmatrix} \begin{bmatrix} \ddot{\underline{q}}_b \\ \ddot{\underline{q}}_s \end{bmatrix} + \Omega \begin{bmatrix} \underline{G}_{bb} & \underline{G}_{bs} \\ \underline{G}_{sb} & \underline{G}_{ss} \end{bmatrix} \begin{bmatrix} \dot{\underline{q}}_b \\ \dot{\underline{q}}_s \end{bmatrix} + \\ \begin{bmatrix} \underline{K}_{bb} & \underline{K}_{bs} \\ \underline{K}_{sb} & \underline{K}_{ss} \end{bmatrix} \begin{bmatrix} \underline{q}_b \\ \underline{q}_s \end{bmatrix} = \begin{bmatrix} \underline{O}_b \\ \underline{O}_s \end{bmatrix} \tag{3.61}$$

In a static reduction process the degrees of freedom which have to be eliminated are written as a function of the remaining degrees of freedom

$$\underline{q} = \begin{bmatrix} \underline{q}_b \\ \underline{q}_s \end{bmatrix} = \begin{bmatrix} \underline{I}_{bb} \\ \underline{T}_{sb} \end{bmatrix} \underline{q}_b = \underline{T}\ \underline{q}_b \tag{3.62}$$

where

$$\underline{I}_{bb} = (b \times b) \text{ unity matrix} \qquad \text{and} \qquad \underline{T}_{sb} = -\underline{K}_{ss}^{-1} \underline{K}_{sb} \tag{3.63}$$

Introducing this coordinate transformation in (3.59) then gives a set of reduced equations of motion for the beam element

$$\underline{M}_b\ \ddot{\underline{q}}_b + \Omega\ \underline{G}_b\ \dot{\underline{q}}_b + \underline{K}_b\ \underline{q}_b = \underline{Q}_b \tag{3.64}$$

where the reduced mass matrix $\underline{M}_b$, gyroscopic matrix $\underline{G}_b$ and stiffness matrix $\underline{K}_b$ (all (8×8)) are given by

$$\underline{M}_b = \underline{T}^T \underline{M}\ \underline{T}; \quad \underline{G}_b = \underline{T}^T \underline{G}\ \underline{T}; \quad \underline{K}_b = \underline{T}^T \underline{K}\ \underline{T} \tag{3.65}$$

Note

Goal: estimating the effect of shear

In this conical **Timoshenko** beam element, the shear deformation as well as the rotational inertia has been taken into account. This

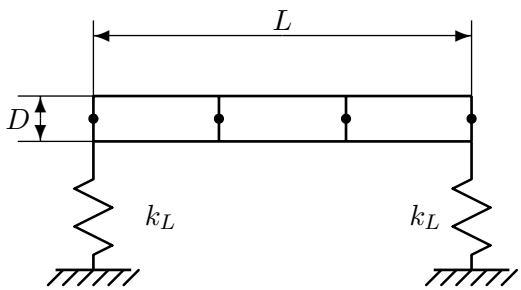
makes it an effectieve element for a rotordynamic analysis. In many cases however (in literature for example) also a more simple type of element is used, based on the **Euler** beam theory. The stiffness- and mass matrix of such an element can be found in [Dimarogonas-92], page 573. For the Stiffness-matrix $\underline{K}$ we find:

$$\underline{K} = \frac{EI}{l^3} \begin{bmatrix} 12 & 0 & 6l & 0 & -12 & 0 & 6l & 0 \\ & 12 & 0 & 6l & 0 & -12 & 0 & 6l \\ & & 4l^2 & 0 & -6l & 0 & 2l^2 & 0 \\ & & & 4l^2 & 0 & -6l & 0 & 2l^2 \\ & & \mathbf{sym} & & 12 & 0 & 6l & 0 \\ & & & & & 12 & 0 & 6l \\ & & & & & & 4l^2 & 0 \\ & & & & & & & 4l^2 \end{bmatrix}$$

and for the Mass-matrix $\underline{M}$ we find:

$$\underline{M} = \frac{\rho A l}{420} \begin{bmatrix} 156 & 0 & 22l & 0 & 54 & 0 & -13l & 0 \\ & 156 & 0 & 22l & 0 & 54 & 0 & -13l \\ & & 4l^2 & 0 & 13l & 0 & -3l^2 & 0 \\ & & & 4l^2 & 0 & 13l & 0 & -3l^2 \\ & & \mathbf{sym} & & 156 & 0 & -22l & 0 \\ & & & & & 156 & 0 & -22l \\ & & & & & & 4l^2 & 0 \\ & & & & & & & 4l^2 \end{bmatrix}$$

Let us look at the simple problem of a non-rotating, uniform beam with length L and diameter D as shown in the next figure. We divide the beam in 3 equal element. The beam is supported at both ends by a very stiff spring to simulate a simple support

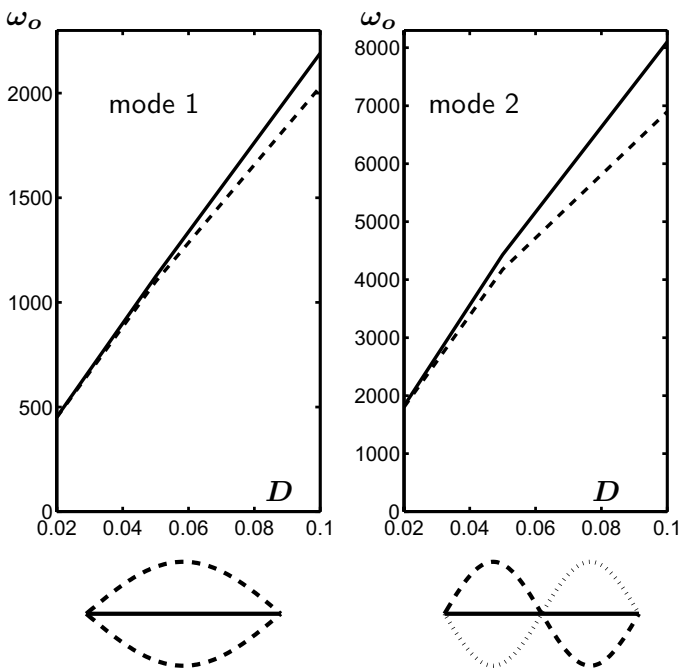


We used the data:

- Length $L = 0.3 \;\; [m]$
- Elasticity modulus $E = 2.1 \; 10^{11} \;\; [N/m^2]$
- Shear modulus $G = 0.8 \; 11^{11} \;\; [N/m^2]$
- Shear coefficient $k = 0.88 \;\; [-]$
- Density $\rho = 7800 \;\; [kg/m^3]$
- 3 values for the Diameter [m]: $D = 0.02 \;\; (\frac{L}{D} = 15)$; $0.05 \;\; (\frac{L}{D} = 6)$; $0.1 \;\; (\frac{L}{D} = 3)$
- Bearing stiffness $k_L = 1.0 \; 10^{12} \;\; [N/m]$. This gives a rigid body mode frequency for a $0.1 \; [m]$ diameter shaft:

$$\omega_{RB} = \sqrt{\frac{2k_L}{M_{shaft}}} = 3.3 \;\; 10^5 \;\; [rad/s]$$

The undamped eigenfrequencies have been calculated and the 2 lowest ones are shown in the next plot as function of the shaft diameter D.



The first 2 undamped eigenfrequencies for Euler beam model (solid) and Timoshenko beam model (dashed)

From these figures we can see:

- for slender shafts ($\frac{L}{D} \gg 6$) both models give almost the same results
- for higher modes the simple Euler-beam model should be used with care

---

End of note

3.4 The Element Assembling

In the preceding sections the important elements for building a rotor dynamic model have been presented. What finally remains is the combination of a (large) number of these elements for modeling a complete structure. In the chosen Finite Element approach this is a straightforward procedure. These basic elements are

- **Beams:** conical (or cylindrical) beam elements with or without shear deformation, characterized by a stiffness matrix, a mass matrix and a damping matrix containing the gyroscopic effects
- **Disks:** rigid bodies with only inertia terms and gyroscopy
- **Bearings:** in general characterized by a stiffness matrix and a damping matrix. They can be dependent on the rotorfrequency or even be excitation frequency dependent (air bearings).

All these elements are fully determined by 4 degrees of freedom (the two translations in x- and z-direction and two rotations around the x- and y-axis) in each nodal point. In a total structure a number of these nodal points

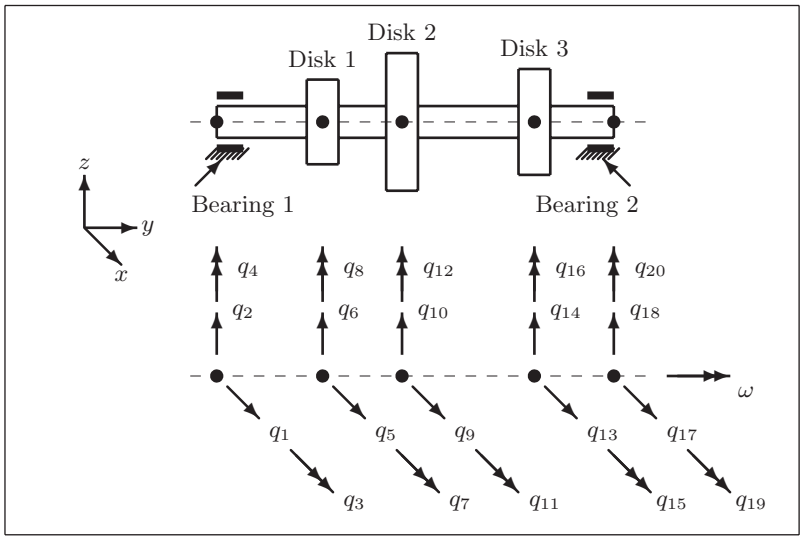


Fig. 3.9. Definition of degrees of freedom for the whole structure

are defined with in each nodal point such a set of 4 degrees of freedom.

Normally they are numbered in a logical way as shown in Fig. 3.9. The (large) rotor speed Ω usually is assumed to be prescribed. In the simple model of Fig. 3.9, we are dealing with 4 cylindrical beam elements with each 8 degrees of freedom, 3 rigid disk elements with each 4 degrees of freedom and finally two bearing elements, also using 4 degrees of freedom. These degree-of-freedom-sets are called the local degrees of freedom. The total system has 5 nodes and consequently 20 degrees of freedom (the so-called global degrees of freedom). The only thing that next has to be done is to define the relation between these local element degrees of freedom and the global structural degrees of freedom. After assembling all the elements, the system equations of motion can be written as

$$\underline{M}\ \underline{\ddot{q}}(t) + \underline{B}\ \underline{\dot{q}}(t) + \underline{K}\ \underline{q}(t) = \underline{f}(t) \tag{3.66}$$

where

$\underline{q}(t)$: column with all the (n) structural degrees of freedom
$\underline{f}(t)$: column with the nodal forces
$\underline{M}$: Mass matrix, symmetric, (n×n)
$\underline{B}$: Damping matrix, (n×n), generally non-symmetric
$\underline{K}$: Stiffness matrix, (n×n), can be non-symmetric

3.5 MDOF Systems with non-symmetric matrices

3.5.1 Summary

For the numerical treatment of a set of n, 2nd-order coupled differential equations as shown in (3.66), the transformation to a state-space formulation consisting of a larger set of 2n, 1st-order equations of motion has some advantages (see also [Kraker/Campen-01]).
This state-space formulation can be written as

$$\underline{C}\ \underline{\dot{x}}(t) + \underline{D}\ \underline{x}(t) = \underline{r}(t) \tag{3.67}$$

where

$$\begin{aligned} \underline{x}(t) &= \begin{bmatrix} \underline{q}(t) \\ \underline{\dot{q}}(t) \end{bmatrix}_{(2n\times 1)} \qquad \underline{r}(t) = \begin{bmatrix} \underline{f}(t) \\ \underline{0} \end{bmatrix}_{(2n\times 1)} \\ \underline{C} &= \begin{bmatrix} \underline{B} & \underline{M} \\ \underline{M} & \underline{0} \end{bmatrix}_{(2n\times 2n)} \qquad \underline{D} = \begin{bmatrix} \underline{K} & \underline{0} \\ \underline{0} & -\underline{M} \end{bmatrix}_{(2n\times 2n)} \end{aligned} \tag{3.68}$$

Having this state-space formulation, the only thing what then rests is the selection of some numerical environment to carry out all kind of analyses (eigenvalue analysis, forced response evaluation, etc) in an easy and flexible way. Also the possibilities of presenting the results of the calculations in many different graphical ways is important to understand the generally complex rotor dynamic behavior of a structure. The general purpose computer programme $\boldsymbol{M_{AT}L_{AB}}$ and its collection of advanced toolboxes have been chosen for this. An additional toolbox $\boldsymbol{R_O D_Y}$ has been developed, containing subroutines for creating a rotor dynamic model in an efficient way, the calculation of critical speeds, plotting of Campbell diagrams, calculation of unbalance responses etc. In the next chapter some examples of rotordynamic problems using this toolbox for the numerical evaluations will be presented and discussed.

3.5.2 State-space formulation

It appears to be advantageous for general non-conservative systems with non-symmetric matrices to reformulate the second-order linearized Lagrange's equations of motion as a set of first-order differential equations

of motion. In view of the absence of symmetry characteristics of the system matrices, no simplification of the associated eigenvalue problem can be achieved anymore and, also, the orthogonality relationships for the eigencolumns, which provide the basis for the decoupling of the equations of motion, do not appear in a straightforward way anymore. We return to the system described by (3.66)

$$\underline{M}\ \underline{\ddot{q}}(t) \ + \ \underline{B}\ \underline{\dot{q}} \ + \ \underline{K}\ \underline{q}(t) \ = \ \underline{f}(t) \tag{3.69}$$

introduce the set of new dependent variables

$$\underline{M}\ \underline{\dot{q}}(t) - \underline{M}\ \underline{z}^*(t) = \underline{0} \tag{3.70}$$

and rewrite (3.69) as a set of first-order differential equations

$$\underline{B}\ \underline{\dot{q}}(t) + \underline{M}\ \underline{\dot{z}}^*(t) + \underline{K}\ \underline{q}(t) = \underline{f}(t) \tag{3.71}$$

Again, it appears to be possible to combine (3.70) and (3.71) to a set of first-order differential equations of motion for the $2n$ variables $\underline{q}(t)$ and $\underline{z}^*(t)$, collected in the state column $\underline{x}(t)$

$$\underline{x}(t) = \begin{bmatrix} \underline{q}(t) \\ \underline{z}^*(t) \end{bmatrix} = \begin{bmatrix} \underline{q}(t) \\ \underline{\dot{q}}(t) \end{bmatrix} \tag{3.72}$$

giving:

$$\underline{C}\ \underline{\dot{x}}(t) + \underline{D}\ \underline{x}(t) = \underline{r}(t) \tag{3.73}$$

where the $(2n \times 2n)$ real matrices

$$\underline{C} = \begin{bmatrix} \underline{B} & \underline{M} \\ \underline{M} & \underline{0} \end{bmatrix}; \qquad \underline{D} = \begin{bmatrix} \underline{K} & \underline{0} \\ \underline{0} & -\underline{M} \end{bmatrix} \tag{3.74}$$

are not symmetric anymore, $\underline{0}$ being the $(n \times n)$ null matrix. In (3.73) the column

$$\underline{r}(t) = \begin{bmatrix} \underline{f}(t) \\ \underline{0} \end{bmatrix} \tag{3.75}$$

contains the $2n$ generalized forces related to the $2n$ generalized state coordinates contained in the column $\underline{x}(t)$.

Note

At this point it is remarked that in system analysis and control engineering generally the state-space equations are written as

$$\underline{\dot{x}}(t) = \underline{\hat{A}}\ \underline{x}(t) + \underline{\hat{B}}\ \underline{u}(t) \tag{3.76}$$

This set of equations is equivalent with (3.73) if we take

$$\underline{\hat{A}} = -\underline{C}^{-1}\underline{D}\ \ ; \qquad \underline{\hat{B}} = \underline{C}^{-1}\begin{bmatrix} \underline{I} \\ \underline{0} \end{bmatrix}\ \ ; \qquad \underline{u}(t) = \underline{f}(t) \tag{3.77}$$

We can easily express the system matrices $\underline{\hat{A}}$ and $\underline{\hat{B}}$ in terms of the matrices $\underline{M}$, $\underline{B}$ and $\underline{K}$ of the original second-order system as follows

$$\underline{\hat{A}} = -\begin{bmatrix} \underline{0} & ; & -\underline{I} \\ \underline{M}^{-1}\underline{K} & ; & \underline{M}^{-1}\underline{B} \end{bmatrix}; \qquad \underline{\hat{B}} = \begin{bmatrix} \underline{0} \\ \underline{M}^{-1} \end{bmatrix} \tag{3.78}$$

Consequently, the mass matrix $\underline{M}$ should be **non-singular** (i.e. **positive definite**) to be able to determine $\underline{M}^{-1}$.

End of note

3.5.3 The non-symmetric eigenvalue problem

For free motions the set of linear first-order differential equations of motion (3.73) reduces to

$$\underline{C}\ \underline{\dot{x}}(t) + \underline{D}\ \underline{x}(t) = \underline{0} \tag{3.79}$$

Because (3.79) represents a linear homogeneous system with constant matrices, it admits a solution of the exponential form

$$\underline{x}(t) = \underline{v}\ e^{st} \tag{3.80}$$

where s is a constant scalar and $\underline{v}$ is a constant column, both in general complex. Inserting (3.80) into (3.79) and dividing through by e^{st}, we obtain the *general algebraic eigenvalue problem*

$$[s\ \underline{C} + \underline{D}]\ \underline{v} = \underline{0} \tag{3.81}$$

Equation (3.81) represents the eigenvalue problem associated with the non-symmetric matrices $\underline{C}$ and $\underline{D}$ and it possesses a nontrivial solution if and only if the determinant of the coefficients of the elements $v_i = v\,[i]\,(i = 1, 2, ..., 2n)$ contained in $\underline{v}$ vanishes, i.e. if

$$\det\,(s\ \underline{C} + \underline{D}) = 0 \tag{3.82}$$

The characteristic equation (3.82) of degree $2n$ in s possesses in general $2n$ roots $s_r(r = 1, 2, ..., 2n)$, referred to as eigenvalues. The associated eigencolumns $\underline{v}_r(r = 1, 2, ..., 2n)$ can be obtained from (3.81) and, hence, they satisfy the equations

$$s_r\ \underline{C}\ \underline{v}_r = -\underline{D}\ \underline{v}_r \tag{3.83}$$

Directly can be seen that each (complex) eigencolumns $\underline{v}_r$ can be written as:

$$\underline{v}_r = \begin{bmatrix} \underline{u}_r \\ s_r\ \underline{u}_r \end{bmatrix} \quad ; \qquad \underline{u}_r = \begin{bmatrix} u_r(1) \\ u_r(2) \\ ..... \\ u_r(n) \end{bmatrix} \quad (r = 1, 2, ..., 2n) \tag{3.84}$$

So, the (generally complex) subcolumns $\underline{u}_r$ contain all the relevant information for the free vibration.
The question arises naturally as to whether the eigencolumns are orthogonal in some sense and whether the solution for $\underline{x}(t)$ can be expanded in a series of eigencolumns, providing the basis for decoupling of the equations of motion. We confine ourselves to the case in which all the eigenvalues $s_r(r = 1, 2, ..., 2n)$ are distinct, from which it follows that all the eigencolumns $\underline{v}_r(r = 1, 2, ..., 2n)$ are independent. To serve as a basis for the problem at hand, the eigencolumns must be orthogonal with respect to the matrices $\underline{C}$ and $\underline{D}$. The eigencolumns cannot be orthogonal to $\underline{C}$ and $\underline{D}$, however, because $\underline{C}$ and $\underline{D}$ are not symmetric anymore. But, whereas the eigencolumns are not orthogonal in the ordinary sense, they are orthogonal in some fashion. Before we explore the nature of the orthogonality, we recall from the theory of linear algebra that

$$\det\,(s\ \underline{C} + \underline{D})^T = \det\,(s\ \underline{C} + \underline{D}) \tag{3.85}$$

because the value of the determinant of a matrix is the same, regardless of whether the determinant is expanded by a row or a column. Hence,

we conclude from (3.85) that the characteristic equation (3.82) and the characteristic equation

$$\det\,(s\ \underline{C} + \underline{D})^T = 0 \tag{3.86}$$

possess the same eigenvalues. We can write the eigenvalue problem associated with the characteristic equation (3.86) in the form

$$[s\ \underline{C} + \underline{D}]^T\ \underline{w} = \underline{0} \tag{3.87}$$

This eigenvalue problem is referred to as the *adjoint eigenvalue problem* of (3.79). It admits solutions in the form of the eigenvalues s_s ($_s = 1, 2, ..., 2n$) and the eigencolumns $\underline{w}_s (s = 1, 2, ..., 2n)$, where $\underline{w}_s$ are called the *adjoint eigencolumns* of the eigencolumns $\underline{v}_r$. They satisfy the equations

$$s_s\ \underline{C}^T\ \underline{w}_s = -\underline{D}^T\ \underline{w}_s \qquad (_s = 1, 2, ..., 2n) \tag{3.88}$$

Taking the transposed of (3.88), we obtain

$$s_s\ \underline{w}_s^T\ \underline{C} = \underline{w}_s^T\ \underline{D} \tag{3.89}$$

Because of their position to the left of the matrices $\underline{C}$ and $\underline{B}$, the adjoint eigencolumns $\underline{w}_s$ are known as the *left eigencolumns* of the eigenvalue problem associated with (3.79). Consistent with this, the eigencolumns $\underline{v}_r$ are called the *right eigencolumns* of the eigenvalue problem associated with (3.79). It is of interest to note that when $\underline{C}$ and $\underline{D}$ are real symmetric matrices, i.e. $\underline{C} = \underline{C}^T$, $\underline{D} = \underline{D}^T$, then the adjoint eigencolumns $\underline{w}_s$ coincide with the eigencolumns $\underline{v}_s (s = 1, 2, ..., 2n)$ in which case the eigenvalue problem is said to be *self-adjoint*. Now, we premultiply (3.83) by $\underline{w}_s^T$ and postmultiply (3.89) by $\underline{v}_r$. Subtracting the second result from the first, we obtain

$$(s_r - s_s)\ \underline{w}_s^T \underline{C}\ \underline{v}_r = 0 \tag{3.90}$$

Hence, because we have assumed all eigenvalues to be distinct, we must have

$$\underline{w}_s^T \underline{C}\ \underline{v}_r = 0 \quad , \quad s_r \neq s_s \quad , \quad r, s = 1, 2, ..., 2n \tag{3.91}$$

Then, in view of (3.83), we must also have

$$\underline{w}_s^T\ \underline{D}\ \underline{v}_r = 0 \quad , \quad s_r \neq s_s \quad , \quad r, s = 1, 2, ..., 2n \tag{3.92}$$

Equations (3.91) and (3.92) state that the right eigencolumns and the left eigencolumns of the real non-symmetric eigenvalue problem associated with (3.79) are so-called *bi-orthogonal* with respect to the matrices $\underline{C}$ and $\underline{D}$ if the related eigenvalues are distinct.
The pairs of right and left eigencolumns can be normalized by letting

$$\underline{w}_r^T \underline{C}\ \underline{v}_r = c_r \quad , \quad r = 1, 2, ..., 2n \tag{3.93}$$

where $c_r (r = 1, 2, ..., 2n)$ are normalization parameters to be selected as desired and being generally complex. An alternative normalization scheme consists of setting

$$\underline{w}_r^T \underline{D}\ \underline{v}_r = d_r \quad , \quad r = 1, 2, ..., 2n \tag{3.94}$$

where $d_r (r = 1, 2, ..., 2n)$ constitute another set of complex normalization parameters. It can be concluded from (3.83), (3.93) and (3.94) that the parameters c_r and d_r depend on one another via the relationship

$$d_r = -s_r\ c_r \tag{3.95}$$

The preceding bi-orthogonality conditions and normalization schemes can be expressed conveniently in a compact matrix form. To this end, we introduce the matrices of right and left eigencolumns

$$\underline{V} = [\underline{v}_1 \quad \underline{v}_2 \quad ... \quad \underline{v}_{2n}] \tag{3.96}$$

$$\underline{W} = [\underline{w}_1 \quad \underline{w}_2 \quad ... \quad \underline{w}_{2n}] \tag{3.97}$$

Then, (3.91) and (3.93) can be written in terms of $\underline{V}$ and $\underline{W}$ as

$$\underline{W}^T \underline{C}\ \underline{V} = \begin{bmatrix} c_1 & & 0 \\ & \ddots & \\ 0 & & c_{2n} \end{bmatrix} := \lceil c_r \rfloor \tag{3.98}$$

whereas (3.92) and (3.94) can be written in terms of $\underline{V}$ and $\underline{W}$ as

$$\underline{W}^T \underline{D}\ \underline{V} = \begin{bmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_{2n} \end{bmatrix} := \lceil d_r \rfloor \tag{3.99}$$

In view of (3.95) we have

$$\begin{bmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_{2n} \end{bmatrix} = - \begin{bmatrix} s_1 & & 0 \\ & \ddots & \\ 0 & & s_{2n} \end{bmatrix} \begin{bmatrix} c_1 & & 0 \\ & \ddots & \\ 0 & & c_{2n} \end{bmatrix} \tag{3.100}$$

or, in abbreviated form

$$\lceil d_r \rfloor = - \lceil s_r \rfloor \lceil c_r \rfloor \tag{3.101}$$

It follows from (3.98) that

$$\underline{W}^T = \lceil c_r \rfloor (\underline{C}\ \underline{V})^{-1} \tag{3.102}$$

so that, in stead of solving the adjoint eigenvalue problem (3.87), it is possible to obtain the left eigencolumns by inverting the matrix $\underline{C}\ \underline{V}$ and premultiplying the resulting matrix by $\lceil c_r \rfloor$. However for small systems, it will be easier just to solve the left-eigenvalue problem in itself.

Note

Let us look at the simple problem, presented in Section 1.5. For that system we found:

$$\underline{M} = m \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \underline{B} = \Omega c \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad \underline{K} = k \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{3.103}$$

We again use the values $m = 1.0$, $c = 0.5$ and $k = 1.0$. We select an arbitrary value for the rotorspeed Ω, namely $\Omega = 2.0$.
We can build the $(4 * 4)$ matrices $\underline{C}$ and $\underline{D}$ and solve the eigenvalueproblem:

$$[s\ \underline{C} + \underline{D}]\underline{v} = \underline{0} \tag{3.104}$$

This can easily be done by using the Matlab-statement:

```
<<      [V,Lr]=eig(D,-C]  (see: help eig~~)
```

The $(4{*}4)$ matrix $\underline{V}$ contains the right-eigencolumns $\underline{v}_i$ and the $(4{*}4)$ diagonalmatrix $\underline{Lr}$ the eigenvalues s_i. The result is:

$$diag(Lr) = \begin{bmatrix} 0.00 + 1.62i \\ 0.00 - 1.62i \\ 0.00 + 0.62i \\ 0.00 - 0.62i \end{bmatrix}$$

$$V = \begin{bmatrix} 1.00; & 1.00; & 1.00; & 1.00 \\ -0.00+1.00i; & -0.00-1.00i; & -0.00-1.00i; & -0.00+1.00i \\ 0.00+1.620i; & 0.00-1.620i; & 0+0.62i; & 0-0.62i \\ -1.62-0.00i; & -1.62+0.00i; & 0.62-0.00i; & 0.62+0.00i \end{bmatrix}$$

It can also be seen that each eigencolumn $\underline{v}_i$ can be written as

$$\underline{v}_i = \begin{bmatrix} \underline{u}_i \\ s_i \ \underline{u}_i \end{bmatrix}$$

The real parts are zero because there is no energy dissipation in the system. The "damping matrix", coming from the gyroscopic effect is a conservative term. The imaginary parts match with the eigenfrequencies from Fig. 1.14 for the specific rotorfrequency $\Omega = 2.0$ and the eigencolumns are scaled such that $\underline{v}_i(1) = 1.0$
We find two pairs of complex-conjugate eigenvalues and corresponding complex-conjugate eigencolumns:

$$\begin{array}{ll} s_1 = 1.62\ i & \underline{u}_1^T = [1, i] \\ s_2 = -1.62\ i & \underline{u}_2^T = [1, -i] \\ s_3 = 0.62\ i & \underline{u}_3^T = [1, -i] \\ s_4 = -0.62\ i & \underline{u}_4^T = [1, i] \end{array}$$

(check that these values correspond with the analytical outcomes given in Section 1.5).

If we also solve the adjoint problem:

```
<<      [W,Ll]=eig(D.',-C.')
```

(Pay attention to the use of .' for the regular transpose of a complex matrix or column.
We see that this leads to the same eigenvalues $Lr = Ll$, and the left-eigencolumns:

$$W = \begin{bmatrix} 1.00; & 1.00; & 1.00; & 1.00 \\ 0.00-1.00i; & 0.00+1.00i; & 0.00+1.00i; & 0.00-1.00i \\ 0.00+1.62i; & 0.00-1.62i; & 0+0.62i; & 0-0.62i \\ 1.62+0.00i; & 1.62-0.00i; & -0.62+0.00i; & -0.62-0.00i \end{bmatrix}$$

(It is also always relevant to check the sorting of the eigenvalues of the two sets because sometimes the order for the one set can differ from the other. Then a comparison of eigencolumns or use of the

orthogonality property will make no sense.)
If we calculate $\underline{V}^T\underline{CV}$ we get:

$$V.' * C * V = \begin{bmatrix} 0.00 - 0.00i \ ; \ 0.00 + 2.00i \ ; \ -0.00 + 6.47i \ ; \ \ 0.00 + 0.00i \\ 0.00 - 2.00i \ ; \ 0.00 + 0.00i \ ; \ \ 0.00 - 0.00i \ ; \ -0.00 - 6.47i \\ 0.00 + 2.47i \ ; \ 0.00 + 0.00i \ ; \ -0.00 + 0.00i \ ; \ -0.00 - 2.00i \\ 0.00 - 0.00i \ ; \ 0.00 - 2.47i \ ; \ -0.00 + 2.00i \ ; \ -0.00 - 0.00i \end{bmatrix}$$

This is not a symmetric matrix and certainly not a diagonal matrix as we normally would find for systems with symmetric matrices $\underline{C}$ and $\underline{D}$.
For the product $\underline{W}^T\underline{CV}$ we get:

$$W.' * C * V = \begin{bmatrix} -0.00 + 4.47i \ ; \ \ 0.00 + 0.00i \ ; \ 0.00 - 0.00i \ ; \ 0.00 + 0.00i \\ 0.00 + 0.00i \ ; \ -0.00 - 4.47i \ ; \ 0.00 - 0.00i \ ; \ 0.00 + 0.00i \\ 0.00 - 0.00i \ ; \ \ 0.00 - 0.00i \ ; \ 0.00 + 4.47i \ ; \ 0.00 - 0.00i \\ 0.00 + 0.00i \ ; \ \ 0.00 + 0.00i \ ; \ 0.00 + 0.00i \ ; \ 0.00 - 4.47i \end{bmatrix}$$

which is indeed a diagonal matrix. The sam holds for $\underline{W}^T\underline{DV}$.

End of note

3.5.4 Decoupling of the equations of motion; the transfer function matrix

We return to the set of first-order differential equations of motion (3.73) for the generalized state coordinates contained in the column $\underline{x}(t)$. These equations can be decoupled by expanding $\underline{x}(t)$ in terms of the right eigencolumns $\underline{v}_r$ in the form

$$\underline{x}(t) = \sum_{r=1}^{2n} \underline{v}_r \ \eta_r(t) = \underline{V} \ \eta(t) \tag{3.105}$$

where $\underline{V}$ is the square matrix of right eigencolumns defined by (3.96), while the column $\underline{\eta}(t)$ contains the functions $\eta_1(t), \eta_2(t), ..., \eta_{2n}(t)$

$$\underline{\eta}(t) = \begin{bmatrix} \eta_1(t) \\ \eta_2(t) \\ ... \\ \eta_{2n}(t) \end{bmatrix} \tag{3.106}$$

The functions contained in the column $\underline{\eta}(t)$ can be considered to constitute a new set of $2n$ state coordinates. Introducing the linear transformation (3.105) into the state equations of motion (3.73) and premultiplying the resulting set of equations by the transposed of the matrix of left eigencolumns, $\underline{W}^T$, we obtain

$$\underline{W}^T \underline{C}\ \underline{V}\ \underline{\dot{\eta}}(t) + \underline{W}^T \underline{D}\ \underline{V}\ \underline{\eta}(t) = \underline{W}^T \underline{r}(t) \tag{3.107}$$

Using (3.98), (3.99), (3.101) and (3.106), we arrive at the following set of uncoupled equations

$$c_r \left\{ \dot{\eta}_r(t) - s_r\ \eta_r(t) \right\} = N_r(t) \quad , \quad r = 1, 2, ..., 2n \tag{3.108}$$

where

$$N_r(t) = \underline{w}_r^T\ \underline{r}(t) \tag{3.109}$$

are the generalized forces in the decoupled state equations associated with the state coordinates $\eta_r(t)$. The differential equations of motion (3.108) have to be supplemented with the initial conditions for $\eta_r(t)$. These can be obtained from the initial conditions for $\underline{x}(t)$, i.e. $\underline{x}(0) = \underline{x}_0 = \left[\underline{q}^T(0) \quad \underline{\dot{q}}^T(0)\right]^T$, by premultiplying (3.105) for $t = 0$ by $\underline{W}^T \underline{C}$ and utilizing the orthogonality conditions and normalization scheme (3.98). This results in

$$\eta_r(0) = \frac{1}{c_r}\ \underline{w}_r^T\ \underline{C}\ \underline{x}(0) \quad , \quad r = 1, 2, ..., 2n \tag{3.110}$$

The decoupled state equations of motion (3.108) for systems with non-symmetric matrices completely resemble the decoupled equations for systems with symmetric matrices (see [Kraker/Campen-01]). The only difference being the appearance of the left eigencolumns $\underline{w}_r$ in the normalization parameters c_r and in the generalized forces $N_r(t)$. In view of this, the discussion of the free response is similar to the discussion carried out in [Kraker/Campen-01] for systems with symmetric matrices and it will not be repeated here. Also, the solution of (3.108) for general external excitation can be obtained by the methods for single-degree-of-freedom linear systems. The transfer function matrix $\underline{H}(\Omega)$ for systems with symmetric matrices, given in [Kraker/Campen-01]) can be easily generalized to systems with non-symmetric matrices. Collecting the first n elements of the right eigencolumn $\underline{v}_r$ in the column $\underline{v}_r^u$

$$\underline{v}_r^u = [v_{1r} \quad v_{2r} \quad .... \quad v_{nr}]^T \tag{3.111}$$

and collecting the first n elements of the left eigencolumn $\underline{w}_r$ in the column $\underline{w}_r^u$

$$\underline{w}_r^u = [w_{1r} \quad w_{2r} \quad .... \quad w_{nr}]^T \tag{3.112}$$

we can express the transfer function matrix for systems with non-symmetric matrices as

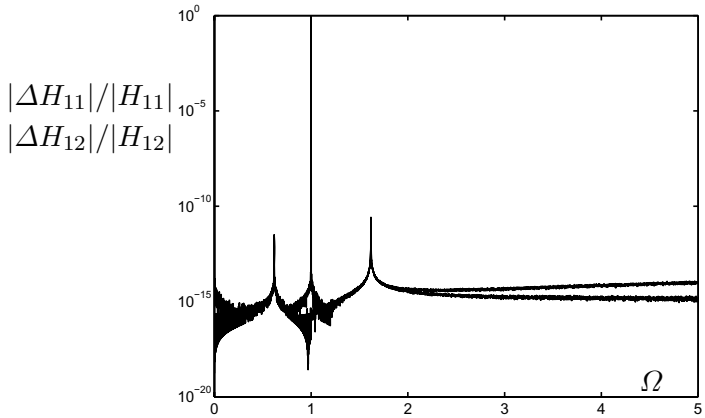
$$\underline{\boldsymbol{H}}(\boldsymbol{\Omega}) = \sum_{\boldsymbol{r=1}}^{\boldsymbol{2n}} \frac{\underline{\boldsymbol{v}}_{\boldsymbol{r}}^{\boldsymbol{u}} \left(\underline{\boldsymbol{w}}_{\boldsymbol{r}}^{\boldsymbol{u}}\right)^{\boldsymbol{T}}}{\boldsymbol{c_r}\left(\boldsymbol{j\Omega} - \boldsymbol{s_r}\right)} \qquad (\boldsymbol{n} * \boldsymbol{n}) \tag{3.113}$$

Of course, this should give the same result as solving:

$$\underline{H}(\Omega) = \left[-\Omega^2 \underline{M} + j\Omega \underline{B} + \underline{K}\right]^{-1} \tag{3.114}$$

Note

To check the equivalency between the two methods we calculate the Frequency Respons Function for the 2-dof system giving by equation (3.103). Using the expressions (3.113) respectively (3.114) the functions $H_{11}(\Omega)$ and $H_{12}(\Omega)$ have been calculated and the relative error between them is plotted in the next figure.



Relative error between the two methods

It can be seen that the difference is nothing more than the numerical accuracy of the computer. At the two resonance peaks and the anti-resonance the error is larger due to "divisions by zero" or calculating the inverse of a singular matrix. Remember we have a system with purely imaginary eigenvalues s_r for which $(j\Omega - s_r)$ easily can become zero for a discrete Ω-range.

End of note

4

Illustrative Examples

The history of rotordynamics is replete with the interplay of theory and practice. W. J. M. Rankine first performed an analysis of a spinning shaft in 1869, but his model was not adequate and he predicted that supercritical speeds could not be attained. In 1895 Dunkerley published an experimental paper describing supercritical speeds. Carl Gustaf De Laval, a Swedish engineer, ran a steam turbine to supercritical speeds in 1889, and Kerr published a paper showing experimental evidence of a second critical speed in 1916. Henry Jeffcott was commissioned by the Royal Society of London to resolve the conflict between theory and practice. He published a paper now considered classic in the Philosophical Magazine in 1919 in which he confirmed the existence of stable supercritical speeds. August Fppl published much the same conclusions in 1895, but history largely ignored his work. Between the work of Jeffcott and the start of World War II there was much work in the area of instabilities and modeling techniques culminating in the work of Prohl and Myklestad which led to the Transfer Matrix Method (TMM) for analyzing rotors. The most prevalent method used today for rotordynamics analysis is the Finite Element Method.
****

Modern computer models have been commented on in a quote attributed to Dara Childs, "the quality of predictions from a computer code has more to do with the soundness of the basic model and the physical insight of the analyst. ... Superior algorithms or computer codes will not cure bad models or a lack of engineering judgment."
http://en.wikipedia.org/wiki/Rotordynamics

4.1 Introduction

In this chapter some (simple) examples of rotordynamic problems will be presented and some of the most interesting rotordynamic phenomena will be evaluated. If possible, the results will be compared with results from literature. The general procedure for carrying out a rotordynamic analysis is schematically shown in Fig. 4.1. Starting from a rotor system as present in

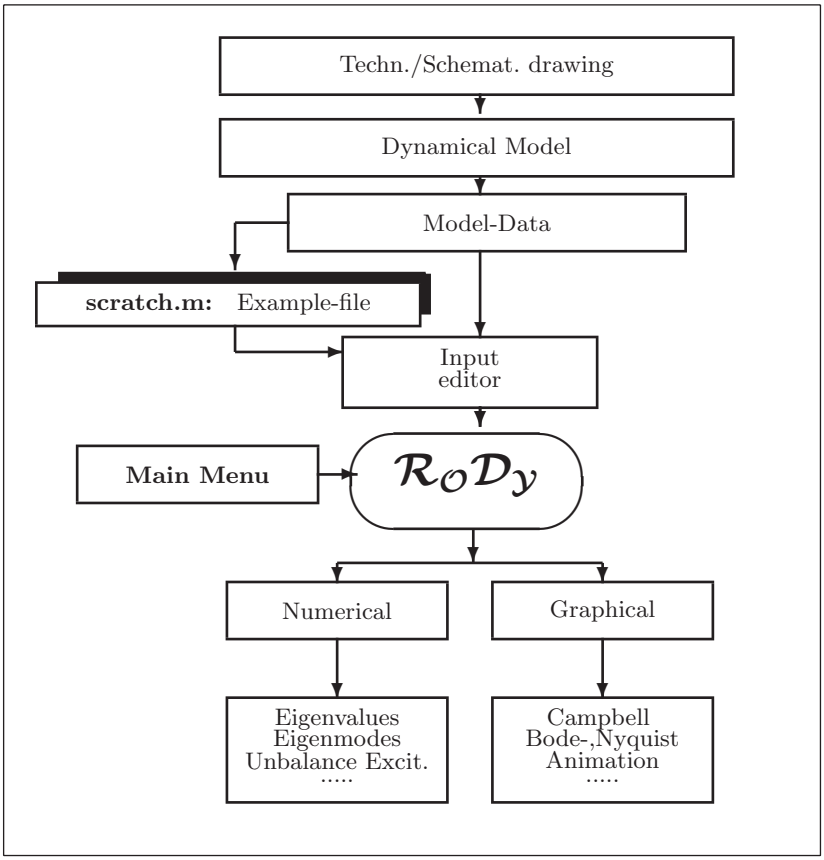


Fig. 4.1. General structure of a rotordynamic analysis using the $\boldsymbol{R_O D_Y}$ -toolbox

a real practical situation, the first (and generally also the most important) step will be to generate a model for the system. In some cases this has to be done by carefully looking at the real machine, in other cases a set of

technical drawings and related documentation will be the main source of information. This modeling proces starts with the conclusion that a linear model should be able to represent the reality, meaning that non-linear effects might be neglected in this first analysis step. In the following step, discrete elements (shaft-sections, disks, bearings, springs, dampers, etc) in the rotor system have to be defined. In this step also the excitations acting on the system (imbalances, disk skewness, fluid-structure interaction in for example a pump-impeller, etc) have to be considered. Also this is not a trivial step in the procedure and it will mainly be governed by the present geometrical situation (bearing locations, shaft diameter changes, etc) on one hand and by the pursued accuracy of the results on the other hand. In these first steps of the analysis the practical experience of the researcher will be of major importance for the quality of the results.
Once the (Finite Element) model has been defined the input file for the programme has to be generated and the relevant numerical analysis can be selected. The results can be presented graphically in order to be able to understand what type of phenomena are dominating the system's behavior. The final and again one of the most important steps in the whole procedure is to evaluate these results and to generate ideas how to solve the rotor dynamic problem or to improve the dynamic behavior of the proposed design. In the next sections we will skip the important pre- and post-modeling phases but directly start from a given model and only show some important results.

4.2 Symmetric Rotor in two Hydrodynamic Bearings

In this first example we will look at the model, shown in Fig. 4.2. It consists of a flexible shaft supported by two identical fluid film bearings. For the

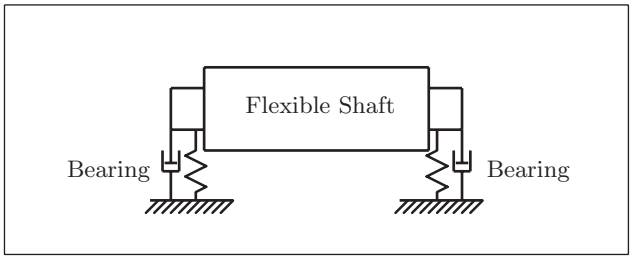


Fig. 4.2. Rotor model in two fluid film bearings

two bearings the method from Section 2.4 has been used to generate the rotorspeed dependent bearing-stiffness and -damping characteristics. This method was based on an impedance description of the bearing forces and is implemented in the $\boldsymbol{R_O D_Y}$ -procedure volcyl2m.m. We use the following system parameters:

Shaft length:	$L_{shaft} = 0.12\ [m]$
Shaft Diameter:	$D_{shaft} = 0.04\ [m]$
Shaft Material:	$E = 2.1\ 10^{11}\ [Nm^{-2}],\ \rho = 7800\ [kgm^{-3}]$
Bearing Diameter:	$D_{bearing} = 0.012\ [m]$
Bearing Length:	$L_{bearing} = 0.008\ [m]$
Radial Bearing Clearance:	$C = 2.0\ 10^{-5}\ [m]$
Viscosity	$\eta = 5.0\ 10^{-3}\ [Nsm^{-2}]$
Shaft Speed	$\Omega_{shaft}\ [rads^{-1}]$ will be chosen later

We assume that the shaft will only be loaded by its own weight ($11.5\ [N]$, so for each bearing $5.75\ [N]$. For each rotorspeed Ω the equilibrium position will be calculated and also the stiffness- and damping terms for small vibrations around this equilibrium position. For the shaft we use a single cylindrical beam element so the system will have 8 degrees of freedom. This means that for the analysis of the free response of the system (calculation of the complex eigenvalues and corresponding modes) we have 16 eigenvalues.

Due to the gyroscopic effect in the shaft, but even more because of the fluid film bearing model these eigenvalues will be dependent of the rotorspeed Ω. For a rotorspeed $50 \leq \Omega \leq 750$ [Hz], the imaginary parts of the most important eigenvalues are plotted in Fig. 4.3; the so-called Campbell plot. These imaginary parts are associated with the damped eigenfrequencies so they can give some information about the possibility of resonance problems. In this graph the line for which the damped frequency is equal to the rotorspeed is indicated by a dashed line. For small rotorspeeds all the

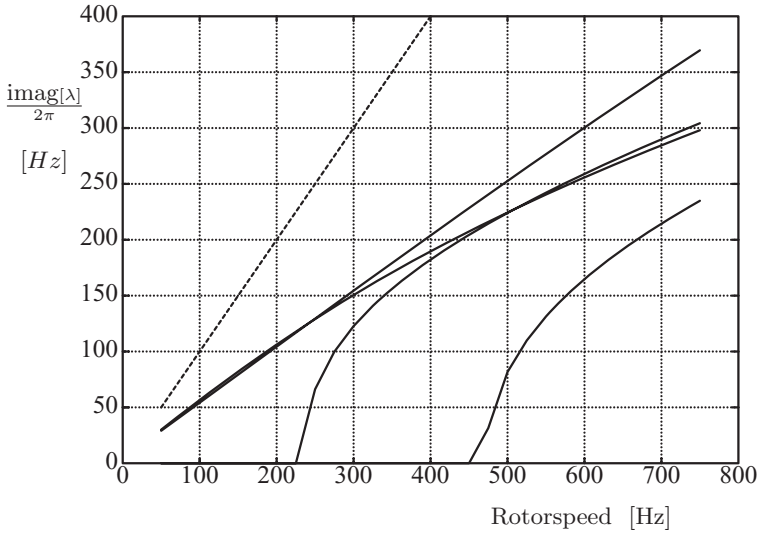


Fig. 4.3. Campbell plot for a rotor supported in two fluid-film bearings

values go to zero. This is due to the fact that this fluid film bearing has no stiffness for very small rotorspeeds. Then there is a pair of eigenvalues which for small rotorspeeds are almost equal. It can be shown that one of them is a purely translation mode and the other is a purely tilting mode. Another remarkable phenomenon in this graph is that two curves have a zero imaginary part up to a rotorspeed of 230 respectively 450 [Hz]. This means that for small rotorspeeds these modes are supercritically damped (so we are dealing with a pair of two real eigenvalues). At the rotorspeeds mentioned before, they become critically damped (two equal real eigenvalues) and for higher rotorspeeds we get a pair of two complex conjugate eigenvalues. This can also be concluded from Fig. 4.4, where the real parts

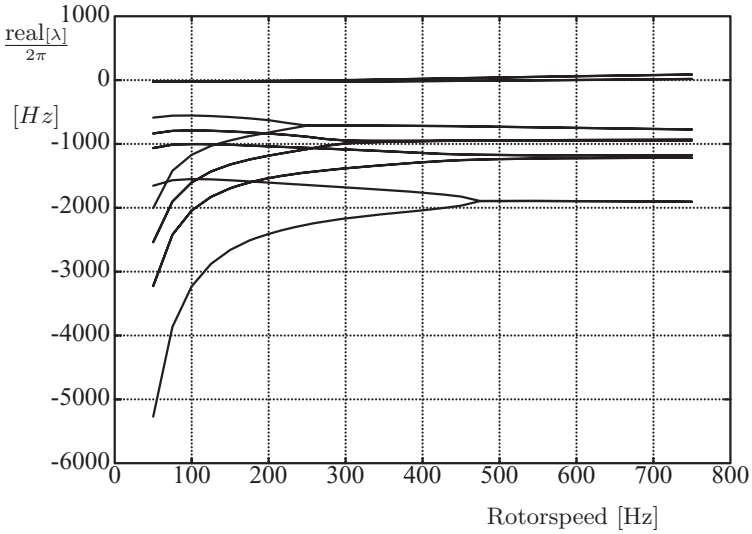


Fig. 4.4. Real parts of the eigenvalues

of all the 16 eigenvalues are shown as function of the rotorspeed. There we can see that pairs of real eigenvalues come together and change into a pair of complex conjugate eigenvalues.

In Fig. 4.3 we can see that none of the eigenvalue curves has a point of intersection with the dotted line. So we cannot speak about the existence of a critical speed for which an unacceptable response might be expected. But there is another potential source of problems in this case. If we plot the real parts of the eigenvalues for the translational mode and the tilting mode as mentioned before we get the picture shown in Fig. 4.5. There we can see that for a rotorspeed of $\Omega \approx 290$ [Hz], one of the real parts becomes positive. This means that the rotor will become unstable above this speed. This is called the **onset speed of instability.** We can read from Fig. 4.3 that the imaginary part of the eigenvalue corresponding to this mode at this rotorspeed of $\Omega = 290\ [Hz]$ is approximately $150\ [Hz]$. So again we are dealing with the well-known $\frac{1}{2}\omega$-whirl. At $\Omega \approx 590$ [Hz], even a second mode will become unstable. The investigation of the real- and imaginary parts of the system's eigenvalues and the visualization of the corresponding modes (see later), for changing rotorspeeds is one of the most important tasks in a rotor dynamic analysis.

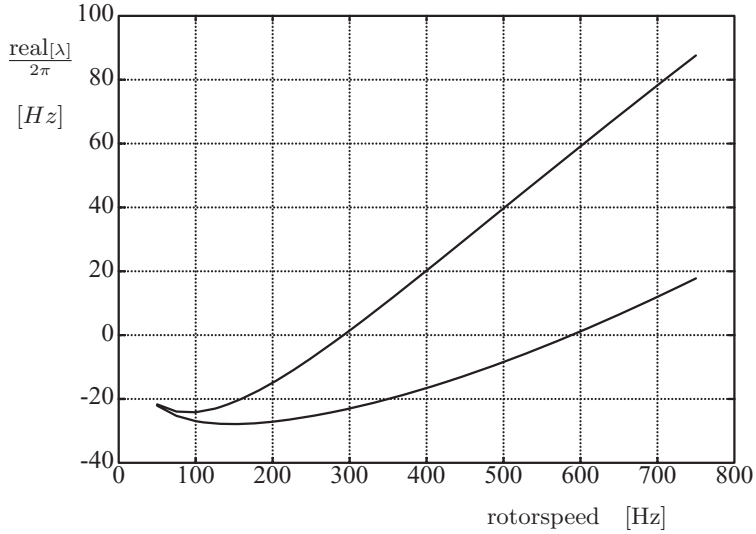


Fig. 4.5. Real parts of the eigenvalues of the two dominant modes

4.3 Industrial Fan

The second example we will investigate is an industrial fan as shown in Fig. 4.6. The system consists of a shaft with at one end a large disk (the fan) and supported by two bearings (also called an overhung rotor). The problem is taken from [Krämer-83]. For modeling this system, 4 shaft el-

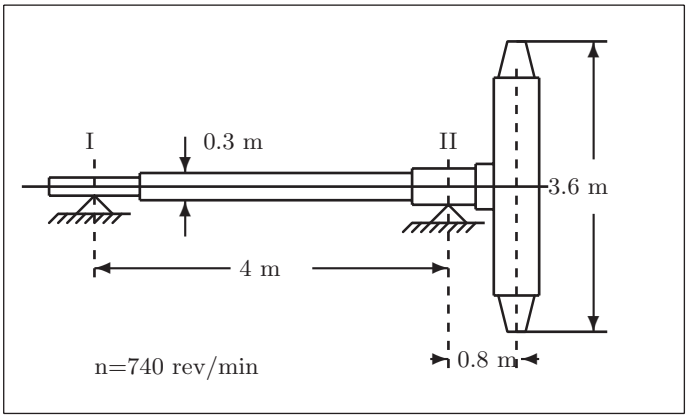


Fig. 4.6. Schematic impression of the fan

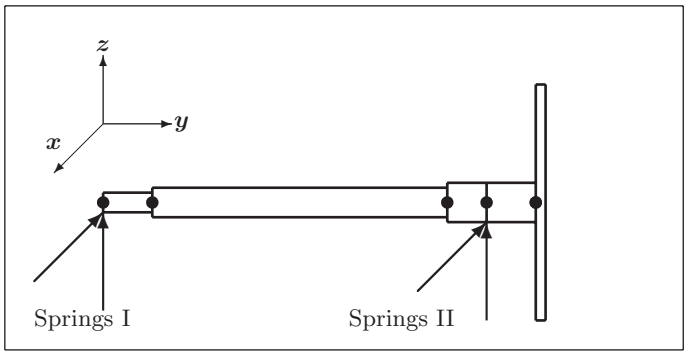


Fig. 4.7. The Rotor model

ements with constant cross section and no shear deformation have been defined. The shear deformation has been ignored because in this slender rotorsystem it will not play an important role and also because in [Krämer-83] an Euler beam model was used. Additionally a single disk element has been chosen. This leads to a system with 20 degrees of freedom. The element

Disk-element	M (kg)	I_p (kgm^2)	I_t (kgm^2)		
1	8000	8520	4260		
Shaft-element	L (m)	D_u (m)	D_i (m)	E (Pa)	ρ (kgm^{-3})
1	0.7	0.10	0	$2.1\ 10^{11}$	1
2	2.9	0.30	0	$2.1\ 10^{11}$	1
3	0.4	0.32	0	$2.1\ 10^{11}$	1
4	0.8	0.34	0	$2.1\ 10^{11}$	1

Table 4.1. Model-data for industrial fan problem

data for this system is given in Table 4.1. As can be seen from the table the material density ρ for the shaft has been chosen 1.0 $[kg/m^3]$, to simulate a massless shaft.
The system is supported by two ball bearings. We assume that they can be modeled by linear springs without damping. We also assume that there is no cross-coupling effect between the stiffnesses in x- and z-direction and that the (direct) stiffnesses in these directions are different due to the support of the bearings. The bearing data is given in Table 4.2. For bearing I

	Bearing I	Bearing II
k_{zz}	$(1/6) * 10^9$ $[Nm^{-1}]$	$(2/3) * 10^9$ $[Nm^{-1}]$
k_{xx}	$(1/12) * 10^9$ $[Nm^{-1}]$	$(1/3) * 10^9$ $[Nm^{-1}]$

Table 4.2. Bearing stiffness data

the stiffness in z-direction is twice the stiffness in x-direction. For bearing number II we can see the same asymmetry, but the stiffnesses are 4 times as large. This bearing has to support the fan so it has been chosen for a higher load carnying capacity, which is reflected in the higher stiffness. In general, one can say that roller-bearing type supports can be modeled as linear springs without cross-coupling. The lack of damping in these structural elements is one of the main disadvantages. Therefore special damping measures might have to be introduced to reduce the vibration level such as so-called *Squeeze Film Dampers*.

First we study the free response. If we calculate the state-space eigenvalues, we will only get purely imaginary eigenvalues (in complex conjugate pairs)

without any real parts. So it looks like we are dealing with an undamped system. However, in the derivation of the equation of motion for the single (rigid) disk we have seen that the gyroscopic effect will lead to a skew symmetric damping matrix contribution. However, it can be shown that these gyroscopic forces are conservative forces (reflected by the skew-symmetric property). This means that they will not lead to energy dissipation and that the fact that the gyroscopy leads to a damping matrix contribution, in fact is misleading. Because we do not have any additional dampers in our system this means that we will end up with only imaginary eigenvalues.
In Fig 4.8 the Campbell plot is shown for the 4 smallest eigenvalues for a rotorspeed range of $0 \leq \Omega \leq 80$ [Hz]. In the Campbell-plot also the line

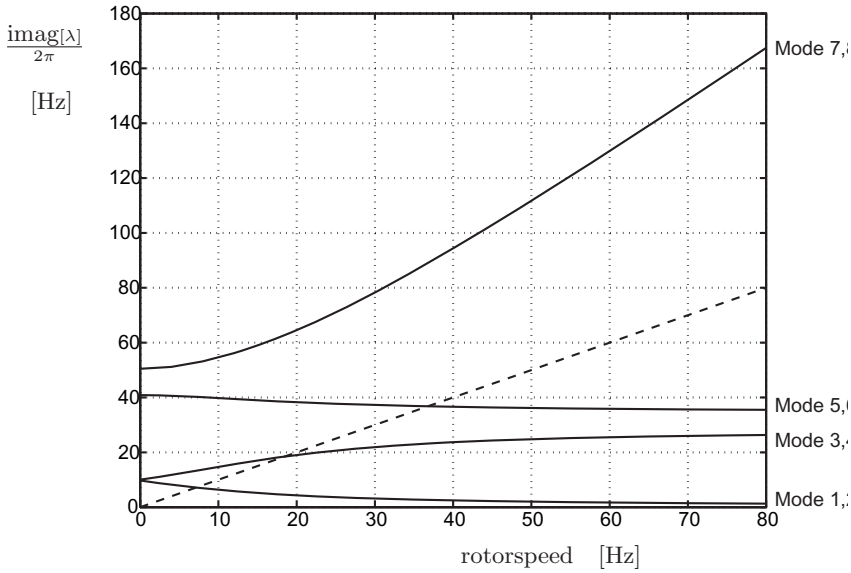


Fig. 4.8. Campbell plot for Industrial Fan

where the damped eigenfrequency equals the rotorspeed is shown. The intersections of this line with the eigenvalue curves are the critical speeds of the system. These critical speeds can be extracted and compared with the values from Krämer as shown in table 4.3 The differences between the results from $\boldsymbol{R_O D_Y}$ and given in Krämer are due to the fact that the $\boldsymbol{R_O D_Y}$ model probably is not exactly the same as Krämer's model. In Krämer's book not all the details are given so some parameters had to be chosen as good as possible.

RODY	Krämer	Whirl type
7.1 Hz	7.68 Hz	Forward whirl
18.4 Hz	13.1 HZ	Backward whirl
36.8 Hz	31.7 Hz	Forward whirl

Table 4.3. Critical speeds according to $\boldsymbol{R_O D_Y}$ and Krämer

In a free vibration analysis not only the eigenvalues are important but also the corresponding eigenvectors or eigenmodes. These eigenmodes in general are complex modes which means that the best way to look at a specific eigenmode is to create an animation in a 3-dimensional representation. In the Figures 4.9 and 4.10, two of the eigenmodes, calculated for a rotorspeed of 12 [Hz] are shown. One of the modes is a backward whirl (eigenfrequency is 5.826 [Hz]), the second is a forward whirl mode with an eigenfrequency of 15.66 [Hz]. The central axis (the Y-axis) is shown and the positive direction is from bottom left to upper right. The positive rotorspeed is clockwise around this axis. In these figures, first the rotor center line at $t = 0$ is shown and starting from that line also the orbits of the shaft center at the nodal positions of the model. These orbits are not completely closed which allows for a determination of the whirl direction of the mode. For both modes the orbits at the bearing positions are very small and the orbits at the disk position but also the nodal orbit close to the left bearing are relatively large. In Section 1.5, page 35, a systematic procedure is given to determine the whirl type automatically from the complex eigencolumn.

For zero rotorspeed, we can recognize two eigenmodes with almost the same eigenvalue of approximately 10 [Hz]. That the bearing stiffnesses are very nonsymmetrical has almost no effect because the smaller shaft bending stiffness is dominating the deformation for these modes. For the higher eigenvalues we can see that the nonsymmetric bearing stiffnesses leads to differences for the two directions. One of the modes for zero rotorspeed is a mode in the X-direction and the other mode a mode in Z-direction. When the system starts rotating, the gyroscopic effect will change one mode in a forward whirl mode with increasing eigenvalue and the other mode in a backward whirl mode with a decreasing eigenvalue. We have seen in Chapter 1 that the forward whirl mode will be excited by a mass

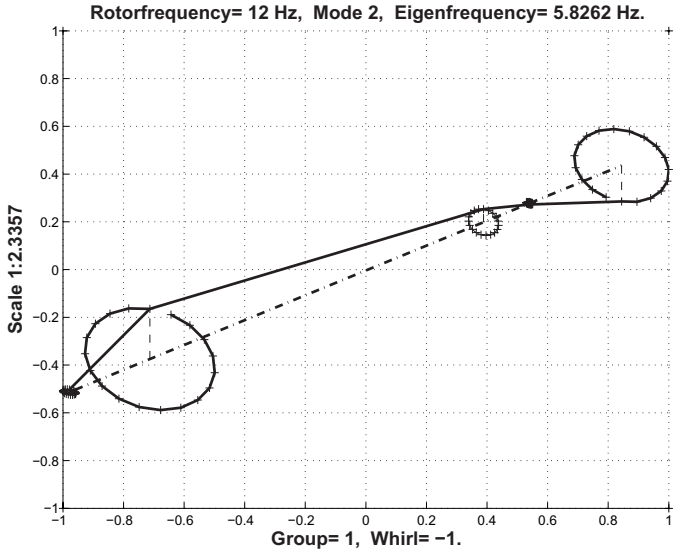


Fig. 4.9. Mode 2, Eigenfreq. 5.826 Hz, Backward whirl

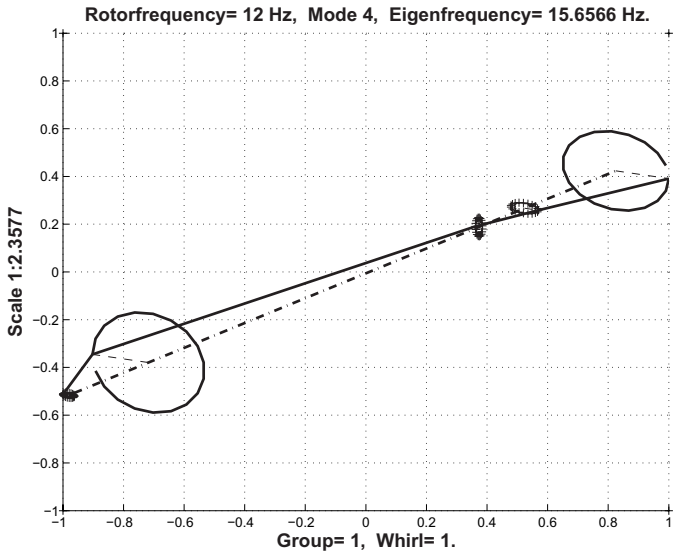
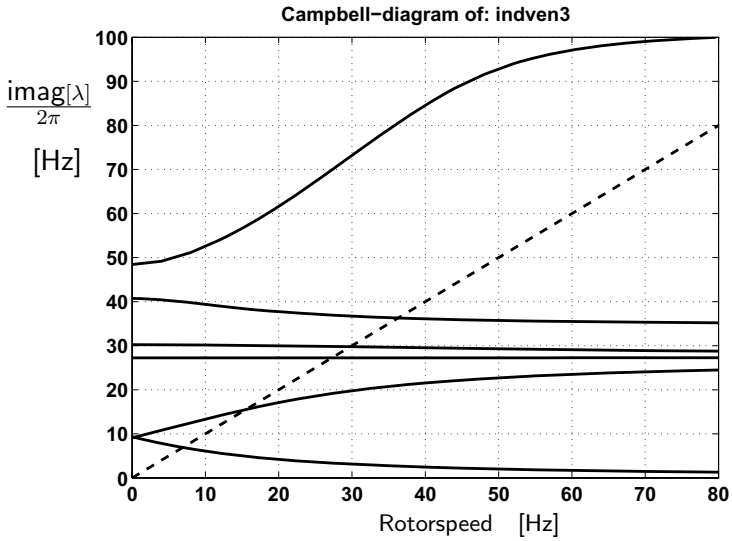


Fig. 4.10. Mode 4, Eigenfreq. 15.66 Hz, Forward whirl

unbalance and therefore will clearly be seen in practical rotor responses. The backward whirl mode theoretically cannot be excited by a mass unbalance and therefore it is frequently overlooked in rotor dynamics literature.

Note

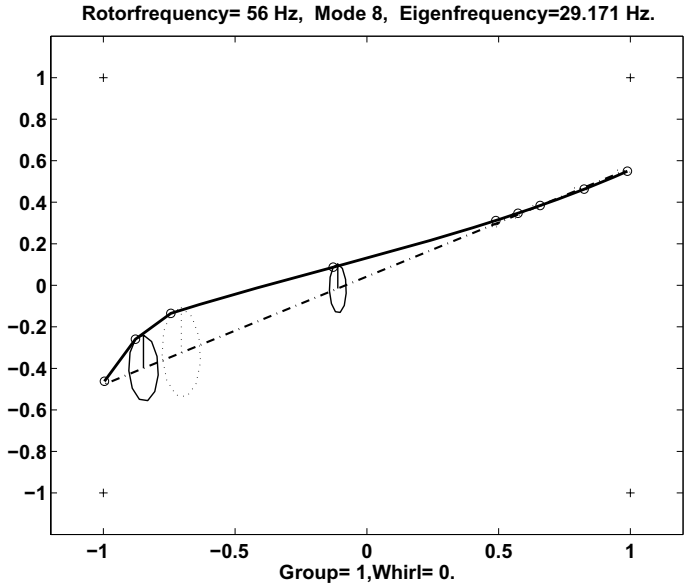
In the analysis above, the simple SHAFT8 element has been used and we also assumed that the mass of the shaft might be neglected (by taking a density of $\rho = 1.0\ [kg/m^3]$). Here we first will redo the analysis by assuming a real value for the mass of the shaft ($\rho = 7800\ [kg/m^3]$) and secondly, by also taking into account the option of shear by using the CONE8 element.
In the next figure, the Campbell plot for the first case is shown.



Campbell plot for Industrial Fan,
taking mass of the shaft in account

If we compare this plot with Figure 4.8 we see that some general tendencies are the same but that certainly also are some clear differences. The most striking difference is that we now have 2 extra modes with a frequency of $\approx 30\ [Hz]$, which appear to be almost independent of the Rotorspeed Ω. Taking a look at the corresponding mode shapes (one of them is given below) illustrates that in case of these modes we are dealing with a more or less transverse displacement of only the part of the shaft between the two bearings.
So, obviously by making the mass of the shaft zero, these modes (in the two transverse directions will go to infinity (and disappear from

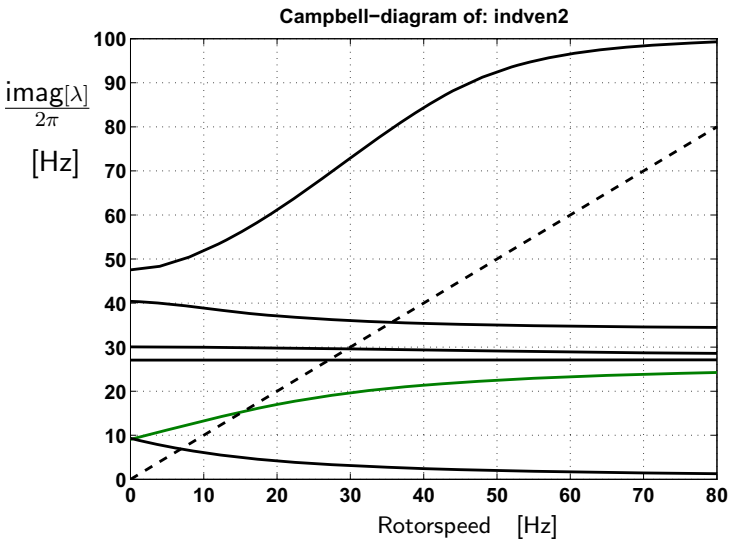
the plot). So, just doing the analysis by neglecting the mass of the shaft with respect to the large mass of the disk, makes that we are missing some essential dynamic information.



One of the surprising modes when using a realistic shaft-mass

Secondly, we will look at the effects of also taking into account shear-deformation and rotary-inertia of the shaft. Therefore the same analysis has been done again and the resulting Campbell plot is shown next.

At first sight the additional effect of using the more advances CONE8 element is very small.



Campbell plot for Industrial Fan,
using CONE8 elements

If we look at the calculated Critical Speeds for the 3 models we get the following:

SHAFT8 $\rho = 1$	SHAFT8 $\rho = 7800$	CONE8 $\rho = 7800$
7.1 Hz	6.92 Hz	6.92 Hz
18.4 Hz	15.51 Hz	15.36 HZ
not	27.24 Hz	27.07 HZ
not	29.77 Hz	29.59 HZ
36.8 Hz	36.29 Hz	35.65 Hz

In general we can see that the Critical Speeds for the model including shear are somewhat smaller, which is obvious, due to the additional flexibility of the element. However the effect is small for this relatively slender shaft.

End of note

4.4 Three-Disk Rotor

The third example has been taken from [Lalanne/Ferraris-90], pp 68-76. The rotor consists of a solid shaft with uniform diameter. On the shaft three rigid disks are mounted and the shaft is supported by two bearings as shown in Fig.4.11. We will use a very rough model based on only 4 shaft

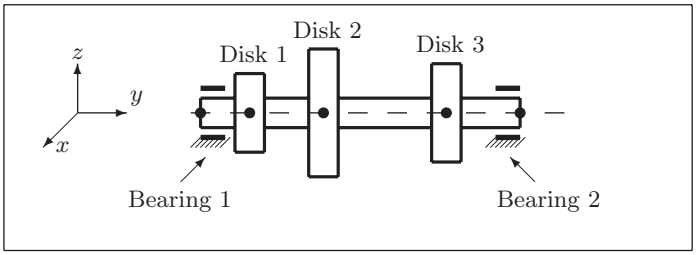


Fig. 4.11. Three-disk Rotor

elements but the shear deformation will be taken into account. So we will end up with a model of 20 degrees of freedom. In the calculations, the following parameter values are used

- **Disk elements**

nr.	Mass $[kg]$	I_p $[kgm^2]$	I_t $[kgm^2]$
1	14.58	0.1232	0.06464
2	45.95	0.9763	0.4977
3	55.14	1.172	0.6023

- **Shaft elements**

nr.	Length $[m]$	Diam. $[m]$
1	0.2	0.1
2	0.3	0.1
3	0.5	0.1
4	0.3	0.1

- **Bearings**

Stiffness $[Nm^{-1}]$ $k_{xx} = 5.0\ 10^7$ $k_{zz} = 7.0\ 10^7$ $k_{xz} = k_{zx} = 0$
Damping $[Nsm^{-1}]$ $c_{xx} = 5.0\ 10^2$ $c_{zz} = 7.0\ 10^2$ $c_{xz} = c_{zx} = 0$

- **Material data**
  - Modulus of Elasticity $E = 2.0\ 10^{11}\ [Pa]$
  - Shear modulus $G = 7.6923\ 10^{10}\ [Pa]$
  - Shear factor $\kappa = 0.9\ [-]$
  - Density $\rho = 7800\ [kgm^{-3}]$

So, we see that the bearings have a direct stiffness- as well as a direct damping contribution but that there is no cross-coupling effect. The two bearings are identical but also symmetrical (equal in x- and z-direction).
The 8 smallest eigenvalues have been calculated as function of the rotorspeed Ω in the range of $0 \leq \Omega \leq 500\ [Hz]$. The results are shown in the Campbell plot, shown in Fig. 4.12. If we look at the intersections between

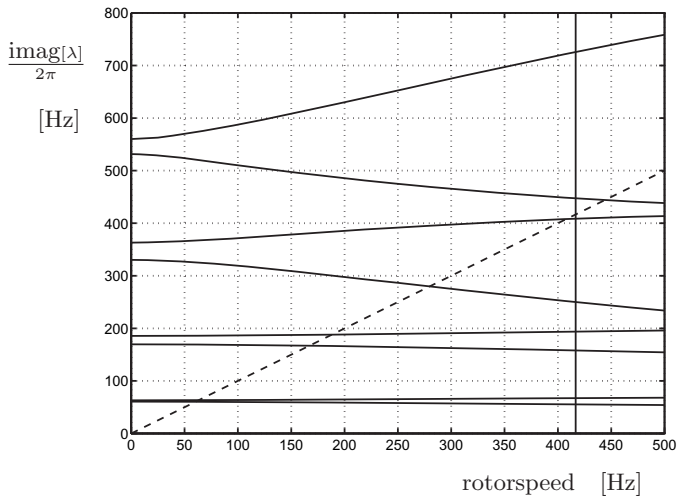


Fig. 4.12. Campbell plot for three-disk rotor

the given curves and the line for which the damped eigenfrequency equals the rotorspeed (dashed line) we can recognize 7 critical speeds. They appear to be 60.34, 63.32, 167.00, 188.08, 279.77, 407.85 and 444.00 [Hz]. These critical speeds will later show up again if we look at the unbalance response of the system.
Next we will look at the eigenvalues for a fixed rotorspeed of 25000 [RPM] or 416.67 [Hz]. In the Campbell plot this rotorspeed is indicated by the vertical line. In Table 4.4 the results for the 10 smallest eigenvalues are shown as given by Lalanne on page 71, and the results derived with $\boldsymbol{R_O D_Y}$

for two situations: including shear deformation and rotary inertia (3rd and 4th column) and without these effects (5th and 6th column). The results

mode nr	Lalanne (52 dofs)	$\boldsymbol{R_O D_Y}$ CONE8 (20 dofs)	% error	$\boldsymbol{R_O D_Y}$ SHAFT8 (20 dofs)	% error
1	55.408	55.417	0.016	55.575	0.3
2	67.209	67.213	0.006	67.554	0.5
3	157.90	157.95	0.03	159.18	0.8
4	193.71	193.75	0.02	196.23	1.3
5	249.90	250.04	0.06	251.16	0.5
6	407.62	408.58	0.24	415.45	1.9
7	446.62	447.36	0.17	459.12	2.8
8	715.03	725.67	1.4	734.23	2.7
9	622.65	624.15	0.24	640.76	2.9
10	1093.0	1091.5	0.13	1120.6	2.5

Table 4.4. Results for the 10 smallest eigenvalues at a fixed rotorspeed of 10.000 [RPM]

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of Lalanne are based on a model with 13 shaft elements (52 degrees of freedom). From these results we can conclude that a model based on only 4 shaft elements already gives very nice results, certainly if we take shear deformation and rotary inertia of the shaft elements into account. Additionally, the eigenmodes corresponding to the 4 smallest eigenvalues for the fixed rotor speed of 10000 [RPM] are shown in the Figs. 4.13, 4.14, 4.15 and 4.16. First we can recognize a pair of modes, corresponding to the

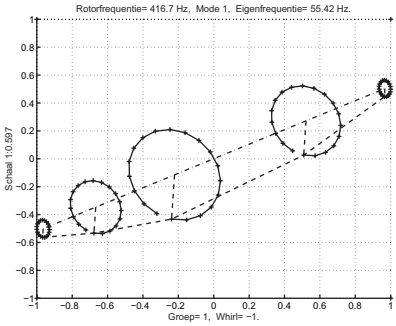


Fig. 4.13. Mode 1, Backward whirl

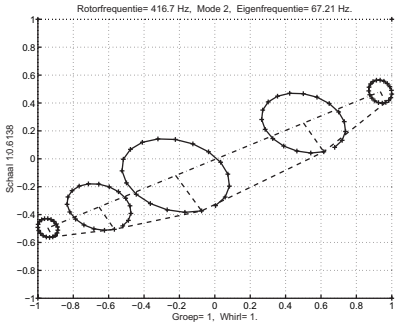


Fig. 4.14. Mode 2, Forward whirl

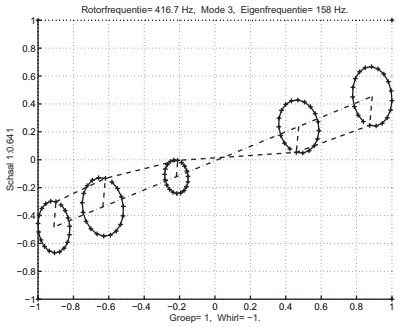


Fig. 4.15. Mode 3, Backward whirl

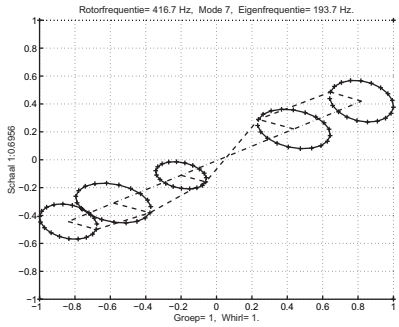


Fig. 4.16. Mode 4, Forward whirl

lowest bending mode of the rotor, one in backward whirl and the other in forward whirl. The next pair is related to the second bending mode, and we can see again a forward whirl- and a backward whirl version.

Finally we will look at the mass unbalance response of this rotor. Therefore we assume that we have a mass unbalance on the middle disk of 0.2 [kg] at a radius of $\epsilon = 0.001\ [m]$. For the response we will also look at the translational degrees of freedom of this middle disk, namely the dof's q_9 and q_{10}. In Fig. 4.17 the amplitudes of these degrees of freedom are shown as function of the rotorspeed. We can see that at the critical speeds, mentioned before we are dealing with resonance. Because the system is only weakly damped (which could be verified by plotting the real parts of the lower eigenvalues as function of the rotorspeed), we meet very sharp peaks resulting in very high stresses in the system when running close to one of the critical speeds. So, the damping in the system should be increased (for example by adding a squeeze film damper) or the passage of, or operation close, to a critical speed should be avoided.
Finally we can look at the orbits of points of the shaft center in the x-z plane. They can for example give us information about the motion of the shaft at the bearing location(s). If we compare these orbit-sizes with the bearing clearances we can decide whether the linear bearing model which has been applied in the analysis will be valid or not. In Fig. 4.18 the orbits of

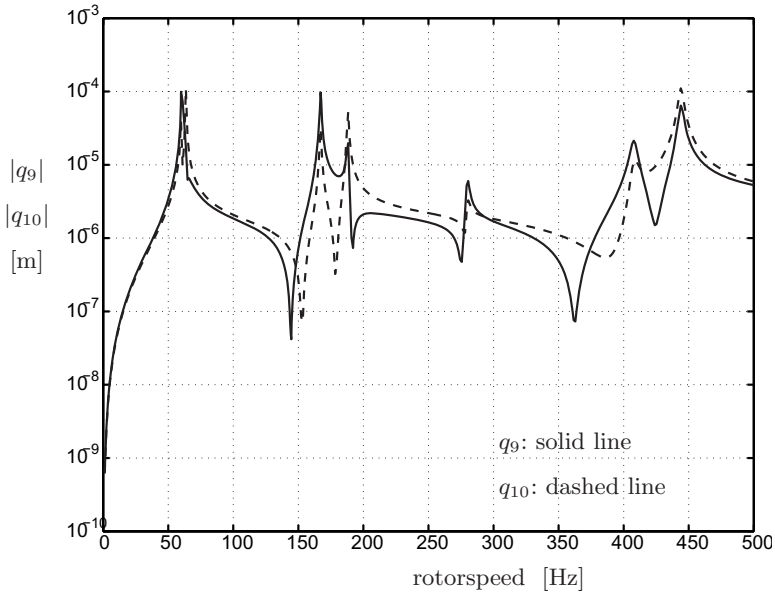


Fig. 4.17. Amplitude responses for unbalance excitation of three-disk rotor; ($m = 0.2$ $[kg]$, $\epsilon = 1.0\ 10^{-3}$ $[m]$)

4 point of the shaft center (the two bearing locations and the nodal points corresponding to the disks 1 and 2) are shown for a rotorspeed of 57 [Hz]. In the upper left plot we see the orbit of the shaft in bearing number 1. The dof-nr. 1 is the displacement in x-direction and dof-nr. 2 the displacement in z-direction. This means that the y-axis is perpendicular to and pointing into the paper and consequently that the rotorspeed is positive in clock-wise direction. The dot and the gap in the orbit indicate the whirl direction so we see that all the orbits are forward whirl orbits (what we already expected because an unbalance excitation only excites the forward whirl). We also can see that for all the orbits the displacement in x-direction is larger then the displacement in z-direction. This is due to the fact that the stiffness in z-direction is 40 % larger then the stiffness in x-direction for both the bearings. Finally it also is obvious that the displacements at the disk locations are roughly 3 times as large as the bearing location displacements.

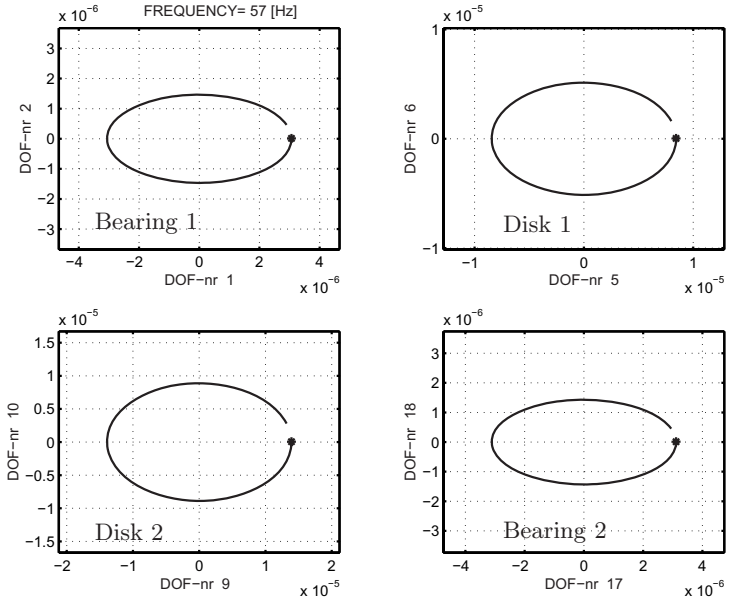


Fig. 4.18. Orbits of 4 shaft center points for a rotorspeed of 57 [Hz]

5

Balancing

How a balancing machine works

With the rotating part resting on the bearings, a vibration sensor is attached to the suspension. In most soft-bearing machines, a velocity sensor is used. Accelerometers, which measure acceleration of the vibration, can also be used. A photocell (sometimes called a phaser), proximity sensor, or encoder is used to determine the rotational speed, as well as the relative phase of the rotating part. This phase information is then used to filter the vibration information to determine the amount of movement, or force, in one rotation of the part. Also, the time difference between the phase and the vibration peak gives the angle at which the unbalance exists. Amount of unbalance and angle of unbalance give an unbalance vector. Calibration is performed by adding a known weight at a known angle. In a soft-bearing machine, trial weights.....

http://en.wikipedia.org/wiki/Balancing_Machine

5.1 Introduction

Perhaps one of the most important sources of excitation of rotorsystems is mass unbalance. This means that the center of mass of a disk, a shaft or a combinations of disks and shaft is not exactly on the rotation axis. When the system starts rotating, this center of mass offset will generate dynamic forces (so-called centrifugal forces) which are proportional to the square of the rotorspeed. We have seen in the derivation of the equations of motion for the rigid disk, that also some disk skewness has the same effect, namely it will generate an exciting moment, which again is proportional to the rotorspeed squared. In both cases the excitation rotates with the same speed and in the same direction as the rotorspeed. Therefore this will lead to a so-called **synchronous** motion.
In general, ons can distinguish three measures to reduce the effect of such an unbalance on the response of a rotorsystem

(1) Balancing the rotorsystem
(2) Changing the operational rotorspeed range of the system
(3) Adding damping to the system

In this chapter we will concentrate on the first option in which we will try to reduce the source of the vibration whereas in the other two measures only the resulting vibration will be made acceptable.
In a practical situation, perfectly balanced rotorsystems do not exist, so any rotorsystem will have some (hopefully small) unbalance. This is due to limited fabrication tolerances, material inhomogeneities, system flexibility, non-linearities (for example in shrink-fits) time-dependent material effect (such as creep in non-ferro flywheels, etc). In spite of all these problems it will be possible in many practical situations to reduce these synchronous vibrations to an acceptable level by balancing the rotorsystem.

5.2 Some Basic Principles

If a rotor might be assumed to be completely rigid in the relevant rotorspeed range, the balancing is not such a difficult task. In general this rigidity assumption will never be completely correct but in many situations it will be an acceptable starting point. This is for example the case for relatively low rotorspeeds or a stiff rotor in flexible supports (bearings). We will therefore first look at balancing techniques which are fundamentally based on this assumption of a rigid rotor.

A rigid, rotating body can be defined to be perfectly balanced when one of the principal axis of inertia exactly coincides with the rotation axis. This rotation axis in general can be seen as the geometrical connection line between the bearing centers which support the rotor. An alternative definition which is based on the effect of a possible unbalance is that a rotorsystem is called perfectly balanced when the synchronous vibrations are completely eliminated.

A rigid rotor can be balanced in principle by adding mass to the rotor in two arbitrary planes perpendicular to the rotation axis (similar to the balancing of new car-tyres). An alternative for adding mass is removing mass at the opposite position by drilling small holes at proper places. These mass corrections should do two things. First they should move the center of mass of the complete system to a point on the rotation axis and secondly they should lead to a rotation of the relevant principal axis of inertia in the direction of this rotation axis. To perform this, it will also be necessary to carry out vibration measurements, also in two planes perpendicular to the rotation axis. These planes however can be different from the mass-correction planes.

5.3 Single Plane Balancing

The first step in balancing a rotor is removing the static unbalance. To illustrate this we look a the rigid rotor with mass M (shown in Fig. 5.1) which in itself is assumed to be perfectly balanced. A coordinate frame

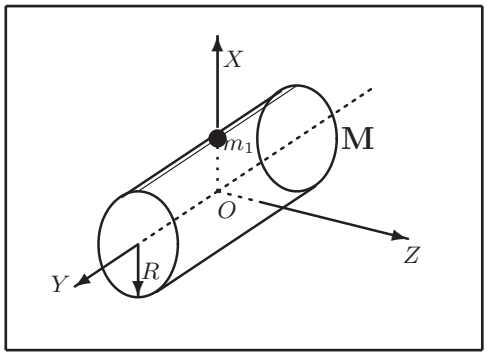


Fig. 5.1. Static rotor unbalance

$(O - XYZ)$ is fixed to the rotor. The origin coincides with the center of mass of the rotor and the Y-axis with the rotation axis. A small unbalance mass m_1, lying on the $X - axis$, on a distance R from the origin, will generate a so-called static unbalance which can be translated to a mass-eccentricity u for the whole rotor

$$u = \frac{m_1 \; R}{M + m_1} \qquad [m] \tag{5.1}$$

If the unbalanced rotor is supported in two low friction bearings, or is placed on two knife-edge supports the gravity effect will rotate the rotor until the X-axis is pointing downward exactly. This defines the direction of the unbalance and also the place where some additional mass should be added or removed. Using some trial and error this can be done until the rotor has no preferential position anymore. A more direct approach consists of rotating the rotor with a known rotorspeed in two bearings which are equipped to measure accurately the bearing load. The unbalance will create a centrifugal force $F_c = m \; R \; \Omega^2$. If the rotor for example is supported symmetrically, the bearing forces will be equal and each will be $F_c/2$. Then directly the quantity $m \; R$ which is necessary to balance the rotor can be calculated. So,

choosing the correction radius R gives the necessary m, or vice versa. This technique is called *single-plane balancing* or *static balancing*.

A second procedure for single plane balancing is the so-called dynamic balancing based on the evaluation of the centrifugal forces due to first the unknown, and additionally a known unbalance at a fixed rotorspeed. In a fixed reference frame this will lead to a harmonic excitation and -response. First the principle of this technique will be considered. The unknown unbalance will generate an unknown harmonic force $f_o(t) = F_o\ e^{j\Omega t}$ and this force will generate a measurable response $x_o(t) = X_o\ e^{j\Omega t}$. So we can write

$$X_o = H(\Omega)\ F_o \tag{5.2}$$

where $H(\Omega)$ is the unknown transfer function of the system and X_o respectively F_o are the **complex** amplitudes of respons respectively the excitation. If we next add a **known** test-mass we get the total excitation $F_o + F_m$ and the resulting total measurable response X_{om}, for which holds

$$X_{om} = H(\Omega)\ [F_o + F_m] = X_o + H(\Omega)\ F_m \tag{5.3}$$

This gives directly

$$F_o = F_m\ \frac{X_o}{X_{om} - X_o} \tag{5.4}$$

From the value and the position of the additional (known) test-mass we know F_m and from the measured responses X_o and X_{om} (magnitude and phase) the value and position of the necessary balancing mass (F_o) can be calculated.

The practical set-up to carry out this type of balancing is schematically shown in Fig 5.2. The structural part which has to be balanced (called the rotor) is mounted on a balancing machine. This machine consists of a motor and a bearing to support the rotor. A fixed rotorspeed Ω is selected. The accelerations of the bearing housing can be measured using for example an accelerations and this harmonic signal with frequency Ω is send to a frequency-analyzer which can measure the amplitude of the signal but which also can generate a trigger signal (some signal which is just in phase with the measured signal). This trigger signal is send to a stroboscope. So, for example when the measured acceleration will reach its maximum value (or pass through zero) the stroboscope gives a short light-flash. The procedure starts with a non-rotating system and creating a mark on the rotor

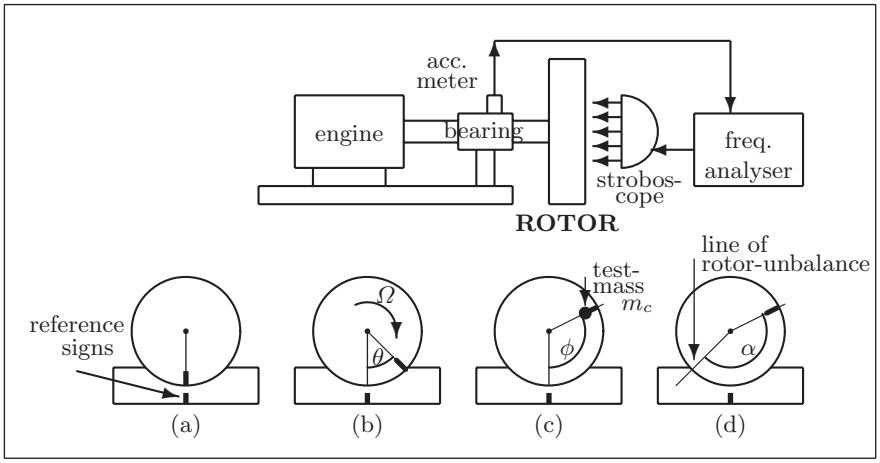


Fig. 5.2. Test-mass-balancing technique

and a mark on the balancing machine as shown in Fig 5.2(a). Next the rotation with fixed rotorspeed Ω starts. Due to the unknown unbalance, the rotor-support-system will be excited and the stroboscope will lighten some position (defined by the angle θ) of the rotor as shown in Fig 5.2(b). This angle θ and the measured amplitude A_o determine the complex respons X_o. In the third step a **known test-mass** m_c is mounted on the rotor and the stroboscope will give the picture shown in Fig 5.2(c). The corresponding phase ϕ and measured amplitude A_{om} determine the complex respons X_{om}. So in this last situation we are dealing with an excitation due to the sum of the (unknown) initial unbalance and the (known) additional unbalance.[1] Using the vector diagram shown in Fig. 5.3, the necessary correction mass (weight and position) which is needed to balance the rotor can be determined. In this diagram the initial response vector X_o is drawn in an arbitrary direction. The (combination) response vector X_{om} is drawn at an angle $\phi-\theta$ with X_o and a length of A_{om}. The difference vector $X_c = X_{om} - X_o$ gives the complex unbalance response of only the test mass. The amplitude and phase of this response are

$$A_c = [A_o^2 + A_{om}^2 - 2A_oA_{om}\cos(\phi - \theta)]^2 \tag{5.5}$$

[1] Instead of marks and a stroboscope, also some electronic trigger signal (indicating a fixed position of the rotor shaft) can be used to extract the phase information from the digitized measurement signals

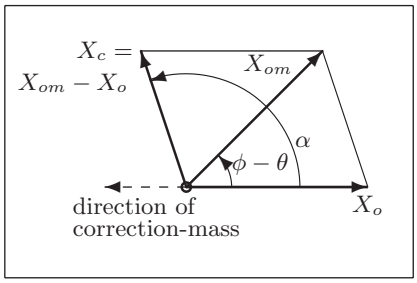


Fig. 5.3. Vector-diagram for estimating the balancing-mass

$$\alpha = \arccos(\frac{A_o^2 + A_c^2 - A_{om}^2}{2A_oA_c}) \tag{5.6}$$

The angle α gives the direction of the rotor unbalance with respect to the test mass position as shown in Fig. 5.2(d). Also the size (m_c) of the test mass in known. The size of the rotor unbalance then follows from

$$m_o = m_c\ (\frac{A_0}{A_m}) \tag{5.7}$$

If, for the balancing the same radius is used as the test mass radius, this m_o directly gives the necessary balancing masses. The angle α determines the position of this mass.

5.4 Two Plane Balancing

Single plane balancing can typically be used for (thin) disk-type rotors with an eccentric center of mass as unbalance. In practice however, we frequently are dealing with rotors of a type as shown in Fig. 5.4. The unbalance can be distributed along the axis of the rotor. The balancing now can take place in two planes perpendicular to the rotation axis. For these planes the end-planes of the rotor might be used. A first important remark is that

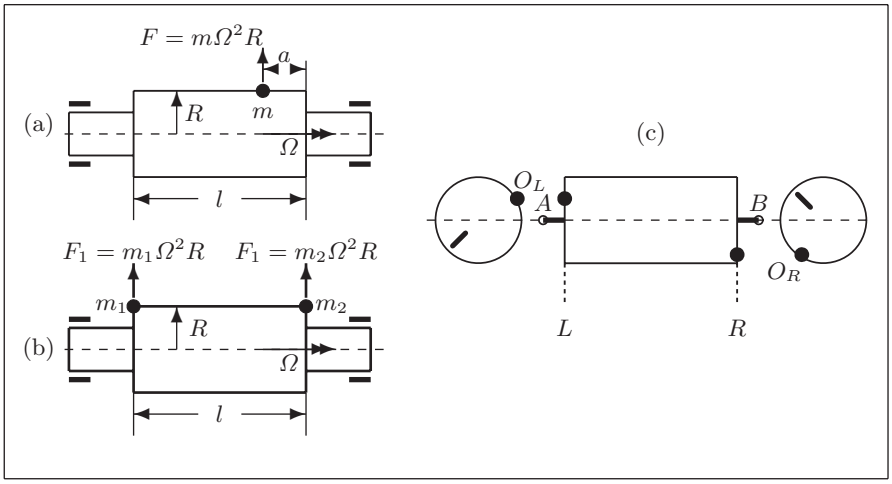


Fig. 5.4. General set-up for two-plane balancing

for a rigid rotor any distributed unbalance can be replaced by two discrete unbalance masses. For example, the arbitrary unbalance mass m shown in Fig. 5.4 (a) on a radius R can be replaced by two masses m_1 and m_2 (as shown in Fig. 5.4 (b), both at the radius R in the same direction if we take $m_1 = m\ a/l$ and $m_2 = m\ (1 - a/l)$. So we can also use the inverse of this principle which means that any unbalance of the rotor can always be balanced by adding two masses in two different planes.

In this method again the frequency analyzer is used. We assume that the total, unknown rotor unbalance can be modeled as a combination of two fictitious unbalance masses O_L and O_R, in de endplanes R and L at some points at a circle with radius R. This circle can be any circle. It is just the place where the fictitious imbalances are thought and later it will be

used to attach the necessary balancing masses. So the unbalance masses are unknown and their positions on the circle is unknown. For a fixed and known rotorspeed, the amplitude and phases of two response signals (for example the vertical accelerations of the bearings at A and B) are measured for the initial, unknown unbalance. This gives the complex numbers V_A en V_B. Then we can write

$$\begin{aligned} V_A &= Z_{AL}\ O_L + Z_{AR}\ O_R \\ V_B &= Z_{BL}\ O_L + Z_{BR}\ O_R \end{aligned} \tag{5.8}$$

where Z_{ij} are the so-called *dynamic influence coefficients* which define the effect of an unbalance in plane $j(j = L, R)$ on the bearing-responses $i(i = A, B)$. All these factors including O_L and O_R are unknowns in these equations.
In the second step we attach some known test mass M_L to the left plane of the rotor (at an arbitrary point on the chosen circle and measure the amplitudes and phases of the bearing responses again. This gives

$$\begin{aligned} V_A^L &= Z_{AL}\ (O_L + M_L) + Z_{AR}\ O_R \\ V_B^L &= Z_{BL}\ (O_L + M_L) + Z_{BR}\ O_R \end{aligned} \tag{5.9}$$

Combination of (5.8) and (5.9) gives

$$Z_{AL} = \frac{V_A^L - V_A}{M_L}, \qquad Z_{BL} = \frac{V_B^L - V_B}{M_L} \tag{5.10}$$

After removing this test mass the procedure will be repeated with another test mass M_R attached to the right plane of the rotor, giving

$$\begin{aligned} V_A^R &= Z_{AL}\ (O_L + M_R) + Z_{AR}\ O_R \\ V_B^R &= Z_{BL}\ (O_L + M_R) + Z_{BR}\ O_R \end{aligned} \tag{5.11}$$

Combining (5.8) and (5.11) gives

$$Z_{AR} = \frac{V_A^R - V_A}{M_R}, \qquad Z_{BR} = \frac{V_B^R - V_B}{M_R} \tag{5.12}$$

Now that all the dynamic influence coefficients Z_{ij} are known, the unknown unbalance masses O_L and O_R (size as well as position or phase) can be calculated from

$$
\begin{aligned}
O_L &= (Z_{BR}\ V_A - Z_{AR}\ V_B)/(Z_{BR}\ Z_{AL} - Z_{AR}\ Z_{BL}) \\
O_R &= (Z_{BL}\ V_A - Z_{AL}\ V_B)(Z_{BL}\ Z_{AR} - Z_{AL}\ Z_{BR})
\end{aligned}
\tag{5.13}
$$

The balancing itself now consists of attaching two masses O_L en O_R in the two rotor planes on the circles , but 180 degrees out of phase with the fictitious unbalance mass positions.
In commercial balancing machines a procedure like the one indicated here is programmed in some automatic way. Very large en complicated machines like turbine and generator units or compressors normally are a combination of cooperating rotating parts. The usual procedure then is to balance each part after its manufacturing on special balancing machines. After assembling all the components on the final foundation and bearings, a second balancing operation has to be carried out for the total system for the following reasons

- The mutual fitting of the components will never be perfect. We always have clearances, manufacturing tolerances, some pre-loading etc.,
- Certainly in case of rotors with fluid film bearings, the bearing stiffness and damping plays an important role in the dynamic behavior and therefore it will influence the balancing,
- The dynamic behavior of a machine can (strongly) change in time due to for example wear, small damages, etc.

The two-plane balancing technique as outlined before is therefore very frequently used in practical situations. It can be done on a dedicated balancing machine or in situ using the installed rotorsystem.

5.5 Effect of Rotorflexibility

If in a practical situation none of the balancing techniques mentioned before will be successful in reducing the synchronous vibrations in some operating range of a rotating system, then the problem might be that the rotor may not be assumed to be rigid in that speed range. This certainly will be the case when one or more of the critical speeds is close to or even in the speed range and if in the modes, corresponding to these critical speeds considerable bending of the rotor is involved. In that case the balancing of the rotor can change with the rotorspeed. The elastic deformation of the rotor will lead to a redistribution of mass relative to the rotation axis. This will result in a change of the center of mass or a rotation of the direction of the principal moment of inertia. In Fig 5.5 this is illustrated on a simple example. verduidelijkt. In the upper part (a), a symmetric rotor is shown

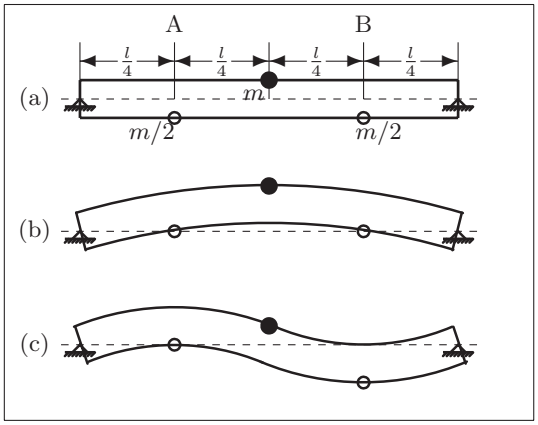


Fig. 5.5. Effect of rotorflexibility on the balancingprocess

supported in two bearings with some rotor unbalance m in the middle. The rotor will be balanced as explained in the previous section resulting in a pair of two correction masses $m/2$, one in each of the balancing planes A and B. At a speed close to the first critical speed and for relatively stiff bearings, the mode shape will be as plotted in the middle part (b). Now the centrifugal force due to the unbalance is much larger, but the compensating effect of the two balance masses has disappeared. So we will end up with very high bearing loads in this situation. Making the balance masses much

larger will destroy the balanced situation for lower rotorspeeds.
The lower part (c) shows the rotor close to the second critical speed. Now especially a moment excitation of the rotor will take place again leading to high bearing loads. Intuitively, it looks to be possible to give the optimal balancing state for each situation (a), (b) and (c) separately, but it is not possible to do this for all cases simultaneously. In practice the *rule of thumb* is used that by applying only two balancing planes a flexible rotor can be balanced for one single rotorspeed. If additionally, the system appears to be very sensitive for speeds close to some critical speed and the corresponding mode has a clear elastic deformation then balancing will be very difficult anyhow.
In theory the number of balancing planes should be equal to the number of critical speeds that is present in the operating range of the system and these planes should be chosen carefully by looking at the mode shapes corresponding to the critical speeds. In the next section this will be discussed.

5.6 Balancing Flexible Rotorsystems

Theoretically the number of balancing planes should be equal to the number of critical speeds in the operating range of the rotorsystem. In practical situations however a large number of arbitrary balancing planes is not available. This means that with a limited number of balancing planes which cannot be chosen anywhere, some optimal situation has to be created. A well-known technique to do this is called *Least Squares Balancing*. We will only look at the principle of this technique (see also [Lund/Tonnesen-72]). We start from

- B Balancing planes
- T Different rotorspeeds or load situations
- R Response measurements

So we assume to have available $S = T * R$ measurements (each measurement being a complex number having an amplitude and a phase). The problem now is to select the (B) correction masses in the (B) balancing planes in such a way that the amplitudes of the S measurements will be minimized. Then two situations can be distinguished

I If $B \geq S$, the amplitudes theoretically can be made zero

II If $B < S$, not enough balancing planes are available for a complete elimination of the dynamic responses. In this case we can only search for a minimum in a least squares sense.

Let us suppose that

* $\underline{V}$, $(S \times 1)$ is the column with all the S measurements,
* $\underline{Z}$, $(S \times B)$ is the generally non-square matrix of so-called influence coefficients,
* $\underline{O}$, $\underline{M}$ and $\underline{\hat{M}}$ are the $(B \times 1)$ complex columns with respectively the unknown imbalances, the known added test masses and the optimal balance masses in each of the balancing planes which we are searching for.

The initial problem with the unknown, uncorrected rotor imbalances can be written as

$$\underline{V}_o = \underline{Z}\ \underline{O} \tag{5.14}$$

After adding a set of test masses we can write

$$\underline{V}_m = \underline{Z}\ [\underline{O} + \underline{M}] = \underline{V}_o + \underline{Z}\ \underline{M} \tag{5.15}$$

The columns of $\underline{V}_m$ are determined by attaching a test mass in one of the balancing planes and no test masses in all the other planes and monitoring all the measurement devices. If we for example attach a test mass m_i to plane i, then the jth equation of (5.15) reads

$$\underline{V}_m(j) = \underline{V}_o(j) + \underline{Z}(j,i)\ m_i \tag{5.16}$$

which directly gives

$$\underline{Z}(j,i) = \frac{\underline{V}_m(j) - \underline{V}_o(j)}{m_i} \tag{5.17}$$

By repeating this procedure for all the balancing planes, all the dynamic influence coefficients $\underline{Z}(j,i)$ can be determined.
When this dynamic influence matrix $\underline{Z}$ is known, the optimal set of balancing masses can be calculated. For situation I where the number of test masses is greater or equal then the number of measurements, a square matrix can be extracted from $\underline{Z}$, which can be inverted, leading to

$$\hat{\underline{M}} = -\underline{Z}^{-1}\underline{V}_o \tag{5.18}$$

Substitution of this result for $\underline{M}$ in (5.15) obviously gives $\underline{V}_m = \underline{0}$. In the more general situation where we have more measurements then balancing planes, we can minimize the **real** scalar function

$$\begin{gathered} \Sigma = [\underline{V}_o - \underline{Z}\ \underline{O}]^H [\underline{V}_o - \underline{Z}\ \underline{O}] = \\ \underline{V}_o^H \underline{V}_o - \underline{O}^H \underline{Z}^H \underline{V}_o - \underline{V}_o^H \underline{Z}\ \underline{O} + \underline{O}^H \underline{Z}^H \underline{Z}\ \underline{O} \end{gathered} \tag{5.19}$$

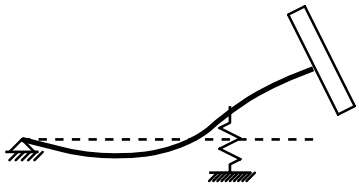
Here $\underline{A}^H$ is the Hermitic transpose of the complex column $\underline{A}$. It means $\underline{A}^H = \bar{\underline{A}}^T$, so we not only must take the (normal) transpose but also the complex conjugate. This error function Σ will have an extremum (minimum) for

$$\frac{\partial \Sigma}{\partial \underline{O}} = \underline{0} \quad \Longrightarrow \quad \hat{\underline{M}} = -\underline{O} = -\{[\underline{Z}^H \underline{Z}]^{-1} \underline{Z}^H\}\ \underline{V}_o \tag{5.20}$$

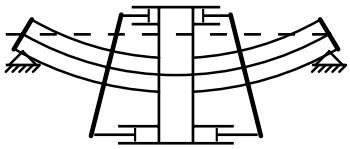
The matrix $[\underline{Z}^H \underline{Z}]^{-1} \underline{Z}^H$ is called the **pseudo-inverse** of the non-square, complex matrix $\underline{Z}$.

6

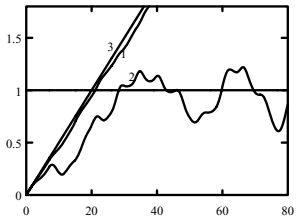
Special Topics



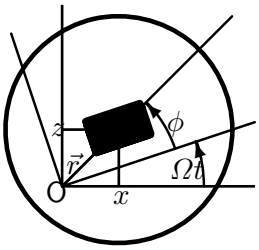
Ritz-Approach



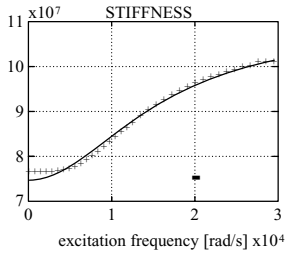
Internal Damping

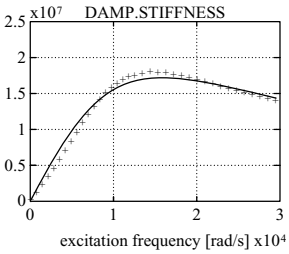


Critical Speed Crossing



Non Rotational Symmetry





Gas Bearings

6.1 Pitfalls of a Rayleigh-Ritz approach

Let us look at the **industrial fan problem** from Section 4.3 and shown in Fig. 4.6. We will try to calculate an approximation for the critical speeds by using the Rayleigh-Ritz approach as presented in Section 1.5. The system which was analyzed there is shown in Fig. 1.11. For the industrial fan problem we make the following approximations:

- due to the fact that the left bearing is not so heavily loaded and the bearing stiffness is relatively high, we assume this point to be rigid and even may be seen as a clamped.
- we assume the shaft diameter is constant over the whole shaft length, and take $d_{shaft} = 0.3\ [m]$

At first sight, this seems to be a situation which might lead to sufficiently accurate results.
Using a quadratic deformation shape for the shaft as motivated in section 1.5 the following equations of motion are valid:

$$\begin{bmatrix} m^* & 0 \\ 0 & m^* \end{bmatrix} \begin{bmatrix} \ddot{u}_l \\ \ddot{w}_l \end{bmatrix} + \Omega \begin{bmatrix} 0 & -m_p \\ m_p & 0 \end{bmatrix} \begin{bmatrix} \dot{u}_l \\ \dot{w}_l \end{bmatrix} + \begin{bmatrix} k_x^* & 0 \\ 0 & k_z^* \end{bmatrix} \begin{bmatrix} u_l \\ w_l \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{6.1}$$

or

$$\underline{M}\ \underline{\ddot{q}} + \underline{B}\ \underline{\dot{q}} + \underline{K}\ \underline{q} = \underline{o} \tag{6.2}$$

With the data from the Fan problem we can find:

- $m^* = 8.7396\ 10^3$
- $k_x^* = 1.6361\ 10^8$
- $k_z^* = 3.2420\ 10^8$
- $m_p = 1.4792\ 10^3$

Next, we can calculate the eigenvalues λ_i for this set of equations for a range of rotorfrequencies $\Omega = 0...80\ \ [Hz]$. For each rotorfrequency Ω this will lead to a set of 2 pairs of 2 purely imaginary, complex conjugate eigenvalues $[\lambda_1 = j\omega_1; \lambda_2 = -j\omega_1; \lambda_3 = j\omega_2; \lambda_4 = -j\omega_2]$, with real ω_i.
The frequencies $f_i = \omega_i/(2\pi)$ are plotted in the following figure as function of Ω.
If we compare these results with the results from the Finite Element calculation as shown in Fig. 4.8 we should conclude that they do not match at all. In the Rayleigh-Ritz approach we only have 2 dof's, so only two

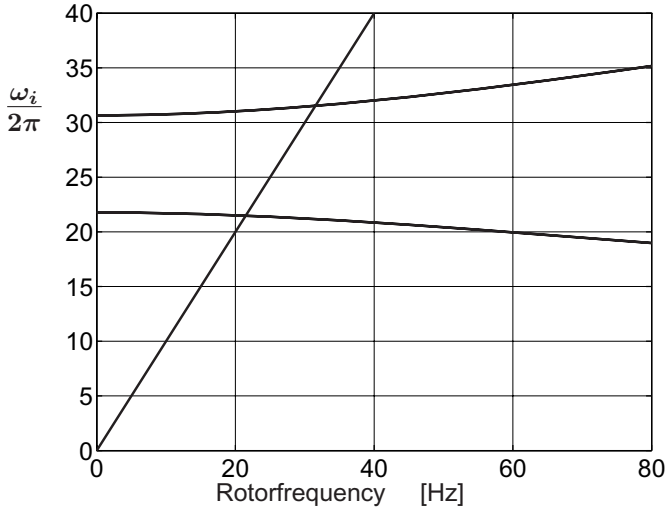


Fig. 6.1. Campbell plot

eigenfrequencies and corresponding modes. But they should at least show a tendency as shown by the 2 smallest eigenfrequencies of the FEM model. In Fig. 4.8 we can see two modes, both starting for $\Omega = 0$ from almost the same point ($\approx 10Hz$), a stiffening forward whirling mode and a softening backward whirling mode (due to the effect of the gyroscopy of the disk at increasing rotorfrequency Ω).
In the Rayleigh-Ritz approach we see 2 modes, a stiffening one starting from 31 Hz at $\Omega = 0$ and a softening one, starting from 22 Hz at $\Omega = 0$. This is an unacceptable discrepancy.

What might be wrong?
For the deformation of the shaft in the Rayleigh-Ritz approach we have assumed a quadratic function. If we look at Fig. 1.11 then at a first glance, this seems to be reasonable starting point.But if we look at the transversal disc- displacement in x- (or z-) direction we can see that

$$u(l,t) = \frac{l^2}{a^2}\, u(a,t)$$

So the disc displacement is just dominated by the bearing stiffness (with only a scale factor l^2/a^2). The much smaller bending stiffness of the shaft has no influence at alle, whereas is should dominate the smallest modes extremely. (for the effective stiffness k_x^* we can find:

$$k_x^* = (4 * EI)/(l^3) + k_x(a^4)/(l^4) \approx 3.0210^6 \ + \ 1.6110^8$$

So, it is logical that in the FEM analysis the (symmetric, shaft bending stiffness dominates whereas in the Rayleigh-Ritz approach the non-symmetric (high) bearing stiffness dominates.
So, we should at least have a model where the bending stiffness of the shaft can play its own role.

Second attempt

Let us redo the analysis with again a quadratic displacement function which fulfils:

- for $y = 0$ the shaft is simply supported $u(0,t) = w(0,t) = 0$
- we will treat the displacements $u(a,t)$ and $u(l,t)$ as independent degrees of freedom (the same for $w(a,t)$ and $w(l,t)$

his is illustrated in Fig. 6.2. We again want to apply the Rayleigh-Ritz

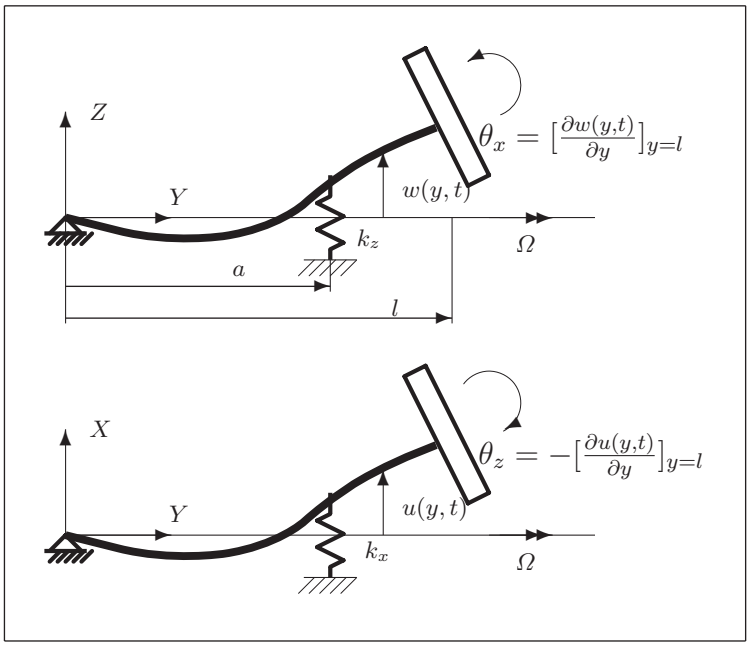


Fig. 6.2. Fan problem with alternative deformation shape

method for the creation of a discrete model for this rotor. For the displacement-functions in x-direction (and z-direction) we choose now the quadratic function:

$$u(y,t) = w(y,t) = a_0 + a_1 y + a_2 y^2 \tag{6.3}$$

As the final degrees of freedom for the problem we choose:

- $u_a(t) = u(y = a, t)$
- $u_l(t) = u(y = l, t)$
- $w_a(t) = w(y = a, t)$
- $w_l(t) = w(y = l, t)$

Together with the zero condition at $y = 0$ we then can write the displacement functions as:

$$det = (al^2 - a^2 l) \tag{6.4}$$

$$u(y,t) = \frac{1}{det}[(l^2 y - l y^2)u_a + (a y^2 - a^2 y)u_l] \tag{6.5}$$

$$w(y,t) = \frac{1}{det}[(l^2 y - l y^2)w_a + (a y^2 - a^2 y)w_l] \tag{6.6}$$

If we first look at the displacement-function $u(y,t)$. If we define the sub-column of degrees of freedom $\underline{q}_u^T = [u_l, u_a]$, we can write:

$$u(y,t) = \frac{1}{det}[a y^2 - a^2 y \ , \ l^2 y - l y^2] \ \underline{q}_u \tag{6.7}$$

$$\frac{\partial u}{\partial y} = \frac{1}{det}[2ay - a^2 \ , \ l^2 - 2ly] \ \underline{q}_u \tag{6.8}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{1}{det}[2a \ , \ -2l] \ \underline{q}_u \tag{6.9}$$

$$\theta_z = -[\frac{\partial u(y,t)}{\partial y}]_{y=l} = \frac{1}{det}[a^2 - 2al \ , \ l^2] \ \underline{q}_u \tag{6.10}$$

With the definition of $\underline{q}_w^T = [w_l, w_a]$, we can write directly:

$$w(y,t) = \frac{1}{det}[a y^2 - a^2 y \ , \ l^2 y - l y^2] \ \underline{q}_w \tag{6.11}$$

$$\frac{\partial w}{\partial y} = \frac{1}{det}[2ay - a^2 \ , \ l^2 - 2ly] \ \underline{q}_w \tag{6.12}$$

$$\frac{\partial^2 w}{\partial y^2} = \frac{1}{det}[2a \ , \ -2l] \ \underline{q}_w \tag{6.13}$$

$$\theta_x = [\frac{\partial w(y,t)}{\partial y}]_{y=l} = \frac{1}{det}[2al - a^2 \ , \ -l^2] \ \underline{q}_w \tag{6.14}$$

For the **Elastic energy** we get

$$U = \tfrac{1}{2}EI\int_0^l[\tfrac{\partial^2 u}{\partial y^2}]^2dx + \tfrac{1}{2}k_x u^2(a,t) + \\ \tfrac{1}{2}EI\int_0^l[\tfrac{\partial^2 w}{\partial y^2}]^2dx + \tfrac{1}{2}k_z w^2(a,t) \tag{6.15}$$

Substitution of the given displacement functions gives

$$U = \frac{1}{2}\,\underline{q}_u^T\left[\frac{EI}{det^2}\int_0^l\begin{bmatrix}2a\\-2l\end{bmatrix}\begin{bmatrix}2a & -2l\end{bmatrix}dx\right]\underline{q}_u + \frac{1}{2}k_x\,u_a^2 \\ + \frac{1}{2}\,\underline{q}_w^T\left[\frac{EI}{det^2}\int_0^l\begin{bmatrix}2a\\-2l\end{bmatrix}\begin{bmatrix}2a & -2l\end{bmatrix}dx\right]\underline{q}_w + \frac{1}{2}k_w w_a^2 \tag{6.16}$$

which can be written as:

$$U = \frac{1}{2}\,\underline{q}_u^T\left[\frac{EI}{det^2}\begin{bmatrix}4a^2l & -4al^2\\-4al^2 & 4l^3\end{bmatrix} + \begin{bmatrix}0 & 0\\0 & k_x\end{bmatrix}\right]\underline{q}_u + \\ +\frac{1}{2}\,\underline{q}_w^T\left[\frac{EI}{det^2}\begin{bmatrix}4a^2l & -4al^2\\-4al^2 & 4l^3\end{bmatrix} + \begin{bmatrix}0 & 0\\0 & k_w\end{bmatrix}\right]\underline{q}_w \\ = \frac{1}{2}\,\underline{q}_u^T\,\underline{K}_u\,\underline{q}_u + \frac{1}{2}\,\underline{q}_w^T\,\underline{K}_w\,\underline{q}_w \tag{6.17}$$

We define the total column of dof's as $\underline{q}^T = [\underline{q}_u^T, \underline{q}_w^T]$. Then we can write the elastic energy as:

$$U = \frac{1}{2}\,\underline{q}^T\,\underline{K}\,\underline{q} \quad \text{with} \quad \underline{K} = \begin{bmatrix}\underline{K}_u & \underline{0}\\ \underline{0} & \underline{K}_w\end{bmatrix} \tag{6.18}$$

For the **Kinetic energy** of the rigid disk we can write (see Section 3.2)

$$T = \frac{1}{2}[m(\dot{u})^2 + m(\dot{w})^2 + J_t(\dot{\theta}_x)^2 + J_t(\dot{\theta}_z)^2 + J_p(\Omega^2 - 2\Omega\theta_z\dot{\theta}_x)]_{y=l} \tag{6.19}$$

where Ω is the constant rotorspeed around the y-axis, m the mass of the disk, J_t the mass-moment of inertia of the disk around an axis **in** the plane of the disk and J_p the mass-moment of inertia around an axis **perpendicular** to the disk-plane. Substitution of the displacement-functions now leads to

$$
\begin{aligned}
T = \frac{1}{2}[m(\dot{u}_l)^2 + m(\dot{w}_l)^2] + \frac{1}{2}J_p\Omega^2 \\
+ \frac{1}{2}\ \underline{\dot{q}}_w^T \left[\frac{J_t}{det^2}\begin{bmatrix} 2al - a^2 \\ -l^2 \end{bmatrix}[2al - a^2, -l^2]\right]\ \underline{\dot{q}}_w \\
+ \frac{1}{2}\ \underline{\dot{q}}_u^T \left[\frac{J_t}{det^2}\begin{bmatrix} -2al + a^2 \\ l^2 \end{bmatrix}[-2al + a^2, l^2]\right]\ \underline{\dot{q}}_u \\
-\Omega\ \underline{q}_u^T \left[\frac{J_p}{det^2}\begin{bmatrix} -2al + a^2 \\ l^2 \end{bmatrix}[2al - a^2, -l^2]\right]\ \underline{\dot{q}}_w
\end{aligned} \tag{6.20}
$$

This can also be written as:

$$
\begin{aligned}
T = \frac{1}{2}\underline{\dot{q}}_w^T \left[\frac{J_t}{det^2}\begin{bmatrix} (2al - a^2)^2; & l^2(a^2 - 2al) \\ l^2(a^2 - 2al); & l^4 \end{bmatrix} + \begin{bmatrix} m & 0 \\ 0 & 0 \end{bmatrix}\right]\ \underline{\dot{q}}_w \\
+ \frac{1}{2}\underline{\dot{q}}_u^T \left[\frac{J_t}{det^2}\begin{bmatrix} (2al - a^2)^2; & l^2(a^2 - 2al) \\ l^2(a^2 - 2al); & l^4 \end{bmatrix} + \begin{bmatrix} m & 0 \\ 0 & 0 \end{bmatrix}\right]\ \underline{\dot{q}}_u \\
+ \Omega\ \underline{q}_u^T \left[\frac{J_p}{det^2}\begin{bmatrix} (2al - a^2)^2; & l^2(a^2 - 2al) \\ l^2(a^2 - 2al); & l^4 \end{bmatrix}\right]\ \underline{\dot{q}}_w \\
+ \frac{1}{2}J_p\Omega^2
\end{aligned} \tag{6.21}
$$

If we introduce some sub-mass-matrices $\underline{M}_u$, $\underline{M}_w$ and $\underline{M}_p$ we can write this shortly as:

$$
\begin{aligned}
T = \frac{1}{2}\ \underline{\dot{q}}_w^T\ \underline{M}_w\ \underline{\dot{q}}_w \quad + \quad \frac{1}{2}\ \underline{\dot{q}}_u^T\ \underline{M}_u\ \underline{\dot{q}}_u \\
- \ \Omega\ \underline{q}_u^T\ \underline{M}_p\ \underline{\dot{q}}_w \quad + \quad \frac{1}{2}\ J_p\Omega^2
\end{aligned} \tag{6.22}
$$

We can now apply the Lagrange equation:

$$
\left[\ \frac{d}{dt}\left(T,_{\underline{\dot{q}}}\right)\ -\ T,_{\underline{q}}\ +\ V,_{\underline{q}}\ \right]^T\ =\ \underline{0} \tag{6.23}
$$

This will lead to:

$$
\begin{aligned}
\begin{bmatrix} \underline{M}_u & \underline{0} \\ \underline{0} & \underline{M}_w \end{bmatrix}\begin{bmatrix} \underline{\ddot{q}}_u \\ \underline{\ddot{q}}_w \end{bmatrix} + \Omega\ \begin{bmatrix} \underline{0} & \underline{M}_p \\ -\underline{M}_p & \underline{0} \end{bmatrix}\begin{bmatrix} \underline{\dot{q}}_u \\ \underline{\dot{q}}_w \end{bmatrix} \\
+ \quad \begin{bmatrix} \underline{K}_u & \underline{0} \\ \underline{0} & \underline{K}_w \end{bmatrix}\begin{bmatrix} \underline{q}_u \\ \underline{q}_w \end{bmatrix} = \underline{0}
\end{aligned} \tag{6.24}
$$

Again, we can calculate the eigenvalues λ_i for this set of equations for a range of rotorfrequencies $\Omega = 0...80 \quad [Hz]$. For each rotorfrequency Ω this will lead to a set of 2 pairs of purely imaginary eigenvalues (and their complex conjugate ones) $\lambda_i = \pm j\omega_i \quad (i = 1, ...4)$, with real ω_i.
The frequencies $f_i = \omega_i/(2\pi)$ are plotted in the following figure as function of Ω. As we can see the result is much better in this case. The smallest

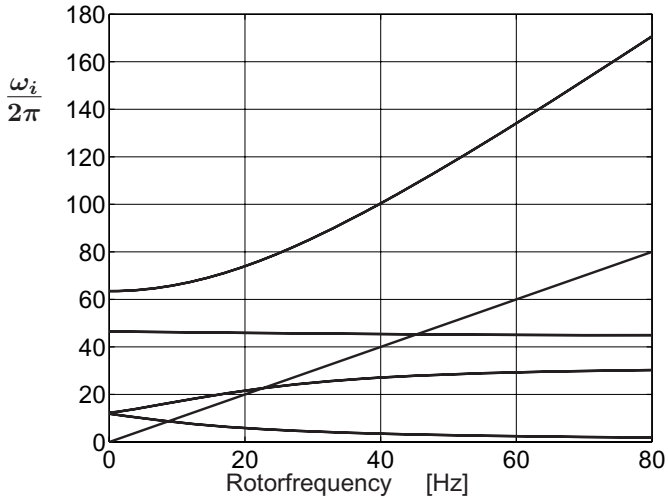


Fig. 6.3. Campbell plot

(shaft bending dominated) frequencies (based on a 4-dof model) are very close to the FEM results for which a 20-dof model was used. Also the 3rd and 4th eigenfrequency show the same tendency as the corresponding FEM-ones.
So, we may conclude that the Rayleigh-Ritz approach is a nice technique to generate an approximate model with a small number of degrees of freedom. But the accuracy in general will be limited and the basic functions should be selected with great care (one should have a good idea of the mode shape(s) which are to be calculated; experience counts)

6.2 Internal Damping

In general, the presence of any form of damping in a rotorsystem will be appreciated, because it has a positive effect on the stability and the response at or near a critical speed can be reduced. However, in case of so-called internal damping the presence of damping can have a negative effect instead of a positive one. Generally we can say that we are dealing with internal damping if there is some form of energy dissipation during a deformation with respect to a rotating frame which is fixed to the rotor. Simple examples of systems with internal damping are shown in Fig. 6.4. In the left

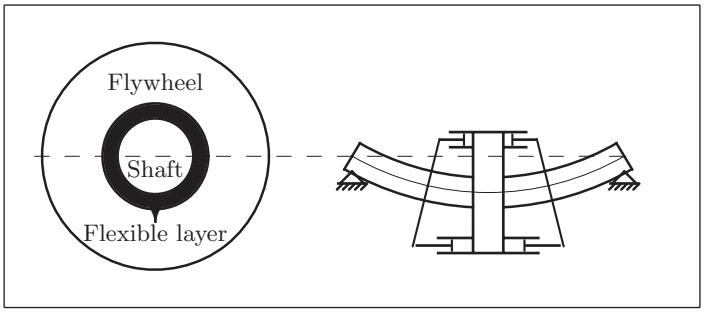


Fig. 6.4. Examples of internal damping in a rotor

system we are dealing with a (fiber reinforced plastic) flywheel which has been connected to a rotating shaft by means of a flexible layer. The reason to introduce this flexible layer is to reduce the bearing forces due to the always present and also time-dependent unbalance in such a system. In the system on the right, a representative viscous damping model is shown which should take account for the energy dissipation in for example an imperfect shrink-fit of the disk on the shaft. For the analysis of the effects due to internal damping we will use the model for the Jeffcott rotor from Chapter 2. The only modification is that we assume that the mass-eccentricity e will be zero, so the center of mass M and the geometrical center G are the same points. For the introduction of the internal damping force we look at two orthonormal bases, one fixed in space $\{X, Z\}$and the basis $\{X_L, Y_L\}$, fixed to the rotating system (with rotor speed Ω) as shown in Fig. 6.5 and appendix A. For the matrix representation $\underline{x}$ of an arbitrary vector $\vec{x}$ with respect to the fixed frame and the matrix representation $\underline{x}_L$ with respect to the rotating frame we can write

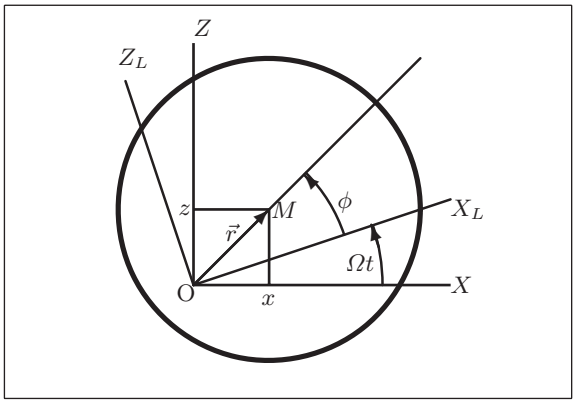


Fig. 6.5. Rigid rotor with fixed and rotating bases

$$\underline{x} = \underline{R}\ \underline{x}_L, \qquad \text{and} \qquad \underline{x}_L = \underline{R}^T \underline{x} \tag{6.25}$$

where $\underline{R}$ is the matrix representation of the rotation tensor

$$\underline{R} = \begin{bmatrix} \cos(\Omega t) & -\sin(\Omega t) \\ \sin(\Omega t) & \cos(\Omega t) \end{bmatrix} \tag{6.26}$$

In case of internal damping we are dealing with damping forces which are the result of deformations in the rotating frame. So, in case of a synchronous motion this type of damping will not dissipate any energy. This means that internal damping will only result in energy dissipation when $\dot{\underline{r}}_L \neq 0$, where $\underline{r}_L$ is the matrix representation of the position vector $\vec{r}$ of the center of mass with respect of the rotating frame. We assume that the internal demping has a linear and viscous character, so

$$\underline{F}_{bi_L} = -b_i\ (\dot{\underline{r}}_L) \tag{6.27}$$

with $\underline{F}_{bi_L}$ the matrix representation of the damping force vector $\vec{F}_{bi}$ (acting on the disk) with respect to the rotating frame. Then we can write

$$\underline{F}_{bi_L} = -b_i\ \frac{d}{dt}(\underline{R}^T \underline{r}) = -b_i\ (\dot{\underline{R}}^T \underline{r} + \underline{R}^T \dot{\underline{r}}) \tag{6.28}$$

which gives

$$\underline{F}_{bi} = \underline{R}\ \underline{F}_{bi_L} = -b_i\ (\underline{\Omega}^T \underline{r} + \dot{\underline{r}}) \tag{6.29}$$

using

$$\underline{\Omega} = \dot{\underline{R}}\ \underline{R}^T = \begin{bmatrix} 0 & -\Omega \\ \Omega & 0 \end{bmatrix} \tag{6.30}$$

For the matrix representation of the internal damping force vector in the fixed frame ($\underline{F}_{bi}$) we find

$$\underline{F}_{bi} = -\begin{bmatrix} b_i\ \Omega z + b_i\ \dot{x} \\ -b_i\ \Omega x + b_i\ \dot{z} \end{bmatrix} \tag{6.31}$$

where we used $\underline{r}^T = [x, z]$. If we add this damping force to the model for the Jeffcott rotor without eccentricity, (see (1.9), we get the set of equations of motion in the fixed-frame coordinates

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{z} \end{bmatrix} + \begin{bmatrix} b + b_i & 0 \\ 0 & b + b_i \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix} + \begin{bmatrix} k & b_i\Omega \\ -b_i\Omega & k \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{6.32}$$

A surprising result from the introduction of this internal damping effect in the Jeffcott-rotor is not the presence of the damping factor b_i in the damping matrix but the skew-symmetric, rotor speed dependent contribution $\pm b_i\ \Omega$ in the stiffness matrix.
For the evaluation of the dynamic stability of this system we try a solution of the type

$$\underline{r} = [x, z]^T = \underline{r}_o\ e^{\lambda t}$$

So, for a non-trivial solution we should demand

$$\mathbf{Det} \begin{bmatrix} \lambda^2 m + \lambda(b + b_i) + k & b_i\ \Omega \\ -b_i\ \Omega & \lambda^2 m + \lambda(b + b_i) + k \end{bmatrix} = 0 \tag{6.33}$$

leading to

$$\lambda^4 + a_3\ \lambda^3 + a_2\ \lambda^2 + a_1\ \lambda + a_0 = 0 \tag{6.34}$$

with

$$\begin{aligned} a_3 &= 2(b + b_i)/m \\ a_2 &= (2km + (b + b_i)^2)/m^2 \\ a_1 &= (2k(b + b_i))/m^2 \\ a_0 &= (k^2 + b_i^2\Omega^2)/m^2 \end{aligned} \tag{6.35}$$

If we are interested in purely imaginary eigenvalues (real part zero, onset of instability), we can apply the Routh-Hurwitz stability criterion

$$\begin{aligned} 1: &\quad a_0, a_1, a_2, a_3 > 0 \\ 2: &\quad a_1 a_2 a_3 - a_0 a_3^2 - a_1^2 > 0 \end{aligned} \tag{6.36}$$

The first requirement is fulfilled if m, k and $(b + b_i)$ are > 0, which in general will be true. The second requirement is fulfilled if

$$\frac{k}{m}(b + b_i)^2 - b_i^2 \; \Omega^2 > 0 \tag{6.37}$$

With $\omega_o^2 = k/m$ this leads to the stability-limit ω_s for the rotor speed Ω

$$\omega_s^2 = \omega_o^2 (1 + b/b_i)^2 \tag{6.38}$$

For $\Omega < \omega_s$ the system is stable and for very little internal damping b_i (with respect to the always existing (external) damping b), we can use $\omega_s^2 \gg \omega_o^2$. So, the onset speed of instability then will be far above the systems critical speed. For increasing internal damping b_i, the onset speed of instability ω_s will go down to ω_o. If the internal damping b_i is larger then the regular damping b, the system practically cannot run supercritically (at least based on this linear rotor dynamic analysis).

For $\Omega = \omega_s$ the system is on the boundary of instability. The question then is which eigenfrequencies and eigenmodes will be possible in that situation. To investigate this we use $\Omega = \omega_s$ and for the eigenvalues λ of (6.33) we put $\lambda_g = j\omega_g$ with $\omega_g =$ real. Then (6.34) leads to the complex equation

$$\omega_g^4 - a_3 j \omega_g^3 - a_2 \omega_g^2 + a_1 j \omega_g + a_0 = 0 \tag{6.39}$$

or the two real equations

$$\omega_g^4 - a_2 \omega_g^2 + a_0 = 0 \qquad \text{and} \qquad -a_3 \omega_g^3 + a_1 \omega_g = 0 \tag{6.40}$$

For the solutions of these equations we find

$$\omega_g^2 = \frac{a_2 \pm \sqrt{a_2^2 - 4a_0}}{2} \qquad \text{respectively} \qquad \omega_g^2 = \frac{a_1}{a_3} \tag{6.41}$$

Using (6.35), the only solution can be

$$\omega_g^2 = \omega_o^2 = \frac{k}{m} \tag{6.42}$$

So, when a rotor is running with rotor speed ω_s on the onset speed of instability, a free motion is possible with the eigenfrequency ω_g, where in general $\omega_s \neq \omega_g$. For deriving the corresponding eigenmode $\underline{u}_g$ we go back to (6.33). This gives

$$[\lambda_g^2 m + \lambda_g(b + b_i) + k \ ; \ b_i \ \omega_s] \ \underline{u}_g = 0 \tag{6.43}$$

If we substitute the results for ω_s and $\lambda_g = j \ \omega_g$ we get

$$\underline{u}_g = \alpha \begin{bmatrix} 1 \\ -j \end{bmatrix} \tag{6.44}$$

with α=arbitrary (complex) constant. So for the free vibration we find

$$\underline{r} = \begin{bmatrix} x \\ z \end{bmatrix} = Re \ [\underline{u}_g \ e^{j\omega_g t}] \tag{6.45}$$

In 6.6 the motion of a disk with a marker-line is illustrated for the special

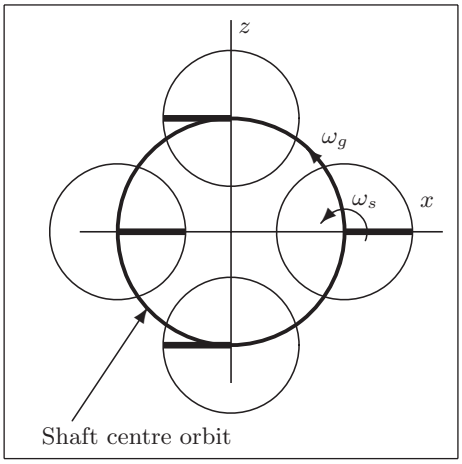


Fig. 6.6. Eigenmode on the stability boundary for $b_i = b$

case that $b_i = b$, so $\omega_g = \frac{1}{2}\omega_s$. The shaft-center is rotating with half the rotor speed in the same direction as this rotor speed. Therefore it is called a **forward whirl**. It might also be named a $\frac{1}{2}\Omega$-whirl in this case.

6.3 Non-Rotational-Symmetric Rotors

Again we look at the Jeffcott rotor from the previous section without mass eccentricity. We also make use again of two orthonormal frames, one fixed in space $\{X, Z\}$ and a rotating frame (rotating with the constant rotor speed Ω) $\{X_L, Z_L\}$, see Fig. 6.7. However we assume that the rotor is not perfectly rotational symmetric anymore because the bending stiffness k_x with respect to the X_L-axis is assumed to be different from the bending stiffness k_z with respect to the Z_L-axis. This is for example the case for a rotor with a rectangular cross section or for a symmetrical rotor with a (small) fatigue crack. The rotor will be modeled as a rigid body in the

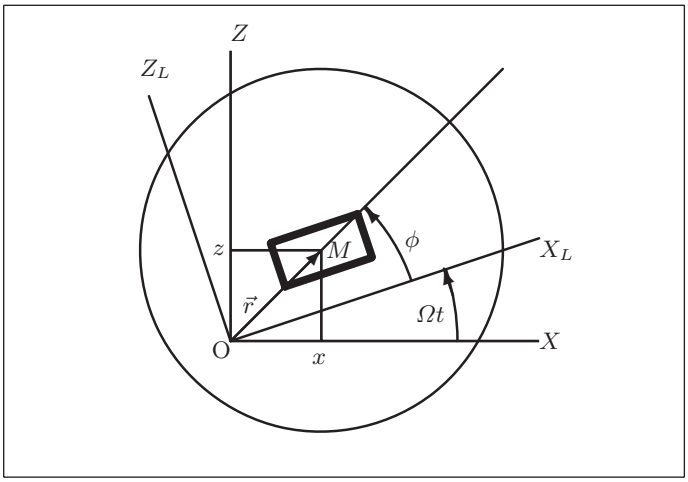


Fig. 6.7. Non-rotationally symmetric rotor

(X_L, Z_L)-plane with mass m (center of mass M). It is supported by the linear springs k_x respectively k_z. For the kinetic energy we can write

$$T = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}m\dot{z}^2 = \frac{1}{2}\underline{\dot{x}}^T \underline{M}\ \underline{\dot{x}} \tag{6.46}$$

where

$$\underline{x} = \begin{bmatrix} x \\ z \end{bmatrix}; \qquad \underline{M} = \begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \tag{6.47}$$

Next we use

$$\underline{x} = \underline{R}\ \underline{x}_L \Longrightarrow \underline{\dot{x}} = \underline{R}\ \underline{\dot{x}}_L + \underline{\dot{R}}\ \underline{x}_L \tag{6.48}$$

so

$$T = \tfrac{1}{2}\{\underline{x}_L^T \dot{\underline{R}}^T \underline{M}\ \dot{\underline{R}}\ \underline{x}_L + \underline{x}_L^T \dot{\underline{R}}^T \underline{M}\ \underline{R}\ \dot{\underline{x}}_L + \\ \dot{\underline{x}}_L^T \underline{R}^T \underline{M}\ \dot{\underline{R}}\ \underline{x}_L + \dot{\underline{x}}_L^T \underline{R}^T \underline{M}\ \underline{R}\ \dot{\underline{x}}_L\} \tag{6.49}$$

It is easy to show that

$$\dot{\underline{R}}^T \dot{\underline{R}} = \Omega^2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; \quad \dot{\underline{R}}^T \underline{R} = \Omega \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \\ \underline{R}^T \dot{\underline{R}} = \Omega \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}; \quad \underline{R}^T \underline{R} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{6.50}$$

Then we can write for the kinetic energy

$$T = \frac{1}{2}\left[m\Omega^2(x_L^2 + z_L^2) + m(\dot{x}_L^2 + \dot{z}_L^2) + 2m\Omega(z_L\dot{x}_L - x_L\dot{z}_L)\right] \tag{6.51}$$

For the elastic energy we get

$$U = \frac{1}{2}\left[k_x\ {x_L}^2 + k_z\ {z_L}^2\right] \tag{6.52}$$

With Lagrange's equations this results in

$$\begin{bmatrix} m & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} \ddot{x}_L \\ \ddot{z}_L \end{bmatrix} + 2m\Omega \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} \dot{x}_L \\ \dot{z}_L \end{bmatrix} + \\ \begin{bmatrix} k_x - m\Omega^2 & 0 \\ 0 & k_z - m\Omega^2 \end{bmatrix} \begin{bmatrix} x_L \\ z_L \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{6.53}$$

We define the two (different) fundamental frequencies

$$\omega_x^2 = k_x/m \qquad \text{and} \qquad \omega_z^2 = k_z/m \tag{6.54}$$

so we can write for (6.54)

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \ddot{x}_L \\ \ddot{z}_L \end{bmatrix} + \begin{bmatrix} 0 & 2\Omega \\ -2\Omega & 0 \end{bmatrix} \begin{bmatrix} \dot{x}_L \\ \dot{z}_L \end{bmatrix} + \\ \begin{bmatrix} (\omega_x^2 - \Omega^2) & 0 \\ 0 & (\omega_z^2 - \Omega^2) \end{bmatrix} \begin{bmatrix} x_L \\ z_L \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \tag{6.55}$$

For this coupled set of homogeneous equations of motion we look for a solution of the type

$$\begin{aligned} x_L &= x_o\ e^{\lambda t} \\ z_L &= z_o\ e^{\lambda t} \end{aligned} \tag{6.56}$$

leading to the characteristic equation

$$\lambda^4 + (\omega_x^2 + \omega_z^2 + 2\Omega^2)\lambda^2 + (\omega_x^2 - \Omega^2)(\omega_z^2 - \Omega^2) = 0 \tag{6.57}$$

The solution can be written as

$$2\lambda_1^2 = -(\omega_x^2 + \omega_z^2 + 2\Omega^2) + \sqrt{\Delta} \tag{6.58}$$

$$2\lambda_2^2 = -(\omega_x^2 + \omega_z^2 + 2\Omega^2) - \sqrt{\Delta} \tag{6.59}$$

where the discriminant Δ reads

$$\begin{aligned} \Delta = (\omega_x^2 + \omega_z^2 + 2\Omega^2)^2 - 4(\omega_x^2 - \Omega^2)(\omega_z^2 - \Omega^2) = \\ (\omega_x^2 - \omega_z^2)^2 + 8\Omega^2(\omega_x^2 + \omega_z^2) \end{aligned} \tag{6.60}$$

So, Δ will always be a positive number and consequently the roots λ_1^2 and λ_2^2 will always be real. Additionally, $(\omega_x^2 - \Omega^2)(\omega_z^2 - \Omega^2)$ can become negative which means that λ_1^2 can be positive.

$$\{\lambda_1^2 = a; \quad 0 < a\} \Longrightarrow \lambda_{1,1} = +\sqrt{a}; \quad \lambda_{1,2} = -\sqrt{a} \tag{6.61}$$

The fundamental solution $e^{\lambda_{1,1} t}$ will go to ∞ for increasing time t, so the solution is unstable. This is the case for

$$\omega_x < \Omega < \omega_z \tag{6.62}$$

So, in the interval between the two resonance frequencies ω_x and ω_z the rotor behavior will be unstable. As an illustration the flow of the 4 eigenvalues $\lambda_i,\ i = 1..4$ is shown in Fig. 6.8 for the following situation $\omega_x = 1.0;\ \omega_z = 2.0$. We see that in general we are dealing with 4 imaginary eigenvalues (so each disturbance to this undamped system will exist forever) except for the interval $[\omega_x < \Omega < \omega_z]$ where we get two real eigenvalues, a positive one and a negative one. Each disturbance will lead to a response which will grow unlimited.

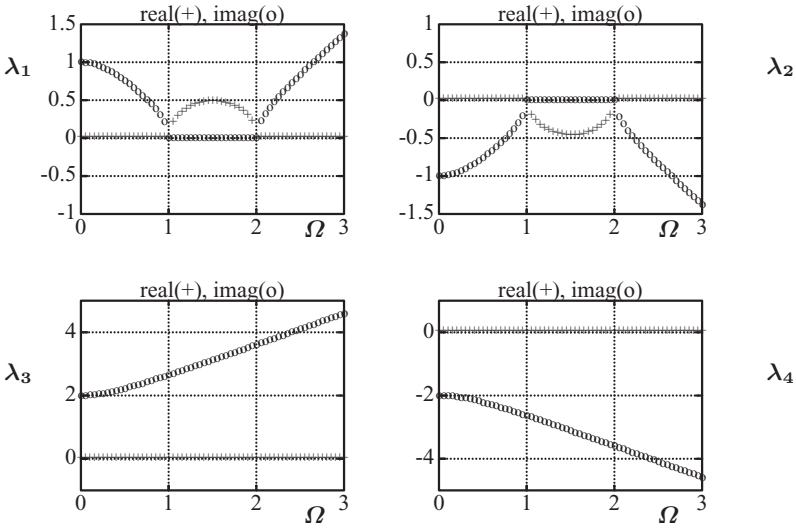


Fig. 6.8. Eigenvalues as function of rotor speed for non-symmetric rotor ($\omega_x = 1; \omega_z = 2$). Real parts (+), imaginary parts (o)

6.4 Critical Speed Crossing

In general, when a rotor is operating close to one of its critical speeds, the responses can be severe and can cause damage to the system. So this situation has to be avoided as much as possible. But in many cases it will be necessary to pass one or more of these critical speeds to reach the operating range of the machine. This is certainly the case when a system has to run supercritically. If the power of the driving motor of the machine is too small, the passage of the critical speed can take too much time (leading to a dangerous situation) or it will even be possible that at some rotorspeed just below the critical speed the rotor will settle down with high vibrations and no further rotorspeed increase. This of course is a situation which should be avoided at al times. In this paragraph this phenomenon will be illustrated and the dominant parameters will be identified using a very simple rotor model.

For the derivation of the relevant equations of motion we return to the Jefftcott rotor model as shown in Fig. 6.9 We only look at planar motions

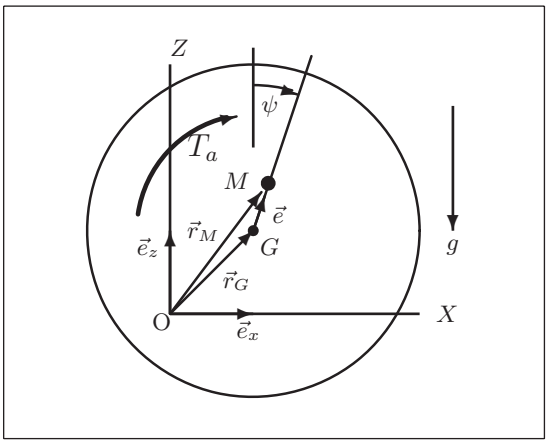


Fig. 6.9. Jeffcott Rotor Model with Driving moment T_a

in the $X-Z$-plane. The shaft-connection point G is marked by the position vector $\vec{r}_G$, whereas the center of mass of the disk is marked by the vector $\vec{r}_M$ and the mass eccentricity is marked by the eccentricity vector $\vec{e}$ (with

magnitude e). The angle between the fixed Z-axis and this eccentricity vector is called ψ. We assume that the external damping, acting on the disk, can be taken into account by introducing a force in point M which is proportional with the absolute velocity of this point with proportionality factor (damping constant) b. Then the following forces are acting of the disk

- In M the gravity force $\vec{F}_g = -mg\ \vec{e}_z$,
- In M the external damping force $\vec{F}_b = -b\ \dot{\vec{r}}_M$,
- In G the elastic shaft force due to bending $\vec{F}_r = -k\ \vec{r}_G$.

Applying Newton's second law for the center of mass M gives

$$m\ \ddot{\vec{r}}_M = \vec{F}_b + \vec{F}_g + \vec{F}_r \tag{6.63}$$

and the angular momentum equation with respect to this center of mass

$$-\vec{e} \star \vec{F}_r + T_a\ \vec{e}_y = J\ \ddot{\psi}\ \vec{e}_y \tag{6.64}$$

where J is the mass moment of inertia of the disk with respect to M. Using

$$\begin{aligned} \vec{r}_M &= x_M\ \vec{e}_x + z_M\ \vec{e}_z \\ \vec{r}_G &= x_G\ \vec{e}_x + z_G\ \vec{e}_z \\ \vec{e} &= e\ \sin\psi\ \vec{e}_x + e\ \cos\psi\ \vec{e}_z \end{aligned} \tag{6.65}$$

the equations (6.63) can be written as

$$\begin{aligned} m\ \ddot{x}_M &= -b\ \dot{x}_M - k\ x_G \\ m\ \ddot{z}_M &= -b\ \dot{z}_M - k\ z_G - mg \end{aligned} \tag{6.66}$$

while (6.64) leads to

$$J\ \ddot{\psi} = -k\ e\ (z_M\ \sin\psi - x_M\ \cos\psi) + T_a \tag{6.67}$$

If we also use

$$\vec{r}_G = \vec{r}_M - \vec{e} \tag{6.68}$$

we can get

$$\begin{aligned} m\ \ddot{x}_M + b\ \dot{x}_M + k\ x_M &= k\ e\ \sin\psi \\ m\ \ddot{z}_M + b\ \dot{z}_M + k\ z_M &= k\ e\ \cos\psi - mg \\ J\ \ddot{\psi} - k\ e\ (x_M\ \cos\psi - z_M\ \sin\psi) &= T_a \end{aligned} \tag{6.69}$$

We introduce the following parameters

- eigenfrequency of the undamped system, $\omega_o = \sqrt{k/m}, \quad [s^{-1}]$
- dimensionless damping constant, $\xi = b/2m\omega_o, \quad [-]$
- the inertia radius, $i = \sqrt{J/m}, \quad [-]$

The equations (6.69) can now be written as

$$\begin{aligned}
&\ddot{x}_M + 2\ \xi\ \omega_o\ \dot{x}_M + \omega_o^2\ x_M = \omega_o^2\ e\ \sin\psi \\
&\ddot{z}_M + 2\ \xi\ \omega_o\ \dot{z}_M + \omega_o^2\ z_M = \omega_o^2\ e\ \cos\psi - g \\
&\ddot{\psi} - \frac{\omega_o^2\ e}{k^2}\ (x_M\ \cos\psi - z_M\ \sin\psi) = \frac{T_a}{mk^2}
\end{aligned} \tag{6.70}$$

To reduce the number of relevant parameters further, we define the dimensionless quantities

- $\tau = \omega_o\ t$
- $X_M = x_M/e$ and $Z_M = z_M/e$
- $G = g/(e\ \omega_o^2)$
- $T = T_a/(m\ k^2\ \omega_o^2)$
- $\dot{X} = dX/d\tau$

This leads us to the equations of motion in dimensionless form

$$\begin{aligned}
&\ddot{X}_M + 2\ \xi\ \dot{X}_M + X_M = \sin\psi \\
&\ddot{Z}_M + 2\ \xi\ \dot{Z}_M + Z_M = \cos\psi - G \\
&\ddot{\psi} - (e/k)^2\ (X_M\ \cos\psi - Z_M\ \sin\psi) = T
\end{aligned} \tag{6.71}$$

So, we are dealing with a set of 3, nonlinear, coupled equations of motion. In general a simple, closed form solution will not exist. So the only way will be to perform a numerical procedure to evaluate the characteristic rotordynamic behavior for a specific set of parameters and to investigate the effect of changing certain parameters such as for example the eccentricity or the driving moment of the motor. In Fig. 6.10 an example is shown of the dynamic behavior of a rotor (starting from rest) and two different values for the eccentricity. Further, the following parameter values have been used, see also (6.71): $T = 0.05$; $G = 5$; $\xi = 0.02$; $(e/k)^2 = 0.002$ respectively 0.01.

In this figure the normalized rotor speed $\dot{\psi}$ is plotted as function of the dimensionless time τ. Also the line $(\dot{\psi} = T\ \tau)$ is shown to indicate the situation of a rotor with a uniformly accelerating motion. From the figure we can see that in case of the small eccentricity (curve 1), the rotor will spin

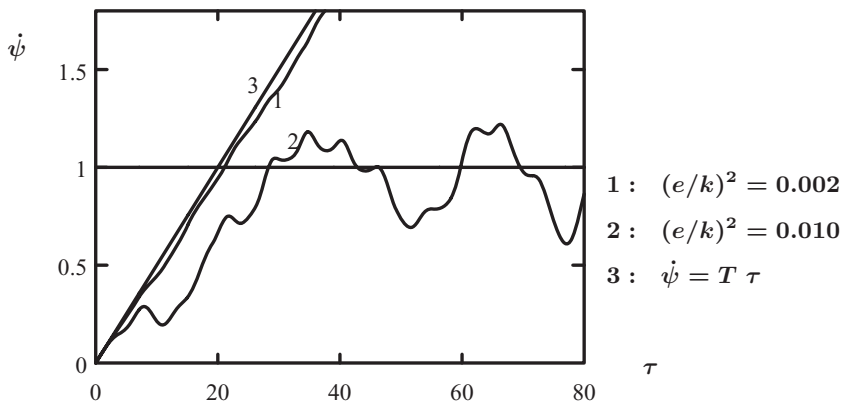


Fig. 6.10. Rotor start-up behavior for 2 different excentricities

up with an almost constant acceleration. The angular velocity $\dot{\psi}$ for this situation is just a little bit smaller than the angular velocity of only the rigid disk loaded by a constant driving moment (line 3). This small difference is due to the presence of the two translational degrees of freedom X_M and Z_M in the rotational equation of motion (6.71). From the figure we can see that the rotor can pass the critical speed ($\omega_{critical} = 1.0$) rather smoothly. We can also see that in case of the larger eccentricity (curve 2), the rotor (with the used value for the driving moment T), will note be able to pass this critical speed definitely. What after start-up remains as some kind of steady state motion, is a situation with very strong vibrations of the rotor. It will be obvious that this situation is very undesirable. The solution will be to reduce the eccentricity (by rotor balancing) and/or to increase the driving moment (a stronger motor).

In many cases such a passage of a critical speed has to be carried out only incidentally (for example once a day at start-up). Once the rotor is running supercritically, the driving moment can be relatively small. Then choosing a very strong motor only for passing the critical speed will not be very efficient. So, it will be worth while to have some estimate for the necessary moment for a critical speed crossing as presented in the next section.

6.4.1 Analytical Approximation of Necessary Moment

We start again from the equations of motion (6.71). Next, we will assume that the rotor spin-up will happen very slowly, so that we may ignore the inertia effect and investigate a more or less quasi-static situation at some specific rotor speed. To generate an approximation for the *moment of resistance* on the disk for this stationary speed we apply a perturbation analysis. We assume that the driving moment (T) as well as the eccentricity (e) are sufficiently small. Additionally we neglect any free vibration which is reasonably because there will always be some damping in the system.
We apply a Taylor series representations

$$\begin{aligned} X_M &= X_M^{(o)} + (e/k)^2 X_M^{(1)} + ... \\ Z_M &= Z_M^{(o)} + (e/k)^2 Z_M^{(1)} + ... \\ \psi &= \psi^{(o)} + (e/k)^2 \psi^{(1)} + ... \end{aligned} \tag{6.72}$$

Substitution in (6.71) leads to

$$\begin{gathered} \ddot{X}_M^{(0)} + (e/k)^2 \ddot{X}_M^{(1)} + 2\xi[\dot{X}_M^{(0)} + (e/k)^2 \dot{X}_M^{(1)}] + X_M^{(0)} + (e/k)^2 X_M^{(1)} \\ = \sin(\psi^{(0)} + (e/k)^2 \psi^{(1)}) \\ \ddot{Z}_M^{(0)} + (e/k)^2 \ddot{Z}_M^{(1)} + 2\xi[\dot{Z}_M^{(0)} + (e/k)^2 \dot{Z}_M^{(1)}] + Z_M^{(0)} + (e/k)^2 Z_M^{(1)} \\ = \cos(\psi^{(0)} + (e/k)^2 \psi^{(1)}) - G \\ \ddot{\psi}^{(0)} + (e/k)^2 \ddot{\psi}^{(1)} - (e/k)^2 [X_M^{(0)} + (e/k)^2 X_M^{(1)}] \cos(\psi^{(0)} + (e/k)^2 \psi^{(1)}) \\ +(e/k)^2 [Z_M^{(0)} + (e/k)^2 Z_M^{(1)}] \sin(\psi^{(0)} + (e/k)^2 \psi^{(1)}) = T \end{gathered} \tag{6.73}$$

We also use

$$\begin{aligned} \cos(\psi^{(0)} + (e/k)^2 \psi^{(1)}) &= \cos(\psi^{(0)}) - (e/k)^2 \psi^{(1)} \sin(\psi^{(0)}) \\ \sin(\psi^{(0)} + (e/k)^2 \psi^{(1)}) &= \sin(\psi^{(0)}) - (e/k)^2 \psi^{(1)} \cos(\psi^{(0)}) \end{aligned} \tag{6.74}$$

where we applied the assumption that $\|(e/k)^2 \psi^{(1)}\| \ll 1$. From (6.73) we then can write for the stationary situation

$$\begin{aligned} \ddot{X}_M^{(0)} + 2\xi \dot{X}_M^{(0)} + X_M^{(0)} &= \sin(\psi^{(0)}) \\ \ddot{Z}_M^{(0)} + 2\xi \dot{Z}_M^{(0)} + Z_M^{(0)} &= \cos(\psi^{(0)}) - G \\ \ddot{\psi}^{(0)} &= 0 \end{aligned} \tag{6.75}$$

This is the first order approximation. The second order approximation can be derived by equalization of the terms which are linear in $(e/k)^2$, leading to

$$\begin{aligned}
&\ddot{X}_M^{(1)} + 2\xi\dot{X}_M^{(1)} + X_M^{(1)} = -\psi^{(1)}\sin(\psi^{(0)}) \\
&\ddot{Z}_M^{(1)} + 2\xi\dot{Z}_M^{(1)} + Z_M^{(1)} = \psi^{(1)}\cos(\psi^{(0)}) \\
&\ddot{\psi}^{(1)} - X_M^{(0)}\cos(\psi^{(0)}) + Z_M^{(0)}\sin(\psi^{(0)}) = \frac{T}{(e/k)^2}
\end{aligned} \tag{6.76}$$

From the third equation of (6.75) it follows that we can take for $\psi^{(0)}$

$$\psi^{(0)} = \eta\tau + \nu; \qquad \text{so} \qquad \dot{\psi}^{(0)} = \eta \tag{6.77}$$

Next we only look at the particular solutions $X_M^{(0)}$ and $Z_M^{(0)}$ from (6.75). These can be written as

$$\begin{aligned}
X_M^{(0)} &= c_1\cos\psi^{(0)} + c_2\sin\psi^{(0)} + c_3 \\
Z_M^{(0)} &= d_1\cos\psi^{(0)} + d_2\sin\psi^{(0)} + d_3
\end{aligned} \tag{6.78}$$

Substitution of these expressions in (6.75), while using (6.77), leads to the coefficients c_1 up to c_3 and d_1 up to d_3

$$\begin{aligned}
c_1 &= \tfrac{-2\xi\eta}{(1-\eta^2)^2+(2\xi\eta)^2}; \quad c_2 = \tfrac{1-\eta^2}{(1-\eta^2)^2+(2\xi\eta)^2}; \quad c_3 = 0 \\
d_1 &= \tfrac{1-\eta^2}{(1-\eta^2)^2+(2\xi\eta)^2}; \quad d_2 = \tfrac{2\xi\eta}{(1-\eta^2)^2+(2\xi\eta)^2}; \quad d_3 = -G
\end{aligned} \tag{6.79}$$

This gives for the stationary situation

$$\begin{aligned}
X_M^{(0)} &= \frac{-2\xi\eta\cos(\psi^{(0)}) + (1-\eta^2)\sin(\psi^{(0)})}{(1-\eta^2)^2 + (2\xi\eta)^2}, \\
Z_M^{(0)} &= \frac{2\xi\eta\sin(\psi^{(0)}) + (1-\eta^2)\cos(\psi^{(0)})}{(1-\eta^2)^2 + (2\xi\eta)^2} - G, \\
\psi^{(0)} &= \eta\tau + \nu
\end{aligned} \tag{6.80}$$

This stationary situation will next be used for the elaboration of the first order equation for $\psi^{(1)}$, leading to

$$\begin{aligned}
&\ddot{\psi}^{(1)} - \{\frac{-2\xi\eta\cos(\psi^{(0)}) + (1-\eta^2)\sin(\psi^{(0)})}{(1-\eta^2)^2 + (2\xi\eta)^2}\}\cos(\psi^{(0)}) \\
&+\{\frac{2\xi\eta\sin(\psi^{(0)}) + (1-\eta^2)\cos(\psi^{(0)})}{(1-\eta^2)^2 + (2\xi\eta)^2}\}\sin(\psi^{(0)}) - G\sin(\psi^{(0)}) = \\
&\qquad \frac{T}{(e/k)^2}
\end{aligned} \tag{6.81}$$

We look for a stationary situation for $\dot{\psi}^{(1)}$, so we may apply $\ddot{\psi}^{(1)} = 0$, giving

$$\frac{2\xi\eta}{(1-\eta^2)^2 + (2\xi\eta)^2} - G\sin(\psi^{(0)}) = \frac{T}{(e/k)^2} \tag{6.82}$$

On the left side of the = sign, we recognize a term which can be seen as the reduced moment of resistance T_w of the rotor, divide by $(e/k)^2$

$$\frac{T_w}{(e/k)^2} = \frac{2\xi\eta}{(1-\eta^2)^2 + (2\xi\eta)^2} - G\sin(\psi^{(0)}) \tag{6.83}$$

This moment has its maximal value for $\sin(\psi^{(0)}) = -1$, so $\psi^{(0)} = -\pi/2 + 2i\pi$. Within a period $(\psi^{(0)}, \psi^{(0)} + 2\pi)$ we are dealing with a positive but also a negative contribution of G to T_w, so in the average it wil be zero. Therefore we ignore this effect (or we assume to be dealing with a vertical rotor). For a relatively weakly damped system (ξ=small), the maximum for T_w will then be situated very close to $\eta = 1$, (corresponding to $\Omega = \omega_o$). This maximum is

$$T_{w_{max}} = (\frac{1}{2\xi})(\frac{e}{k})^2 \tag{6.84}$$

In Fig. 6.11, the value for the moment of resistance (6.83), is shown graph-

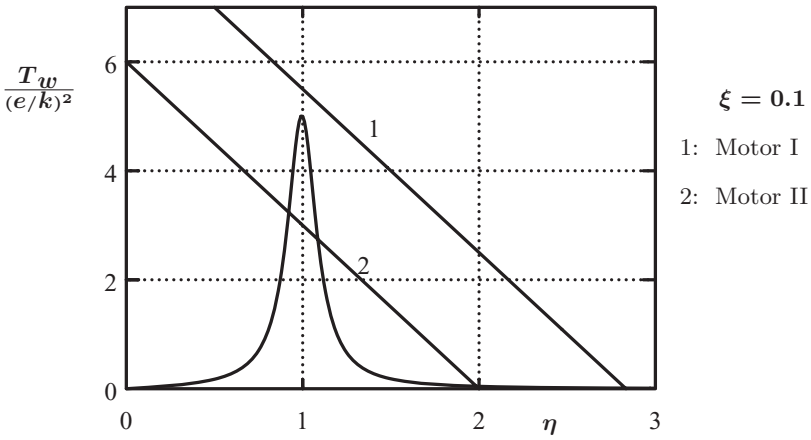


Fig. 6.11. Reduced moment of resistance and two possible motor characteristics

ically for $G = 0$ (no gravity) and the specific dimensionless damping choice $\xi = 0.1$. On the horizontal axis the stationary rotor speed $\eta = \dot{\psi}/\Omega$ is shown and on the vertical axis the reduced moment of resistance T_w. In the same

figure also two driving motor characteristics as possibly met in practice are shown, indicated by $motor\ I$ and $motor\ II$. In case of motor I, the available moment will always be larger then the moment of resistance (also in the resonance area), so the speed of the rotor will continuously increase until at approximately $\eta \approx 2.8$ some stationary situation will be reached.
In case of the less powerful drive (motor II), at $\eta \approx 0.9$ the moment of resistance will become larger then the available moment, leading to a stationary situation at this speed. The critical speed will not be passed. For speeds between $\eta \approx 0.9$ and $\eta \approx 1.1$, the available driving moment will be too small, so the $\eta \approx 0.9$ will be a stable situation. For a speed $\eta > 1.1$ the available moment will again be sufficiently large for a further acceleration of the rotor up to the stationary situation $\eta \approx 2$.

Finally we go back to the system from page 178 which has been analyzed numerically. We now take $G = 0$ and the approximated moment of resistance T_w is shown for two values for the eccentricity in Fig. 6.12. We

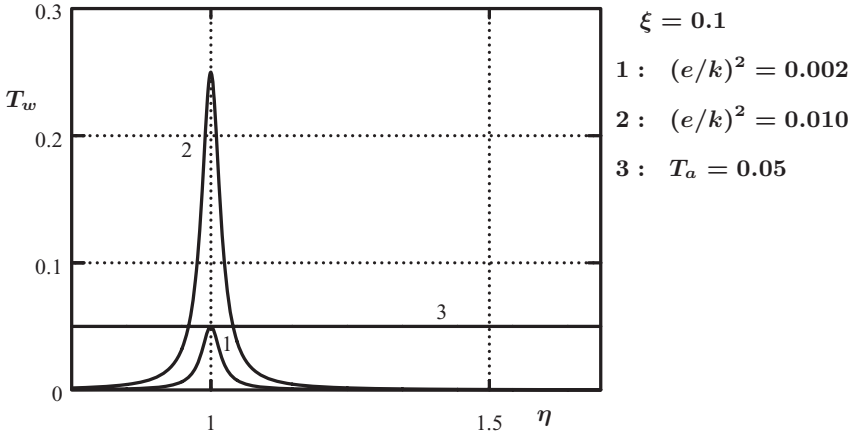


Fig. 6.12. Moment of resistance for two excentricities

assume to be dealing with a (constant) driving moment of $T_a = 0.05$. We can see from this figure that this driving moment is just large enough to accelerate the system with the small eccentricity ($(e/k)^2 = 0.002$) through the critical speed. This will not be the case for the system with the larger eccentricity ($(e/k)^2 = 0.01$). For this system the driving moment will be too small and it will not pass its critical speed. Finally, it will settle down

at a dimensionless rotor speed $\eta_{hang} \approx 0.85$. This frequency can also be recognized in Fig. 6.10 as the high frequency component in signal 2.

Resume

The minimum necessary driving moment for passing a critical speed can be estimated from

$$\boldsymbol{T_{min} = \frac{e^2\ m^{0.5}\ k^{1.5}}{b} + m\ g\ e \quad [Nm]} \tag{6.85}$$

where

e = eccentricity $[m]$
m = disk mass $[kg]$
k = bending stiffness of the shaft $[Nm^{-1}]$
b = viscous damping constant $[Nsm^{-1}]$
g = acceleration due to gravity $[ms^{-2}]$

6.5 Rotorsystems with Gas Bearings

6.5.1 Problem Formulation

In Chapter 3 we have seen that in case of compressible (gas) bearings the bearing stiffness- and damping are not only a function of the rotor speed Ω, but also a function of the excitation frequency ω. The set of equations of motion, governing the rotor behavior in general can then be written as

$$\underline{M}\ \underline{\ddot{q}} + \underline{B}(\Omega, \Omega)\ \underline{\dot{q}} + \underline{K}(\Omega, \omega)\ \underline{q} = \underline{F} \tag{6.86}$$

We assume that our model has n degrees of freedom, so the matrices in this expression are of order $(n \times n)$ and the columns have n elements.
If, in a certain situation we are dealing with a specific rotor speed Ω and additionally also a specific excitation frequency ω there will be no problem. This for example is the case in the calculation of the synchronous unbalance response for a given rotor speed. The bearing data is completely known and the response can be calculated directly.
However, if we want to investigate the rotor-bearing stability for a fixed rotor speed the situation is more complex. Then we are investigating the free response and this implies that the excitation frequency ω will be unknown. In fact, this frequency is implicitly depending on the response which has to be calculated.
In the following discussion we will assume that the rotor speed Ω will be fixed and known, so the symbol Ω will be ignored. This means that the bearing data will be only depending on the excitation frequency ω. For the analysis of rotor bearing instability we look at the homogeneous part of (6.86). We substitute a solution of the type $\underline{q}(t) = \underline{u}\ e^{\lambda t}$ leading to the eigenvalue problem

$$[\lambda_k^2\ \underline{M} + \lambda_k\ \underline{B}(\omega) + \underline{K}(\omega)]\ \underline{u}_k = \underline{0} \tag{6.87}$$

We write the complex eigenvalue λ_k as

$$\lambda_k = \mu_k + j\ \omega_k \tag{6.88}$$

Now, the excitation frequency ω is assumed to be coupled directly to the imaginary part ω_k of the eigenvalue λ_k and the real part μ_k of this eigenvalue will determine the stability of the solution. This means that in this procedure

we assume that the eigen mode of which we check the stability is relatively weakly damped and consequently that the corresponding eigenvalue will not be real (undercritically damped). This is because the bearing data principally has been determined for a stationary harmonic situation, so they are only valid for a *vibrating* system, generally with a (slowly) decreasing amplitude due to damping.
The following problem has to be solved

$$\left[(\mu_k + j\omega_k)^2 \ \underline{M} + (\mu_k + j\omega_k) \ \underline{B}(\omega_k) + \underline{K}(\omega_k)\right] \ \underline{u}_k = \underline{0} \tag{6.89}$$

This should be a standard eigenvalue problem when the matrices $\underline{B}$ and $\underline{K}$ would not be a function of the imaginary part of the unknown eigenvalue. In the following we will look at two possible procedures to tackle this problem.

6.5.2 Intersection Method

This is a simple and straightforward method to generate a solution for the eigenvalue problem (6.89). First a number of discrete excitation frequencies ($\omega_p, \ \ p = 1..m$) is chosen and these discrete values are used in the stiffness- and damping matrix. Now we end up with an eigenvalue problem with constant parameters which can be solved using standard procedures, so we solve

$$\left[(\lambda_k)^2 \ \underline{M} + (\lambda_k) \ \underline{B}(\omega_p) + \underline{K}(\omega_p)\right] \ \underline{u}_k = \underline{0}; \qquad k = 1...n, \ p = 1, ...m \tag{6.90}$$

The imaginary parts $\omega_k = \omega_k(\omega_p)$ of the calculated eigenvalues λ_k can now be plotted as a function of the discrete values $\omega_p, \ \ p = 1..m$. In such a plot also the line $\omega = \omega_p$ can be drawn. The points of intersection are solution of the problem (6.88). The corresponding real parts are giving information about the stability of the particular solution.
An advantage of this procedure is that it is very simple and that only standard (rotordynamic) solution procedures are applied. The disadvantage is that for a large number of excitation frequencies the total eigenvalue problem has to be solved which for realistic rotorsystems will need a lot of computer time but also will generate a large amount of data which will never be used.

Note

To illustrate this procedure we look at a simple mass-spring-damper system for with the spring-stiffness and viscous damping-constant are assumed to be a function of the excitation frequency as shown in Fig. 6.5.2. For

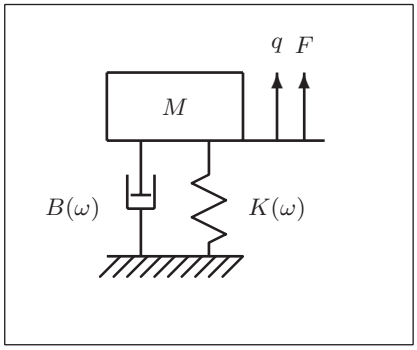
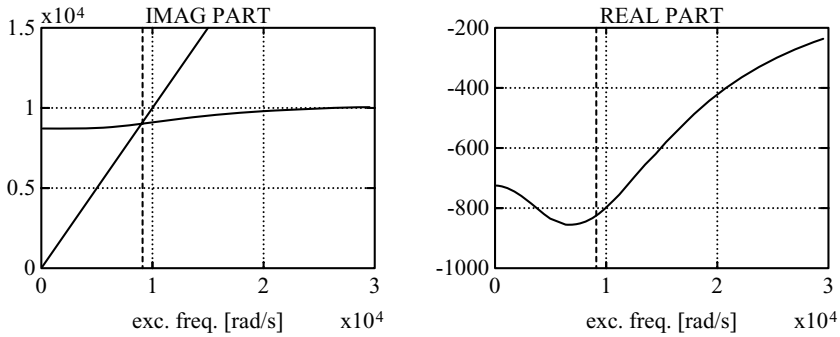


Fig. 6.13. Simple mass-spring-damper system with frequency dependent spring-stiffness and damping-constant

the spring-stiffness and damping-constant data we apply the data for the bearing, evaluated in Chapter 3, see Fig. 2.13. We will use the values K_{xx} and B_{xx} for a rotor speed of 100000 [RPM]. For the mass M we take $M = 1.0$ [kg].

It appears that for the excitation range $\{0 \leq \omega \leq 3.0\ 10^4\ [rad/s]\}$ we always are dealing with a pair of two complex conjugate eigenvalues, so the system is undercritically damped for this range. The (positive) imaginary part and the real part of the calculated eigenvalues are plotted in the next figure as function of the excitation frequency.



Application of the point of intersection method for the

evaluation of free vibrations and their stability

In the figure for the imaginary part also the line $y = \omega$ is drawn. This gives the intersection point(s) for which imag$[\lambda] = \omega$. In both figures this intersection point is indicated by the dotted line. So we find the homogeneous solution $q(t) = u\ e^{\lambda\ t}$ with $\lambda \approx -825 + 9018\ j$. The real part is negative, so the solution is **stable**. For more heavily damped systems (dimensionless damping $\xi \geq 0.5$, this point of intersection method will give problems or might even become impossible. A possible alternative can be found in [Geerts-95a]).

End of note

6.5.3 A Frequency Dependent Formulation

In general it will be difficult to deal with (excitation) frequency dependent elements in standard rotordynamic calculations. The points of intersection method of the previous paragraph offers a possible solution but normally it is rather inefficient. An alternative method will be to replace the frequency dependent elements (stiffness and/or damping) in the model by a combination of **constant** masses, springs and dampers with an effective dynamic stiffness which will be an accurate approximation for the original frequency dependent system, a so-called representative system approach. An example is shown in Fig. 6.14. In part (A) of this figure the model with frequency dependent elements is shown. In part (B) the replacement model is shown with constant elements $k_o,\ k_1$ and b_1. The problem now is to select parameter values for these constant elements such that we get a corresponding dynamic stiffnesses $Z = \hat{F}/\hat{q}$ for both models. For model (B) we can easily derive

$$Z(\omega) = \frac{k_o k_1 + b_1(j\omega)(k_o + k_1)}{b_1(j\omega) + k_1} = \frac{k_o k_1^2 + b_1^2\omega^2(k_o + k_1) + b_1 j\omega k_1^2}{k_1^2 + b_1^2\omega^2} \tag{6.91}$$

So, we can see that both models are identical if

$$K(\omega) = \text{real}[Z(\omega)] = \frac{k_o k_1^2 + b_1^2\omega^2(k_o + k_1)}{k_1^2 + b_1^2\omega^2} \tag{6.92}$$

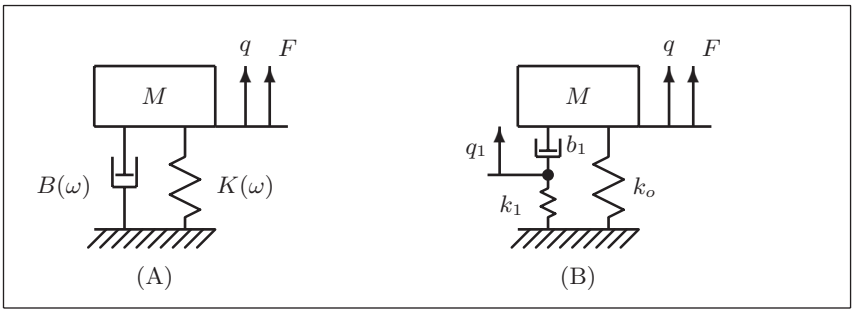


Fig. 6.14. System with frequency dependent elements (A) and a possible replacement model (B)

$$B(\omega) = \text{imag}[Z(\omega)]/\omega = \frac{b_1 k_1^2}{k_1^2 + b_1^2\omega^2} \tag{6.93}$$

This means that we are dealing with a parameter estimation procedure in which we have to calculate an optimal (in some sense) set of parameters (k_o, k_1 and b_1) from (measured) frequency dependent stiffness $K(\omega)$ and damping $B(\omega)$. A possible approach is to apply a procedure (see [Geerts-95a]) based on the $\boldsymbol{M_{AT}L_{AB}}$ -procedure **"invfreqs.m"**, which stars from a given, discrete series of functionvalues Z_i and corresponding angular frequencies ω_i. Then the procedure generates (in some least squares fit) the coefficients of the numerator and denominator polynomial of the general expression for the transferfunction in the form

$$Z(\omega) = \frac{c_1(j\omega)^{n-1} + c_2(j\omega)^{n-2} ... + c_{n-1}(j\omega) + c_n}{a_1(j\omega)^{m-1} + a_2(j\omega)^{m-2} ... + a_{n-1}(j\omega) + a_m} \tag{6.94}$$

The integers n and m determine the order of numerator respectively denominator polynomial and they can be chosen arbitrary. For the replacement system from Fig. 6.14 we in fact are dealing with a 1st-order approximation. In Fig. 6.15 the results are shown of fitting the bearing data (stiffness $K(\omega)$ and damping stiffness $\omega B(\omega)$) as mentioned before in Sections 2.6 and 6.5.2. The two top figures show the fit results using a 1st-order approach ($n = 2,\ m = 2$) and the two bottom figures the fit results using a general 2nd-order model ($n = 3, m = 3$). It can be concluded from these plots that the higher order model will be more accurate of course but that also the first order model already produces an acceptable representation of the transfer function. Practical experience with this fit procedure for a certain class of compressible bearings showed that a 1st-order model generally

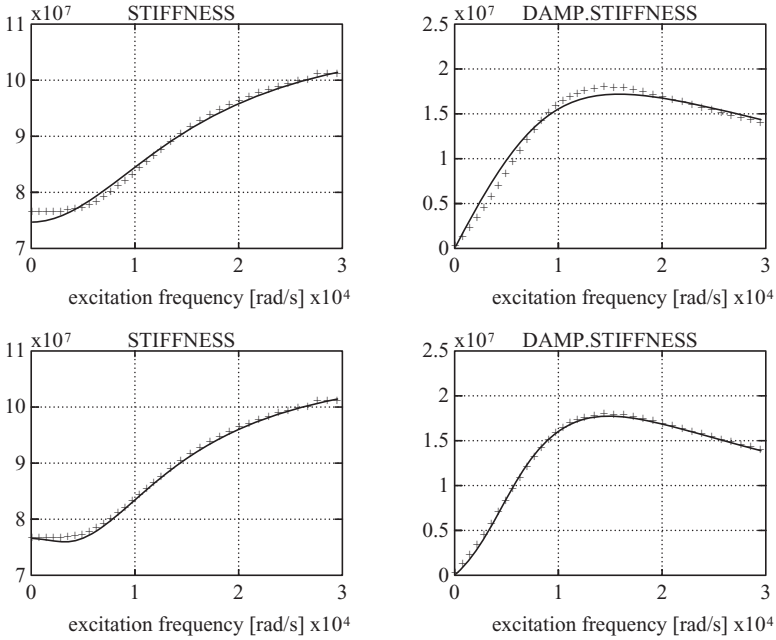


Fig. 6.15. Fit-results (solid lines) for a 1st-order model (top) and 2nd-order model (bottom) compared with original (measured) data (+++)

will not produce sufficiently accurate results, but that a 2nd-order model in many practical situations will meet the needs.

By adding such a replacement model to a rotor bearing system the frequency dependency will be removed and standard analysis routines can be applied. However the number of degrees of freedom of the model will increase. For example in the case of the model from Fig. 6.14, we have to introduce the additional (massless) degree of freedom q_1. The introduction of massless degrees of freedom has the disadvantage that the mass matrix will become singular. An attractive approach to deal with this is by introducing the bearing force f_l as additional degree of freedom. If we want to apply a 2nd-order model for the system from Fig. 6.14, we then will get the set of equations of motion

$$M\ddot{q} + f_l = F \tag{6.95}$$

$$c_1\ddot{q} + c_2\dot{q} + c_3 q = a_1\ddot{f}_l + a_2\dot{f}_l + a_3 f_l \tag{6.96}$$

or in matrix form

$$\begin{bmatrix} M & 0 \\ -c_1 & 1 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \ddot{f}_l \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ -c_2 & a_2 \end{bmatrix} \begin{bmatrix} \dot{q} \\ \dot{f}_l \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ -c_3 & a_3 \end{bmatrix} \begin{bmatrix} q \\ f_l \end{bmatrix} = \begin{bmatrix} F \\ 0 \end{bmatrix} \tag{6.97}$$

(where the coefficients have been scaled such that $a_1 = 1$).
For the frequency dependent stiffness and -damping from Fig. 6.15 we found the parameter values $a_i,\ c_i,\ i = 1..3$

i	a_i	c_i
1	1.0	$1.08\ 10^8$
2	$1.84\ 10^4$	$1.49\ 10^{12}$
3	$7.22\ 10^7$	$5.53\ 10^{15}$

If we next assume the system mass M to be: $M = 1.0$ [kg], a solution of the eigenvalue problem corresponding to (6.97) leads to

i	λ_i
1	$-926.28 + 9040.7\ j$
2	$-926.28 - 9040.7\ j$
3	-6966.2
4	-9603.7

So we get a pair of 2 complex conjugate eigenvalues λ_1 and λ_2 which are practically in good agreement with the result found with the points of intersection method, namely $\lambda = -825 + 9018j$. Only the real part shows some deviation because we are dealing with a system which is not very weakly damped (dimensionless damping approx. 10%).

However, we also find two additional eigenvalues (due to the extra degree of freedom f_l) which appear to be purely real and negative values. In fact we are using a polynomial expression for the transferfunction $Z(s)$ with $s = \nu + j\omega$, but for the calculation of the coefficients we only apply data for purely imaginary s, so for $\nu = 0$. Once the model for the transferfunction has been added to the rotor model this independent variable s will in general be complex or even real as follows from the value $s = \lambda_3$. For this value we may not expect anymore that real$[Z(s)] = K$ and imag$[Z(s)] = \omega B$. If we evaluate the transferfunction for $s = \lambda_3$ we namely see that real$(Z) = -4.85\ 10^7\ [Nm^{-1}]$ and imag$(Z) = -2.26\ 10^{-7}\ [Nm^{-1}]$. For the 4th eigenvalue $s = \lambda_4$ we get real$(Z) = -9.22\ 10^7\ [Nm^{-1}]$ and

imag$(Z) = -7.80\ 10^{-8}\ [Nm^{-1}]$.
So in both cases we get a **negative** stiffness and a neglectable damping stiffness. This means that not any physical conclusions may be drawn from these additional (numerical) eigenvalues. It can be shown that these additional eigenvalues are related to the roots of the numerator and denominator polynomials of the transferfunction (6.94). These roots appear to be for the denominator $s_1 = -12774,\ s_2 = -5648$ and for the numerator $s_{1,2} = -6907 \pm 1853j$.
More information about these additional **numerical** eigenvalues, their background, how to recognize them and the possibility to avoid unrealistic, positive real eigenvalues can be found in [Geerts-95a]).

6.5.4 A 2-Dimensional Example

In this section the tools for the analysis of the dynamic behavior of rotorsystems with compressible bearings will be applied to the 2-dimensional system as shown in Fig. 6.16. In this figure a frequency dependent stiffness

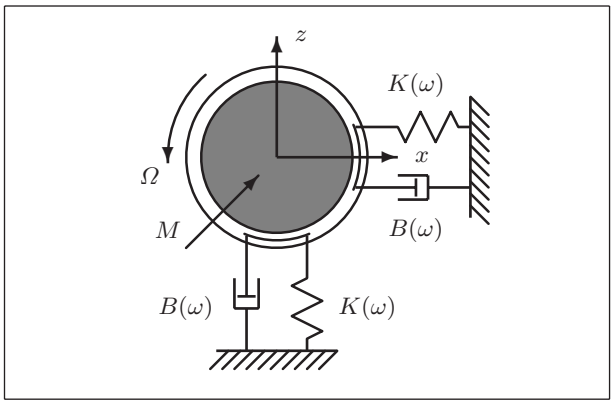


Fig. 6.16. Rotorsystem in a gasbearing. Cross coupling stiffness and damping influences are not shown

and damping are shown. Additionally we will be dealing with cross coupling terms. We assume a zero static bearing eccentricity, so $K_{xx} = K_{zz}$, $K_{xz} = -K_{zx}$, $B_{xx} = B_{zz}$, $B_{xz} = -B_{zx}$. We will use again the representative bearing data for the fixed rotor speed of $\Omega = 100000\ [RPM]$ for

the rotor shown in Fig. 2.13, paragraph 3.6, and for the rotormass we take $M = 1.0$ [kg].
First we apply the points of intersection method to estimate the eigenvalues of the free vibrating system. The two eigenvalues with positive imaginary part are shown in Fig. 6.17 together with the intersection points with the line imag$(\lambda) \equiv \omega$. The resulting eigenvalues appear to be

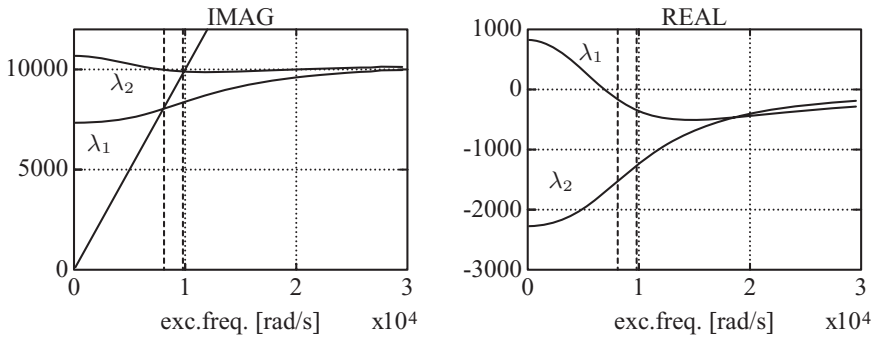


Fig. 6.17. Application of intersection method for 2-D rotor

$$\begin{aligned} \lambda_1 &\approx \ -115 \pm 7976j \\ \lambda_2 &\approx -1266 \pm 9900j \end{aligned} \tag{6.98}$$

So, we find two stable eigenvalues. Additionally we can see that by adding the cross coupling terms in the bearing model these eigenvalues deviate considerably from the eigenvalue for the related 1-D problem from paragraph 6.4.2, namely $\lambda = -825 \pm 9018j$.
Secondly we will apply the method based on a constant parameter representative system. For the complex transfer functions $H_{xx} = H_{zz}$ and $H_{zx} = -H_{xz}$ we will use a 2nd-order polynomial. For the coefficients we then find

	H_{xx}		H_{xz}	
i	a_i	c_i	a_i	c_i
1	1.0	$1.08\ 10^8$	1.0	$-2.41\ 10^5$
2	$1.84\ 10^4$	$1.49\ 10^{12}$	$2.16\ 10^4$	$9.35\ 10^{10}$
3	$7.22\ 10^7$	$5.53\ 10^{15}$	$1.35\ 10^8$	$3.46\ 10^{15}$

The reconstructed polynomial fits for the transfer functions H_{xx} and H_{xz} are shown in Fig. 6.18. If we next add the polynomial formulation for the

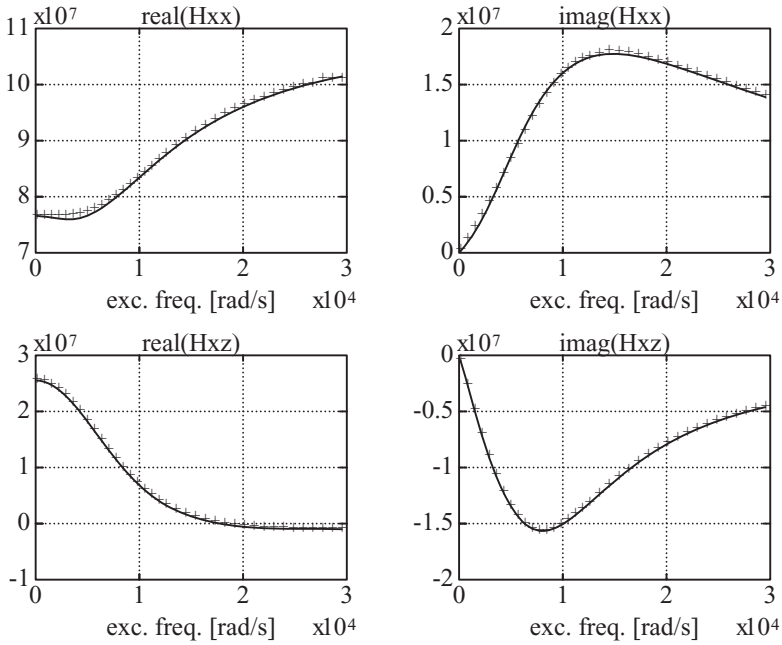


Fig. 6.18. Original bearing data (**) and the polynomial fits (solid lines) for H_{xx} and H_{xz}

bearing forces to the 2-D model we get the set of equations

$$\underline{M}\ \underline{\ddot{q}} + \underline{B}\ \underline{\dot{q}} + \underline{K}\ \underline{q} = \underline{f} \tag{6.99}$$

where

$$\underline{q} = \begin{bmatrix} x \\ z \\ f_{xx} \\ f_{xz} \\ f_{zx} \\ f_{zz} \end{bmatrix} \ ; \ \underline{f} = \begin{bmatrix} F_x \\ F_z \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \ ; \ \underline{M} = \begin{bmatrix} M & 0 & 0\,0\,0\,0 \\ 0 & M & 0\,0\,0\,0 \\ -c_1^{xx} & 0 & 1\,0\,0\,0 \\ 0 & -c_1^{xz} & 0\,1\,0\,0 \\ c_1^{xz} & 0 & 0\,0\,1\,0 \\ 0 & -c_1^{xx} & 0\,0\,0\,1 \end{bmatrix} \tag{6.100}$$

$$\underline{B} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -c_2^{xx} & 0 & a_2^{xx} & 0 & 0 & 0 \\ 0 & -c_2^{xz} & 0 & a_2^{xz} & 0 & 0 \\ c_2^{xz} & 0 & 0 & 0 & a_2^{xz} & 0 \\ 0 & -c_2^{xx} & 0 & 0 & 0 & a_2^{xx} \end{bmatrix} \tag{6.101}$$

$$\underline{K} = \begin{bmatrix} 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ -c_3^{xx} & 0 & a_3^{xx} & 0 & 0 & 0 \\ 0 & -c_3^{xz} & 0 & a_3^{xz} & 0 & 0 \\ c_3^{xz} & 0 & 0 & 0 & a_3^{xz} & 0 \\ 0 & -c_3^{xx} & 0 & 0 & 0 & a_3^{xx} \end{bmatrix} \tag{6.102}$$

Evaluating the free response for a rotormass of $M = 1.0$ [kg] gives the eigenvalues

i	λ_i
$1,2$	$-6.19\ 10^3 \pm 5.65\ 10^2\ j$
$3,4$	$-2.07\ 10^2 \pm 7.99\ 10^3\ j$
$5,6$	$-1.19\ 10^3 \pm 1.01\ 10^4\ j$
$7,8$	$-9.25\ 10^3 \pm 5.89\ 10^3\ j$
$9,10$	$-1.11\ 10^4 \pm 1.18\ 10^3\ j$
$11,12$	$-1.21\ 10^4 \pm 4.44\ 10^3\ j$

So, we can recognize 2 pairs of complex conjugate eigenvalues $-207 \pm 7990\ j$ respectively $-1190 \pm 10060\ j$, which correspond rather well with the values from the intersection method: $-115 \pm 7976\ j$ respectively $-1266 \pm 9900\ j$. Additionally we get 4 pairs of complex conjugate, heavily damped eigenvalues with dimensionless damping factors ξ=-real($\lambda/\|\lambda\|$), of respectively $\xi = 0.995,\ 0.84,\ 0.994,\ 0.94$). Again these eigenvalues have a numerical and not physical background so they can be ignored.

If we calculate the roots of the numerator- respectively denominator polynomials for H_{xx} and H_{xz} we find

$$\begin{array}{lll} H_{xx}: & s_1 = -12774 \quad s_2 = -5648 & \text{respectively} \quad s_{1,2} = -6907 \pm 1853\ j \\ H_{xz}: & s_{1,2} = -10782 \pm 4416\ j, & \text{respectively} \quad s_1 = 421481 \quad s_2 = -34041 \end{array}$$

Finally we look at the evolution of the two relevant eigenvalue pairs as function of the rotormass in the range $0.1 \leq M \leq 10\ [kg]$. This is shown in Fig. 6.19. We see that for an increasing rotormass both the imaginary parts of the eigenvalues will decrease, as might be expected. The real parts first are strongly negative but are getting less negative for increasing rotormass. At approximately a rotormass of $M = 1.2\ [kg]$, one of the real parts will become positive, meaning that from that moment we are dealing with an unstable system. Therefore this mass sometimes is called the **critical rotormass**.

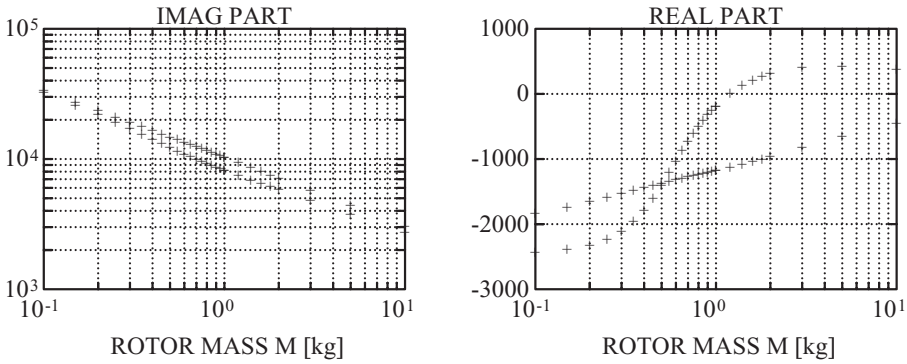
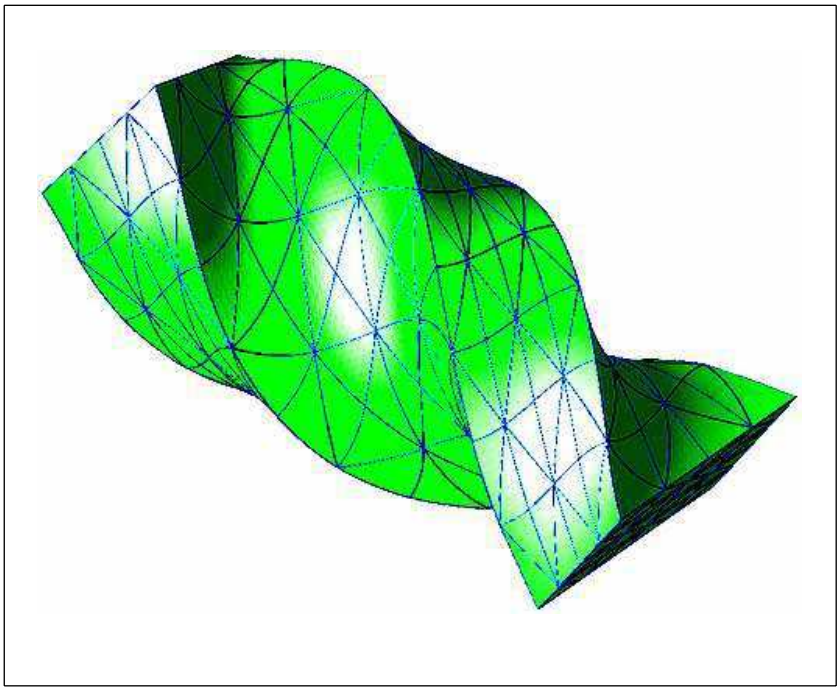


Fig. 6.19. Real and imaginary parts of the relevant eigenvalues as function of the rotormass M

7

Torsional Problems



7.1 Introduction

In many mechanical systems like gearboxes, turbine-generator systems and car drive-line systems, we are dealing with a number of coupled shafts which will mainly show torsional vibrations. The prediction of the torsional eigen-frequencies and corresponding vibration modes in such systems can be an effective tool to avoid resonance problems in the operational speed range. Additionally it is important in many cases to have information about the transient behavior of these systems (for example the start-up of the unit). In such cases the torsional moments and consequently also the shear stresses in structural components can be much larger than under normal operation. In models to analyse torsional vibrations the damping generally will be ignored because it is assumed to be small and therefore it will not have a large effect on the free vibrational behavior. In many practical situations also the interaction-effects between bending, torsion and axial deformation will be assumed to be negligible. Fig 7.1 shows an example of a simple rotorsystem. It consists of 3 (idealized) rigid disks and two shaft elements which can also add some inertia to the system but are mainly introducing the flexibility (torsion) in the system As generalized coordinates we use the disk rotations

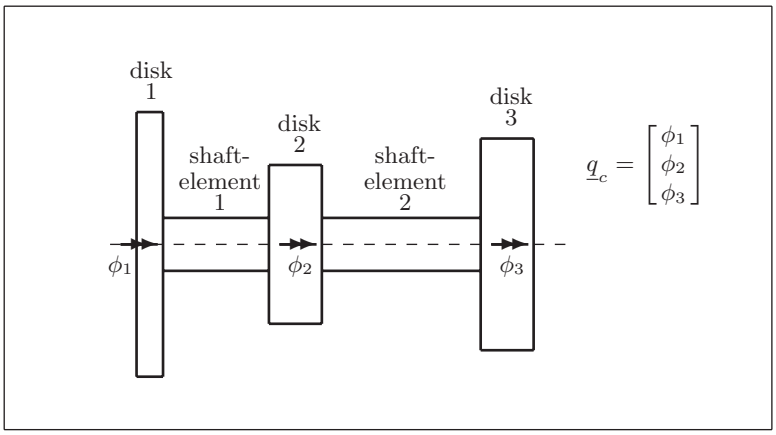


Fig. 7.1. A simple 3 degree-of-freedom torsional rotor model

around the central axis, namely ϕ_1, ϕ_2 and ϕ_3.

In the following sections we first will specify some simple basic elements of torsional analysis models.

7.2 Basic Elements

7.2.1 Disk Element

The disk element consists of a rigid body rotating around the fixed rotor-axis with rotational frequency $\dot{\phi}$. The mass moment of inertia around this axis is called J_p. So the relevant dynamic equation reads

$$M = J_p \, \ddot{\phi} \tag{7.1}$$

7.2.2 Shaft Element

For the (torsional) shaft element we only consider a circular shaft with uniform cross section (radius R and length l), as shown in Fig. 7.2. For the

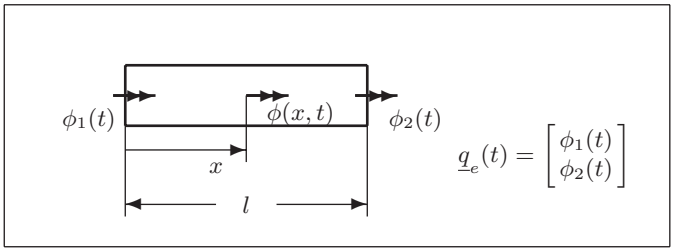


Fig. 7.2. Torsional shaft element

kinetic energy T of this element we can write

$$T = \frac{1}{2} \int_{x=0}^{l} \rho \, I_p \left[\frac{\partial \phi(x,t)}{\partial t} \right]^2 dx \tag{7.2}$$

with I_p the polar moment of inertia for the cross section. Next we assume a **linear interpolation** function for the rotation $\phi(x,t)$ as function of the axial coordinate x, so we write:

$$\phi(x,t) = [1 - \frac{x}{l} \; , \; \frac{x}{l}] \begin{bmatrix} \phi_1(t) \\ \phi_2(t) \end{bmatrix} = \underline{X} \; \underline{q}_e(t) \tag{7.3}$$

where $\underline{q}_e(t)$ is the column of degrees of freedom for the element e. If we substitute this expression in (7.2), we get:

$$T = \frac{1}{2}\,\underline{\dot{q}}_e^T \left[\int_{x=0}^{l} \rho I_p \underline{X}^T \underline{X} dx \right] \underline{\dot{q}}_e = \frac{1}{2}\underline{\dot{q}}_e^T\ \underline{M}_e\ \underline{\dot{q}}_e \tag{7.4}$$

So we find for the mass matrix $\underline{M}_e$ of the element

$$\underline{M}_e = \int_{x=0}^{l} \rho\ I_p\ \underline{X}^T \underline{X}\ dx = \frac{J_e}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad J_e = \rho I_p l \tag{7.5}$$

where J_e is the axial mass moment of inertia of the shaft element.
For the elastic energy U we can write

$$U = \frac{1}{2} \int_{x=0}^{l} G\ I_p \left[\frac{\partial \phi(x,t)}{\partial x} \right]^2\ dx \tag{7.6}$$

where G is the shear modulus. Substitution of (7.3) in (7.6) gives:

$$U = \frac{1}{2}\ \underline{q}_e^T \left[\int_{x=0}^{l} G\ I_p\ \frac{\partial \underline{X}^T}{\partial x} \frac{\partial \underline{X}}{\partial x}\ dx \right]\ \underline{q}_e = \frac{1}{2}\ \underline{q}_e^T\ \underline{K}_e\ \underline{q}_e \tag{7.7}$$

giving for the element stiffness matrix $\underline{K}_e$:

$$\underline{K}_e = \int_{x=0}^{l} G\ I_p\ \frac{\partial \underline{X}^T}{\partial x} \frac{\partial \underline{X}}{\partial x}\ dx = \frac{G\ I_p}{l} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \tag{7.8}$$

7.2.3 Gear Transmission Element

If the rotation of two or more shafts is coupled by a gear transmission, the deformations in this transmission (especially in the teeth) can be ignored in some situations. In that case the rotations of the relevant shafts can then be coupled directly in a kinematic way by using the transmission ratio. In our Finite Element Model this can simply be done by taking the rotation of one of the shafts as generalized coordinate and translating all the inertia to this coordinate. If we, for example, take the tooth-stiffness of the two gears, shown in Fig 7.3 infinitely large (rigid), and take the rotation ϕ_1 as the generalized coordinate, we can write for the kinetic energy

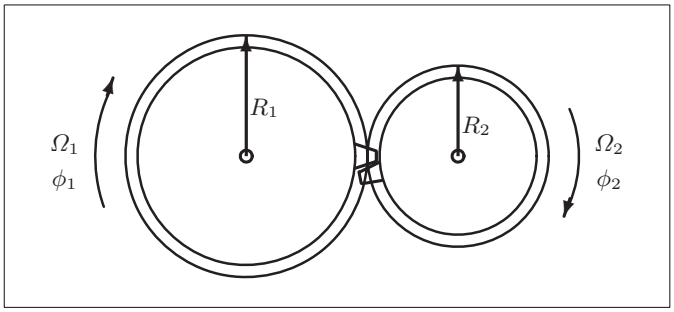


Fig. 7.3. Gear Transmission Element

$$T = \frac{1}{2}\ J_1\ \dot{\phi}_1^2 + \frac{1}{2}\ J_2\ \dot{\phi}_2^2 = \frac{1}{2}\ \left[J_1 + J_2 \left(\frac{-R_1}{R_2}\right)^2\right] \dot{\phi}_1^2 = \frac{1}{2}\ J_1^*\ \dot{\phi}_1^2 \quad (7.9)$$

This means that the generalized coordinate ϕ_1 has an associated (effective) mass moment of inertia of

$$J_1^* = J_1 + J_2\ i^2 \quad (7.10)$$

where the transmission ratio $i = \frac{R_1}{R_2}$ is introduced.
If the transmission may not be assumed to be rigid (teeth flexibility, effect of lubrication, see also [Roosmalen-94]), both the rotations ϕ_1 and ϕ_2 have to be accepted as generalized coordinates. Then we also have to introduce an additional elastic energy term

$$U = \frac{1}{2}\ k_T\ \ (R_1\ \phi_1 + R_2\ \phi_2)^2 \quad (7.11)$$

where k_T is the global stiffness ($[N/m]$) of the gear transmission. If we define the column of degrees of freedom for this element as $\underline{q}_e^T = [\phi_1, \phi_2]$, we can also write for U

$$U = \frac{1}{2}\ \underline{q}_e^T\ \underline{K}_e\ \underline{q}_e \quad (7.12)$$

with a transmission element stiffness matrix $\underline{K}_e$

$$\underline{K}_e = k_T \begin{bmatrix} R_1^2 & R_1 R_2 \\ R_1 R_2 & R_2^2 \end{bmatrix} \quad (7.13)$$

7.3 A Practical Example

To illustrate the general procedure in a torsional analysis we look at the system shown in Fig 7.4. The system consists of two shafts and 5 disks. The nominal rotational speed of shaft 1 is Ω_1 and of shaft 2 it is Ω_2. The torsional deformations which are superimposed on these nominal rotations are described by the angles $\phi_i,\ i = 1..5$. It can be shown that the torsional

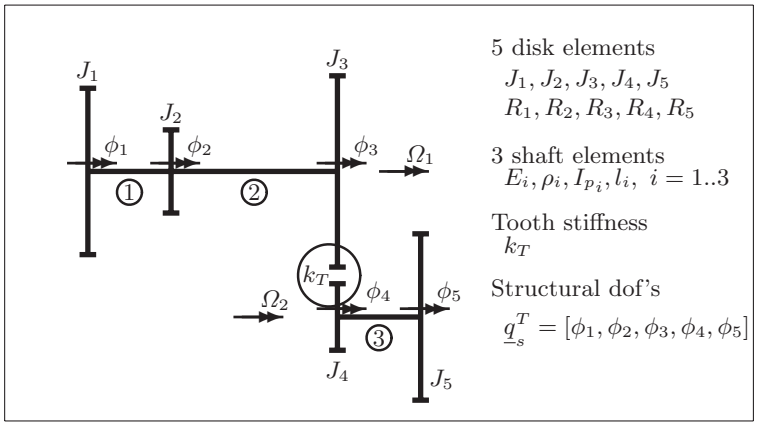


Fig. 7.4. Model of a torsional system

vibrations in this system are described by the set of equations of motion

$$\underline{M}_s\ \underline{\ddot{q}}_s\ +\ \underline{K}_s\ \underline{q}_s = \underline{T}_s \tag{7.14}$$

For the structural mass matrix $\underline{M}_s$, -stiffness matrix $\underline{K}_s$ and -column of external loads $\underline{T}_s$ we can find

$$\underline{M}_s = \begin{bmatrix} \frac{J_{s1}}{3} + J_1; & \frac{J_{s1}}{6}; & 0; & 0; & 0 \\ \frac{J_{s1}}{6}; & \frac{J_{s1}+J_{s2}}{3} + J_2; & \frac{J_{s2}}{6}; & 0; & 0 \\ 0; & \frac{J_{s2}}{6}; & \frac{J_{s2}}{3} + J_3; & 0; & 0 \\ 0; & 0; & 0; & \frac{J_{s3}}{3} + J_4; & \frac{J_{s2}}{6} \\ 0; & 0; & 0; & \frac{J_{s3}}{6}; & \frac{J_{s3}}{3} + J_5 \end{bmatrix} \tag{7.15}$$

$J_{si} = \rho_i\ I_{p_i}\ l_i,$ the mass moment of inertia of shaft i

$$\underline{K}_s = \begin{bmatrix} k_1 & ; & -k_1 & ; & 0 & ; & 0 & ; & 0 \\ -k_1 & ; & k_1+k_2 & ; & -k_2 & ; & 0 & ; & 0 \\ 0 & ; & -k_2 & ; & k_2+k_T R_3^2 & ; & k_T R_3 R_4 & ; & 0 \\ 0 & ; & 0 & ; & k_T R_3 R_4 & ; & k_3+k_T R_4^2 & ; & -k_3 \\ 0 & ; & 0 & ; & 0 & ; & -k_3 & ; & k_3 \end{bmatrix} \tag{7.16}$$

$$k_i = \frac{G_i \; I_{p_i}}{l_i}$$

$$\underline{T}_s^T = [T_1, T_2, T_3, T_4, T_5] \tag{7.17}$$

For the numerical evaluation we apply

- Shear modulus $G_i = 8.0 \; 10^{10} \; [Nm^{-2}]$
- Specific mass $\rho_i = 7.8 \; 10^3 \; [kgm^{-3}]$
- Shaft lengths $l_1 = 0.1, \; l_2 = 0.2, \; l_3 = 0.1 \; [m]$
- Shaft diameters $d_1 = d_2 = 0.08, \; d_3 = 0.06 \; [m]$
- Disk diameters $D_1 = 0.4, D_2 = 0.2, D_3 = 0.48, D_4 = 0.16, D_5 = 0.2 \; [m]$
- Mass moments of inertia $J_1 = 0.2, J_2 = 0.05, J_3 = 0.4, J_4 = 0.05, J_5 = 0.05 \; [kgm^2]$
- Tooth stiffness $k_T = 5.0 \; 10^8 \; [Nm^{-1}]$

The analysis of the free vibrational behavior leads to the following (torsional) angular eigenfrequencies

eigenfrequency	[rad/s]
ω_1	0.0
ω_2	2326.23
ω_3	5409.37
ω_4	10100.69
ω_5	12263.27

So, we can recognize one mode with eigenfrequency 0.0 [rad/s], which appears to be a rigid body mode as might be expected. The background is that the stiffness matrix $\underline{K}_s$ is not positive definite. This mode has been illustrated graphically in Fig. 7.5. From this figure we can conclude that the rigid body mode consists of a uniform rotation of shaft 1, and also a uniform rotation of shaft 2, with a relative magnitude which fits with the gear ratio (i=3) of the gear-pair of gears 3 and 4. For this mode the elastic energy will be zero leading to a non-positive definite stiffness matrix.

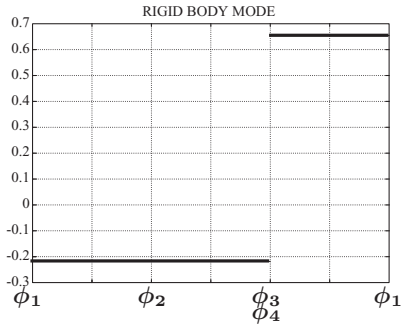


Fig. 7.5. Mode 1, the rigid body mode

The remaining (elastic) modes are shown in Fig 7.6. For mode 2 we can see that $\phi_4 \approx -3\ \phi_3$ still holds, so the gear transmission may still be assumed to be rigid. For the higher modes (for example the modes 4 and 5) this is not true anymore. The transmission stiffness k_T appears to be an essential element to describe the vibrational behavior of this structure.

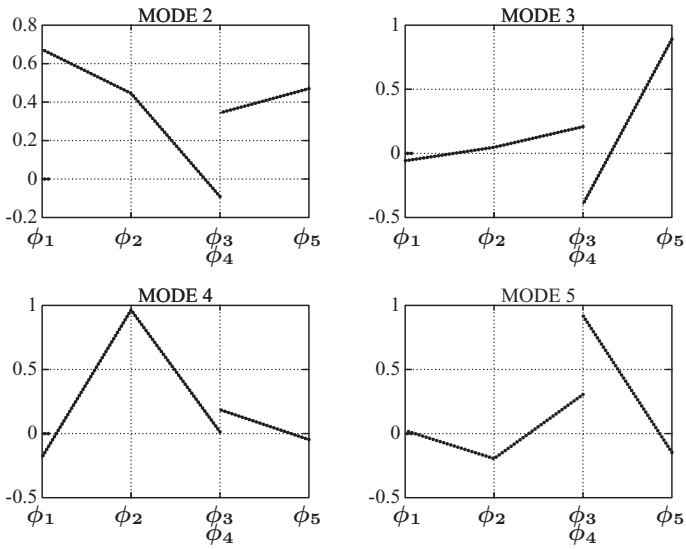
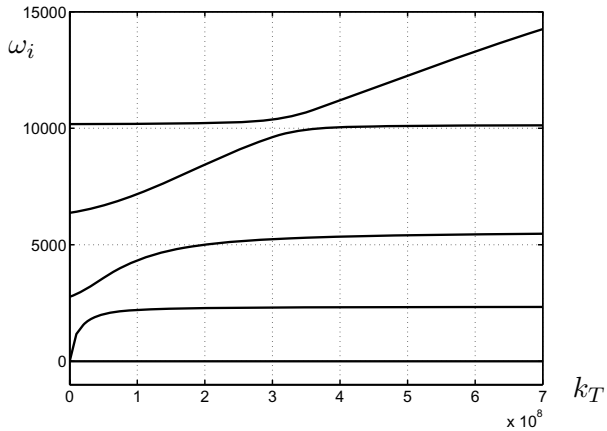


Fig. 7.6. The elastic modes 2,3,4 and 5

Note

Let us look at the effect of the Tooth-stiffness k_T on the eigenfrequencies of this torsional system. In the next figure these (5) eigenfrequencies are shown for the range $0 \leq k_T \leq 7.0\ 10^8\ [N/m]$.



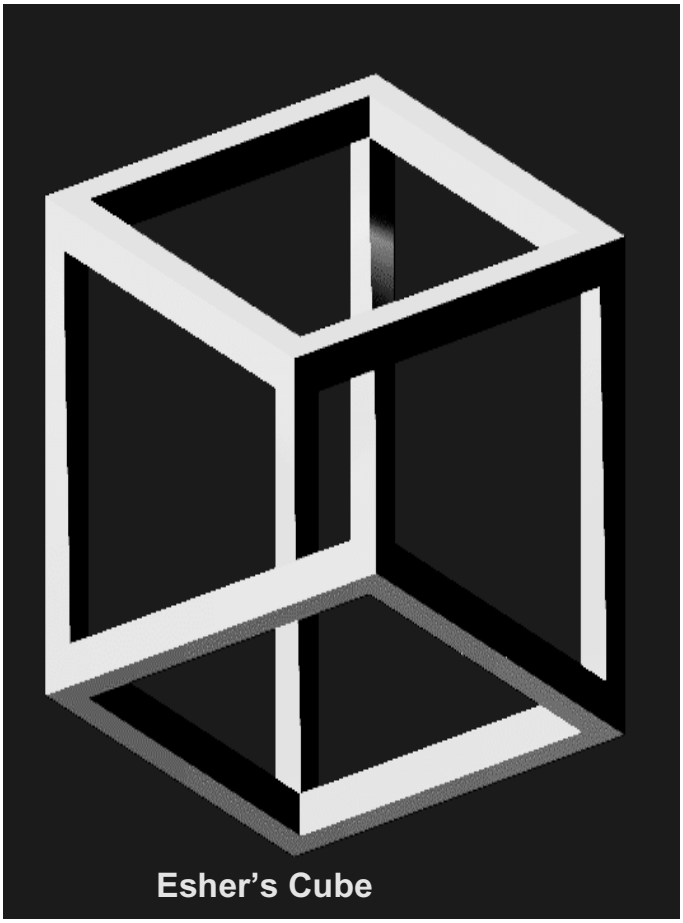
The 5 eigenfrequencies as function of the Tooth-stiffness k_T

It is clear that the tooth-stiffness is important for the dynamic behavior. It is up to the reader to look at the regions for very small and very large k_T (and think about simple approximate models). Also looking at the change of vibration modes can give interesting information.

End of note

8

Problems



8.1 Problem overview

Nr.	Subject	Page	Sol.
1	Gyroscopic effect in a rotating shaft	209	222
2	Hummingtop	210	225
3	A rotor on elastic foundation	211	229
4	Rotor operating in a magnetic field	213	238
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8.2 Problem descriptions

Problem 1 Gyroscopic effect in a rotating shaft

We look at a simple uniform shaft with outer- and inner diameter D_u, D_i and length L. The shaft is supposed to be supported in a very stiff bearing at the left side. So, we assume that the shaft is clamped at one side, but can still rotate around the shaft axis with rotational speed Ω. For the properties

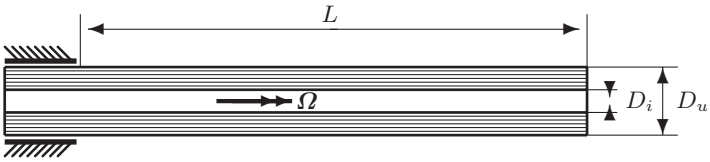


Fig. 8.1. simple, one-sided clamped shaft

we use:

- $L = 0.2$, $D_u = 0.02$, $D_i = 0.01$ $[m]$
- $E = 2.1\ 10^{11}$, $G = 0.8\ 10^{11}$ $[N/m^2]$, shearfactor=0.88 $[-]$
- $\rho = 7800$ $[kg/m^3]$
- Speed range $0 \leq \Omega \leq 500$ $[Hz]$, $(30.000$ $[RPM])$

Assignments:

* Find an (approximate) analytical value for the lowest bending frequencies for the case $\Omega = 0$. Assume the bearing stiffness to be rigid.

* Use $\boldsymbol{R_O D_Y}$ to study the impact of the gyroscopic effects on the lowest bending frequencies in the given Ω-range

Problem 2 Hummingtop

The Hummingtop is one of the most well-known gyroscopic systems. If we give it a sufficiently large rotational spin speed Ω, the upright position will be stable. In that stable position, a small perturbation will introduce two motions, both with their own specific eigenfrequency, namely a **Precession** mode and a **Nutation** mode.

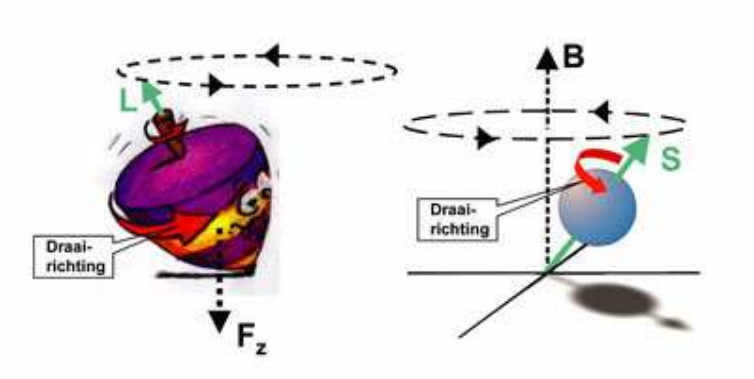


Fig. 8.2. Hummingtop

In *Engineering Mechanics, Volume Two: DYNAMICS* from J.L. Meriam and L.G. Kraige, 3rd edition, ISBN 0-471,59273-0, pp 572-576 we can find an approximation for the frequency of a steady-state precession:

$$\omega_p = \frac{g\ a}{r^2 \Omega}$$

where

- g is the gravity constant
- a is the distance between the center of gravity and the ground-contact point
- r is the radius of gyration: $I_p = m\ r^2$
- Ω is the spin speed

Assignment:

Use the $\boldsymbol{R_O D_Y}$ program to simulate the dynamics of the Hummingtop and look at the correlation between spin-speed and Precession speed.

Problem 3 A rotor on elastic foundation

We look at a simple rotor, consisting of a uniform slender shaft with 2 equal rigid discs. The rotor is supported by two identical bearings which can be modeled by a set of linear springs (no damping). The problem in this case is that the bearing housings are not rigidly connected to the ground, but are mounted in a supporting structure. This structure consists of a very stiff foundation plate with two pedestals for the bearings and supported by 4 (rubber) vibration isolators. These isolators can be modeled as having only a deformation in one direction (the vertical one), so shear is excluded. The general picture is shown in figure 8.3.

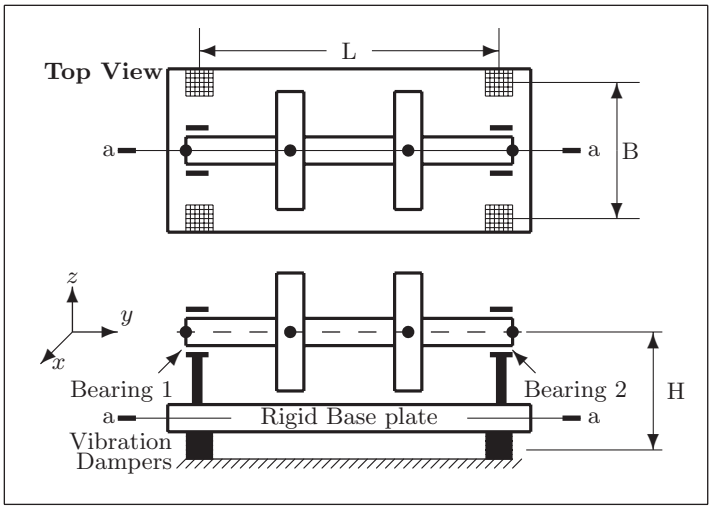


Fig. 8.3. Two-disc rotor on base-plate

For the rotor-data we take:

- Shaft sections each: length 0.25 [m], diameter 0.012 [m], material Steel
- Disks: diameter 0.24 [m], thickness 0.02 [m], material Brass
- Bearings: $k_{xx} = k_{zz} = 1.0\ 10^6$ [N/m], $k_{xz} = k_{zx} = 0$.

The main excitation source will be an imbalance on one or both of the discs.

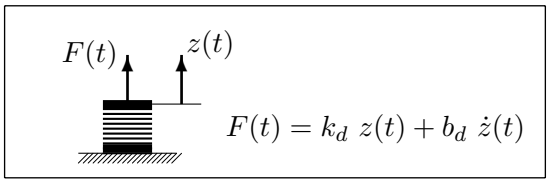
Assignments:

A Assume that the bearing housings are rigidly supported. This means that we assume that the foundation structure is so heavy and stiff (including the isolators) that the displacements of these bearing housings can be ignored with respect to the rotor displacement (bearing displacement and shaft bending).

A1 Analyse the free dynamical behavior

A2 Assume the disc have some imbalance. 1st case: same imbalance but 90 degrees out-of-phase, 2nd case: same imbalance at opposite sides. Evaluate the imbalance responses.

B Next we assume that the supporting frame can have some displacement, but that its stiffness is so high that it only moves as a rigid body on the elastic vibration isolators. For the frame (see again figure 8.3) we use $L/B/H = 0.8/0.4/0.14\ [m]$. Total mass is $70\ [kg]$. Mass moment of Inertia with respect to the axis a-a is $1.1\ [kgm^2]$. For the vibration isolators we use a simple uni-axial model as shown in next figure.



For the stiffness constant k_d we use $k_d = 2.0\ 10^4\ [N/m]$, for the damping constant we use $b_d = 2.0\ 10^2\ [Ns/m]$.

Repeat the analysis steps as done under item A, but now for the flexible supported rotor-system.

Problem 4 Rotor operating in a magnetic field

We look at a simple model of an electric engine. It consists of a shaft which at both end is supported in two regular ball-bearings (linear springs) and in the middle a single rigid disc. (see figure 8.4. The disc is operating in

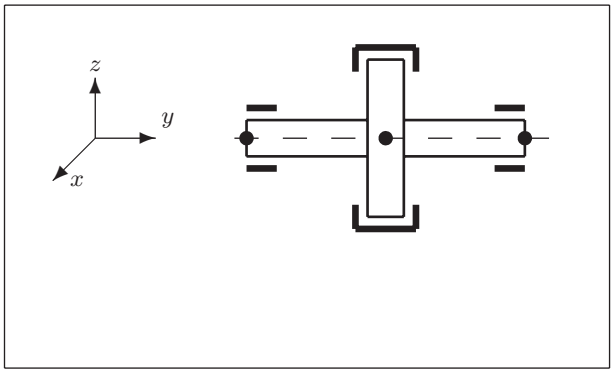


Fig. 8.4. Rotor operating in a magnetic field

an electric field which may be modeled as a bearing-stiffness of which the coefficients are linearly depending on the rotor-speed Ω.
The relevant data is:

- Shaft: Total length 0.5 [m], diameter 0.02 [m], steel
- Disc: mass 6.5 [kg], diameter 0.3 [m]
- Ball bearings $k_{xx} = k_{zz} = 1.0\ 10^4, \quad k_{xz} = k_{zx} = 0$ [N/m]
- "Disc-bearing" $k_{xx} = k_{zz} = \Omega * 1.0\ 10^3, \quad k_{xz} = k_{zx} = 0$ [N/m]

Assignment:
Analyse this problem using $\boldsymbol{R_O D_Y}$. (The element-type ECK2M can model a rotor-frequency dependent bearing stiffness matrix)

Problem 5 Plate Balancing

A well-known theater act is the balancing of a plate on a very thin flexible spear by giving it enough rotational speed, and then not just a single one but as many as possible simultaneously, (see figure 8.5. The plate can be

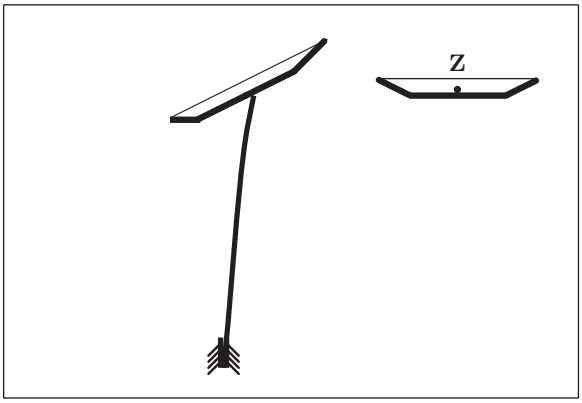


Fig. 8.5. Plate balancing

seen as a rigid body with mass m=0.45 [kg] and Polar mass-moment of inertia $I_p = 3.2\ 10^{-3}\ [kgm^2]$. The center of gravity Z is lying 0.01 [m] above the bottom-plane of the plate. For the flexible spear we take a thin steel beam with a length L=1.0 [m] and a diameter of d=0.002 [m].

Assignment:

Analyze the problem.

Problem 6 Rotor Balancing

In this problem we will try to identify the necessary balancing-mass(es) (the quantity and position) which is necessary to reduce the vibration amplitudes of a badly balanced rotor-system.
The (calculation) model of the system is shown in figure 8.6. The rotor-data

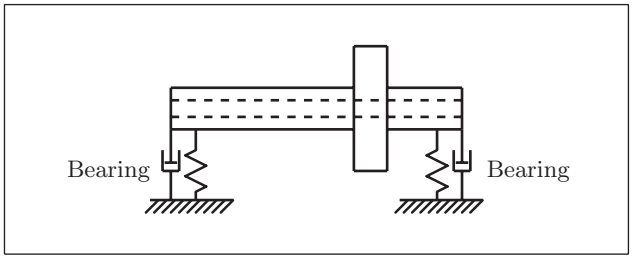


Fig. 8.6. Single disc rotor in two ball bearings

has been translated into a $\boldsymbol{R_O D_Y}$ input file. The essential elements of this file are:

```
*********************************************************
CONE8
Nr. L (m)         Do(m)  Di(m)  E(N/m2) rho(kg/m3) connect   group
  1 2.e-1  left:  4.e-2  2.e-2  2e+11  7800       q 1  q 6   1
           right: 4.e-2  2.e-2
  2 2.e-1  left:  4.e-2  2.e-2  2e+11  7800       q 6  q11   1
           right: 4.e-2  2.e-2
  3 2.e-1  left:  4.e-2  2.e-2  2e+11  7800       q11  q16  1
           right: 4.e-2  2.e-2
DISK4
Nr.  m(kg) Ip(kgm2) It(kgm2) connect group
  1  29    3.e-1    1.5e-1   q11      1
SPRING1
Nr.  k (N/m, Nm/rad)         connect
  1  3e+7                    q 1
  2  5e+7                    q 2
  3  3e+7                    q16
  4  5e+7                    q17
DAMP1
Nr.  d (Ns/m, Nms/rad)       connect
  1  1e+5                    q 1
  2  1e+5                    q 2
```

```
  3  1e+5                    q16
  4  1e+5                    q17
 NUMBER OF DEGREES OF FREEDOM: 20
*********************************************************
```

The "measured" respons of the system, particularly at the disc-location and around the first critical speed is shown in figure 8.7. The displacement in x- (dof11) and z-direction (dof12) is almost the same. It seems therefore that the shaft-bending stiffness is the most relevant here and not the non-symmetrical bearing stiffness. The system also looks to be moderately damped. The orbit at the disc location for the specific rotor-speed of Ω=67

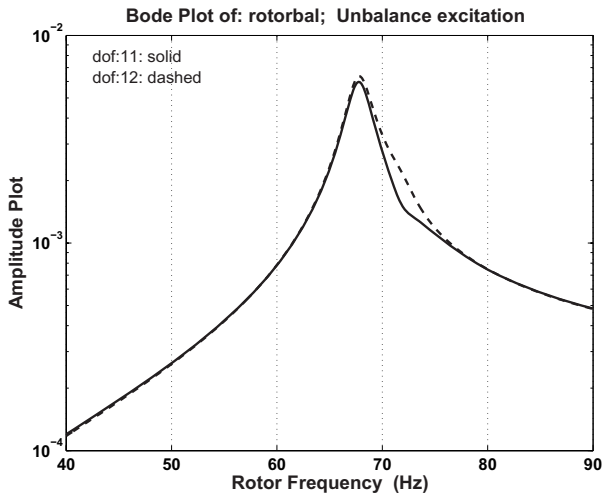


Fig. 8.7. Part of the imbalance respons at the disc location

[Hz] is shown in figure 8.8. The point for t=0 is indicated by a star and it clearly is a forward whirl: dof11=x, dof12=z and rotor-speed is in positive y-direction. (We have seen before -see Laval-rotor- that the backward whirl in these systems generally will not be triggered by an unbalance excitation). In figure 8.9 the orbits at the rotor-speeds of 62 and 72 [Hz] are shown.

Assignment:

- Try to estimate the imbalance of the system from the given response data.

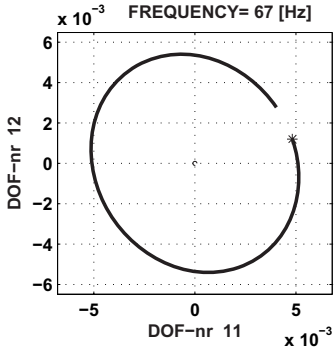


Fig. 8.8. Orbit at rotor-speed of $\Omega = 67$ Hz

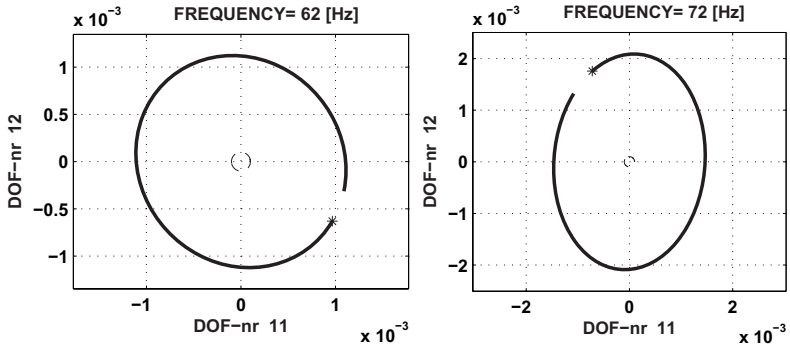


Fig. 8.9. Orbits at rotor-speeds of Ω =62 and 72 Hz

- Assume the disc has some small skewness. Generate comparable "measured" orbits and investigate whether this imperfection can also be compensated by adding imbalance weights.

Problem 7 Windmill Dynamics

A windmill operates at the top of a steel tube as shown in figure 8.10. The uniform tube has a diameter of 800 [mm], a wall-thickness of 15 [mm] and a length of 40 [m]. The mill consists of a housing with bearings which

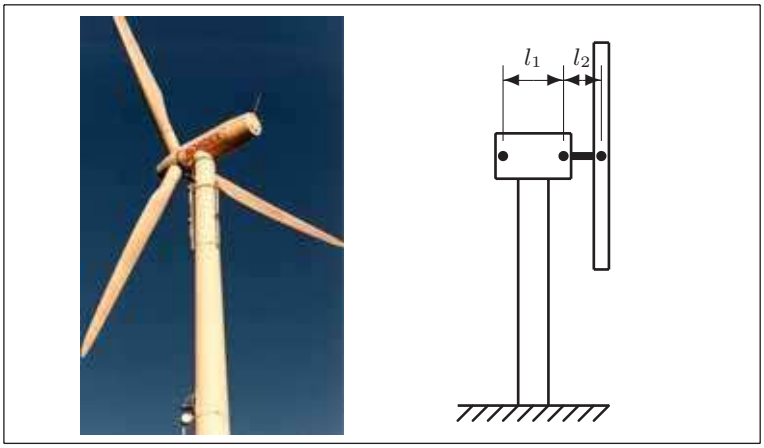


Fig. 8.10. Windmill model

has a weight of 141 [kg] with a distance l_1=0.6 [m] between the two rotor bearings. The rotor and blades may be (very simplified) seen as a rigid disc with a diameter of 30 [m], a total mass of 80 [kg] and a mass-moment of inertia with respect to the rotation axis of $I_p = 3000\ [kgm^2]$. It is mounted on a shaft with a diameter of 30 [mm] and the distance l_2 between the rotor-midplane and the closest bearing is 0.4 [m]. The 2 roller bearings of the mill may be seen as (uncoupled)pairs of linear springs with a stiffness constant of $1.0\ 10^6$ [N/m] with no damping.

Assignments:

- Evaluate the dynamic behavior and calculate the critical speeds
- Assume that one of the blades has been damaged and that a piece of material with a mass of 2 [kg] has been broken from the blad-tip. Calculate the respons for the rotor-speed range $\Omega \leq 1.5$ [Hz].

Problem 8 Gyro-Coach

To enhance the riding comfort of a coach it might be attractive to build in a heavy flywheel for reducing the very unpleasant rolling motions (rotations around the longitudinal axis) as schematically shown in figure 8.11. The

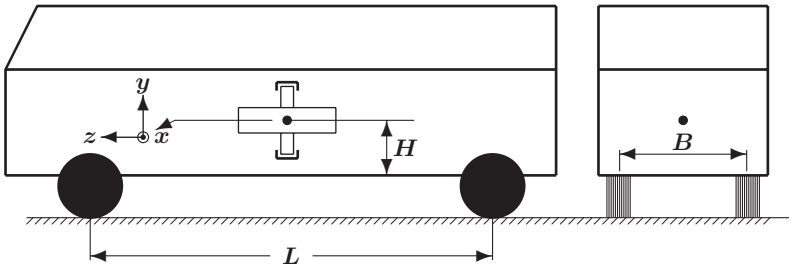


Fig. 8.11. Motor-Coach with gyroscopic stabilization

coach without flywheel may be considered as a rigid body with a mass M_c of 12.000 [kg] with the center of gravity in the middle. The wheelbase $L = 12\ [m]$ and the railway gauge $B = 3.6\ [m]$. The mass moment of inertia with respect to the center of gravity around the x- y- and z-axis ($J_x,\ J_y,\ J_z$) are respectively $1.4\ 10^5,\ 1.4\ 10^5,\ 1.2\ 10^4\ \ [kgm^2]$.
The flywheel has a mass of $m_f = 600\ [kg]$ and has a diameter of $d_f = 1.2\ [m]$. It may be considered as a thin rigid disc and the c.o.g. coincides with that of the coach which is lying $H = 1.4\ [m]$ above the coach-floor. Each of the 4 tyres may be seen as a combination of a linea spring $k_t = 2.0\ 10^4\ [N/m]$ and a viscous damper $b_t = 2.0\ 10^3\ [Ns/m]$.
We assume that the transverse stiffness of tyres and suspension is so high that the main motion of the coach is dominated by:

- a vertical displacement V of the c.o.g
- a rotation φ_z around the z-axis (roll motion)
- a rotation φ_x around the x-axis (jumping motion)

Assignment:
Evaluate the dynamic behavior of the coach as function of the flywheel rotor-speed.

Problem 9 Rotor model for aircraft engine

This problem has been taken from [Vance-88], page 258. It is dealing with a rotordynamic model for a high-pressure rotor in an aircraft engine. A picture of this type of engines is shown in figure 8.12.

The model is shown in figure 8.13. It consists of 6 discs ($D_1 ... D_6$), 2 roller

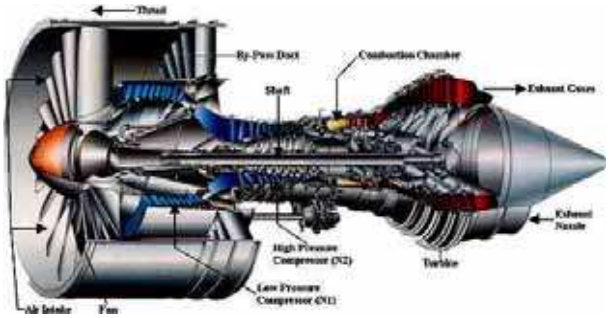


Fig. 8.12. Picture of aircraft turbine engine

bearings (B_1, B_2) and 5 shafts ($S_1, ...S_5$). The shafts S_1 and S_5 are each modeled by 2 shaft-element to allow for the introduction of the bearings. This type of machines has caused many rotor-dynamic difficulties due to

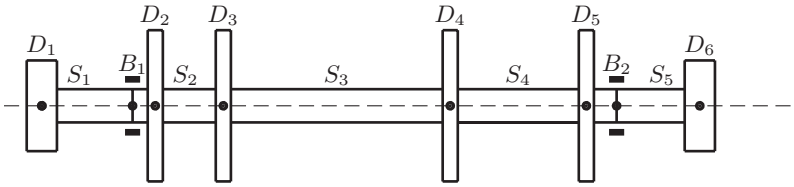


Fig. 8.13. Rotordynamic model for an aircraft engine

dynamic instabilities. The background of these instabilities very often is the presence of large destabilizing forces produced by the medium (fluid) around the impellers.

These forces can in general be modeled as linearized stiffness and damping, especially cross-coupling terms. In this case we only have these cross-coupling terms on each impeller, where $k_{xz} = -k_{zx}$. So at each impeller-

station we additionally assume a "bearing" with stiffness-matrix:

$$\underline{K}_{cc} = \begin{bmatrix} 0 & k_{cc} \\ -k_{cc} & 0 \end{bmatrix}$$

For the analysis two situations will be relevant, a weak- and a strong cross-coupling level.

The rotor is supported by 2 roller bearings (usual in aircraft engines) which generally have only little damping. Therefore it is common to introduce one or more squeeze film dampers in the system. One of the reasons to do this is the reduction of synchronous whirl amplitudes (due to imbalance) but a secondary effect is that they can be used to suppress potential instabilities. In this model a squeeze film damper will be assumed at bearing location B_1. This means that we have a damping matrix at that point with:

$$\underline{B}_{sfd} = \begin{bmatrix} b_{sfd} & 0 \\ 0 & b_{sfd} \end{bmatrix}; \qquad b_{sfd} = \text{variable}$$

The parameters of the model are:

- **Shafts:** General diameter 11.52 [cm], total length 235.5 [cm].
  Parts $S_1..S_5$: (43.5 + 7.7); 23.0; 76.8; 48.6; (7.7 + 28.2) [cm]
  $E_{mod} = 2.0\ 10^{11}\ [N/m^2]$, $G_{mod} = 7.7\ 10^{10}\ [N/m^2]$, shearfactor=0.9 [-] , $\rho = 6680$‘$[kg/m^3]$ (chosen such that total mass (including discs) is 417 [kg]).
- **Discs:** For each disc the values $(M[kg],\ J_p[kgm^2],\ J_t[kgm^2])$:
  D_1, D_6 $[36.28, 0.2107, 0.1054]$; $D_2...D_5$: $[45.36, 1.0535, 0.5268]$
- **Bearings:**
  B_1 : $k_{xx} = k_{zz} = 8.93\ 10^6\ [N/m]$
  $b_{xx} = b_{zz} = b_{sfd} = variable$ $(0...\ 2.0\ 10^5\ [Ns/m])$
  B_2 : $k_{xx} = k_{zz} = 3.57\ 10^6\ [N/m]$; No damping
- **Fluid-structure Cross-coupling terms:**
  at $D_2..D_5$: $k_{xz} = -k_{zx} = 5.36\ 10^4$ **OR** $1.56\ 10^6$ $[N/m]$

Assignment: Analyse the influence of the additional squeeze film damping on the dynamic behavior of the system for the weak- and strong Cross Coupling case.

8.3 Solutions

Problem 1 Gyroscopic effect in a rotating shaft

From: [*Formulas for Natural Frequency and Mode Shape*, Robert D. Blevins, Krieger Publishing, 1995, ISBN 0-89464-894-2], page 107 we find:

$$f_i = \frac{\lambda_i^2}{8\pi L^2}\sqrt{\frac{E\ (D_u^2 + D_i^2)}{\rho}} \qquad [Hz]$$

with for a one-sided, clamped shaft:

- $\lambda_1 = 1.87510407$
- $\lambda_2 = 4.69409113$
- $\lambda_3 = 7.85475744$

'This leads to a value for the first 3 eigenfrequencies:

- $f_1 = 405.79 \ \ [Hz]$
- $f_2 = 2543.0 \ \ [Hz]$
- $f_3 = 7120.5 \ \ [Hz]$

Next we apply the $\boldsymbol{R_O D_Y}$ program. We divide the shaft into 4 equal CONE8 elements. The rigid bearing will be simulated by 2 very stiff translational springs k_t (in the 2 directions x and z) and additionally 2 very stiff rotational springs k_r acting on the rotational degrees of freedom of the left node. We

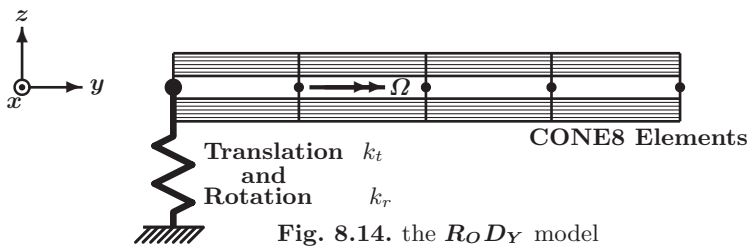


Fig. 8.14. the $\boldsymbol{R_O D_Y}$ model

used the values: $k_t = 1.0\ 10^{14} \ \ [N/m]$ and $k_r = 1.0\ 10^{14} \ \ [Nm/rad]$. Running $\boldsymbol{R_O D_Y}$ gave the following results

Blevins	$\boldsymbol{R_O D_Y}$ $\Omega = 0$	$\boldsymbol{R_O D_Y}$ $\Omega = 500\ [Hz]$	$\boldsymbol{R_O D_Y}$ Critical Speeds
405.79	402.93	401.16	401.51
	402.93	404.71	404.36
2543.0	2432.6	2421.3	
	2432.6	2444.0	
7120.5	6542.3	6518.8	
	6542.3	6565.8	

In the table also the 2 critical speeds in the given Ω-range as produced by $\boldsymbol{R_O D_Y}$ are given.
Looking at these results we can conclude that the numerical ($\boldsymbol{R_O D_Y}$) and analytical (Blevins) results for $\Omega = 0$ match very well. In $\boldsymbol{R_O D_Y}$ also the eigenfrequency-decreasing effects shear and rotary inertia are taken into account which are relatively small effect for this slender bean (L/D=10). The relative errors in the eigenfrequencies are respectively: 0.7; 4.3 and 8.1 %.
We also see that at $\Omega = 0$ the first eigenfrequency of $402.93\ [Hz]$ (which is equal for x- and z-direction) changes for $\Omega = 500\ [Hz]$ in a Backward-Whirl mode of $401.16\ [Hz]$ and a Forward-Whirl mode of $404.71\ [Hz]$. This is due to the gyroscopic effect, which obviously has not much practical relevance for this system.
In Fig. 8.15 the first mode at the rotor-speed of $\Omega = 500\ \ [Hz]$ is plotted. It is a clear $Trompet\text{-}type$ mode which results when a simple lowest static beam mode is rotated around the rotor-axis.

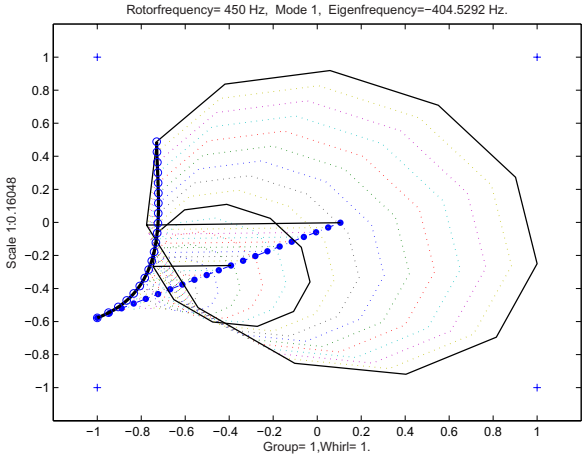


Fig. 8.15. First mode for clamped shaft model

Problem 2 Hummingtop

To analyse the dynamics of the Hummingtop we use the simple model of a rotor consisting of a stiff, massless pin and a rigid disc as shown in figure 8.16.
In order to be able to use $\boldsymbol{R_O D_Y}$ for the analysis of the Hummingtop we have to apply a trick to model the destabilizing effect of the Gravity forces on the upright position of the rotor. This is done by introducing two linear springs with **negative** spring stiffness, as is shown in the figure.

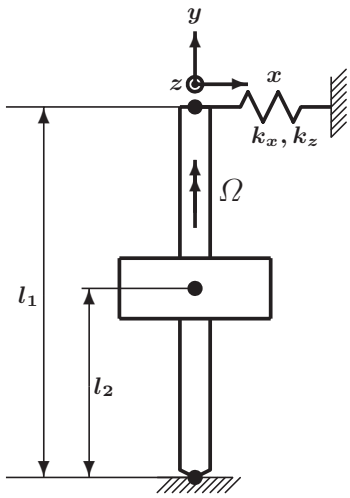


Fig. 8.16. Hummingtop model

We will apply the data:

- Aluminium disc: Diameter $D = 0.08[m]$, height $h = 0.03[m]$, Density $\rho = 2800[kg/m^3]$, Mass $m = 0.42[kg]$, Moments of Inertia $I_p = 3.4\ 10^{-4}$, $I_t = 1.7\ 10^{-4}[kgm^2]$.
- Stiff, (almost) massless shaft: diameter $d = 0.006[m]$, length $l_1 = 0.10$, $l_2 = 0.05[m]$, Elasticity Modulus $E = 2.110^{11}[N/m^2]$, Density $\rho = 1[kg/m^3]$
- Gravity constant $g = 10[m/s^2]$

On the Hummingtop a destabilizing moment due to the gravity force is working which we will replace by a destabilizing moment due to the spring-forces. For a small rotation φ from the central position we may write:

$$-mgl_2\varphi = (kl_1\varphi)l_1 \Longrightarrow k = -\frac{mgl_2}{l_1^2}$$

The $\boldsymbol{R_O D_Y}$ model consists of two shaft elements, one rigid disc, the two gravity replacing springs and two very stiff springs to model the fixed ground contact point. It has 12 degrees of freedom for the bending analysis.

For a spin-speed of $0 \leq \Omega \leq 30 \quad [Hz]$, the (complex) eigenvalues of the model have been calculated. The real parts of the smallest eigenvalues are plotted as function of the spin-speed in Fig. 8.17 We can see that up to a

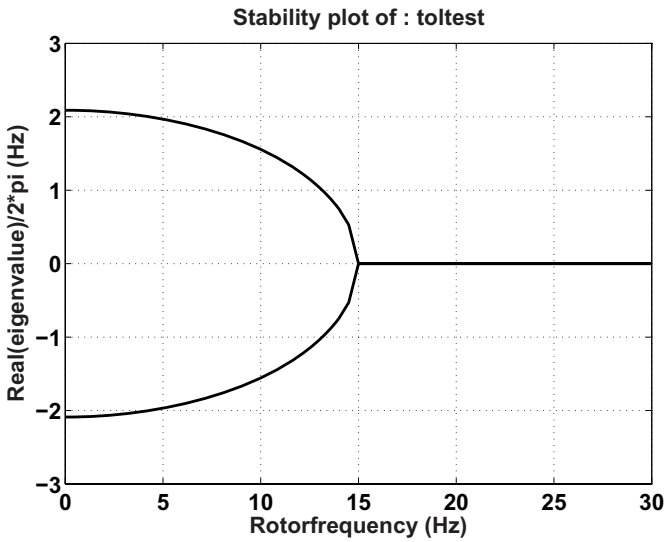


Fig. 8.17. Real parts of the lowest eigenvalues

spin-speed of $\approx 15 \quad [Hz]$, one of the eigenvalue has a positive real part, meaning that the system will be **unstable**. So, the minimum spin-speed for the Hummingtop to keep its upright position is $N = 900 \ [rev/min]$.
The imaginary parts of these eigenvalues are plotted as function of the spin-speed in figure 8.18. So, for higher spin-speeds we have a stable system with two relevant eigenvalues with a zero real part (undamped modes). For a spin-speed $\Omega \geq 15 \ [Hz]$, we see two modes. One small, decreasing mode, the **Precession** mode, and a higher, increasing mode: the **Nutation** mode. So, for example for a spin-speed of $\Omega = 20 \ [Hz]$, we get a precession mode with a frequency of $f_P = 1 \ [Hz]$ and a Nutation mode with a frequency of $f_N = 4.5 \ [Hz]$.

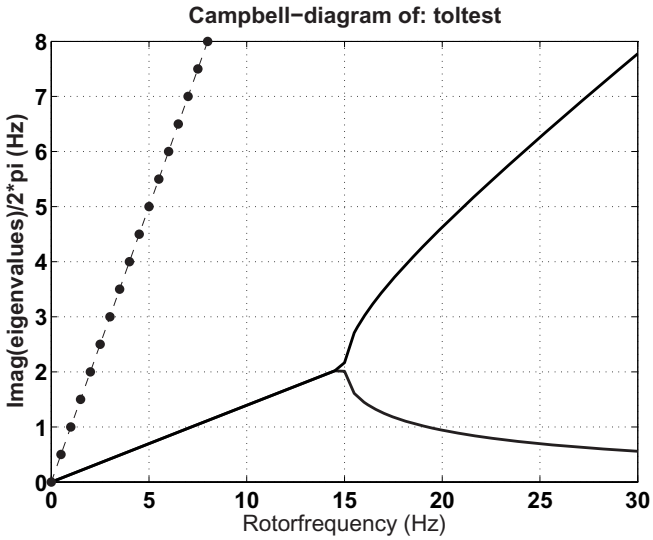


Fig. 8.18. Imaginary parts of eigenvalues

Finally, we can compare the $\boldsymbol{R_O D_Y}$ result for the precession frequency with the approximation given before. This is done in Fig. 8.19.

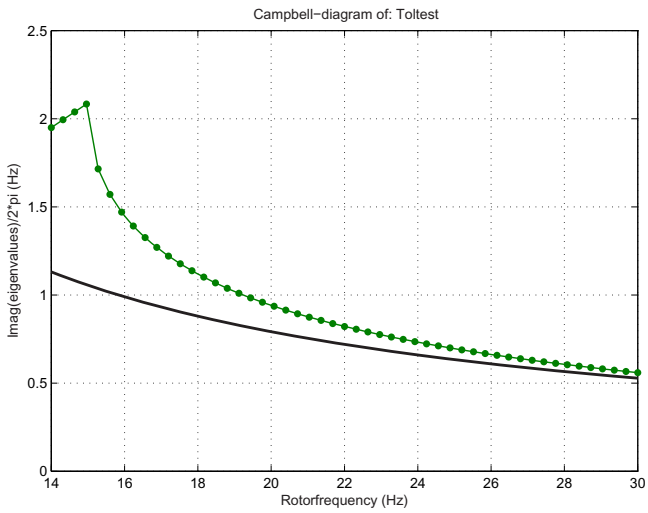


Fig. 8.19. Precession speed: approximation (solid line) and $\boldsymbol{R_O D_Y}$ data (dotted)

We can see that close to the instability point at $\approx 15\ [Hz]$, there is some

difference between the two models but that for higher spin-speeds the correlation is very nice.

Final remark

For this simple system we also can derive easily a direct solution. If we write the equations of motion for this rigid body with respect to the fixed ground-contact point O, we get:

$$\begin{bmatrix} J_O & 0 \\ 0 & J_O \end{bmatrix} \begin{bmatrix} \ddot{\theta}_x \\ \ddot{\theta}_z \end{bmatrix} + \Omega \begin{bmatrix} 0 & -J_p \\ J_p & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_x \\ \dot{\theta}_z \end{bmatrix} + \begin{bmatrix} -k & 0 \\ 0 & -k \end{bmatrix} \begin{bmatrix} \theta_x \\ \theta_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

with: $J_O = J_t + m\ l_2^2$ and $k = m\ g\ l_2$.
We can then calculate the eigenvalues (note: all in [Hz]) for these homogeneous equations as shown in figure 8.20. We can conclude that the results

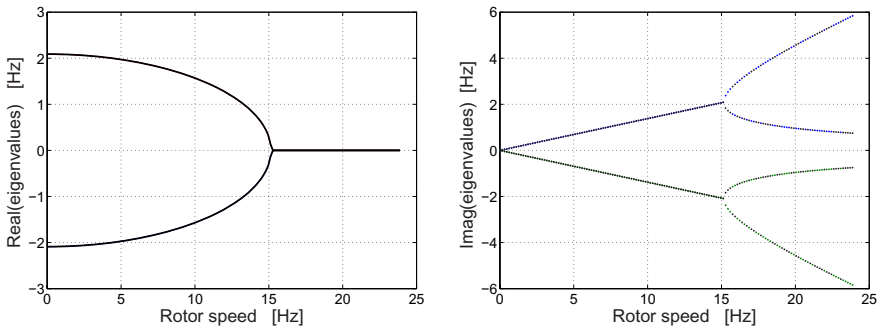


Fig. 8.20. Results from the direct approach

match perfectly.

Problem 3 Rotor on Elastic Foundation

**** Rigidly supported Bearings ****

First we look at the case where the bearing housings are supposed to be the fixed world. The $\boldsymbol{R_O D_Y}$ model contains 3 shaft-elements, 2 discs and 4 springs. The essential rotordata is given next.

```
SHAFT8
Nr.  L (m)      Do      Di   E         rho       connect  group
  1  2.50e-1  1.20e-2  0  2.10e+011  7.80e+3   q 1  q 6  1
  2  2.50e-1  1.20e-2  0  2.10e+011  7.80e+3   q 6  q11  1
  3  2.50e-1  1.20e-2  0  2.10e+011  7.80e+3   q11  q16  1

DISK4
Nr.  m(kg)    Ip(kgm2) It(kgm2)  connect group
  1  7.80e+0  5.60e-2  2.80e-2    q 6      1
  2  7.80e+0  5.60e-2  2.80e-2    q11      1

SPRING1
Nr.  k(N/m, Nm/rad)        connect
  1  1.e+6                 q 1
  2  1.e+6                 q 2
  3  1.e+6                 q16
  4  1.e+6                 q17
NUMBER OF DEGREES OF FREEDOM: 20
```

First we look at the free vibrational behavior in the rotor-speed range $0 \leq \Omega \leq 50 \ \ [Hz]$. The 4 most relevant, lowest eigenvalues are plotted in the Campbell plot in figure 8.21. The critical speeds appear to be : 6.9345, 7.2117, 28.3118 and 24.7017 [Hz]. We can see that we have 2 pairs of modes (one forward-whirl and one backward whirl) of which the smallest ones are only moderately changing with the rotor-speed Ω.
One of the smaller modes (a backward whirl) and one of the higher modes (a forward whirl), -both for a rotor-speed of $\Omega = 30 \ [Hz]$- are shown in figure 8.22. We see that the first one is a symmetrical (beam) bending mode where the discs mainly translate (no much tilting). The gyroscopic effect consequently is small. The second one is an asymmetrical (beam) bending mode with more out-of-plane displacements of the discs (more Ω-dependent). Both modes show a negligible displacement at the bearing locations, meaning that the bearing stiffness is relatively high.

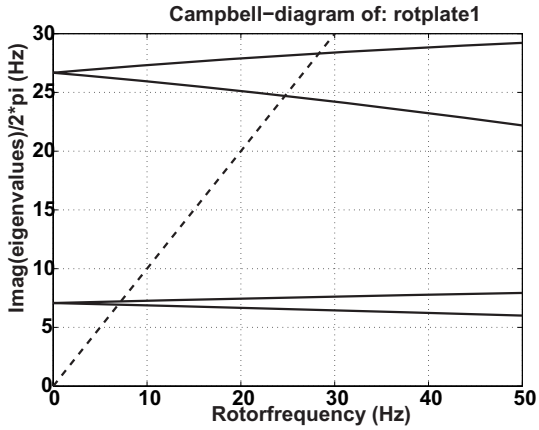


Fig. 8.21. Campbell plot for rigid bearing support

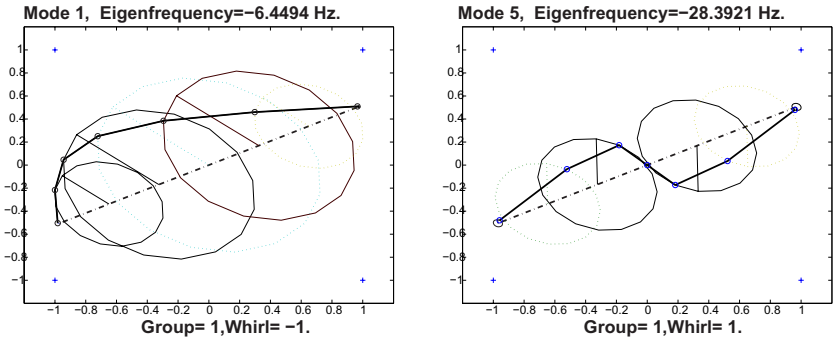


Fig. 8.22. Two typical mode types at the Rotor-speed of $\Omega = 30$ [Hz]

Next we look at the imbalance response. In $\boldsymbol{R_O D_Y}$ an imbalance can be introduced by adding an imbalance mass to one or more of the disc in a system or by assuming that one or more of the discs are not mounted perpendicular to the shaft-axis (disc-skewness).
Here we add some imbalance-masses, namely

- Disc1 m=0.1 [kg] at r=0.125 [m] at φ=0 degrees
- Disc2 m=0.1 [kg] at r=0.125 [m] at φ=90 degrees

So, both discs have the same imbalance, but they are 90 degrees out-of-phase.
Again we look at the rotor-speed range $0 \leq \Omega \leq 50$ $[Hz]$. As output degrees-of-freedom, we look at the translations in x-direction of all the 4 nodes, so the dofs: 1, 6, 11 and 16. The response has been calculated and

the absolute value is plotted in figure 8.23. In the plot, we see only 2 lines

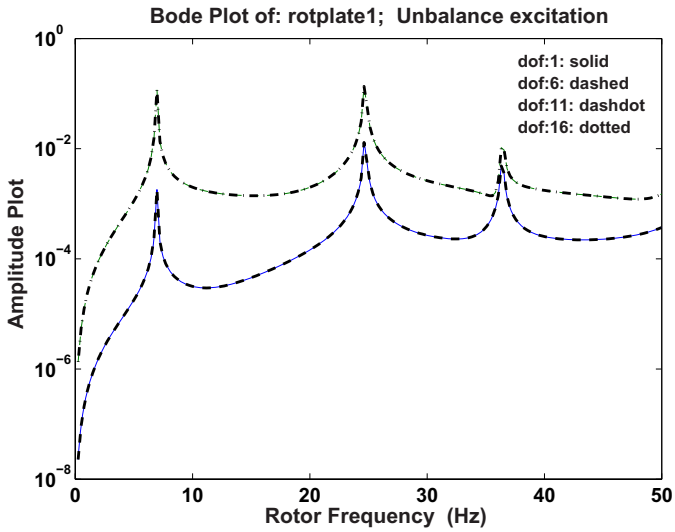


Fig. 8.23. Imbalance response, 90 degrees out-of-phase eccentricity

where we expect to see 4 lines as is also indicated by the legend. The point is that we in fact have a symmetrical system and loading (except for some imbalance phase shift). The result is that the dofs 6 and 11 (the discs, upper line) have the same amplitude (not the same phase) and the same holds for the dofs 1 and 16 (the bearings, bottom line).
We can see that at the bearing locations the displacement are roughly less than 1 % of the mid-beam displacement which is related to the ratio: bearing stiffness versus beam-bending stiffness.

Next we look at the situation where the imbalances are 180 degrees out-of-phase. The result is shown in figure 8.24. We can see that again the bearing dofs (1,16) respectively the disc dofs (6,11) have the same amplitude (but different phases). Striking is that now the first-mode resonance at ≈ 6 [Hz] disappeared completely (also the 3rd one). The reason is that the chosen asymmetrical imbalance will not excite these symmetrical vibration modes (see also [Kraker/Campen-01]).

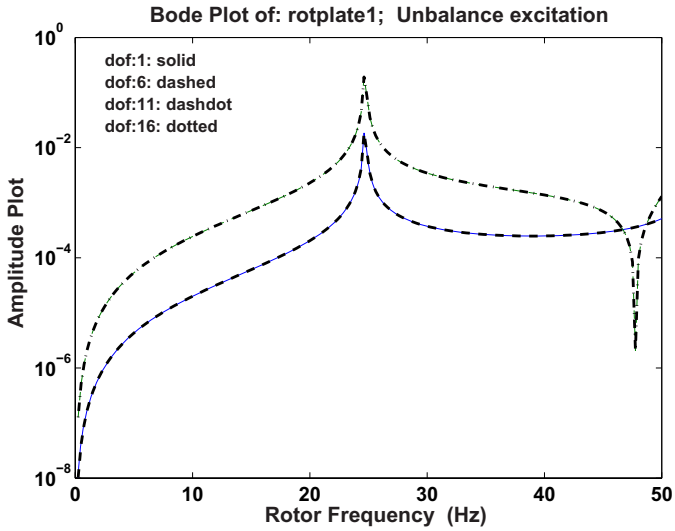


Fig. 8.24. Imbalance response, 180 degrees out-of-phase eccentricity

**** NON-rigidly supported Bearings ****

Next, we assume that the bearing housings also can have some displacement, but this displacement will be controlled by the movement of the foundation structure as a rigid body, and this rigid body is supported by a set of vibration isolators with stiffness and damping.

To be able to model this situation in $\boldsymbol{R_O D_Y}$, we assume that we can

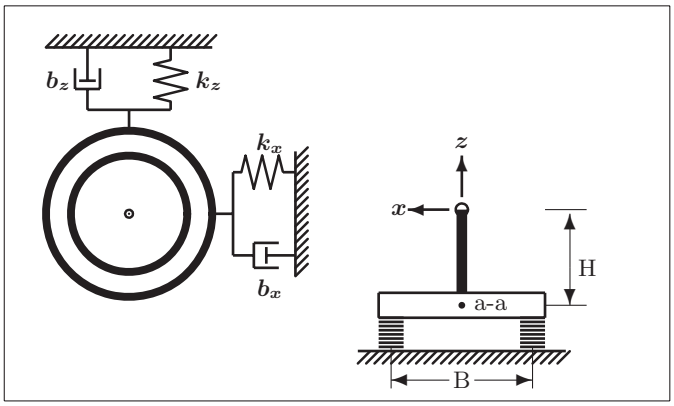


Fig. 8.25. The bearing support model

transform this reality to some (heavy) point mass (at the points of the bearing nodes) which will be supported by a representation of the isolation springs k_d and dampers b_d. This is schematically shown in figure 8.25.
We assume that the displacement of the supporting frame can be seen as a combination of just a vertical displacement and a rotation around the a-a axis which is located in the middle between the vibration isolators. For the properties of the "point mass" for the bearing housing and its support we might have to use a different value for the x- respectively z-direction, namely:

z-direction

- mass $m_z = \frac{1}{2}$ of the total ground plate mass, $m_z = 35$ [kg]
- $k_z = 2\ k_d = 4\ 10^4$ [N/m]; $b_z = 2\ b_d$ (variable)

x-direction

- mass m_x from: $2\ m_x\ H^2 = J_{a-a}$, where J_{a-a} is the mass moment of inertia of the whole frame with respect to the a-a axis. $\Longrightarrow m_x = 28$ [kg].
- $k_x = k_d\ B^2/H^2 = 16.4\ 10^4\ [N/m]$; $b_x = b_d\ B^2/H^2$ (variable).

We see that for the effective masses in both directions we get 35 respectively 28 [kg]. For simplicity we use only one value: $m_x = m_z = 35$ [kg]. We will enter these masses in $\boldsymbol{R_O D_Y}$ by introducing a non=rotating, very stiff shaft with mass M=70 [kg] and a length, equal to the real rotorshaft, so L=0.75 [m]. With a $\rho = 7800$, this gives a diameter of D=0.12 [m].
So we will use a $\boldsymbol{R_O D_Y}$ model with:

- 6 nodes, 30 degrees of freedom (24 for the bending only)
- a non-rotating dummy shaft between the dofs: 1-26
- the 3 shaft elements between the dofs: 6-11, 11-16 and 16-21
- the 2 disc elements
- 4 equal bearing-springs between the dof-combinations: 1-6, 2-7, 21-26 and 22-27
- 2 springs k_x from dofs 1 and 26 to the ground
- 2 springs k_z from dofs 2 and 27 to the ground

The most relevant, lowest eigenvalues are plotted again in the Campbell plot in figure 8.26. For the critical speeds we now find the values: 5.9730, 6.7625, 18.9551, 25.2274 and 30.98 [Hz]. (Rigid housing: 6.9345, 7.2117, 24.7017 and 28.3118 [Hz], but check that the

modes are quite different now).
So in general, the critical speeds are somewhat lower and are not equal anymore for $\Omega = 0$ (different stiffness in x- and z-direction). One of the lowest

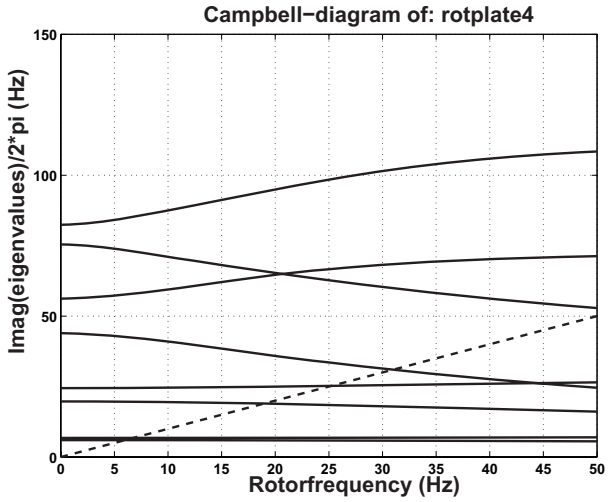


Fig. 8.26. Campbell plot for elastic bearing support

modes is plotted in figure 8.27. On the left (Group 1) we see the rotor, on the right (Group 2) the dummy shaft is shown. The rotor shows also some displacement at its ends (no rigid bearings) combined with a considerable translation of the dummy-shaft (the support structure). One of the higher

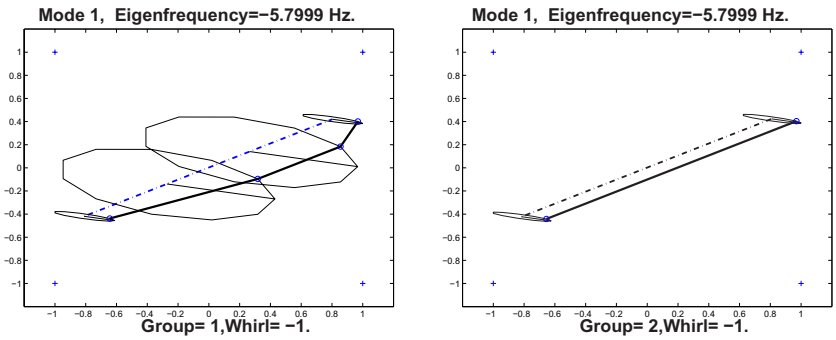


Fig. 8.27. Example of a symmetric mode, Rotor-speed $\Omega = 30$ [Hz]

(asymmetrical) modes is plotted in figure 8.28. Also in this case the vibration amplitude of the support structure is considerable.

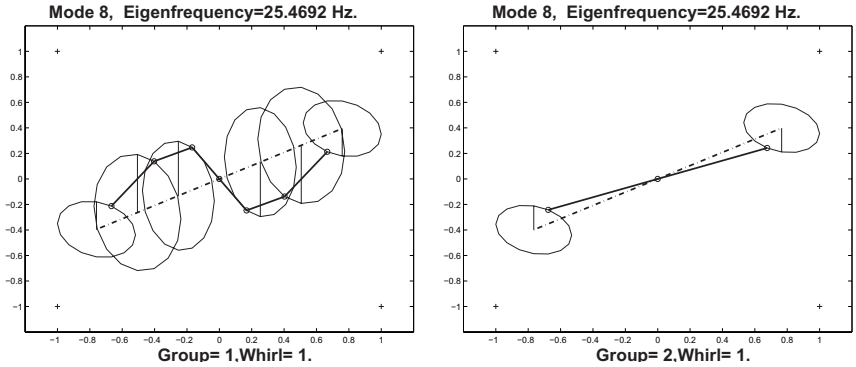


Fig. 8.28. Example of an asymmetrical mode, Rotor-speed $\Omega = 30$ [Hz]

Damping included

Next, we go to the damped situation. We choose a value for the support damping $b_d = 2.0\ 10^2\ [Ns/m]$, leading to the $\boldsymbol{R_O D_Y}$ element values: $b_x = 16.8\ 10^2\ [Ns/m]$ and $b_z = 4.0\ 10^2\ [Ns/m]$.
With damping added to the system, the eigenvalues will no longer be purely imaginary, but will also have e real part. The **real** parts for the first 16 eigenvalues are shown in figure 8.29. All the real parts are negative, so we

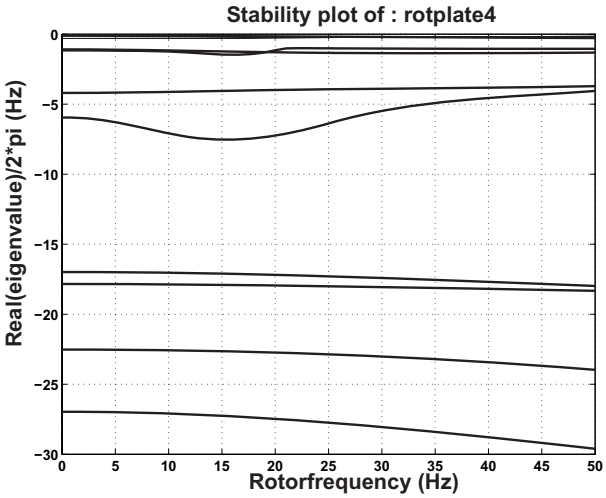


Fig. 8.29. Real parts of eigenvalues for system with elastic bearing support

have a stable system. The effect of the rotor-speed Ω on these real parts in general seems to be very small.

Mode Nr.	$\Omega = 0$ [Hz] f_d [Hz]	ξ %	$\Omega = 50$ [Hz] f_d [Hz]	ξ %
1	-6.05	5.11	-5.59	2.76
2	6.05	5.11	5.59	2.76
3	-6.79	1.62	-7.04	3.96
4	6.79	1.62	7.04	3.96
5	0.00	100	0.00	100
6	0.00	100	0.00	100
7	0.00	100	0.00	100
8	-23.1 .	17.8	-18.3	19.8
9	23.1 .	17.8	18.3	19.8
10	-25.9 .	4.17	-28.0	4.69
11	25.9 .	4.17	28.0	4.69
12	0.00	100	0.00	100
13	-61.2	1.98	-127	0.98
14	61.2	1.98	127	0.98
15	-61.2	8.99	-28.4	13.2
16	61.2	8.99	28.4	13.2

In the table the damped eigen frequencies f_d and the corresponding dimensionless modal damping factors are given for the first 16 modes for the cases $\Omega = 0\ [Hz]$ and $\Omega = 50\ [Hz]$. This shows that we now also have some super-critically damped modes (imaginary part is zero).

The critical speeds (following from the Campbell diagram with the **imaginary** parts as function of the rotor-speed) appear to be: 6.0508, 6.8050, 21.9248, 26.8795 and 35.0392 [Hz].

> *With een exception for the case of weakly damped systems, one should realize that modes for a system with and without significant damping normally cannot be compared with one another directly. The order can have been changed, modes can have been disappeared and/or new modes can have been showed up.*

One of the super-critically damped modes is shown in figure 8.30. It is clearly a mode where the whole system (rotor and support) is moving roughly as a rigid body on the vibration absorbers, while the discs try to keep a fixed position..

Finally we look at the imbalance respons due to the 90 degrees out-of-phase eccentricity excitation. We look at the displacements in x-direction of the 4 rotor-nodes. The result is shown in figure 8.31. The first modes are weakly damped so the effect of adding damping is not so large for a rotor-speed up

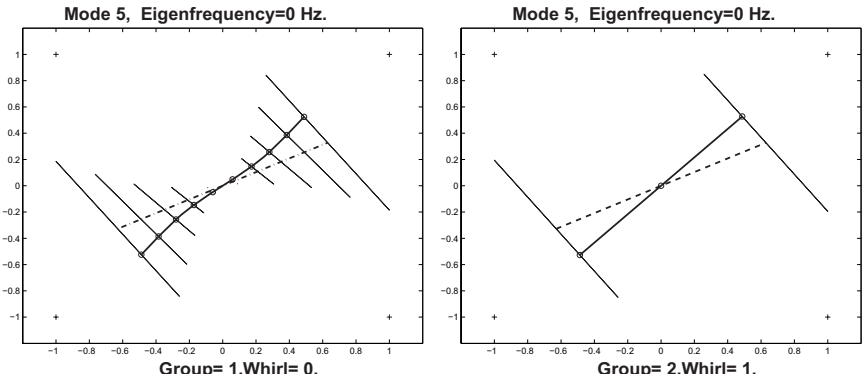


Fig. 8.30. Example of a super-critically damped mode, on the left the rotor, on the right the support frame, Rotor-speed $\Omega = 50$ [Hz]

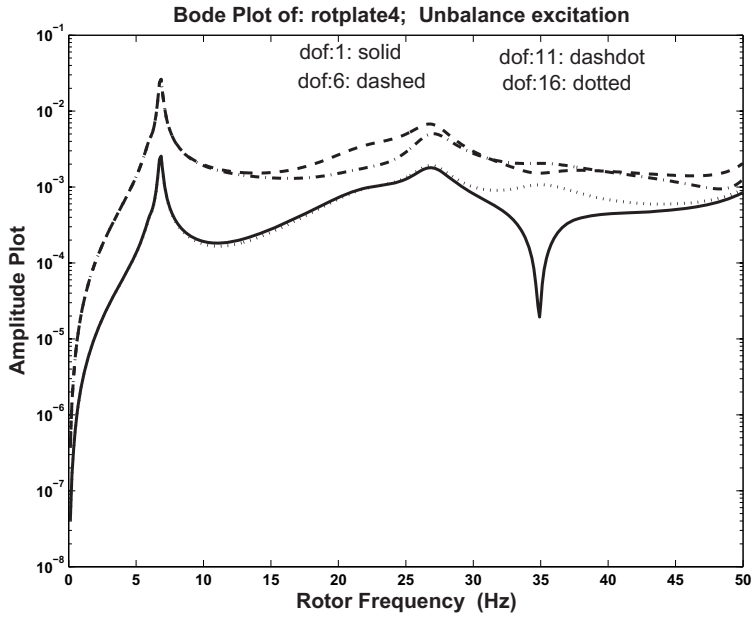


Fig. 8.31. Imbalance response, 90 degrees out-of-phase eccentricity

to 12 [Hz]. Both discs have a large amplitude. For higher rotor-speeds the damping has a large effect. All resonance peaks have been smoothed down.

Problem 4 Rotor operating in a magnetic field

To become familiar with the ECK2M element to model a frequency dependent bearing stiffness, we first analyse a simplified model. It is the same as the given system, but we assume the two bearings with a constant stiffness not to be there. So only the middle bearing is present. In fact we almost have a point mass on one frequency-dependent spring. We analyse the system for a rotor-speed range: $0 \leq f_{rotor} \leq 20$ [Hz]. The relevant stiffness constants are: $k_{xx} = k_{zz} = (2\ \pi\ f_{rotor})\ 1000$ [N/m].
We them should see 3 types of solution:

a A rigid body mode with zero eigenfrequency (the tilting mode).
b A high frequency shaft-bending mode. We see 2 modes of 1540 [Hz]. Each of them is transforming into a forward- and a backward whirl mode due to the gyroscopic effect at higher rotor-speeds.
c A mode starting at zero frequency but increasing with the rotor-speed (symmetrical translation in 2 directions). See figure 8.32.

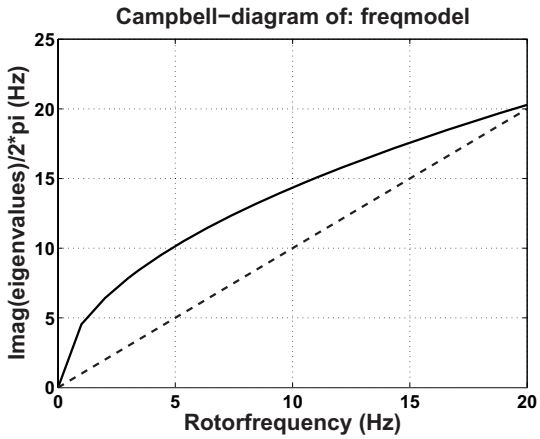


Fig. 8.32. Campbell plot for only a single frequency dependent bearing

We can estimate the eigenfrequency of the translation mode at a rotor-speed of 20 [Hz] as:

$$\omega_{sym} = \sqrt{\frac{k^*\ \Omega}{m_{shaft}\ +\ m_{disc}}}$$

with $k^* = 1000$ [Ns/m], $\Omega = 2\ \pi\ 20$ [rad/s], $m_{shaft} = 1.2252$ [kg] and $m_{disc} = 6.5$ [kg], this gives $\omega_{sym} = 127.5411$ [rad/s].
From $\boldsymbol{R_O D_Y}$ we find the value: $\omega_{sym} = 127.4942$ [rad/s] This is somewhat lower but that is due to the fact that the shaft is not really a rigid body in $\boldsymbol{R_O D_Y}$.

Let us go to the complete system with the linear springs at both ends of the rotor to model the ball bearings and in the middle the rotor-speed depending (uncoupled) set of springs to model the magnetic effects.
In figure 8.33 the Campbell diagram for the 8 lowest eigenvalues is shown.
For zero rotor-speed we have:

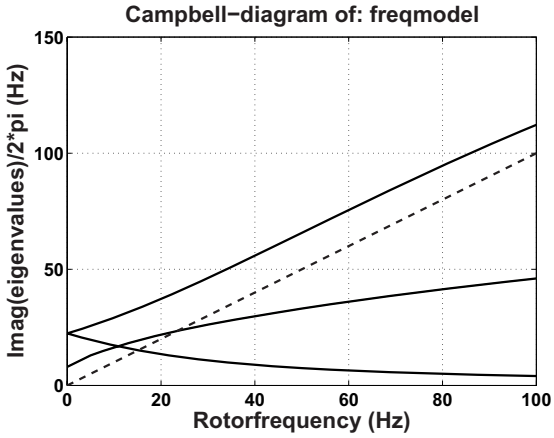


Fig. 8.33. Campbell plot for the system with 2 roller- and a frequency-dependent bearing

* symmetric (translation) modes (almost rigid body, equal in x- and z-direction) with an eigenfrequency of ≈ 8 [Hz]. For increasing rotor-speed the modes remain symmetric (so, no gyroscopic effect) with an increasing frequency due to the middle springs. See left plot of figure 8.34.
The frequency can be estimated as:

$$f = \frac{1}{2\pi}\sqrt{\frac{k_{mid} + 2k_{end}}{m_{total}}}$$

which for Ω=100 [Hz] gives : f=46.1062 [Hz].

* anti-symmetric (tilting) modes (almost rigid body, equal in x- and z-direction) with an eigenfrequency of $\approx$ 22 [Hz]. For increasing rotor-speed the modes remain anti-symmetric (so, no effect of middle springs) but the gyroscopic effect gives a strong divergence into a forward- and a backward whirl mode. See right plot of figure 8.34.
* more or less beam-bending modes with an eigenfrequency of 300 [Hz] and higher.

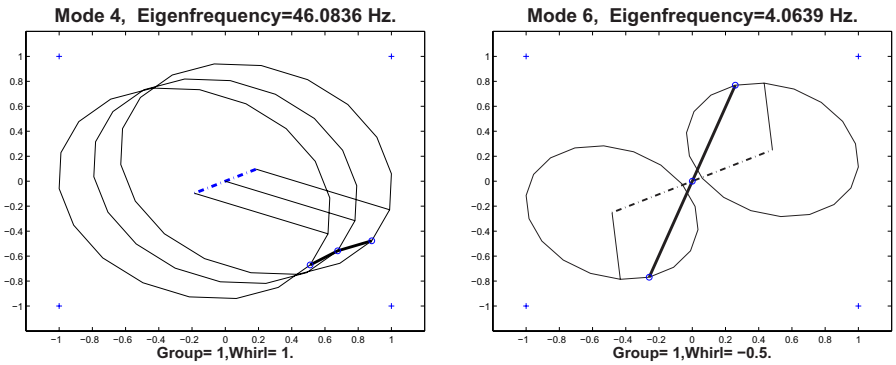


Fig. 8.34. Two typical modes at the rotor-speed of $\Omega = 100$ [Hz]

Finally it might be interesting to look at the mixed case, for example by putting the stiffening bearing not in the middle but at one of the end-nodes of the rotor.

Problem 5 Plate Balancing

For the analysis of this problem with $\boldsymbol{R_O D_Y}$, we assume that the plate may be seen as a thin uniform circular disk with $m = 0.45\ [kg]$, $I_p = 3.2\ 10^{-3}\ [kgm^2]$ and $I_t = 1.6\ 10^{-3}\ [kgm^2]$. It has a center-of-gravity offset of $a = 0.01$ [m]. The offset is taken into account in the model by adding a short (stiff) beam (length 0.01 [m]). On the plate, a destabilizing moment due to the gravity force is acting which we will replace by a (destabilizing) external moment on the disc due to a pair of torsional springs k_φ:

$$-\ m\ g\ a\ \varphi = k_\varphi\ \varphi \quad \Longrightarrow \quad k_\varphi\ =\ -\ m\ g\ a \qquad [Nm/rad]$$

The flexible spear will be replaced by 2 linear springs k_s, directly acting on the displacements of the relevant node of the offset-beam. Their stiffness constant is taken as:

$$k_s = \frac{3\ E\ I}{l^3} \qquad [N/m]$$

(**Remark:** *Only for getting a better picture in mode-animation plots, a second beam with negligible mass is added to the system. It has no effect on the dynamics*).

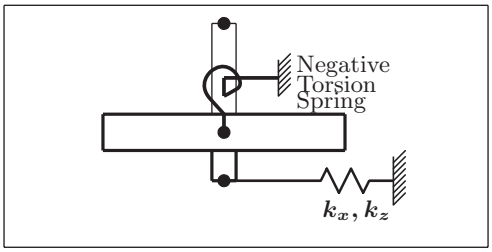


Fig. 8.35. Simplified $\boldsymbol{R_O D_Y}$ model

The $\boldsymbol{R_O D_Y}$ model the look like:

```
SHAFT8
Nr.  L (m) Do(m) Di(m) E(N/m2)  rho(kg/m3) connect  group
  1  1e-2  4e-2  0     2.1e+11  1          q 1  q 6  1
  2  2e-1  4e-2  0     2.1e+11  1          q 6  q11  1
DISK4
Nr.  m (kg)   Ip (kgm2) It (kgm2) connect group
```

```
  1  4.50e-1  3.20e-3   1.60e-3   q 6       1
SPRING1
Nr.  k (N/m, Nm/rad) connect
  1  5.e-1           q 1
  2  5.e-1           q 2
  3  -4.50e-2        q 8
  4  -4.50e-2        q 9
 NUMBER OF DEGREES OF FREEDOM: 15
```

The eigenvalues of this system have been calculated for the speed range $\Omega \leq 3$ [Hz]. The real values (relevant for the stability) are shown in figure 8.36. What we see is that the system for low rotational speeds is unstable

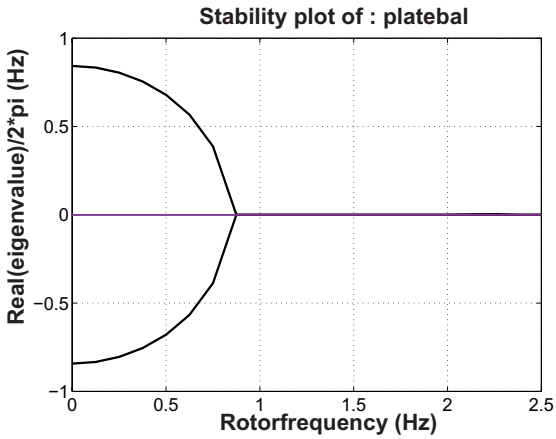


Fig. 8.36. Real parts of eigenvalues

(trivial) but that already for a speed of ≈ 1.0 Hz the system becomes stable. The eigenfrequencies (the imaginary parts of the eigenvalues) are shown in figure 8.37. We can first see one, almost constant, stable eigenfrequency. Next to that one we have a pair of unstable modes (almost synchronous motion with the rotor-speed), starting from zero-eigenfrequency at $\Omega = 0$ which bifurcates at $\Omega = 1$ in two stable modes. One of them is getting large for higher rotor-speeds, the other converges to the constant-frequency mode.

The first (constant-eigenfrequency) mode is simply the translation of the plate on the soft support springs with an eigenfrequency of $\omega_i = \sqrt{k_{spear}/m_{plate}}$.

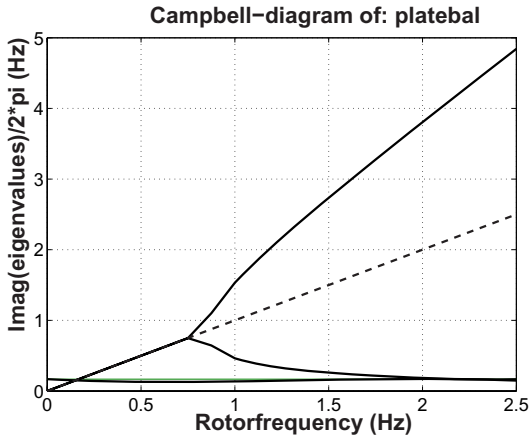


Fig. 8.37. Campbell plot for

It is shown in the left part of figure 8.38.
In the right part one of the stable modes for higher rotor-speed is shown. This is clearly a tilting mode (no or small translation of the plate, mainly a rotation).

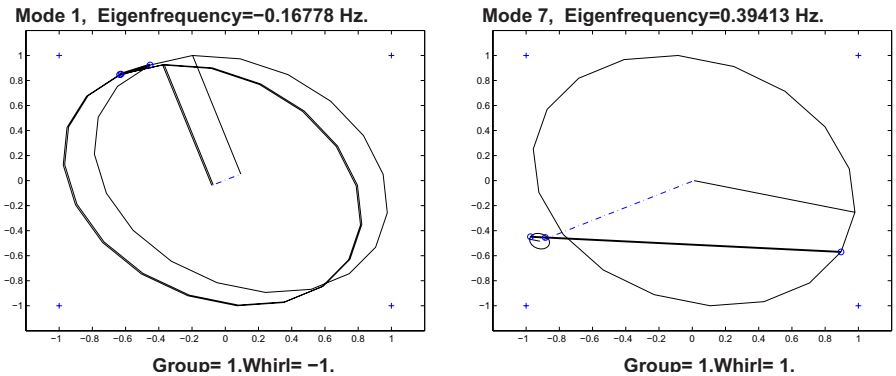


Fig. 8.38. Two typical modes at the Rotor-speed of Ω=1.1 [Hz]

Problem 6 Single disc rotor balancing

In order to identify the unknown disc imbalance we use the given $\boldsymbol{R_O D_Y}$ model to calculate the imbalance respons according to a given, arbitrary imbalance. For this "unity" imbalance we assume an imbalance mass M_{imb} at radius r_{imb} at position angle φ=0 degrees:

$$m_{imb} = 0.01\ [kg], \quad r_{imb} = 0.2\ [m], \qquad \Longrightarrow (mr)_{imb} = 2.0\ 10^{-3}$$

It is the value for $(mr)_{imb}$ which can be calculated and not the separate values for m and r.
The calculated orbits for the disc location for the three relevant rotor-speeds are shown in figure 8.39. We will write the orbit at a specific rotor-speed as

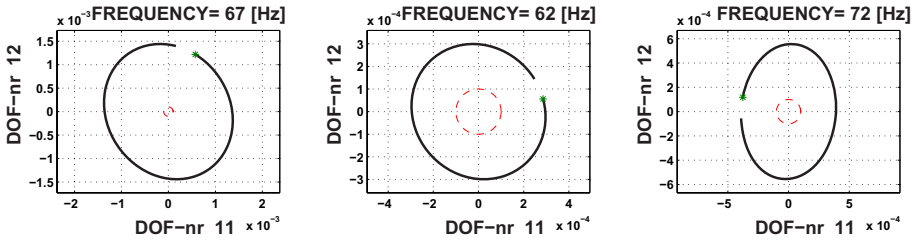


Fig. 8.39. Orbits at rotor-speeds of Ω =62, 67 and 72 Hz

a complex number so, $r_{67} = [x + jz]_{\Omega=67}$
from the given orbits we can find for the positions at $t = 0$:

"unity" imbalance	"measured"
$r^u_{67} = (0.5 + 1.25j)\ 10^{-3}$	$r^m_{67} = (4.9 + 1.5j)\ 10^{-3}$
$r^u_{62} = (2.8 + 0.65j)\ 10^{-4}$	$r^m_{62} = (0.95 - 0.65j)\ 10^{-3}$
$r^u_{72} = (-4.0 + 1.2j)\ 10^{-4}$	$r^m_{72} = (-0.8 + 1.7j)\ 10^{-3}$

First we only look at the central rotor-speed $\Omega = 67$ [Hz]. We also write the imbalance-vector at $t = 0$ as a complex number:

$$f^u_{imb} = (2.0 + 0j)\ 10^{-3}$$

We then can write for the system transfer H_{67}:

$$r^u_{67} = H_{67}\ f^u_{imb} \qquad \Longrightarrow H_{67} = r^u_{67}/f^u_{imb}$$

Now we can easily estimate the unknown imbalance from:

$$f_{imb}^m = [H_{67}]^{-1}\ r_{67}^m$$

giving:

$$f_{imb}^m = (4.8 - 5.9j)\ 10^{-3}$$

So, we find an estimated imbalance with $(mr)_{imb} = 7.6\ 10^{-3}$ [kgm] with an initial position angle $\varphi = +$ 51 degrees (in direction of the rotor-speed definition).
If we also want to take the data for the other rotor-speeds into account to get a more accurate result, we can write:

$$\begin{bmatrix} r_{62}^m \\ r_{67}^m \\ r_{72}^m \end{bmatrix} = \begin{bmatrix} H_{62} \\ H_{67} \\ H_{72} \end{bmatrix} f_{imb}^m \qquad \text{or} \qquad \underline{r}^m = \underline{H}\ f_{imb}^m$$

which gives:

$$f_{imb}^m = \underline{H}^{-1}\ \underline{r}^m$$

where we for the inverse of the non-square matrix $\underline{H}$ the pseudo-inverse should read:

$$\underline{H}^{-1} = [\underline{H}^T \underline{H}]^{-1}\ \underline{H}^T$$

which in fact gives a least-squares-solution.
The result is now:

$$f_{imb}^m = (4.6 - 5.9j)\ 10^{-3}$$

which equals an imbalance with $(mr)_{imb} = 7.5\ 10^{-3}$ [kgm] with an initial position angle $\varphi = +$ 52 degrees.
A reconstruction and comparison from both results with the "measured" orbits" shows that they are both very accurate. (In the second case we use more data but the additional data is not so accurate).

The disc-skewness Case

We assume that the disc has no mass-imbalance but that the disc was not mounted properly on the shaft. It has an initial disc-skewness defined by a skewness-vector with a magnitude of 2 degrees and phase angle of 90 degrees. The respons of the system has been calculated and some results are shown in figure 8.40. We will only use the orbit-data for the rotor-speed

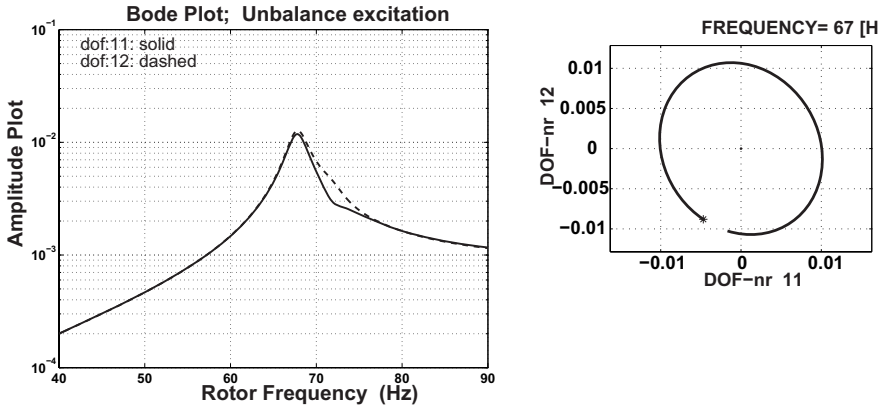


Fig. 8.40. "Measured" results for the disc-skewness case, Ω =67 Hz

of Ω=67 [Hz].

Again we will use:

$$f^u_{imb} = (2.0 + 0j)\ 10^{-3}$$
$$H_{67} = r^u_{67}/f^u_{imb}$$
$$f^m_{imb} = [H_{67}]^{-1}\ r^m_{67}$$

Using $r^u_{67} = (0.5 + 1.25j)\ 10^{-3}$ and $r^m_{67} = (-4.5 - 8j)\ 10^{-3}$ we get:

$$f^m_{imb} = (-13.5 + 1.8j)\ 10^{-3}$$

So, we find a representative imbalance with $(mr)_{imb} = 13.6\ 10^{-3}$ [kgm] with an initial position angle $\varphi = +$ 188 degrees.
To see whether this mass-imbalance can indeed compensate the disc-skewness we recalculate the imbalance respons. The result is shown in figure 8.41. If we compare this result with the respons for the disc-skewness case we can conclude that they match very well.
We should keep in mind that this correlation only holds for the frequency range we used for the estimation of the representative balance weight. From figure 8.42 we can see that for higher frequencies the two models are not comparable anymore.

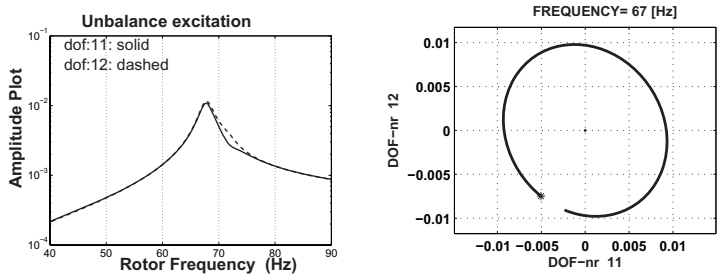


Fig. 8.41. "Reconstructed" results for the disc-skewness case, Ω =67 Hz

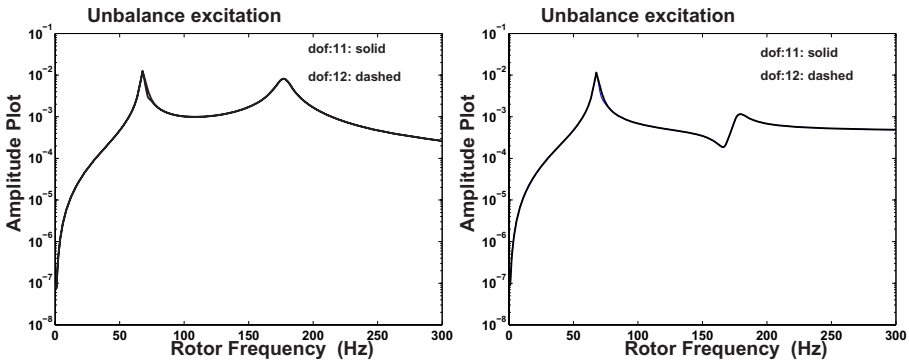


Fig. 8.42. Unbalance respons, disk-skewness (left), balance weight simulation (right)

Problem 7 Windmill Dynamics

To simulate the dynamic behavior of the windmill in $\boldsymbol{R_O D_Y}$ we replace the supporting tube by a combination of 4 spring-elements:

- a transverse linear spring in x-direction k_x
- a torsional spring for rotation around the z-axis k_{tz}
- a torsion spring for rotation around the x-axis k_{tx}
- a transverse linear spring in z-direction k_z

We choose the spring-constants as:

$$k_x = \frac{3EI_t}{l^3}, \quad k_{tz} = \frac{GI_p}{l}, \quad k_{tx} = \frac{EI_t}{l}, \quad k_z = \frac{EA}{l}$$

this gives: $k_x = 2.81 \ 10^4$ [N/m], $k_z = 1.94 \ 10^8$ [N/m], $k_{tz} = 1.15 \ 10^7$ [Nm/rad] and $k_{tx} = 1.50 \ 10^7$ [Nm/rad].
The springs are assumed to act all in the mid-point of the housing structure. The housing will be modeled as a non-rotating hollow shaft with dimensions

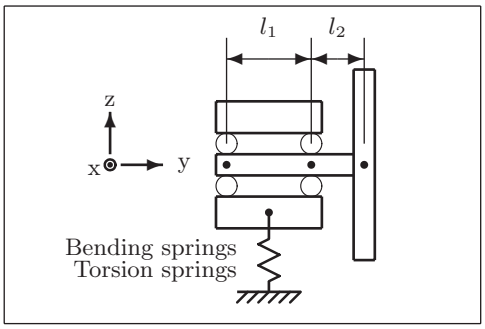


Fig. 8.43. Windmill $\boldsymbol{R_O D_Y}$ model

chosen such that the mass of the housing will be 141 [kg], so:
$L_{housing}$=0.6, D_u=0.5, D_i=0.46 [m]
The housing is modeled by 2 CONE8 elements and the rotor shaft also by 2 CONE8 elements.
First the eigenvalues for the system have been calculated for a rotor-speed range up to 2 [Hz]. The lowest eigenvalues are given in the Campbell diagram in figure 8.44. The critical speeds appear to be 0.2634 and 1.7623 [Hz].
We can see that for a small rotor-speed we have a pair of eigenvalues $\approx$ 0.5 [Hz], of which one is becoming a decreasing backward whirl and the other

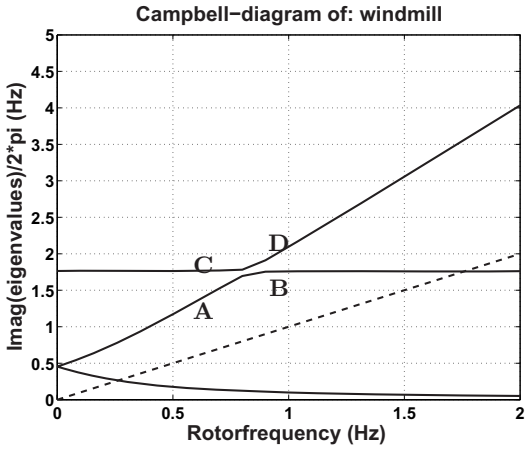


Fig. 8.44. Campbell diagram

an increasing forward whirl. Secondly we have an eigenvalue $\approx$ 1.8 [Hz] which at first is independent of the rotor-speed. For higher rotor-speeds the constant- and the increasing eigenvalue approach one another, but instead of crossing trajectories we see that they just come close and then diverge again. It looks like they changed their character: the constant one has become an increasing one and the increasing one a constant one.
So, one might put the question whether the mode corresponding to eigenvalue B is really the correct continuation of the mode corresponding to eigenvalue A and not eigenvalue C.
The modes (only the rotor-part of the system) corresponding to the eigenvalues indicated by A,B,C and D are shown in figure 8.45. In general the global picture of all the 4 modes is very much the same. It is mainly a translation of the whole system in x-direction (transverse bending of the supporting tube). Only the modes A and D also shown an elliptic orbit for the propeller center. Also from these mode-plots it is not easy to conclude which mode belongs to which one. In order to get more information for the mode-identification, the displacement (u_x and u_z) for the 6 nodes, for all 4 modes have been collected in the next table. (upper dofs rotor, bottom dofs housing).
Again it is disappointing that the only conclusions which can be drawn are that all 4 modes are mainly a rigid body motion (rotor plus housing) in x-direction (almost real modes!) and that only modes A and D show some "real complex" behavior (see propeller-dofs 9 and 10).

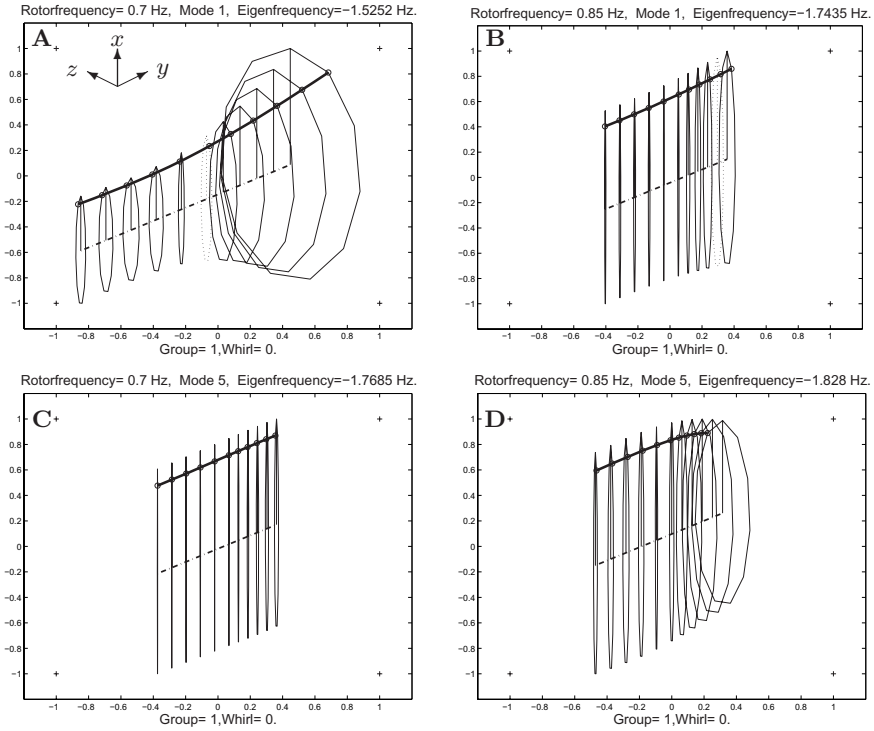


Fig. 8.45. Mode plot of rotor-part for points marked A,B,C and D

dof	A	B	C	D
1	0.2935 i	0.9099 i	0.9887 i	1.0000 i
2	0.0212	0.0042	-0.0011	-0.0137
5	0.3415 i	0.9313 i	1.0000 i	0.9851 i
6	-0.0235	-0.0048	0.0012	0.0158
9	0.6244 i	1.0000 i	0.9999 i	0.8204 i
10	-0.3032	-0.0606	0.0155	0.1960
13	0.3147 i	0.9153 i	0.9890 i	0.9880 i
14	0.0003	0.0001	-0.0000	-0.0002
17	0.3151 i	0.9155 i	0.9890 i	0.9879 i
18	-0.0000	-0.0000	0.0000	0.0000
21	0.3155 i	0.9156 i	0.9891 i	0.9877 i
22	-0.0003	-0.0001	0.0000	0.0002

In $\boldsymbol{R_O D_Y}$ a procedure, based on the orthogonality principle has been used to check the correct numbering of the eigencolumns after each frequency step (see the $\boldsymbol{R_O D_Y}$ subroutine **ordenen.m**). It is based on the

eigenvalue problem version: $\underline{A}\ \underline{u} = \lambda\ \underline{u}$.
If we have for some rotor-speed Ω_i the matrices with right- and left eigencolumns $\underline{U}_i$ respectively $\underline{V}_i$, we can write this property as:

$$\underline{V}_i^T\ \underline{U}_i = \text{diagonal}$$

If we solve the eigenvalue problem again for a new $\Omega_j = \Omega_i + \delta\Omega$, and calculate:

$$\underline{V}_i^T\ \underline{U}_j$$

this will not be a diagonal matrix anymore. But if we take $\delta\Omega$ small enough it will still be mainly diagonal or can be made mainly diagonal by renumbering the eigencolumns $\underline{U}_j$. This procedure resulted in the continuation shown in figure 8.44.

In the normal operating range ($\Omega \leq 1.5$ [Hz]), we only have 1 critical speed at 0.26 [Hz]. The relevant mode shape for this speed is shown in figure 8.46. It is a backward whirl with mainly a deformation from the rotor-shaft. Next

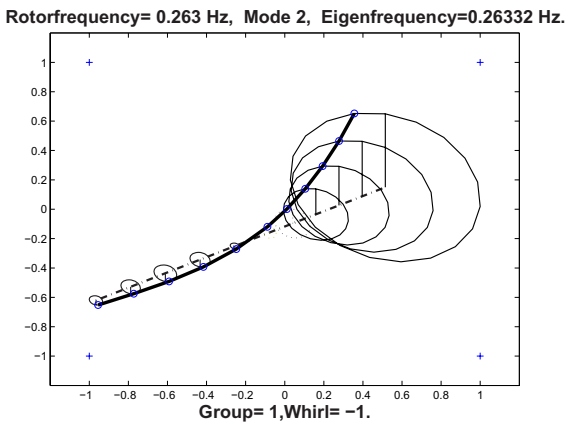


Fig. 8.46. Mode shape at first critical speed (only rotating part)

we look at the imbalance respons due to a fracture in one blade of the mill giving a loss of a part of the tip with a mass of 2 [kg]. In $\boldsymbol{R_O D_Y}$ we can only add an imbalance weight, not removing material. But instead of reducing the mass of one of the blades, we can also add an imbalance weight to the other 2 blades. These two imbalance forces under an angle of 120 degrees then give the same result. But we then also can add 2 [kg] to the top of

only the relevant blade but give it a phase shift of 180 degrees. The only point left is that by loosing the tip-part also the rotor model changed a little bit (moments of inertia for example). We assume that we can ignore this effect and just run the "undamaged" model.
So, we just calculate the respons due to an imbalance of 2 [kg] on a radius of 15 [m] at an angle of 180 degrees. The result is shown in figure 8.48. Wa can see that there is a dominant displacement in x-direction (dof 11),

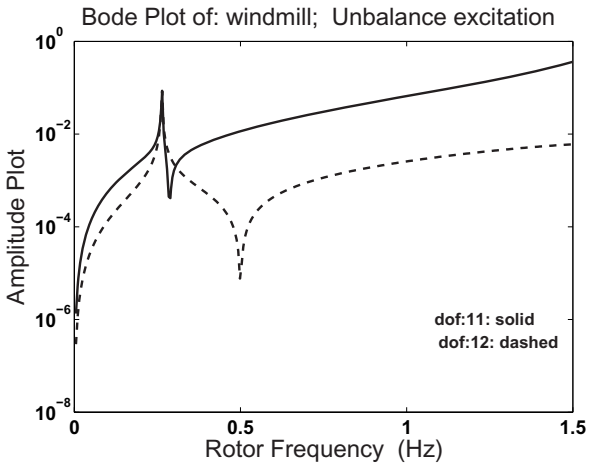


Fig. 8.47. Imbalance respons due to a blade-tip damage

growing to $\approx$ 0.5 [m]! for the higher rotor-speeds. This probably will not be safe anymore in a real situation. At the first critical speed also a sharp resonance peak can be seen which might be surprising because we know that this is a backward whirl which in theory does not show up. But in this case the rotor-center will not carry out a perfectly cylindrical motion (see figure 8.48), giving this secondary respons effect.
The displacement of the rotor-shaft center is larger in x- than in z-direction (see left orbit). The displacement of the bearing housing at the rotor-side (see right picture) is practically only in horizontal direction due to the high axial stiffness of the supporting tube.

Final remark

For the imbalance respons we can distinguish two limit cases:

a The imbalance force equals the windmill head inertia force:

$$m_{imb}\Omega^2 R_{blade} \quad = \quad m_{total}\Omega^2 x_{ampl}$$

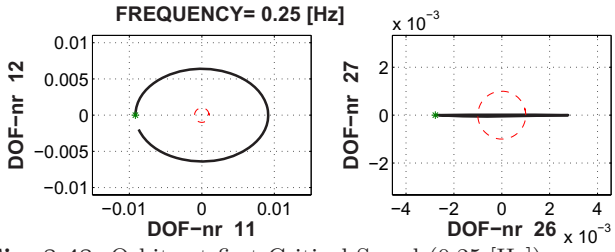


Fig. 8.48. Orbits at first Critical Speed (0.25 [Hz])

with gives $x_{ampl} \approx 0.15$ [m] (constant, valid for $\Omega > 1.5$ [Hz])

b The imbalance force equals the tube bending force:

$$m_{imb}\Omega^2 R_{blade} \quad = \quad k_{xx}\ x_{ampl}$$

giving $x_{ampl} \approx 1.1\ 10^{-3}\Omega^2$. For $\Omega = 2\pi$ [rad/s], this gives x_{ampl} = 0.04 [m]. (Quadratic function; obviously valid for $\Omega < 1.5$ [Hz]).

We can see that both estimates are fairly accurate, and can support our confidence in complicated numerical outcomes.

The reader should investigate the effect of adding damping to the system. This can be natural damping such as material damping or air damping, but also artificial damping by means of a specially designed passive or even active damping device.

Problem 8 Gyro-Coach

The generalized coördinates of the problem are

$$\underline{q} = \begin{bmatrix} V \\ \varphi_x \\ \varphi_z \end{bmatrix}$$

where V is the vertical displacement of the centre of gravity of the coach. The 4 suspension points will be identified as shown in figure 8.49. The

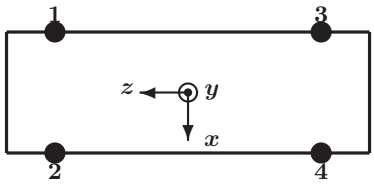


Fig. 8.49. Suspension numbering

elastic energy in the springs can then be written as:

$$T = \frac{1}{2}\,[k_t\ y_1^2 + k_t\ y_2^2 + k_t\ y_3^2 + k_t\ y_4^2]$$

with:

$$\begin{aligned}
y_1 &= V - \tfrac{L}{2}\varphi_x - \tfrac{B}{2}\varphi_z = (1/2)\ [2\ ,\ -L\ ,\ -B\ ]\ \underline{q} \\
y_2 &= V - \tfrac{L}{2}\varphi_x + \tfrac{B}{2}\varphi_z = (1/2)\ [2\ ,\ -L\ ,\ +B\ ]\ \underline{q} \\
y_3 &= V + \tfrac{L}{2}\varphi_x - \tfrac{B}{2}\varphi_z = (1/2)\ [2\ ,\ +L\ ,\ -B\ ]\ \underline{q} \\
y_4 &= V + \tfrac{L}{2}\varphi_x + \tfrac{B}{2}\varphi_z = (1/2)\ [2\ ,\ +L\ ,\ +B\ ]\ \underline{q}
\end{aligned}$$

So, we get:

$$V = \frac{1}{2}\ \underline{q}^T \{ \frac{k_t}{4} \begin{bmatrix} 2 \\ -L \\ -B \end{bmatrix} [2\ ,\ -L\ ,\ -B\ ]\ \}\ \underline{q}\ +\ \ .....$$

This gives:

$$V = \frac{1}{2}\ \underline{q}^T \{ k_t \begin{bmatrix} 4 & 0 & 0 \\ 0 & L^2 & 0 \\ 0 & 0 & B^2 \end{bmatrix} \}\ \underline{q}\ =\ \frac{1}{2}\ \underline{q}^T\ \underline{K}\ \underline{q}$$

This gives the (3*3) stiffness-matrix $\underline{K}$. The damping-matrix $\underline{B}$ has the same structure, so:

$$\underline{B} = b_t \begin{bmatrix} 4 & 0 & 0 \\ 0 & L^2 & 0 \\ 0 & 0 & B^2 \end{bmatrix}$$

For the Kinetic Energy of the coach we may write:

$$T_C = \frac{1}{2}\ M_c\ \underline{v}_C^T \underline{v}_C\ +\ \frac{1}{2}\ J_x\ \dot{\varphi}_x^2\ +\ \frac{1}{2}\ J_z\ \dot{\varphi}_z^2$$

where (see also figure 8.50:

$$\underline{v}_C^T = [-H\dot{\varphi}_z\ ,\ \dot{V}\ ,\ H\dot{\varphi}_x]$$

For the Kinetic Energy of a non-rotating flywheel we may write:

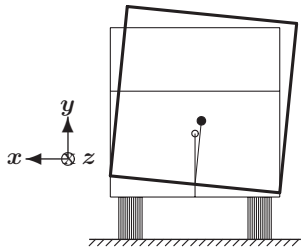


Fig. 8.50. Transverse motion of the Coach

$$T_f = \frac{1}{2}\ m_f\ \underline{v}_C^T \underline{v}_C\ +\ \frac{1}{2}\ J_t\ \dot{\varphi}_x^2\ +\ \frac{1}{2}\ J_t\ \dot{\varphi}_z^2$$

This gives for the total kinetic energy:

$$T = \frac{1}{2}\ \dot{\underline{q}}^T \underline{M}\ \dot{\underline{q}}$$

with the system Mass-matrix $\underline{M}$:

$$\underline{M} = \begin{bmatrix} M_T & 0 & 0 \\ 0 & J_x + J_t + H^2 M_T & 0 \\ 0 & 0 & J_z + J_t + H^2 M_T \end{bmatrix}$$

$M_T = M_C + m_f$ is the total mass (coach plus flywheel).

The gyroscopic effect of the flywheel rotation will introduce a coupling between the rotation φ_x and φ_z, giving an additional "Damping-matrix" $\underline{B}_G$:

$$\underline{B}_G = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -\Omega\ J_p \\ 0 & \Omega\ J_p & 0 \end{bmatrix}$$

First we look at the situation $\Omega = 0$. Then we are dealing with 3 uncoupled equations of motion (matrices $\underline{K}$, $\underline{M}$ and $\underline{B}$ all diagonal). The system also is proportionally damped because $\underline{B} = \alpha\ \underline{K}$, so we have **REAL** eigenmodes. The eigenvalues λ_s and corresponding dimensionless damping factors ξ_s appear to be:

$$\begin{array}{lll} \lambda_{1,2} = -0.0505 \pm 0.3978i\ \ [Hz],\ 12.6\%, & \text{pure vertical translation} \\ \lambda_{3,4} = -0.1391 \pm 0.6507i\ \ [Hz],\ 20.9\%, & \text{pure rotation } \varphi_x \\ \lambda_{5,6} = -0.0561 \pm 0.4189i\ \ [Hz],\ 13.3\%, & \text{pure rotation } \varphi_z \end{array}$$

Next, the eigenvalues for the rotating situation have been calculated for the rotor-speed range $0 \leq \Omega \leq 800\ \ [Hz]$ (which practically is rather high, namely 48.000 [rev/min]!). The imaginary parts (the damped eigenfrequencies) have been plotted in figure 8.51. We see that the vertical mode will

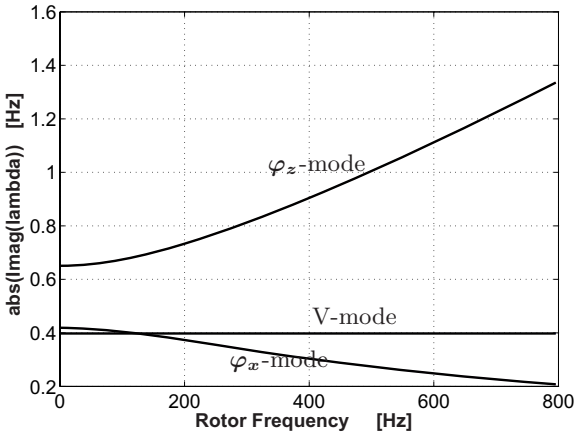


Fig. 8.51. Damped eigenfrequencies as function of flywheel speed

not be influenced by the rotation of the flywheel). The Roll-mode frequency is increasing and can almost become twice as high. The Jumping mode

however is decreasing from 0.4 to 0.2 [Hz].
So the "transverse stiffness" of a coach can be enlarged without changing the "vertical comfort" by adding such a flywheel but then at the same time, the "jumping frequency" will be smaller.
To see whether the (conservative) gyroscopic damping matrix $\underline{B}_G$ also influences the damping in the system, the real parts of the eigenvalues are plotted in figure 8.52. We see that the real parts will also be influenced by

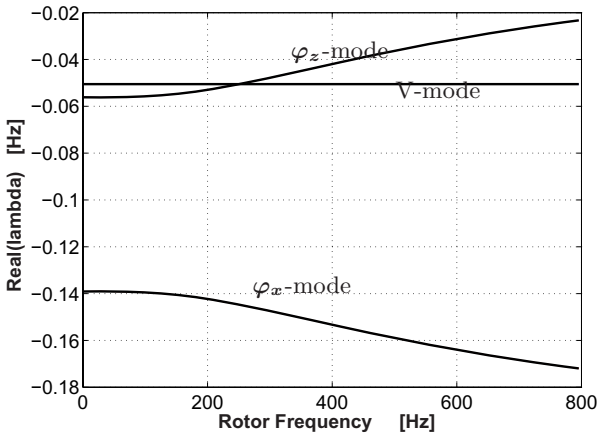


Fig. 8.52. Real parts of eigenvalues as function of flywheel speed

the flywheel-effect (except the pure vertical mode). For the modal dimensionless damping factors with and without rotating flywheel we find:

Ω [Hz]	0	800
mode 1	12.6%	12.6%
mode 2	20.9%	12.8%
mode 3	13.3%	11.1%

So, looking at these damping factors one might get the impression that adding the (Ω-depending) "damping" matrix $\underline{B}_G$ has resulted in a more weakly damped system, which is surprising.
We should however realize that the matrix $\underline{B}_G$ mainly changes the "modal stiffness" because there will be a conservative energy exchange between the rotations φ_x and φ_z.

The vibration modes for $\Omega = 800$ [Hz] are shown below. (they are scaled

such that $max(|u_s|) = 1.0$).
The modes now are really complex modes (not a proportionally damped system anymore). What we also can see is that roughly, the rotations φ_x and φ_z for modes 3 and 5 are 90 degrees out of phase.

dof	λ_1= -0.05 + 0.40i	λ_3=-0.17 + 1.34i	λ_5= -0.02 + 0.21i
V	1	0	0
φ_x	0	0.03 - 0.51i	0.003+0.275i
φ_z	0	1	1

From these modes one could get the idea that only rotation φ_z counts. But we should realize that the coach length L is much larger then its width B. To get a better picture we calculate the vertical displacements of both front-suspension points for the modes 3 (and 5):

$$y_1 = real[\{-\tfrac{L}{2}u3(2) - \tfrac{B}{2}u3(3)\}\ e^{j\omega_{d3}t}]$$

$$y_2 = real[\{-\tfrac{L}{2}u3(2) + \tfrac{B}{2}u3(3)\}\ e^{j\omega_{d3}t}]$$

where $u3 = u(:,3)$ and $\omega_{d3} = imag(\lambda_3)$. The results are shown in figure 8.53. What we see is that for mode 3, displacement y_2 is running ahead

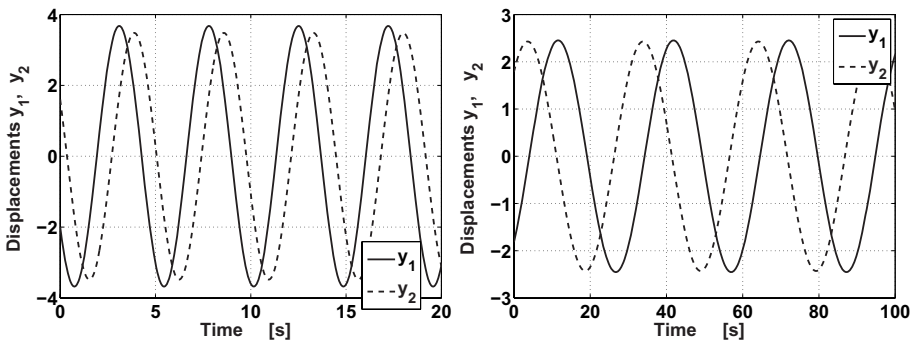


Fig. 8.53. Frontwheel displacements. Left: mode 3, $\lambda = -0.17 + 1.34i$, Right: mode 5, $\lambda = -0.02 + 0.21i$

and for mode 5, the displacement y_1.

Finally we assume that the coach is driving over a road with a long-wavelength, harmonic, anti-symmetric disturbance. For each spring-support point we introduce a prescribed displacement:

$$s(t) = a\ \cos(2\pi\frac{z}{L_R}) = a\ \cos(2\pi\frac{V_C t}{L_R}) = a\ \cos(\omega_R t)$$

with L_R is the (long) wavelength, V_C the coach-speed and $\omega_R = 2\pi V_C/L_R$ the (low) excitation frequency.
So, we get:

$$\begin{array}{rl} \text{right side} & s_1 = s_3 = a\ \cos(\omega_R t) \\ \text{left side} & s_2 = s_4 = -a\ \cos(\omega_R t) \end{array}$$

Then we can calculate the complex amplitudes ($\hat{y}_1$ and $\hat{y}_2$) of the respons for the front-wheel points from:

$$\underline{\hat{q}} = [-\omega_R^2 \underline{M} + j\omega_R(\underline{B} + \underline{B}_G) + \underline{K}]^{-1}\ \underline{\hat{Q}}$$

with

$$\underline{\hat{Q}} = \begin{bmatrix} 0 \\ 0 \\ 2a(k_t + j\omega_R b_t)B \end{bmatrix}$$

and

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \end{bmatrix} = \begin{bmatrix} 1\ ; & -L/2\ ; & -B/2 \\ 1\ ; & -L/2\ ; & +B/2 \end{bmatrix}\ \underline{\hat{q}}$$

It is clear that adding a flywheel to a passenger bus can considerably

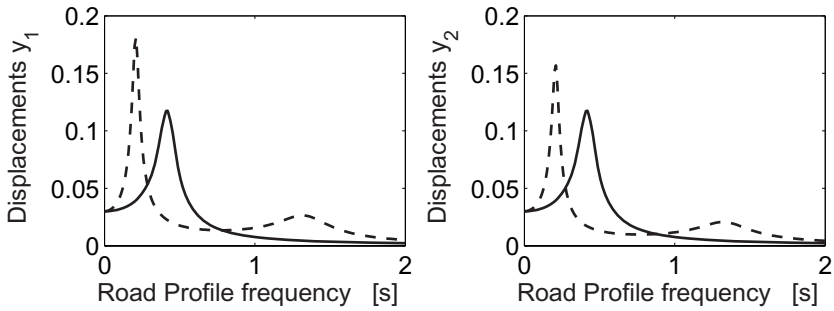


Fig. 8.54. Response to an asymmetrical, harmonic road-profile with amplitude of 0.03 [m].With (dashed) and without (solid) flywheel

change its dynamic performance. Whether a designer might be happy with the outcomes of this particular situation is another question.

Problem 9 Rotor model for aircraft engine

First we look at the situation with no additional damping from the squeeze film damper and weak cross-coupling. The (lowest) damped eigenfrequecies for the rotor-speed $0 \leq \Omega \leq 250 \quad [Hz]$ are shown in the Campbell plot in figure 8.55. In general we get a sequence of diverging eigenvalue-pairs (a

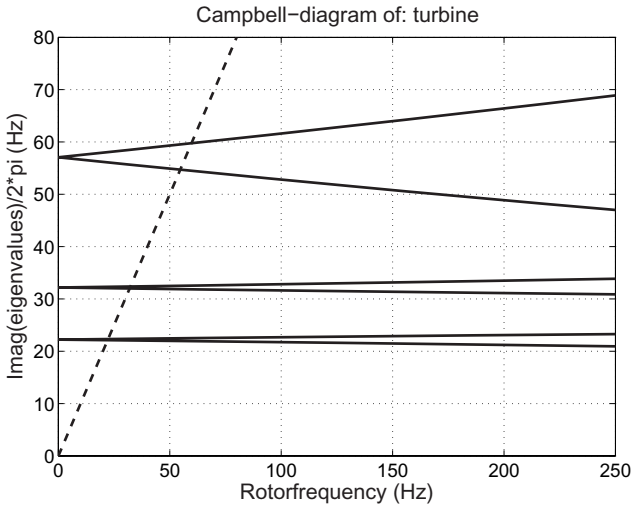


Fig. 8.55. Campbell plot with lowest eigenfrequencies

forward- and a backward whirl). The **Critical speeds** appear to be:
22.1246; 22.3339; 32.3742; 31.9933; 59.7502 and 54.7040 [Hz].
Next we concentrate on the dynamic behavior for the fixed rotor-speed of $\Omega = 250$ [Hz] (15.000 [RPM]). In the next table the data for the first 20 modes is given, each row containing: the mode nr.; the complex eigenvalue λ in [Hz]; the logarithmic decrement δ in % and the type of whirl.
Remember that for weakly damped systems (δ=small):

$$\delta = 2\pi\xi; \qquad \xi = \frac{-Re(\lambda)}{|\lambda|}$$

where ξ is the dimensionless damping factor. A negative value for δ or ξ is pointing at an **unstable** system.

We know that normally only a forward whirl will be excited by an unbalance excitation (so the backward whirl does practically not show up). Therefore

we will only look at the **unstable, forward whirling** modes which have been marked in the table by an arrow. Similar to Vance, we can identify them as the 1400 [CPM], 2030 [CPM], 4100 [CPM] and 9100 [CPM]-mode. In Vance's book the second one is not mentioned and there are some small differences, because some parameters have been chosen by guess.

	Nr.	$\boldsymbol{\lambda}$	$\boldsymbol{\delta}$ %	Whirl type
	1	-0.23619 - 20.94 i	7.1	Backward
	2	-0.23619 + 20.94 i	7.1	Backward
	3	0.26189 - 23.277 i	-7.1	Forward
$\Longrightarrow$	4	0.26189 + 23.277 i	-7.1	Forward
	5	-0.16656 - 30.885 i	3.3	Backward
	6	0.14012 - 33.849 i	-2.6	Forward
$\Longrightarrow$	7	0.14012 + 33.849 i	-2.6	Forward
	8	-0.16656 + 30.885 i	3.3	Backward
	9	0.045343 - 68.884 i	-0.41	Forward
$\Longrightarrow$	10	0.045343 + 68.884 i	-0.41	Forward
	11	-0.039518 - 47.011 i	0.53	Backward
	12	-0.039518 + 47.011 i	0.53	Backward
	13	-0.046219 - 128.01 i	0.23	Backward
	14	-0.046219 + 128.01 i	0.23	Backward
	15	0.043035 - 151.58 i	-0.18	Forward
$\Longrightarrow$	16	0.043035 + 151.58 i	-0.18	Forward
	17	0.018175 - 335.7 i	-0.03	Forward
$\Longrightarrow$	18	0.018175 + 335.7 i	-0.03	Forward
	19	-0.015454 - 266.37 i	0.04	Backward
	20	-0.015454 + 266.37 i	0.04	Backward

The mode shapes of these relevant modes are shown in figure 8.56. The first two are clearly mainly rigid-shaft modes with only bearing displacements. In modes 3..5 also significant shaft bending is playing a role.

The next step is to evaluate the effect of a squeeze film bearing at bearing station B_1 on the dynamical behavior (especially the instability). This can not be done automatically with the $\boldsymbol{R_O D_Y}$ toolbox. The data after calculating the Campbell diagram has been exported in a .mat file. This gives us the system-matrices for the rotor-speed $\Omega = 250$ [Hz], being the mass-matrix $\underline{M}$, Damping-matrix $\underline{B}$ and Stiffness-matrix $\underline{K}$, all of order (32*32).

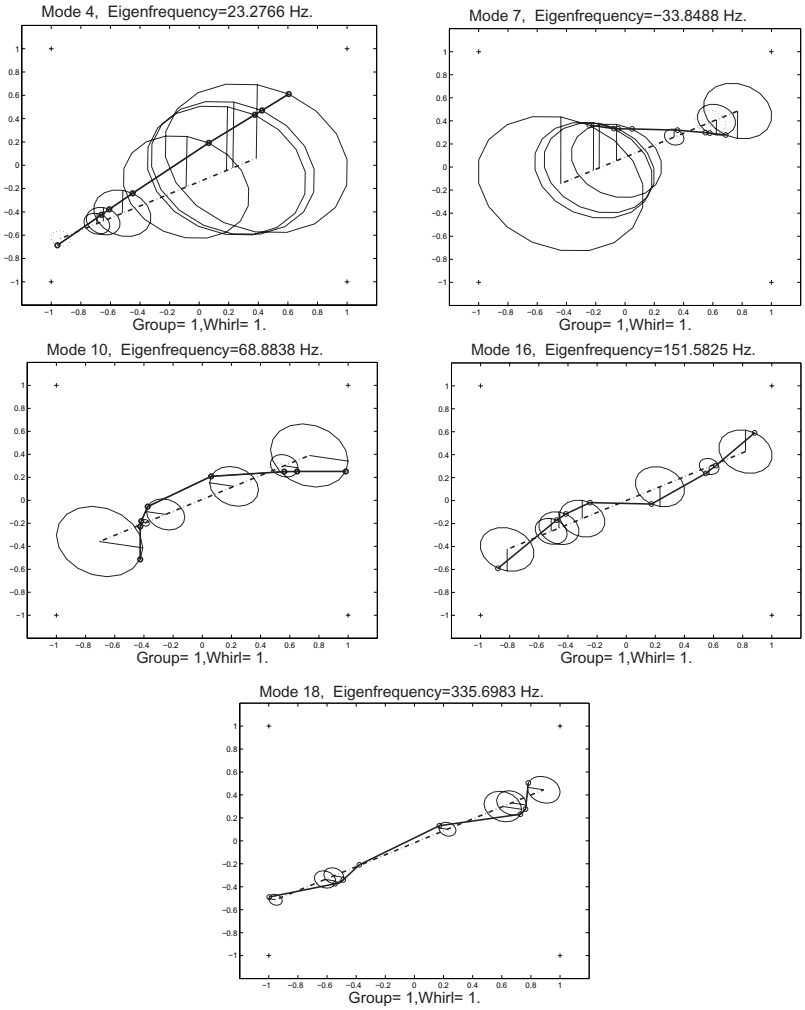


Fig. 8.56. Mode shapes of the relevant modes. Rotor-speed = 250 [Hz], weak coupling situation

In a separate programm the dampers $b_{xx} = b_{zz}$ were added to the degrees of freedom 5 and 6 (the translation of bearing B_1) and the eigenvalues calculated for the range $0 \leq (b_{xx} = b_{zz}) \leq 2.0\ 10^5 \quad [Ns/m]$.
Here we will focus on the afore-mentioned Vance-modes. The evolution of the imaginary part is shown in figure 8.57. Normally, adding (some) damping to a system will not change the eigenfrequencies dramatically. They will become somewhat smaller, depending on the modal displacements at the

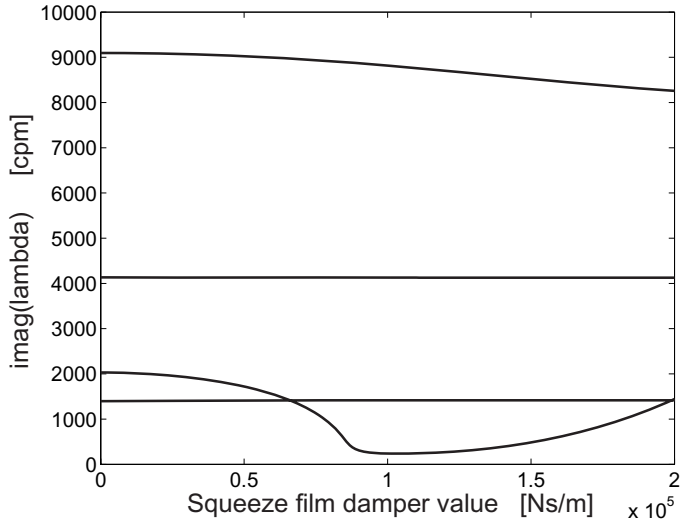


Fig. 8.57. Damped eigenfrequencies as function of damping value

damper location. This is true as long as the damping will remain undercritically.
For 3 of the 4 modes we can see this effect. An exception is the 2000[cps]-mode, which is strongly influenced by the damping increase.
Looking at the mode-shape we might understand this. Surprising however is the kink at $\approx b = 0.9\ 10^5$ after which the damped eigenfrequency is growing again. Calculating the mode-shapes for this area shows that the mode shape is still the same but the corresponding eigenvalue has become almost critically damped.
The logarithmic decrement of the modes is given in figure 8.58. The 1400-mode is first reacting positive to the damping increase but remains negative (unstable), also for large damping value.
The 2030- mode and 9100-mode directly become positive and the damping has a strong influence. This is clearly underlined by the corresponding mode shapes. The 4100-mode is starting negative, becomes slightly positive and later on slightly negative again. Practically the damping seems to have no effect. Because the mode-shape has a node very close to bearing station B_1, this looks logical.

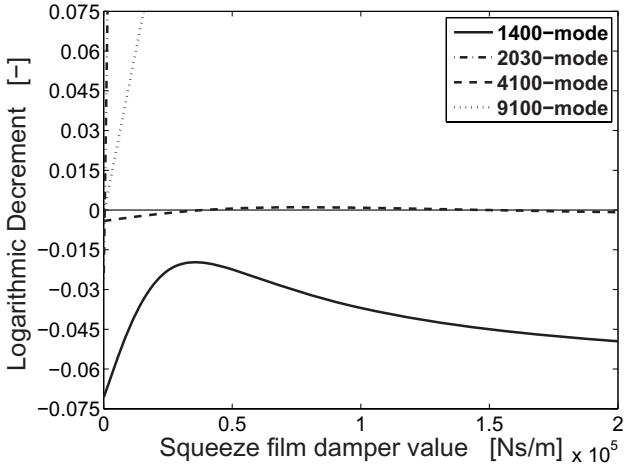


Fig. 8.58. Logarithmic Decrements as function of damping value

Strong Cross-Coupling Case

The two relevant plots for this case are shown in figure 8.59.

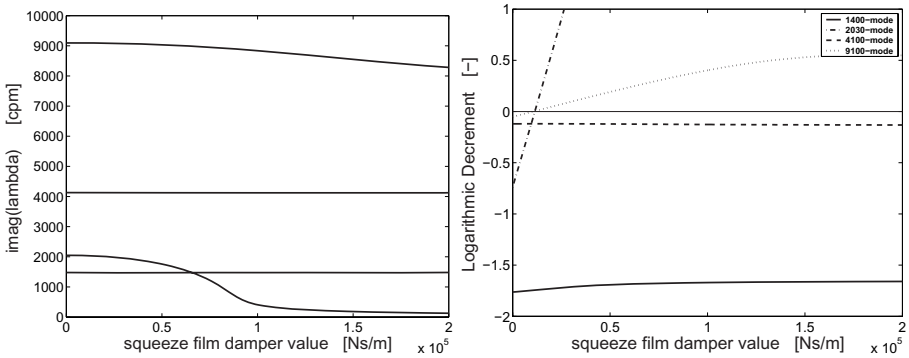
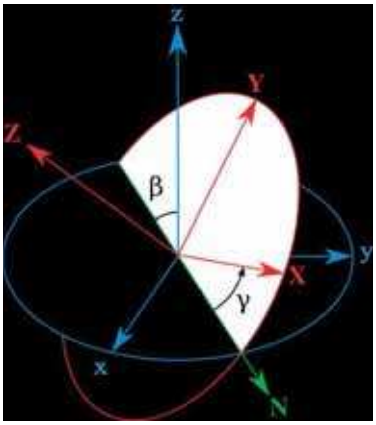


Fig. 8.59. Relevant plots for strong Cross-Coupling

It is up to the reader to elaborate this case further

A

Basic Theory Rigid Body Dynamics



Euler angles

From Wikipedia, the free encyclopedia

This article is about the Euler angles used in mathematics. For the use of the term in aerospace engineering, see Yaw, pitch, and roll. The Euler angles were developed by Leonhard Euler to describe the orientation of a rigid body (a body in which the relative position of all its points is constant) in 3-dimensional Euclidean space. To give an object a specific orientation it may be subjected to a sequence of three rotations described by the Euler angles. This is equivalent to saying that a rotation matrix can be decomposed as a product of three elemental rotations.

Euler angles are one of several ways of specifying the relative orientation of two such coordinate systems. Moreover, different authors may use different sets of angles to describe these orientations, or different names for the same angles. Therefore a discussion employing Euler angles should always be preceded

In this appendix we look at rotations of rigid bodies in 3-dimensional space. In this treatment, transformations of orthonormal vector basis forming right hand systems will be important. It will be necessary to introduce more than one vector base (also called a reference base or simply base). The basis will be identified by an index. First we look at a reference base $\{x_1, y_1, z_1\}$ with the bae vectors $\{\vec{e}_{1x}, \vec{e}_{1y}, \vec{e}_{1z}\}$ and a second base $\{x_2, y_2, z_2\}$ with the bese vectors $\{\vec{e}_{2x}, \vec{e}_{2y}, \vec{e}_{2z}\}$ as shown in Fig A.1. The matrix representation of

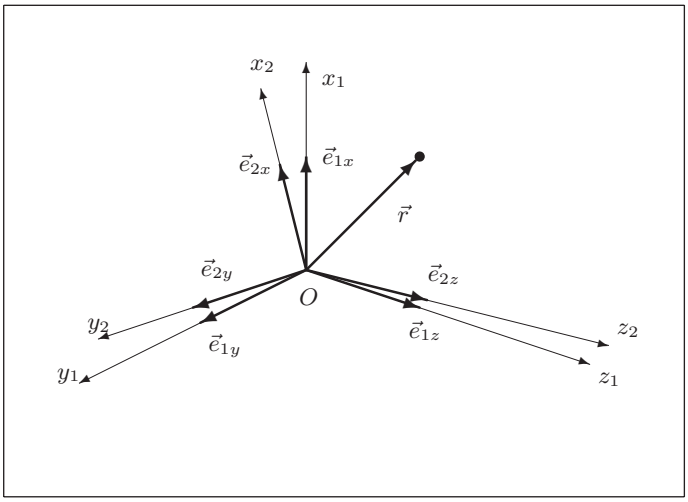


Fig. A.1. Position vector $\vec{r}$ in two reference bases

the position vector $\vec{r}$ with respect to the base $\{x_1, y_1, z_1\}$ is indicated as $\underline{r}_1^T = [r_{1x}, r_{1y}, r_{1z}]$ and the matrix representation with respect to the base $\{x_2, y_2, z_2\}$ as $\underline{r}_2^T = [r_{2x}, r_{2y}, r_{2z}]$. So we can write

$$\vec{r} = \begin{bmatrix} \vec{e}_{1x} & \vec{e}_{1y} & \vec{e}_{1z} \end{bmatrix} \begin{bmatrix} r_{1x} \\ r_{1y} \\ r_{1z} \end{bmatrix} = \begin{bmatrix} \vec{e}_{2x} & \vec{e}_{2y} & \vec{e}_{2z} \end{bmatrix} \begin{bmatrix} r_{2x} \\ r_{2y} \\ r_{2z} \end{bmatrix} \tag{A.1}$$

giving

$$\underline{r}_1 = \underline{R}\ \underline{r}_2, \text{ en } \underline{r}_2 = \underline{R}^T \underline{r}_1 \tag{A.2}$$

Here $\underline{R}$ is the matrix representation of the rotation tensor with respect to the base $\{x_1, y_1, z_1\}$. We can write

$$\underline{R} = \begin{bmatrix} (\vec{e}_{1x}.\vec{e}_{2x}) & (\vec{e}_{1x}.\vec{e}_{2y}) & (\vec{e}_{1x}.\vec{e}_{2z}) \\ (\vec{e}_{1y}.\vec{e}_{2x}) & (\vec{e}_{1y}.\vec{e}_{2y}) & (\vec{e}_{1y}.\vec{e}_{2z}) \\ (\vec{e}_{1z}.\vec{e}_{2x}) & (\vec{e}_{1z}.\vec{e}_{2y}) & (\vec{e}_{1z}.\vec{e}_{2z}) \end{bmatrix} \tag{A.3}$$

It can easy be shown that the matrix $\underline{R}$ is an orthonormal matrix with the properties

$$\underline{R}^{-1} = \underline{R}^T \quad \text{and} \quad det\{\underline{R}\} = +1 \tag{A.4}$$

In handling vectors and their matrix representations we apply the following basic relations

vector notation	matrix representation
$\vec{c} = \lambda\vec{a}$	$\underline{c} = \lambda\underline{a}$
$\vec{c} = \vec{a} + \vec{b} = \vec{b} + \vec{a}$	$\underline{c} = \underline{a} + \underline{b} = \underline{b} + \underline{a}$
$\alpha = (\vec{a}.\vec{b}) = (\vec{b}.\vec{a})$	$\alpha = \underline{a}^T\underline{b} = \underline{b}^T\underline{a}$
$\vec{c} = (\vec{a} * \vec{b}) = -(\vec{b} * \vec{a})$	$\underline{c} = (\underline{a})\underline{b} = -(\underline{b})\underline{a}$

We also will use the **skew-symmetric** matrix $(\underline{a})$ respectively $(\underline{b})$ according to the columns $\underline{a}$ respectively $\underline{b}$, defined as

$$\vec{a} \Longrightarrow \underline{a} = \begin{bmatrix} a_2 \\ a_2 \\ a_3 \end{bmatrix} \Longrightarrow (\underline{a}) = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix} \tag{A.5}$$

For a column $\underline{a}$ we have to aply a vector transformations which means that we can write $\underline{a}_1 = \underline{R}\ \underline{a}_2$ whereas for a matrix $(\underline{a})$ we should apply a tensor transformation, meaning $(\underline{a})_1 = \underline{R}\ (\underline{a})_2\ \underline{R}^T$.
If a tensor $\tilde{T}$ is operating on a vector $\vec{b}$, giving the vector $\vec{a}$, so if

$$\vec{a} = \tilde{T}\ \vec{b} \tag{A.6}$$

we can write for the matrix representations

$$\underline{a}_1 = \underline{T}_1\ \underline{b}_1, \qquad \text{en} \quad \underline{a}_2 = \underline{T}_2\ \underline{b}_2 \tag{A.7}$$

giving directly

$$\underline{T}_1 = \underline{R}\ \underline{T}_2\underline{R}^T, \qquad \text{en} \quad \underline{T}_2 = \underline{R}^T\underline{T}_1\underline{R} \tag{A.8}$$

A.1 Rotation of a rigid body around a fixed point

We consider a rigid body with a vector base $\{X_2, Y_2, Z_2\}$ which is fixed to the body and a material point of that body. In the reference (starting) position this body-fixed frame coincides with a second, fixed vector base $\{x_1, y_1, z_1\}$. We assume an arbitrary rotation of this rigid body around the

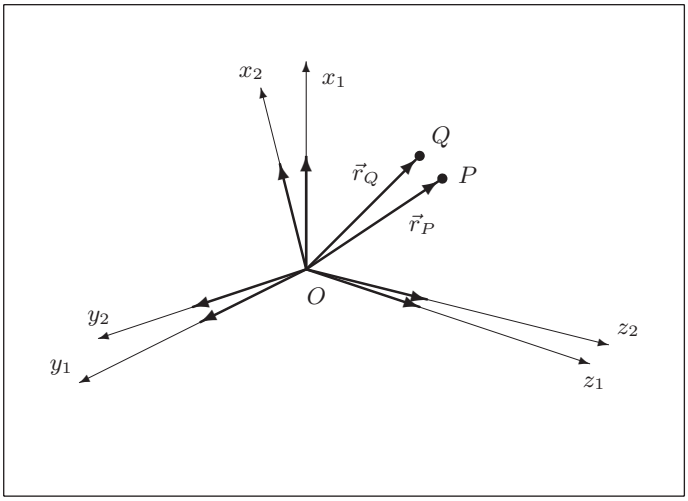


Fig. A.2. Rotation of a rigid body around a fixed point O

origin O of both the bases where the material point will move from the position P to Q. (see Fig A.2
Because the matrix representation of the position vector of the material point with respect to the body fixed basis will not change, we can write

$$\underline{r}_{Q_2} = \underline{r}_{P_1} \tag{A.9}$$

which, using $\underline{r}_{Q_1} = \underline{R}\ \underline{r}_{Q_2}$ gives

$$\underline{r}_{Q_1} = \underline{R}\ \underline{r}_{P_1} \tag{A.10}$$

where $\underline{R} = \underline{R}(t)$ defines the rotation as function of time. The column $\underline{r}_{P_1}$ is no function of time because this is the position of the material point in

the starting position. The velocity of the metarial point can now be written as

$$\underline{v}_1 = \dot{\underline{r}}_{Q_1} = \dot{\underline{R}}\ \underline{r}_{P_1} = \dot{\underline{R}}\ \underline{R}^T \underline{r}_{Q_1} \tag{A.11}$$

For each arbitrary point with position vector $\vec{x}$ at time t we can write

$$\underline{v} = \dot{\underline{R}}\ \underline{R}^T \underline{x} = (\underline{\omega})\ \underline{x} \tag{A.12}$$

Here, $(\underline{\omega})$ is a skew-symmetric matrix, because

$$\frac{d}{dt}(\underline{R}\ \underline{R}^T = \underline{I}) \Longrightarrow \dot{\underline{R}}\ \underline{R}^T + \underline{R}\ \dot{\underline{R}}^T = \underline{O} \tag{A.13}$$

and the matrix $(\underline{\omega})$ will have tensor-properties. According to this matrix $(\underline{\omega})$, a column $\underline{\omega}$ can be uniquely defined such that

$$\underline{v} = (\underline{\omega})\ \underline{x} \qquad \text{and} \qquad \vec{v} = \vec{\omega} * \vec{x} \tag{A.14}$$

where the vector $\vec{\omega}$ is called the **angular velocity vector**.

A.2 Relative, Carrier- and Absolute Motion

We introduce the following vector bases

- $\{x, y, z\}$ fixed in space (the inertial frame)
- $\{x_v, y_v, z_v\}$ rotating frame, for example fixed to a so-called carrier
- $\{x_l, y_l, z_l\}$ rotating frame, fixed to the rigid body

as shown in Fig A.3. Now we will call the motion of

- carrier frame with respect to the fixed frame the **carrier motion**,
- body frame with respect to the carrier frame the **relative motion**,
- body frame with respect to the fixed frame the **absolute motion**.

For a material point of of the body with momentary position vector $\vec{r}$ we can write

$$\underline{r} = \underline{R}_s\ \underline{x}_v \tag{A.15}$$

where $\underline{R}_s$ is the rotation matrix related to the carrier motion. Additionally we have

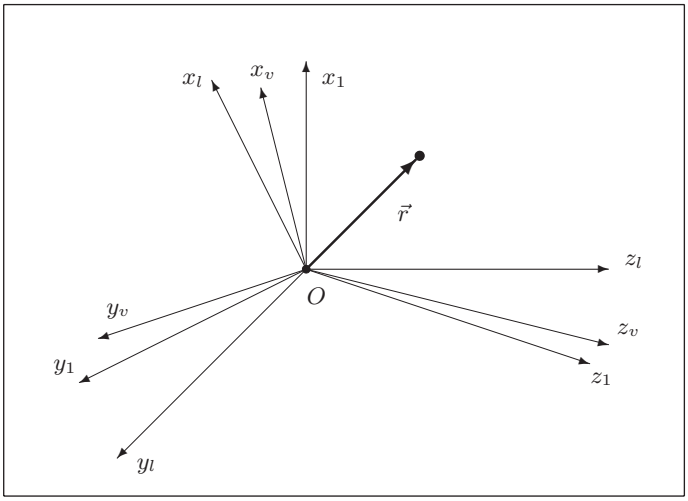


Fig. A.3. Fixed-, carrier- and body fixed frames

$$\underline{r}_v = \underline{R}_{rv}\ \underline{x}_l \tag{A.16}$$

where in this case $\underline{R}_{rv}$ is the rotation matrix related to the rotation of the body frame with respect to the carrier frame which we called the relative motion.
From (A.14) and (A.16) we get

$$\underline{r} = \underline{R}_s\ \underline{R}_{rv}\ \underline{r}_l \tag{A.17}$$

Because the column $\underline{r}_l$ in time-independent, we may write

$$\dot{\underline{r}} = \underline{v} = \dot{\underline{R}}_s\ \underline{R}_{rv}\ \underline{r}_l + \underline{R}_s\ \dot{\underline{R}}_{rv}\ \underline{r}_l \tag{A.18}$$

or

$$\begin{aligned} \underline{v} &= \dot{\underline{R}}_s\ \underline{r}_v + \underline{R}_s\ \dot{\underline{R}}_{rv}\ (\underline{R}_v^T\ \underline{R}_s^T)\ \underline{r} = \\ &= \dot{\underline{R}}_s\ \underline{R}_s^T\ \underline{r} + \underline{R}_s\ (\dot{\underline{R}}_{rv}\ \underline{R}_{rv}^T)\ \underline{R}_s^T\ \underline{r} \\ &= [(\underline{\omega}_s) + (\underline{\omega}_r)]\ \underline{r} \end{aligned} \tag{A.19}$$

From this result we can conclude that

$$\vec{v} = (\vec{\omega}_s + \vec{\omega}_r) * \vec{r} = \vec{\omega}_{abs} * \vec{r} \tag{A.20}$$

In this expression we can recognize $\vec{\omega}_{abs}$ as the absolute angular velocity vector, $\vec{\omega}_s$ as the carrier angular velocity and $\vec{\omega}_r$ as the relative angular velocity vector.

A.3 The Kinetic Energy

We look at the isolated mass particle dm of the rigid body L with position vector $\vec{r}$ and velocity vector $\vec{v}$. For the moment of momentum vector $\vec{D}$ of the body with respect of the fixed point O we can write

$$\vec{D} = \int_L \vec{r} * \vec{v}\ dm; \qquad \text{and} \qquad \underline{D} = \int_L (\underline{r})\ \underline{v}\ dm \tag{A.21}$$

Using (A.14) we can transform this into

$$\underline{D} = \int_L -(\underline{r})\ (\underline{r})\ \underline{\omega}\ dm = -\int_L (\underline{r})\ (\underline{r})\ dm\ \underline{\omega} = \underline{J}\ \underline{\omega} \tag{A.22}$$

or

$$\vec{D} = \tilde{J}\ \vec{\omega} \tag{A.23}$$

where $\tilde{J}$ is the **inertia tensor** of the body L with respect of O and $\underline{J}$ the matrix representation of this tensor with respect to the fixed frame.
For the kinetic energy this results in

$$T = \frac{1}{2}\int_L (\vec{v}.\vec{v})\ dm = \frac{1}{2}\int_L \underline{v}^T \underline{v}\ dm \tag{A.24}$$

or additionally

$$T = \frac{1}{2}\ \underline{\omega}^T \int_L (\underline{r})^T (\underline{r}) dm\ \underline{\omega} = \frac{1}{2}\ \underline{\omega}^T \int_L -(\underline{r})(\underline{r}) dm\ \underline{\omega} = \frac{1}{2}\ \underline{\omega}^T \underline{J}\ \underline{\omega} \tag{A.25}$$

Because the kinetic energy should be positive for each non-zero angular velocity $\underline{\omega} \neq \underline{o}$, the mass moment of inertia matrix $\underline{J}$ should be positive-definite. The eigenvalues of this matrix, so the solutions λ_i from the eigenvalue problem

$$\underline{J}\ \underline{u}_i = \lambda_i\ \underline{u}_i \tag{A.26}$$

are called the **principal moments of inertia** and the corresponding eigencolumns $\underline{u}_i$ define the **principal axes of inertia** with respect to the used frame.

A.4 Generalized Coordinates

Looking at the application of the theory for the rotation of a rigid body with respect to a fixed point we should realise that we need exactly 3 degrees of freedom for completely defining the position of the body uniquely. Several choices for these 3 degrees of freedom can be made. Here we will discuss two practically important selections namely **Euler angles** and **Cardan- or Bryant angles**.

A.4.1 Euler Angles

We consider again two vector bases, namely a fixed basis $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ and a body fixed basis $\{\vec{l}_1, \vec{l}_2, \vec{l}_3\}$. In the starting position the two basis coincide. The total rotation of a body can now be defined by three successive rotations.
The first rotation is a rotation φ_1 around the axis $\vec{e}_3$. The body fixed frame will then move to a position indicated by the frame $\{\vec{f}_1, \vec{f}_2, \vec{f}_3\}$. The second rotation is a rotation φ_2 around the axis $\vec{f}_1$, giving the frame $\{\vec{g}_1, \vec{g}_2, \vec{g}_3\}$. The third and final rotation is a rotation φ_3 around the axis $\vec{g}_3$, giving the final state $\{\vec{l}_1, \vec{l}_2, \vec{l}_3\}$.
For an arbitrary vector $\vec{r}$ with its matrix representations $\underline{r}$, $\underline{r}_f$, $\underline{r}_g$ and $\underline{r}_l$ with respect to the $\{\vec{e}_i\}$, $\{\vec{f}_i\}$, $\{\vec{g}_i\}$ and $\{\vec{l}_i\}$ frames, we can write

$$\begin{aligned} \underline{r} &= \underline{R}\ \underline{r}_f \\ \underline{r}_f &= \underline{R}_f\ \underline{r}_g \\ \underline{r}_g &= \underline{R}_g\ \underline{r}_l \end{aligned} \tag{A.27}$$

where

$$\underline{R} = \begin{bmatrix} c_1 & -s_1 & 0 \\ s_1 & c_1 & 0 \\ 0 & & 1 \end{bmatrix}; \underline{R}_f = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_2 & -s_2 \\ 0 & s_2 & c_2 \end{bmatrix}; \underline{R}_g = \begin{bmatrix} c_3 & -s_3 & 0 \\ s_3 & c_3 & 0 \\ 0 & & 1 \end{bmatrix};$$

In these matrices the following abbreviations have been used

$$s_1 = \sin(\varphi_1); \quad c_2 = \cos(\varphi_2); \quad \text{etc.}$$

From (A.27) follows

$$\underline{r} = \underline{R}\ \underline{R}_f \underline{R}_g\ \underline{r}_l = \underline{R}_E\ \underline{r}_l \tag{A.28}$$

giving for the **Euler rotation matrix** $\underline{R}_E$

$$\underline{R}_E = \begin{bmatrix} c_1c_3 - s_1c_2s_3 & -s_3c_1 - s_1c_3c_2 & s_1s_2 \\ s_1c_3 + c_1c_2s_3 & -s_1s_3 + c_1c_2c_3 & -s_2c_1 \\ s_2s_3 & s_2c_3 & c_2 \end{bmatrix} \tag{A.29}$$

For sufficiently small rotations $\varphi_1, \varphi_2, \varphi_3$ we can simplify this expression by using $\sin(\varphi_i) \approx \varphi_i$ and $\cos(\varphi_i) \approx 1$ and neglecting terms φ_i^2 and higher order terms. Then we get

$$\underline{R}_E \approx \begin{bmatrix} 1 & -\varphi_3 - \varphi_1 & 0 \\ \varphi_1 + \varphi_3 & 1 & -\varphi_2 \\ 0 & \varphi_2 & 1 \end{bmatrix} = \underline{I} + (\underline{\varphi}_E) \tag{A.30}$$

So, the matrix $\underline{R}_E$ can be seen as the sum of a unity matrix $\underline{I}$ and the skew-symmetric matrix $\underline{\varphi}_E$, which can be associated with the column $\underline{\varphi}_E$

$$\underline{\varphi}_E = \begin{bmatrix} \varphi_2 \\ 0 \\ \varphi_1 + \varphi_3 \end{bmatrix} \tag{A.31}$$

This final result is exactly what might be expected for sufficiently small rotations.

A.4.2 Cardan or Bryant Angles

The starting position is identical as for the Euler angles derivation. In this case the rotation protocol is

- the first rotation φ_1 around the $\vec{e}_1$ axis,
- a second rotation φ_2 around the $\vec{f}_2$ axis,
- the final rotation φ_3 around the $\vec{g}_3$ axis.

This gives for the (partial) rotation matrices $\underline{R}$, $\underline{R}_f$ and $\underline{R}_g$

$$\underline{R} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_1 & -s_1 \\ 0 & s_1 & c_1 \end{bmatrix} \quad \underline{R}_f = \begin{bmatrix} c_2 & 0 & s_2 \\ 0 & 1 & 0 \\ -s_2 & 0 & c_2 \end{bmatrix} \quad \underline{R}_g = \begin{bmatrix} c_3 & -s_3 & 0 \\ s_3 & c_3 & 0 \\ 0 & & 1 \end{bmatrix}$$

and the total rotation matrix becomes

$$\underline{R}_B = \begin{bmatrix} c_2c_3 & -c_2s_3 & s_2 \\ c_1s_3 + s_1s_2c_3 & c_1c_3 - s_1s_2s_3 & -s_1c_2 \\ s_1s_3 - c_1s_2c_3 & s_1c_3 + c_1s_2s_3 & c_1c_2 \end{bmatrix} \tag{A.32}$$

For sufficiently small angles $\varphi_1, \varphi_2, \varphi_3$ this can be approximated by

$$\underline{R}_B \approx \begin{bmatrix} 1 & -\varphi_3 & \varphi_2 \\ \varphi_3 & 1 & -\varphi_1 \\ -\varphi_2 & \varphi_1 & 1 \end{bmatrix} = \underline{I} + (\underline{\varphi}_B) \tag{A.33}$$

Now we get a skew symmetric matrix $(\underline{\varphi}_B)$ and its associated column $\underline{\varphi}_B$

$$\underline{\varphi}_B = \begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \end{bmatrix} \tag{A.34}$$

Cardan angles are very well suited for the evaluation of for example rotorsystems or gyroscopic systems when the rotations φ_1 and φ_2 may be assumed to be very small and the rotation φ_3 will be used for the (usually very large) rotation of the body around its own central axis. This normally can lead to a set of *linearized* equations of motion.
In that case we call the vector basis $\{\vec{g}_1, \vec{g}_2, \vec{g}_3\}$ the carrier frame (suspension of the gyroscope). This frame determines the carrier rotation. The final rotation ω around the $\vec{g}_3$ axis is called the relative rotation. For an arbitrary point of the gyroscope with position vector $\vec{r}$ we can the write

$$\underline{r} = \underline{R}\ \underline{R}_f\ \underline{r}_g = \underline{R}_s\ \underline{r}_v \tag{A.35}$$

where $\underline{R}_s$ is the carrier rotation matrix, yielding

$$\underline{R}_s = \begin{bmatrix} c_2 & 0 & s_2 \\ s_1s_2 & c_1 & -s_1c_2 \\ -c_1s_2 & s_1 & c_1c_2 \end{bmatrix} \tag{A.36}$$

With the small rotation assumption for φ_1 and φ_2 this gives

$$\underline{R}_s \approx \begin{bmatrix} 1 & 0 & \varphi_2 \\ 0 & 1 & -\varphi_1 \\ -\varphi_2 & \varphi_1 & 1 \end{bmatrix} = \underline{I} + (\underline{\varphi}_s) \tag{A.37}$$

The skew-symmetric matrix $(\underline{\varphi}_s)$ is associated with the column $\underline{\varphi}_s$

$$\underline{\varphi}_s = \begin{bmatrix} \varphi_1 \\ \varphi_2 \\ 0 \end{bmatrix} \tag{A.38}$$

Finally it should be emphasized that only for small rotations we will get a skew-symmetric matrix which can be associated with a column and consequently with a vector. This means that only small rotations will have **vector properties** (successive rotations may be mixed) and large rotations generally not (sequence of the successive rotations important).

A.5 The Law of Moment of Momentum

For an inertial system we have **Newton**'s second law for translations and as the complement of that for rotations around a fixed point O the so-called axiom of **Euler**

$$\vec{M} = \dot{\vec{D}} \tag{A.39}$$

For the moment of momentum according to (A.21) we can write ($\vec{v} * \vec{v} = \vec{o}$!)

$$\dot{\vec{D}} = \int_L \vec{r} * \dot{\vec{v}} \tag{A.40}$$

For the velocity vector $\vec{v} = \vec{\omega} * \vec{r}$ follows

$$\begin{aligned} \dot{\vec{v}} &= \vec{\omega} * \dot{\vec{r}} + \dot{\vec{\omega}} * \vec{r} \\ &= \vec{\omega} * (\vec{\omega} * \vec{r}) + \dot{\vec{\omega}} * \vec{r} \end{aligned} \tag{A.41}$$

giving

$$\dot{\vec{D}} = \int_L [\vec{r} * (\vec{\omega} * (\vec{\omega} * \vec{r})) + \vec{r} * (\dot{\vec{\omega}} * \vec{r})] \, dm \tag{A.42}$$

Now we use

$$\vec{r} * [\vec{\omega} * (\vec{\omega} * \vec{r})] = \vec{\omega} * [\vec{r} * (\vec{\omega} * \vec{r})] \qquad \text{and}$$
$$\dot{\vec{\omega}} * \vec{r} = -\vec{r} * \dot{\vec{\omega}}$$

to get

$$\dot{\vec{D}} = \int_L [\vec{\omega} * (\vec{r} * \vec{v}) - \vec{r} * (\vec{r} * \dot{\vec{\omega}})] \; dm \tag{A.43}$$

The corresponding matrix representation with respect to a fixed frame reads

$$\begin{aligned} \dot{\underline{D}} &= \int_L [(\underline{\omega})(\underline{r})\underline{v} - (\underline{r})(\underline{r})\dot{\underline{\omega}}] \; dm \\ &= (\underline{\omega})\{\int_L -(\underline{r})(\underline{r}) \; dm\} \; \underline{\omega} + \{\int_L -(\underline{r})(\underline{r}) \; dm\} \; \dot{\underline{\omega}} \end{aligned} \tag{A.44}$$

So, finally we get

$$\underline{M} = (\underline{\omega})\underline{J} \; \underline{\omega} + \underline{J} \; \dot{\underline{\omega}} \tag{A.45}$$

In some cases the application of a body fixed frame should be preferred over a fixed frame because in such a body fixed frame a mass-moment-of-inertia matrix $\underline{J}_l$ generally will note depend on time.
From (A.39), using $\underline{r} = \underline{R} \; \underline{r}_l$ we can see

$$\begin{aligned} \underline{R} \; \underline{M}_l &= \frac{d}{dt}[\underline{R} \; \underline{D}_l] = \frac{d}{dt}[\underline{R} \; \underline{J}_l \underline{\omega}_l] \\ &= \underline{R} \; \underline{J}_l \frac{d}{dt}[\underline{\omega}_l] + \dot{\underline{R}} \; \underline{J}_l \underline{\omega}_l \qquad \text{so} \\ \underline{M}_l &= \underline{J}_l \frac{d}{dt}[\underline{\omega}_l] + \underline{R}^T \dot{\underline{R}} \; \underline{J}_l \underline{\omega}_l \end{aligned} \tag{A.46}$$

We also know that

$$\dot{\underline{R}} \; \underline{R}^T = (\underline{\omega}) \Longrightarrow \underline{R}^T \dot{\underline{R}} \; \underline{R}^T \underline{R} = \underline{R}^T (\underline{\omega}) \underline{R} = (\underline{\omega}_l) = \underline{R}^T \dot{\underline{R}}$$

so, we finally get

$$\underline{M}_l = \underline{J}_l \; \dot{\underline{\omega}}_l + (\underline{\omega}_l) \; \underline{J}_l \; \underline{\omega}_l \tag{A.47}$$

If we choose, in particular, the principal axes of inertia as the directions for the base vectors of this body fixed frame, then the the matrix representation $\underline{J}_l$of the inertia tensor becomes a diagonal matrix

$$\underline{J}_l = \begin{bmatrix} J_1 & 0 & 0 \\ 0 & J_2 & 0 \\ 0 & 0 & J_3 \end{bmatrix} \tag{A.48}$$

where J_1, J_2 and J_3 are the principal moments of inertia. Then (A.47) can be written as

$$
\begin{aligned}
M_{l1} &= J_1\ \dot{\omega}_{l1} - (J_2 - J_3)\ \omega_{l2}\ \omega_{l3} \\
M_{l2} &= J_2\ \dot{\omega}_{l2} - (J_3 - J_1)\ \omega_{l3}\ \omega_{l1} \\
M_{l3} &= J_2\ \dot{\omega}_{l3} - (J_1 - J_2\ )\omega_{l1}\ \omega_{l2}
\end{aligned} \tag{A.49}
$$

These are the well-known **Euler Equations** for the rotation of a single rigid body around a fixed point.

A matrix representation of the law of moment of momentum can not only be derived with respect to a fixed frame or a body fixed frame as shown before, but also with respect to any other vector basis, such as the carrier frame introduced before. We then start from

$$\underline{r} = \underline{R}_s\ \underline{r}^v \tag{A.50}$$

where $\underline{r}$ respectively $\underline{r}^v$ are the matrix representations of the position vector with respect to the fixed- respectively the carrier frame and $\underline{R}_s$ the carrier rotation matrix introduced before. Using the law of moment of momentum $\underline{M} = \underline{\dot{D}}$ gives

$$\underline{R}_s\underline{M}^v = \frac{d}{dt}[\underline{R}_s\underline{D}^v] = \underline{\dot{R}}_s\underline{D}^v + \underline{R}_s\underline{\dot{D}}^v \tag{A.51}$$

giving

$$\underline{M}^v = \underline{R}_s^T\underline{\dot{R}}_s\underline{D}^v + \underline{\dot{D}}^v \tag{A.52}$$

From $(\underline{\omega}_s) = \underline{\dot{R}}_s\underline{R}_s^T$ we get

$$(\underline{\omega}_s^v) = \underline{R}_s^T(\underline{\omega}_s)\underline{R}_s = \underline{R}_s^T\underline{\dot{R}}_s\underline{R}_s^T\underline{R}_s = \underline{R}_s^T\underline{\dot{R}}_s$$

leading to

$$\underline{M}^v = (\underline{\omega}_s^v)\underline{D}^v + \underline{\dot{D}}^v$$

Because we also know that

$$\underline{D}^v = \underline{J}^v\underline{\omega}_{abs}^v$$

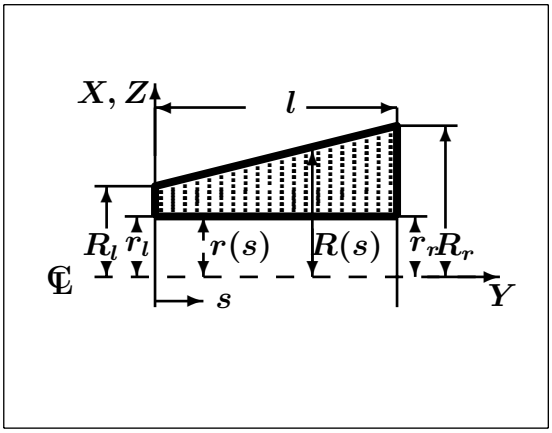
we finally get the matrix representation of the law of moment of momentum with respect to the carrier vector basis

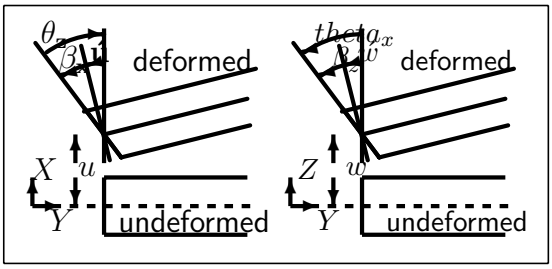
$$\underline{M}^v = (\underline{\omega}_s^v)\underline{J}^v\underline{\omega}_{abs}^v + \frac{d}{dt}[\underline{J}^v\underline{\omega}_{abs}^v] \tag{A.53}$$

If the carrier frame and the body fixed frame are identical, so if $\underline{\omega}_s^v = \underline{\omega}_{abs}^v$ and $\underline{J}^v = \underline{J}_l$=constant, this relation changes into the former relation (A.47).

B

Element Matrices for Conical Beam-element





Inputdata:

- L: Length [m]
- R_l : Outer radius left side $[m]$
- r_l : Inner radius left side $[m]$
- R_r : Outer radius right side $[m]$
- r_r : Inner radius right side $[m]$
- E : Modulus of Elasticity $[Pa]$
- ρ : Density $[kg/m^3]$
- G : Shear modulus $[Pa]$
- k : Shear factor $[-]$
- Ω : Rotor speed $[rad/s]$

Further we will use

$$
\begin{aligned}
&\sigma = R_r/R_l \\
&\nu = r_r/r_l \\
&r = r_l[1 + (\nu - 1)\xi] \\
&R = R_l[1 + (\sigma - 1)\xi] \\
&A = \pi(R^2 - r^2) = A_l(1 + \alpha_1\xi + \alpha_2\xi^2) \\
&A_l = \pi(R_l^2 - r_l^2) \\
&\alpha_1 = 2[R_l^2(\sigma - 1) - r_l^2(\nu - 1)]/(R_l^2 - r_l^2) \\
&\alpha_2 = [R_l^2(\sigma - 1)^2 - r_l^2(\nu - 1)^2]/(R_l^2 - r_l^2) \\
&I = \pi(R^4 - r^4)/4 = I_l[1 + \delta_1\xi + \delta_2\xi^2 + \delta_3\xi^3 + \delta_4\xi^4] \\
&I_l = \pi(R_l^4 - r_l^4)/4 \\
&\delta_1 = 4[R_l^4(\sigma - 1) - r_l^4(\nu - 1)]/(R_l^4 - r_l^4) \\
&\delta_2 = 6[R_l^4(\sigma - 1)^2 - r_l^4(\nu - 1)^2]/(R_l^4 - r_l^4) \\
&\delta_3 = 4[R_l^4(\sigma - 1)^3 - r_l^4(\nu - 1)^3]/(R_l^4 - r_l^4) \\
&\delta_4 = [R_l^4(\sigma - 1)^4 - r_l^4(\nu - 1)^4]/(R_l^4 - r_l^4)
\end{aligned}
$$

and we define the constants

$$
\begin{aligned}
&K1 = \frac{EI_l}{7!\ L^3} \\
&c1 = K1(60480 + 30240\delta_1 + 24192\delta_2 + 21168\delta_3 + 19008\delta_4) \\
&c2 = K1 * L(30240 + 10080\delta_1 + 7056\delta_2 + 6048\delta_3 + 5472\delta_4)
\end{aligned}
$$

$$
\begin{aligned}
c3 &= K1 * L(30240 + 20160\delta_1 + 17136\delta_2 + 15120\delta_3 + 13536\delta 4_{)} \\
c4 &= -K1 * L(30240 + 15120\delta_1 + 12096\delta_2 + 10584\delta_3 + 9504\delta_4) \\
c5 &= K1 * L^2(20160 + 5040\delta_1 + 2688\delta_2 + 2016\delta_3 + 1728\delta_4) \\
c6 &= K1 * L^2(10080 + 5040\delta_1 + 4368\delta_2 + 4032\delta_3 + 3744\delta_4) \\
c7 &= K1 * L^2(15120 + 5040\delta_1 + 3528\delta_2 + 3024\delta_3 + 2736\delta_4) \\
c8 &= K1 * L^2(20160 + 15120\delta_1 + 12768\delta_2 + 11088\delta_3 + 9792\delta_4) \\
c9 &= K1 * L^2(15120 + 10080\delta_1 + 8568\delta_2 + 7560\delta_3 + 6768\delta_4) \\
c10 &= K1 * L^2(15120 + 7560\delta_1 + 6048\delta_2 + 5292\delta_3 + 4752\delta_4)
\end{aligned}
$$

The symmetric stiffness matrix $\underline{K}_{BB}$ can now be written as

$$
\underline{K}_{BB} = \begin{bmatrix} \underline{K}_{bb} & \underline{K}_{bs} \\ \underline{K}_{sb} & \underline{K}_{ss} \end{bmatrix}
$$

with the bending part

$$
\underline{K}_{bb} = \begin{bmatrix}
c1 & 0 & 0 & -c2 & -c1 & 0 & 0 & -c3 \\
0 & c1 & c2 & 0 & 0 & -c1 & c3 & 0 \\
0 & c2 & c5 & 0 & 0 & -c2 & c6 & 0 \\
-c2 & 0 & 0 & c5 & c2 & 0 & 0 & c6 \\
-c1 & 0 & 0 & c2 & c1 & 0 & 0 & c3 \\
0 & -c1 & -c2 & 0 & 0 & c1 & -c3 & 0 \\
0 & c3 & c6 & 0 & 0 & -c3 & c8 & 0 \\
-c3 & 0 & 0 & c6 & c3 & 0 & 0 & c8
\end{bmatrix}
$$

If we also define

$$
\begin{aligned}
K2 &= \frac{kLGA_l}{5!} \\
cs1 &= K2(40 + 10\alpha_1 + 4\alpha_2) \\
cs2 &= K2(20 + 10\alpha_1 + 6\alpha_2) \\
cs3 &= K2(40 + 30\alpha_1 + 24\alpha_2)
\end{aligned}
$$

we get for the shear part $\underline{K}_{ss}$ respectively the coupling parts $\underline{K}_{bs}$ and $\underline{K}_{sb}$ worden:

$$\underline{K}_{ss} = \begin{bmatrix} c10+cs1 & 0 & c10+cs2 & 0 \\ 0 & c10+cs1 & 0 & c10+cs2 \\ c10+cs2 & 0 & c10+cs3 & 0 \\ 0 & c10+cs2 & 0 & c10+cs3 \end{bmatrix}$$

$$\underline{K}_{sb} = \begin{bmatrix} c4 & 0 & 0 & c7 & -c4 & 0 & 0 & c9 \\ 0 & c4 & -c7 & 0 & 0 & -c4 & -c9 & 0 \\ c4 & 0 & 0 & c7 & -c4 & 0 & 0 & c9 \\ 0 & c4 & -c7 & 0 & 0 & -c4 & -c9 & 0 \end{bmatrix} \qquad \underline{K}_{sb} = \underline{K}_{bs}^T$$

For the mass matrix we define

$$\begin{aligned}
M1 &= \frac{\rho A_l L}{9!} \\
c1 &= M1(134784 + 31104\alpha_1 + 10944\alpha_2) \\
c2 &= M1 * L(19008 + 6048\alpha_1 + 2448\alpha_2) \\
c3 &= M1(46656 + 23328\alpha_1 + 13248\alpha_2) \\
c4 &= M1 * L(11232 + 5184\alpha_1 + 2736\alpha_2) \\
c5 &= M1 * L^2(3456 + 1296\alpha_1 + 576\alpha_2) \\
c6 &= -M1 * L(11232 + 6048\alpha_1 + 3600\alpha_2) \\
c7 &= -M1 * L^2(2592 + 1296\alpha_1 + 720\alpha_2) \\
c8 &= M1(134784 + 103680\alpha_1 + 83520\alpha_2) \\
c9 &= -M1 * L(19008 + 12960\alpha_1 + 9360\alpha_2) \\
c10 &= M1 * L^2(3456 + 2160\alpha_1 + 1440\alpha_2)
\end{aligned}$$

For the symmetric, translational part $\underline{M}_T$ of the mass matrix we then can write

$$\underline{M}_T = \begin{bmatrix} \underline{M}_{T_{bb}} & \underline{M}_{T_{bs}} \\ \underline{M}_{T_{sb}} & \underline{M}_{T_{ss}} \end{bmatrix}$$

with the bending contribution

$$\underline{M}_{T_{bb}} \begin{bmatrix} c1 & 0 & 0 & -c2 & c3 & 0 & 0 & c4 \\ 0 & c1 & c2 & 0 & 0 & c3 & -c4 & 0 \\ 0 & c2 & c5 & 0 & 0 & -c6 & c7 & 0 \\ -c2 & 0 & 0 & c5 & c6 & 0 & 0 & c7 \\ c3 & 0 & 0 & c6 & c8 & 0 & 0 & -c9 \\ 0 & c3 & -c6 & 0 & 0 & c8 & c9 & 0 \\ 0 & -c4 & c7 & 0 & 0 & c9 & c10 & 0 \\ c4 & 0 & 0 & c7 & -c9 & 0 & 0 & c10 \end{bmatrix}$$

coupling contributions

$$\underline{M}_{T_{sb}} = \begin{bmatrix} -c2 & 0 & 0 & c5 & c6 & 0 & 0 & c7 \\ 0 & -c2 & -c5 & 0 & 0 & c6 & -c7 & 0 \\ c4 & 0 & 0 & c7 & -c9 & 0 & 0 & c10 \\ 0 & c4 & -c7 & 0 & 0 & -c9 & -c10 & 0 \end{bmatrix} \qquad \underline{M}_{T_{bs}} = \underline{M}_{T_{sb}}^T$$

and shear contribution

$$\underline{M}_{T_{ss}} = \begin{bmatrix} c5 & 0 & c7 & 0 \\ 0 & c5 & 0 & c7 \\ c7 & 0 & c10 & 0 \\ 0 & c7 & 0 & c10 \end{bmatrix}$$

For the rotational contribution $\underline{M}_R$ we define

$$\begin{aligned}
M2 &= \frac{\rho I_l}{9!L} \\
c1 &= M2(435456 + 217728\delta_1 + 124416\delta_2 + 77760\delta_3 + 51840\delta_4) \\
c2 &= M2 * L(36288 + 36288\delta_1 + 25920\delta_2 + 18144\delta_3 + 12960\delta_4) \\
c3 &= M2 * L(36288 - 10368\delta_2 - 12960\delta_3 - 12960\delta_4) \\
c4 &= -M2 * L(217728 + 108864\delta_1 + 62208\delta_2 + 38880\delta_3 + 25920\delta_4) \\
c5 &= M2 * L^2(48384 + 12096\delta_1 + 6912\delta_2 + 4752\delta_3 + 3456\delta_4) \\
c6 &= -M2 * L^2(12096 + 6048\delta_1 + 5184\delta_2 + 4752\delta_3 + 4320\delta_4) \\
c7 &= M2 * L^2(18144 + 18144\delta_1 + 12960\delta_2 + 9072\delta_3 + 6480\delta_4) \\
c8 &= M2 * L^2(48348 + 36288\delta_1 + 31104\delta_2 + 28080\delta_3 + 25920\delta_4) \\
c9 &= M2 * L^2(18144 - 5184\delta_2 - 6480\delta_3 - 6480\delta_4) \\
c10 &= M2 * L^2(108864 + 54432\delta_1 + 31104\delta_2 + 19440\delta_3 + 12960\delta_4)
\end{aligned}$$

The symmetric, rotational part $\underline{M}_R$ of the mass matrix reads

$$\underline{M}_R = \begin{bmatrix} \underline{M}_{R_{bb}} & \underline{M}_{R_{bs}} \\ \underline{M}_{R_{sb}} & \underline{M}_{R_{ss}} \end{bmatrix}$$

with the bending part

$$\underline{M}_{R_{bb}} = \begin{bmatrix} c1 & 0 & 0 & -c2 & -c1 & 0 & 0 & -c3 \\ 0 & c1 & c2 & 0 & 0 & -c1 & c3 & 0 \\ 0 & c2 & c5 & 0 & 0 & -c2 & c6 & 0 \\ -c2 & 0 & 0 & c5 & c2 & 0 & 0 & c6 \\ -c1 & 0 & 0 & c2 & c1 & 0 & 0 & c3 \\ 0 & -c1 & -c2 & 0 & 0 & c1 & -c3 & 0 \\ 0 & c3 & c6 & 0 & 0 & -c3 & c8 & 0 \\ -c3 & 0 & 0 & c6 & c3 & 0 & 0 & c8 \end{bmatrix}$$

the coupling parts

$$\underline{M}_{R_{sb}} = \begin{bmatrix} c4 & 0 & 0 & c7 & -c4 & 0 & 0 & c9 \\ 0 & c4 & -c7 & 0 & 0 & -c4 & -c9 & 0 \\ c4 & 0 & 0 & c7 & -c4 & 0 & 0 & c9 \\ 0 & c4 & -c7 & 0 & 0 & -c4 & -c9 & 0 \end{bmatrix} \qquad \underline{M}_{R_{bs}} = \underline{M}_{R_{sb}}^T$$

and the shear part

$$\underline{M}_{R_{ss}} = \begin{bmatrix} c10 & 0 & c10 & 0 \\ 0 & c10 & 0 & c10 \\ c10 & 0 & c10 & 0 \\ 0 & c10 & 0 & c10 \end{bmatrix}$$

Finally we define for the gyroscopic matrix:

$$\begin{aligned}
M3 &= \frac{\rho I_l}{9!\ L} \\
c1 &= M3(870912 + 435456\delta_1 + 248832\delta_2 + 155520\delta_3 + 103680\delta_4) \\
c2 &= -M3 * L(72576 + 72576\delta_1 + 51840\delta_2 + 36288\delta_3 + 25920\delta_4) \\
c3 &= M3 * L(-72576 + 20736\delta_2 + 25920\delta_3 + 25920\delta_4) \\
c4 &= -M3 * L(435456 + 217728\delta_1 + 124416\delta_2 + 77760\delta_3 + 51840\delta_4) \\
c5 &= M3 * L^2(96768 + 24192\delta_1 + 13824\delta_2 + 9504\delta_3 + 6912\delta_4) \\
c6 &= -M3 * L^2(24192 + 12096\delta_1 + 10368\delta_2 + 9504\delta_3 + 8640\delta_4) \\
c7 &= -M3 * L^2(36288 + 36288\delta_1 + 25920\delta_2 + 18144\delta_3 + 12960\delta_4) \\
c8 &= M3 * L^2(96768 + 72576\delta_1 + 62208\delta_2 + 56160\delta_3 + 51840\delta_4) \\
c9 &= M3 * L^2(-36288 + 10368\delta_2 + 12960\delta_3 + 12960\delta_4) \\
c10 &= M3 * L^2(217728 + 108864\delta_1 + 62208\delta_2 + 38880\delta_3 + 25920\delta_4)
\end{aligned}$$

The **skew-symmetric** gyroscopic matrix $\underline{G}$ then is

$$\underline{G} = \begin{bmatrix} \underline{G}_{bb} & \underline{G}_{bs} \\ \underline{G}_{sb} & \underline{G}_{ss} \end{bmatrix}$$

with its bending part

$$\underline{G}_{bb} = \Omega \begin{bmatrix} 0 & -c1 & c2 & 0 & 0 & c1 & c3 & 0 \\ c1 & 0 & 0 & c2 & -c1 & 0 & 0 & c3 \\ -c2 & 0 & 0 & -c5 & c2 & 0 & 0 & -c6 \\ 0 & -c2 & c5 & 0 & 0 & c2 & c6 & 0 \\ 0 & c1 & -c2 & 0 & 0 & -c1 & -c3 & 0 \\ -c1 & 0 & 0 & -c2 & c1 & 0 & 0 & -c3 \\ -c3 & 0 & 0 & -c6 & c3 & 0 & 0 & -c8 \\ 0 & -c3 & c6 & 0 & 0 & c3 & c8 & 0 \end{bmatrix}$$

the coupling parts

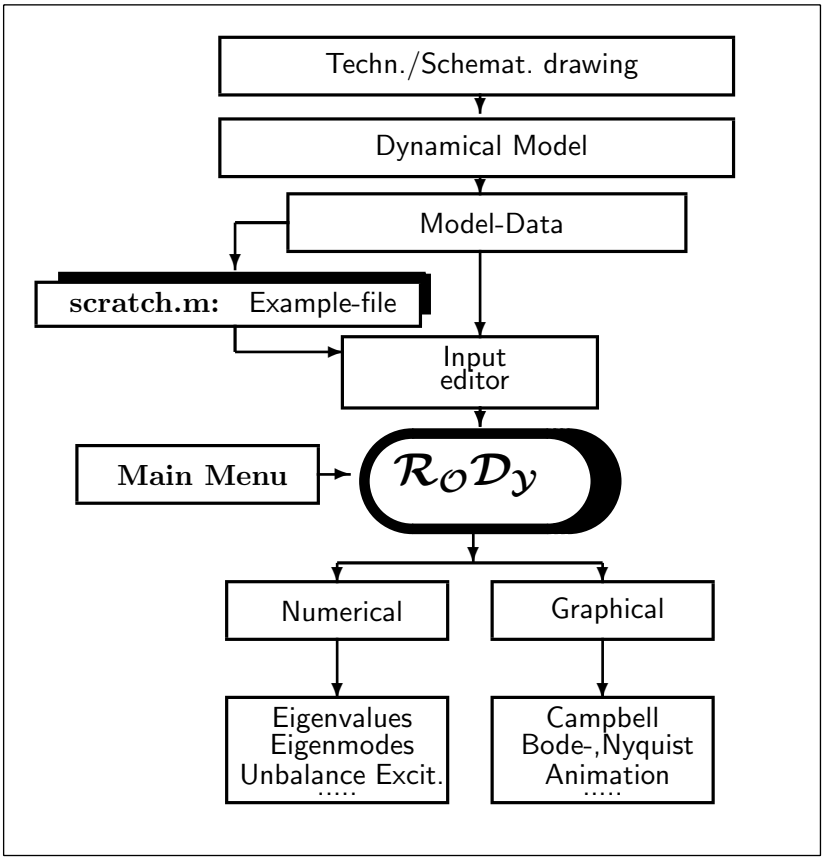
$$\underline{G}_{sb} = \Omega \begin{bmatrix} 0 & -c4 & -c7 & 0 & 0 & c4 & -c9 & 0 \\ c4 & 0 & 0 & -c7 & -c4 & 0 & 0 & -c9 \\ 0 & -c4 & -c7 & 0 & 0 & c4 & -c9 & 0 \\ c4 & 0 & 0 & -c7 & -c4 & 0 & 0 & -c9 \end{bmatrix} \qquad \underline{G}_{bs} = -\underline{G}_{sb}^T$$

and the shear part

$$\underline{G}_{ss} = \Omega \begin{bmatrix} 0 & -c10 & 0 & -c10 \\ c10 & 0 & c10 & 0 \\ 0 & -c10 & 0 & -c10 \\ c10 & 0 & c10 & 0 \end{bmatrix}$$

C

$M_{AT}L_{AB}$ toolbox R_OD_Y



C.1 Introduction

The toolbox $\boldsymbol{R_O D_Y}$ is a collection of so-called **.m** files, to be used within the $\boldsymbol{M_{AT}L_{AB}}$ environment. It can be used to analyse (complex) linear rotordynamic problems. The analysis model can be created easily and straightforward by selecting and combining some of the basic elements, available within the toolbox. These basic element are

- some shaft-elements
- a rigid-disk elements
- a variety of bearing-elements

Not only the standard functions of the toolbox can be used, but it is also possible to edit some of the procedures or add new procedures for special purposes.
From the Degrees of Freedom (DoF)selection and the geometrical data foor all the elements the programme determines the basic matrices $\underline{M}$, $\underline{D}$ and $\underline{K}$ for the set of linear, nonsymmetric equations of motion:

$$\underline{M}\,\underline{\ddot{q}} \;+\; \underline{D}\,\underline{\dot{q}} \;+\underline{K}\,\underline{q} \;=\; \underline{f}(t) \tag{C.1}$$

with the column of dof's: $\underline{q}^T = [q_1, q_2, ..., q_n]$. For the anlusis within $\boldsymbol{M_{AT}L_{AB}}$, this equation is transformed into the state-space form

$$\begin{bmatrix} \underline{\dot{q}} \\ \underline{\ddot{q}} \end{bmatrix} = \begin{bmatrix} \underline{0} & ; & \underline{I} \\ -\underline{M}^{-1}\underline{K} & ; & -\underline{M}^{-1}\underline{D} \end{bmatrix} \begin{bmatrix} \underline{q} \\ \underline{\dot{q}} \end{bmatrix} + \begin{bmatrix} \underline{0} \\ \underline{M}^{-1} \end{bmatrix} \underline{f}(t) \tag{C.2}$$

which , using $\underline{x}^T = [\underline{q}^T, \underline{\dot{q}}^T]$, can be written as:

$$\underline{\dot{\boldsymbol{x}}} \;=\; \underline{\boldsymbol{A}}\,\underline{\boldsymbol{x}} \;+\; \underline{\boldsymbol{B}}\,\underline{\boldsymbol{u}} \tag{C.3}$$

The state-space matrices $\underline{A}$ and $\underline{B}$ now have dimension $(2n * 2n)$. They perfectly fit into the general framework of $\boldsymbol{M_{AT}L_{AB}}$.

With the toolbox $\boldsymbol{R_O D_Y}$, the following basic tasks can be carried out:

- defining and assembling the system
- checking the input
- making plots of the configuration as having in mind
- calculate (complex) eigenfrequencies and corresponding modes for large ranges of rotorfrequencies Ω

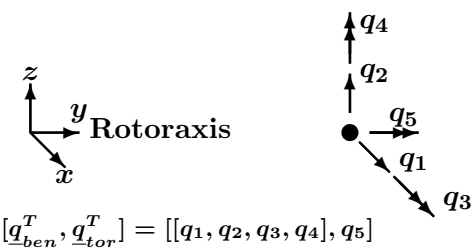
- visualizing modes in 2D and 3D (in motion)
- data-representation in **Campbell plot** with **Critical Speeds**
- evaluation of the stability of the rotorsystem
- calculating Frequency Respons Functions (FRF) in case of harmonic excitation
- calculating mass-unbalance responses
- creating **Bode**- and **Nyquist** plots

Originally the system was designed for the analysis of bending-vibrations. Later, also the possibility of analyzing torsional vibrations was implemented. However, the two "modes" are completely independent. So after making the choice for bending or torsion, two different subsets of degrees of freedom are used.
For the creation of a numerical model of a rotorsystem, two basic steps have to be taken:

- recognize and select the basic elements of the system (shafts, bearings, disks, ..)
- define the global, structural nodes of the system

The global, structural nodes of the system are serving two goals. First they determine the possible deformations of the system.Secondly, they act as steppingstone for hanging the individual elements into the global frame. Each structural node (see next figure) has 5 degrees of freedom (dof):

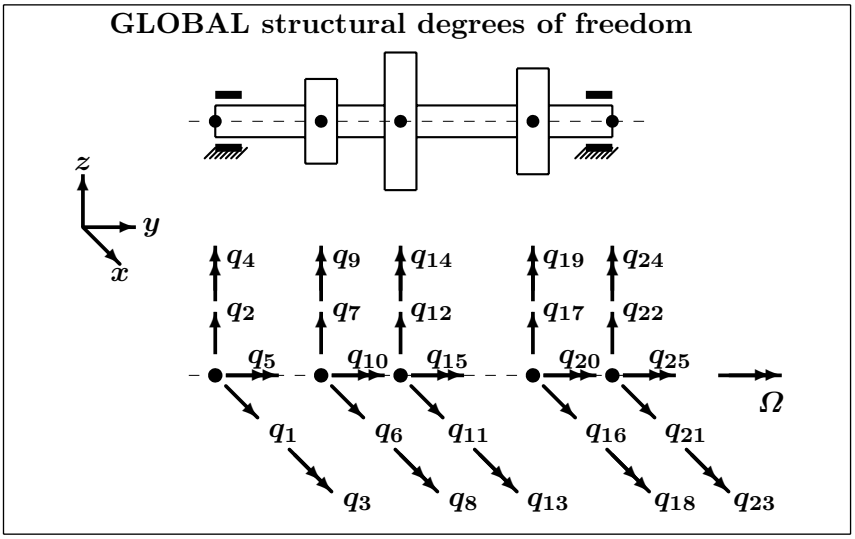
- 4 for bending problems (2 translations, 2 rotations)
- 1 for torsional problems (rotation)



$\underline{q}^T = [\underline{q}_{ben}^T, \underline{q}_{tor}^T] = [[q_1, q_2, q_3, q_4], q_5]$
Coördinate frame and nodal degrees of freedom

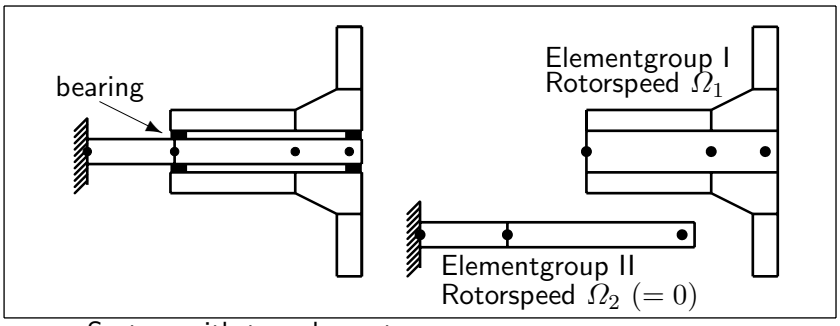
The coördinate frame has been chosen such that the x- and z-axis are in the plane, perpendicular to the rotoraxis, so that the z-axis is in line with

this axis.
In the next figure a simple rotorsystem is given to illustrate some of the features of the programme. It is a rotor which has been modeled by identifying 3 rigid disks, 4 shaft element and 2 bearings. This means that we have to define 5 structural nodes for which the degrees of freedom are automatically, consecutively numbered (from 1 to 25, the total number of dof's).



It is very easy now to build the whole rotorsystems from basic elements. If we for example look at the first (left) disk. For the uploading of this disk in the overall Finite Element Model we only have to give the geometrical parameters like mass and mass-moments of inertia and only the lowest degree of freedom of the relevant node, in this case dof 6. The same simplicity holds for the other elements. If we look at the first (left) beam, we should give things like diameter, length, E-modulus (see further), and the lowest dofs of the relevant nodes, so ik this case the dofs 1 and 6.
In the programme 2 **Element groups** can be distinguished, both with their own rotorfrequency. This option can be helpful for example in the case one is dealing with a rotor, supported by bearings in a flexible housing (which in that case also should be modeled as some simple beam structure. Another example from our experience was a high-speed rotor supported in two spiral-groove bearings which were carried by a clamped beam which

certainly might not be modeled as rigid, compared to the radial stiffness of the bearings. A simple model is shown in the next picture.



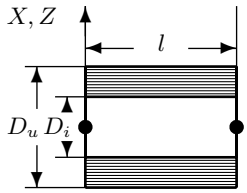
System with two element-groups

So, in general also the elementgroup of a specific element should be given (if relevant).

C.2 Overview of the Basic Elements

As mentioned earlier, the basic elements are beams, disks and bearings. They are available in several specific versions or even can be added to the programm. Here we only present the latest ones (during the expansion of the programme also some specific element types have been and are still present on the background.

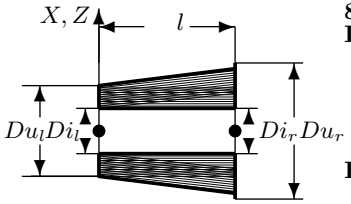
- SHAFT8: Cylindrical (**Euler**) beam (no shear)



8 (bending) degrees of freedom
Input
- Length l [m]
- outer / inner diam. Du and Di [m]
- Modulus of Elasticity E $[N/m^2]$
- Density ρ $[kg/m^3]$

Element topology by giving
- lowest (global) dof of left node
- lowest (global) dof of right node
- element group

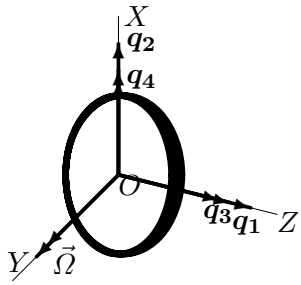
- CONE8: Conical (**Timoshenko**) beam (with shear



8 (bending) degrees of freedom
Input
- Length l [m]
- Left outer/inner diam. Du_l, Di_l [m]
- Right outer/inner diam. Du_r, Di_r [m]
- Mod.of Elast. E $[N/m^2]$, Density ρ $[kg/m^3]$
- Shear Modulus G $[N/m^2]$, Shear factor [-]

Element topology by giving
- lowest (global) dof of left node
- lowest (global) dof of right node
- element group

- DISK4: Rigid disk element

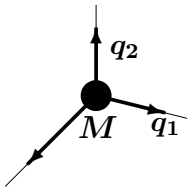


4 (bending) degrees of freedom
Input
- Mass M $[kg]$
- Axial Mass Moment of Inertia J_{ax} $[kgm^2]$
- Transv. Mass Moment of Inertia J_{tr} $[kgm^2]$

Element topology by giving
- lowest (global) dof
- element group

$$\underline{q}^T = [q_1, q_2, q_3, q_4]$$

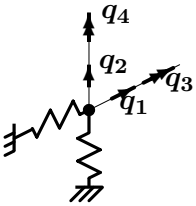
- MASS: Unilateral point mass element



1 degree of freedom
Input
- Mass M $[kg]$

Element topology by giving
- relevant (global) translational dof
- element group

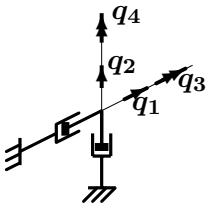
- Different SPRING elements



Elenet types:
SPRING1: Single spring to the ground
SPRING2: Single spring between 2 dofs (same type)
e.g. Elastic Belt
SPRING2M: Coupled spring-pair from 2 dofs to ground
e.g. Ball- or Fluid Film Bearing
SPRING4: Coupled spring between two pairs of dofs
e.g. Bearing between rotor and flexible rotorhousing
ECK2M: Same as SPRING2M but $K_{element} = \Omega * K$
ECK4: Same as SPRING4 but $K_{element} = \Omega * K$
FREQSPR2M: Same as SPRING2M but arbitrary depending on Ω

Type	Properties	Topology
SPRING1	(skalar) stiffness k $[N/m]$ or $[Nm/rad]$	relevant dof
SPRING2	(skalar) stiffness k $[N/m]$ or $[Nm/rad]$ $K_{element} = \begin{bmatrix} k & -k \\ -k & k \end{bmatrix}$	2 relevant dofs
SPRING2M	(stiffnessmatrixelements $\begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix}$	2 relevant dofs
SPRING4	(stiffnessmatrixelements $K = \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix}$ $K_{element} = \begin{bmatrix} K & -K \\ -K & K \end{bmatrix}$	lowest dof node 1 lowest dof node 2

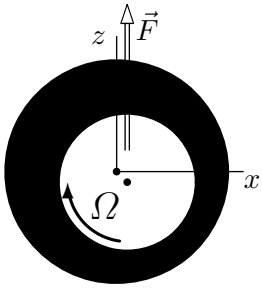
- Different DAMPER elements (viscous damper)



Elenet types:
DAMP1: Single damper to the ground
DAMP2: Single damper between 2 dofs (of same type)
e.g. From elastic Belt
DAMP2M: Coupled damper-pair from 2 dofs to ground
e.g. Ball- or Fluid Film Bearing
DAMP4: Coupled damper between two pairs of dofs
e.g. Bearing between rotor and flexible rotorhousing
FREQDMP2M: Like DAMP2M, arbitrary Ω-depending

Type	Element Properties	Element Topology
DAMP1	(skalar) damping d $[Ns/m]$ or $[Nms/rad]$	relevant dof
DAMP2	(skalar) damping d $[Ns/m]$ or $[Nms/rad]$ $D_{element} = \begin{bmatrix} d & -d \\ -d & d \end{bmatrix}$	2 relevant dofs
DAMP2M	(dampingmatrixelements $\begin{bmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{bmatrix}$	2 relevant dofs
DAMP4	(dampingmatrixelements $D = \begin{bmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{bmatrix}$ $D_{element} = \begin{bmatrix} D & -D \\ -D & D \end{bmatrix}$	lowest dof node 1 lowest dof node 2

- VOLCYL2M: FLUID FILM CYLINDRICAL BEARING



2 translational dofs
Input
- Length, Width and radial Clearance $[m]$
- Viscosity $[Ns/m^2]$
- Static Load $[N]$

Element topology by giving
- the 2 relevant (global) translational dofs
- element group

Important notice

The coördinate frame which has been used in the theory of chapter2 is different from that in the R_OD_Y toolbox.

The effect is that a VOLCYL2-stiffness matrix from R_OD_Y ($\underline{K}_R$) (or Damping matrix) is related to the stiffness matrix $\underline{K}_2$ from Ch. 2 as:

$$\underline{K}_R = \begin{bmatrix} k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix} \Longrightarrow \underline{K}_2 = \begin{bmatrix} k_{22} & -k_{21} \\ -k_{12} & k_{11} \end{bmatrix}$$

C.3 Key Variables

SUMMARY OF ESSENTIAL PROGRAM VARIABLES

Variable	Meaning

user(1) Total number of degrees of freedom
user(2) Number of (**CONE8**) conical shaft elements with
- 8 degrees of freedom for bending problems
- 2 degrees of freedom for torsion problems
user(3) Number of (**SHAFT8**) cylindrical shaft elements with
- 8 degrees of freedom for bending problems
- 2 degrees of freedom for torsion problems
user(4) Not used
user(5) Number of (**DISK4**) rigid disk elements with
- 4 degrees of freedom for bending problems
- 1 degrees of freedom for torsion problems
user(6) Not used
user(7) Number of (**MASS**) mass elements related to a single dof
user(8) Not used
user(9) Not used
user(10) Number of (**SPRING1**) springs with 1 degree of freedom
scalar input: springstiffness k
user(11) number of (SPRING2) springs with 2 degrees of freedom with
scalar input: springstiffness k
user(12) Number of (**SPRING2M**) springs with 2 degrees of freedom and
input all matrix terms k11,k12,k21,k22
$K_{element} = [k11, k12; k21, k22]$
user(13) Number of (**SPRING4**) springs with 4 degrees of freedom with
input all matrix terms k11,k12,k21,k22
$K = [k11, k12; k21, k22]; \quad K_{element} = [K, -K; -K, K];$
user(14) Not used
user(15) Number of (**DAMP1**) dampers with 1 degree of freedom with
input damping factor d
user(16) Number of (**DAMP2**) dampers with 2 degrees of freedom with
input damping factor d
user(17) Number of (**DAMP2M**) dampers with 2 degrees of freedom with
input all matrix terms d11,d12,d21,d22
$D_{element} = [d11, d12; d21, d22]$
user(18) Number of (**DAMP4**) dampers with 4 degrees of freedom with as
input all matrix terms d11,d12,d21,d22
$D = [d11, d12; d21, d22]; \quad D_{element} = [D, -D; -D, D]$
user(19) Not used
user(20) Number of (**ECK2M**) frequency dependent springs with 2 degrees of freedom
input for each element a row with the matrix terms k11,k21,k12, k22
with $K = [k11, k12; k21, k22]$ and $K_{element} = \Omega * K$
user(21) Number of (**ECK4**) frequency dependent springs with 4 degrees of freedom
input the terms k11,k21,k12,k22
with $K = [k11, k12; k21, k22]; \quad K2 = [K, -K; -K, K]$
and $K_{element} = \Omega * K2$
user(22) Number of (**FREQSPR2M**) (veer)elements with 2 degrees of freedom
where the springstiffnesses k11,k12,k21,k22 depend on

the rotorfrequency.

user(23) Number of **(FREQDMP2M)** (demper)elements with 2 degrees of freedom where the damping b11,b12,b21,b22 depend on the rotorfrequency.

user(24) Number of **(VOLCYL2M) cylindrical fluid film bearings** with input: diameter, width, radial clearance, viscosity AND statical load

user(25) number of **(COMP4)** elements for Compressible Bearings See: Geerts, WFW 95.090

omegagroup(1)
Angular velocity ratio of first elementsgroup

omegagroup(2)
Angular velocity ratio of second elementsgroup

References

[Bassani-92] Bassani, R., Piccigallo, B.; *Hydrostatic Lubrication*, Elsevier Science Publ., ISBN 0 444 88498, 1992

[Bartholomeus-93] Bartholomeus, R.P.M., *Ontwikkelen van lagermodellen in het pakket RODY-III*, WFW report 93.033, Vakgroep Fundamentele Werktuigkunde, TU Eindhoven, 1993

[Bot-93] Bot, L. de, *Toepassingen RODY*, WFW report 93.131, Vakgroep Fundamentele Werktuigkunde, TU Eindhoven, 1993

[Childs-77] Childs, D., Moes, H., Leeuwen, H. v.; *Journal Bearing Impedance Description for Rotor Dynamic Applications*, Journal of Lubrication Technology, Vol. 99, No. 2, 1977

[Childs-93] Childs, D., *Turbomachinery Rotordynamics, Phenomena, Modelling and Analysis*, Wiley and Sons, ISBN 0 471 53840-X, 1993

[Crooijmans-87] Crooijmans M.T.M.C.; *On the Computation of Stationary Deterministic Behaviour of Non-linear Dynamic Systems with Application to Rotor-bearing Structures*. Ph.D. thesis, Eindhoven University of Technology, The Netherlands, 1987.

[Crooijmans-90] Crooijmans M.T.M.C., Brouwers, H.J.H, Campen, D.H. van, Kraker, A. de; *Limit-cycle predictions of a non-linear rotor-bearing system*. Jnl. of Eng. for Industry, Vol. 112, No. 2, 1990

[Dimarogonas-92] Dimarogonas, A.D., *Vibration for Engineers*, Prentice Hall, ISBN 0 13 950841-4, 1992

[Doyle et al.-92] Doyle,J.C., Francis, B.A., Tannenbaum, A.R.; *Feedback Control Theory*, Macmillan, N.Y., 1992

[Geerts-95a] Geerts, N.; *Linear Dynamic Analysis of Rotorsystems with Gas Bearings*, WFW report 95.090, Vakgroep Fundamentele Werktuigkunde, TU Eindhoven, 1995

[Geerts-95b] Geerts, N.; *Gebruikershandleiding RODY4, Rotor Dynamica Toolbox voor MATLAB*, WFW report 95.089,

Vakgroep Fundamentele Werktuigkunde, TU Eindhoven, 1995

[Hamilton-87] Hamilton, L.; *Optimalisatie m.b.t. rotordynamica en lagerflensontwerp bij een snel draaiende rotor*, WFW report 87.071, Vakgroep Fundamentele Werktuigkunde, TU Eindhoven, 1987

[Harris-91] Harris,T.; *Rolling Bearing Analysis*, (3rd ed.), Wiley, 1991

[Heeren-86] Heeren, T., Kraker, A. de, Crooijmans, M., Campen, D.H. van; *Development of a geometrically non-linear Timoshenko-beam model for application in rotor dynamics*, Proc. of the Int. Conf. on Rotordynamics, Tokyo, 1986

[Jacobs-89] Jacobs, J.A.H.M., *RODY: A Program for Calculations on Rotordynamics*, NAT-LAB Technical note nr. 101/89, Philips Research Laboratories, Eindhoven, 1989

[Kok-90] Kok, J.J.;*Werktuigkundige Regeltechniek II*, Collegedictaat 4.4594, Faculteit Werktuigbouwkunde, Technische Universiteit Eindhoven 1990

[Kraker-88a] Kraker, A. de, Crooijmans, M.T.M., Campen, D.H. van; *The Dynamics of a Rotor with Rubbing*, Proc. of the 4th. Conf. on Vibrations in Rotating Machinery, paper C 366/078, Edinburgh, UK, 1988.

[Kraker-88b] Kraker, A. de, Campen, D.H. van; *Control of rotor-bearing systems by electro-magnetic bearings*, Proc. of the IMECHE-Post-Conf., Kaiserslautern, 1988

[Kraker-99] Kraker, A. de, Knaapen, R.J.W., Kodde, L. Campen, D.H. van: *Experimental determination of rolling element bearing stiffness*; In: Proc. 1998 Int. Conf. on Noise and Vibration Engineering (ISMA 23), Leuven, Belgium, 16-18 September 1998, Editor P. Sas, pp.433-440.

[Kraker/Campen-01] Kraker, A. de, Campen, D.H. van; *Mechanical Vibrations*, Shaker Publishing B.V., Maastricht, The Netherlands, ISBN 90-423-0165-1,2001.

[Krämer-83] Krämer, E *Maschinendynamik*, Springer-Verlag, ISBN 0 387 12541-8, 1983

[Lalanne/Ferraris-90] Lalanne,M., Ferraris,G., *Rotordynamics Prediction in Engineering*, Wiley and Sons, ISBN 0 471 92633 7, 1990

[Leine-00] Leine,R.I.; *Bifurcations in Discontiuous Mechanical Systems of Filippov Type*. Ph.D. thesis, Eindhoven University of Technology, The Netherlands, 2000.

[Lund/Tonnesen-72] Lund,J., Tonnesen, J.; *Analysis and Experiments on Multi-Plane Balancing of a Flexible Rotor*, Journal of Engineering for Industry, Ser. B, 86(3), pp.273-279 (1964)

[Lund/Saibel-67] Lund,J.W., Saibel,E.; *Oil Whirl Orbits of a Rotor in Sleeve Bearings*, Jnl. of Engineering for Industry, pp813-823, 1967

[$M_{AT}L_{AB}$] *MATLAB User's guide*, The MathWorks, Inc., South Natick, MA 01760

[Muyderman-86] Muyderman, E.A.; *Algebraic formulas for the threshold and mode of instability and the first critical speed of a simple flexibly supported rotor-bearing system*, International Conference on Rotordynamics, Sept. 14-17, 1986, Tokyo

[rody2000] $\boldsymbol{M_{AT}L_{AB}}$ -toolbox $\boldsymbol{R_O D_Y}$; TO BE COMPOSED, Eindhoven University of Technology, Department of Mechanical Engineering, Section of Engineering Dynamics, version 2000, spring 2000.

[Roosmalen-94] v.Roosmalen,A.N.J.; *Design tools for low noise gear transmissions*, Ph.D. thesis, Eindhoven University of Technology, Eindhoven, 1994

[Rops-88] Rops, T.; *Electro-Magnetische Lagers in de Rotordynamica*, WFW report 88.034, Vakgroep Fundamentele Werktuigkunde, TU Eindhoven, 1988

[Schie-91] Schie C. van, *RODY, Rotordynamica binnen MATLAB*, WFW report 91.052, Vakgroep Fundamentele Werktuigkunde, TU Eindhoven, 1991

[Schweitzer at al.-94] Schweitzer,G., Bleuler, H., Traxler, A.; *Active Magnetic Bearings*; vdf Hochschulverlag AG an der ETH Zurich, ISBN 3 7281 2132 0

[Someya-89] Someya Tsuneo; *Journal Bearing Databook*, Springer-Verlag, 1989, ISBN 3-540-17074-X

[Szeri-80] Szeri, A.Z.; *Tribology, Friction, Lubrication and Wear*, 1980, Mc-Graw-Hill Book Company, ISBN 0-07-0626634.

[Thoolen-93] Thoolen, F.J.M.T.; *Develoment of an Advanced High Speed Flywheel Energy Storage System*. Ph.D. thesis, Eindhoven University of Technology, The Netherlands, 1993.

[Vance-88] Vance, J.M., *Rotordynamics of turbomachinery*, John Wiley & Sons, ISBN 0 471 80258-1, 1988

[Vorst et al.-95] Vorst, E.L.B. v.d., Fey, R.H.B., Kraker, A. de, Campen, D.H. v.; *Steady-State Behaviour of Nonlinear Flexible Rotor-Bearing Systems*, Part I- Theory; Part II- Application: Influence of Cavitation Modelling; Machine Vibration, Vol. 3, No. 1, 1995, pp. 75-92

[Vorst-96] Vorst, E.L.B. v.d.; *Long term dynamics and stabilization of nonlinear mechanical systems*. Ph.D. thesis, Eindhoven University of Technology, The Netherlands, 1996.

[Vrande et al.-99] Vrande, B.L. van de; Kraker, A. de: *Influence of bearing compliance on long-term nonlinear rotor dynamics*; Proc. 1999 ASME Design Engineering Technical Conferences: Symposium on Nonlinear Dynamics in Engineering Systems, Las Vegas, Nevada (USA), 12-16 Sept. 1999, Ed. by D.H. van Campen, A. de Kraker and M. Wiercigroch, CD-Rom, paper DETC99/VIB-8043, 6 pp.

[Wang-93] Wang, Jumming; *Design of Gas Bearing Systems for Precision Applications*, Ph.d Thesis, 1993, ISBN 90-386-0112-3

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